Affirmed Networks Charging Data Records Specification

Release 9.4.0.0
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Affirmed Networks Inc.
35 Nagog Park
Acton, MA 01720
USA
Tel: +1-978-268-0800
www.affirmednetworks.com

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Preface

This specification contains information on the Charging Data Records (CDRs) used for offline billing and accounting and HTTP transaction data records (HTDRs) used to provide information about each HTTP transaction that traverses the Mobile Content Cloud (MCC) Muti Protocol Proxy. It also contains information on the offline charging interfaces (Ga and Gz/Rf) used to transport charging information from a gateway (SGW, PGW, GGSN) to an Offline Charging System (OFCS).

For more information on the CLI commands associated with CDRs, see the *Affirmed Networks CLI Reference Guide*. For more information on the Multi Protocol Proxy service, see the *Affirmed Networks Value Added Services Guide*.

Note: *The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide) responsible for charging data. The reader should have a basic knowledge of offline charging, GPRS Tunneling Protocol prime (GTP'), Diameter protocol and HTTP.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, Charging Data Records	Local CDRs and the configuration steps required for storing CDRs in a local file.
Chapter 2, Local CDRs	The SGW, PGW, GGSN, WAG, and ePDG local CDR fields and file formats.
Chapter 3, CDR Descriptions	Each CDR field in alphabetical order.
Chapter 4, HTTP Transaction Data Records	Local HTDRs and the configuration steps required for storing HTDRs in a local file.
Chapter 5, Ga Interface	The message flows, configuration steps and message definitions used to control and monitor the Ga interface.
Chapter 6, Gz/Rf Interface	The message flows, configuration steps, failure handling, and message definitions used to control and monitor the Gz/Rf interface.
Appendix A, Diameter AVP Descriptions	The attribute value pair definitions used in Diameter messages for the Gz/Rf interface.
Appendix B, ASN.1 Definition	A complete ASN.1 definition of the GPRS records based on the ASN.1 definition in 3GPP TS 32.298.

References and Specifications

This document references the following specifications:

- RFC 3539: *Authentication, Authorization and Accounting (AAA) Transport Profile*
- RFC 3588: *Diameter Based Protocol*
- RFC 4006: *Diameter Credit-Control Application*
- RFC 6733: *Diameter Base Protocol*
- 3GPP TS 29.002: *Digital Cellular Telecommunication System; Universal Mobile Telecommunications System (UMTS); Mobile Application Part (MAP)*
- 3GPP TS 29.060 V9.3.0 (2010-06): *General Packet Radio Service (GPRS); GPRS Tunnelling Protocol (GTP) across the Gn and Gp interface*
- 3GPP TS 29.061: *Interworking Between the Public Land Mobile Network (PLMN) Supporting Packet Based Services and Packet Data Networks (PDN)*
- 3GPP TS 29.212 V10.0.0 (2010-09): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 32.251 V9.9.0 (2012-06): *Telecommunication management; Charging management; Packet Switched (PS) domain charging*
- 3GPP TS 29.272 V11.0.0 (2011-09): *Mobility Management Entity (MME) and Serving GPRS Support Node (SGSN) related interfaces based on Diameter protocol*
- 3GPP TS 29.274 V9.3.0 (2010-06): *Evolved GPRS Tunnelling Protocol for Control Plane (GTPv2-C); Stage 3*
- 3GPP TS 32.295 V9.0.0 (2009-12): *Telecommunication management; Charging management; Charging Data Record (CDR) transfer*
- 3GPP TS 32.298 V10.0.0 (2010-03): *Telecommunication management; Charging management; Charging Data Record (CDR) encoding rules description*
- 3GPP TS 32.298 V13.7.0 (2017-03): *Telecommunication management; Charging management; Charging Data Record (CDR) encoding rules description*
- 3GPP TS 32.298 V15.2.0 (2018-03): *Telecommunication management; Charging management; Charging Data Record (CDR) encoding rules description*
- 3GPP TS 32.299 V10.7.0 (2012-09): *Telecommunication management; Charging management; Diameter charging application*

- 3GPP TS 32.299 V13.9.0 (2017-06): *Telecommunication management; Charging management; Diameter charging application*

Features in this Release

This release includes the following new features. Note the following:

- For a complete list of the features supported in this release, see the *Software Release Notes*.
- For a list of additional Affirmed Networks documentation, see the *Affirmed Networks Documentation Library*.
- For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see *Affirmed Networks Terms and Acronyms*.

<table>
<tr><th>Feature</th><th>Description</th></tr>
<tr><td>Added support for new TMO-Transparent Data attribute for the RADIUS Access-Accept message and for the Diameter AVP on the Gy interface, and added support for a new CDR field on the Ga interface

See the <i>Affirmed Networks Gateway Administration Guide</i> and Record Extensions (page 3-31).</td><td>This feature lets the PGW select MCC/MNC (PLMN ID) values in a configured priority order by using the GTP message from the SGSN/SGW, a sig-info-table, or the RADIUS Access-Accept message from the RADIUS authentication server.

This feature also lets you configure the selection of TMO-Transparent-Data values on the PGW when the RADIUS Access-Accept message does not return a transparent data value. The PGW selects a value from a sig-info-table with PLMN ID mappings or from a sig-info-table with SGSN/SGW subnet mappings. This is enabled at the APN level.

Added support for the following:

■ Transparent data AVP in CCR messages. This AVP is configurable in Gy CCR data record templates and by default is set to no, indicating the field is not sent.

<code>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ocs diameter [event | interim | start | stop] <<i>template-name</i>> tmotransparentdata presentFlag [yes|no]</code>

■ Transparent data attribute in RADIUS Access-Accept message from the RADIUS authentication server. This attribute is configurable in RADIUS authorization access-accept data record templates and by default is set to no, indicating the field is not processed.

<code>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw auth radius access-accept <<i>template-name</i>> tmo_transparent_data presentFlag [yes|no]</code>

■ Transparent data subfield in the RecordExtensions field of GTPP CDRs (sent via the Ga interface). This subfield, transparentVSA, is configurable in GTPP data record templates and by default is set to no, indicating the field is not present in CDRs.

<code>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw|sgw] ofcs gtpp [interim | stop] <<i>template-name</i>> extensioninfo presentFlag [yes|no]</code>

Three other subfields have also been added to support the transparentVSA subfield: extensionType, length, and serviceList.

To view the transparent data string for a subscriber on a PGW or SGW:

<code>an3000-mcm-slot7cpu1# show subscriber pdn-session <<i>id</i>></code></td></tr>
</table>

Feature	Description
Factor additional latency for distance for congestion detection See TCP Client Connection Information (page 4-31).	This feature adds a configuration option for Round-trip Time (RTT) offsets when RTT is used in the congestion detection's throughput formula. RTT offsets can be configured in data centers for both *nearby* and *far away* subscribers to avoid misidentifying *far away* subscribers as congested. This feature reports the length of time that the client connection remains in each congestion level in the HTDR/EDR and also reevaluates policies for flows where the congestion levels change. This feature adds the following new HTDR/EDR fields in tcpClientConnectionInfo: ■ CongestionLevelNoneTime ■ CongestionLevelMildTime ■ CongestionLevelModerateTime CongestionLevelSevereTime
Added a new configuration option under the gateway service to configure the Node ID value used in SGW/PGW/GGSN CDRs See Configuring the Node ID in CDRs (page 1-53).	This feature adds a new configuration option used to configure the Node ID value under the gateway service, which will be used as the Node ID value in local (ASN.1 and CSV), GTPP and Gz/Rf SGW/PGW/GGSN CDRs. If this parameter is not configured, the gateway service name is used as the Node ID field in SGW/PGW/GGSN CDRs. `an3000-mcm-slot7cpu1(config)#` **`zone default gateway [sgw|pgw|ggsn]`** ***`<gateway-service-name>`*** **`node-id`** ***`<node-id-string>`*** ***Note:*** *The* `node-id` *parameter under the gateway service should not be configured or changed mid-session. If it is, the node-id in subsequent CDRs will contain the newly configured value.* There is an existing configuration parameter under the offline-charging interface that allows you to specify whether the Node ID value in CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface offline-charging`** ***`<interface-name>`*** **`[sgw|pgw|ggsn] unique-cdr-nodeid [enabled|disabled]`** See Table 1-2 on page 1-53 for information on the MCC behavior when using the `node-id` parameter under the gateway service and the `unique-cdr-nodeid` parameter under the offline-charging interface together.

Feature	Description
Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50 See Storing CDRs in a Local File (page 1-19), Configuring the Gz/Rf Offline Charging Interface (page 6-7), and Sending Charging Records over the Ga Interface (page 5-15).	This feature adds two new event-triggers to the `event-trigger-template`, which is used to specify the events that trigger an interim update on offline/online charging interfaces and to trigger the closure of an interim SGW/PGW/GGSN CDR. *Applies to the Gy, Gz/Rf and Ga (GTPP), local (ASN.1) interfaces.* For all event-triggers in the `event-trigger-template`, the maximum service data container (SDC) limit per CDR closure event is increased from 20 to 50. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface-trigger-template event-trigger-template <template-name> [cgi-sai-change-close-trigger| ecgi-change-close-trigger]`** ■ **`cgi-sai-change-close-trigger`** – Specifies whether a change in the Cell Global Identification- Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close. ■ **`ecgi-change-close-trigger`** – Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close. This feature also increased the maximum number of SDCs that the gateway maintains per CDR at the bearer-level from 20 to 50 if interim-updates are enabled on the interface. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface offline-charging <interface-name> [pgw|sgw|ggsn|wag|epdg] num-service-containers [1..50]`** Added the following change in charging conditions to the Change-Condition AVP within the Service-Information –> PS-Information AVP sent on the Gz/Rf interface. **`14 CGI-SAI Change`** **`16 ECGI Change`** Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Service Condition field sent on the Ga interface. **`cGI-SAIChange(21)`** **`eCGIChange(29)`** Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Charging Condition field sent on the Ga interface. **`cGI-SAIChange(6)`** **`eCGIChange(10)`**

<table>
<tr><th>Feature</th><th>Description</th></tr>
<tr><td>Added support for the CGI-SAI change and ECGI change event-triggers and increased the maximum number of SDCs that the gateway maintains per CDR from 20 to 50

<i>Continued</i>

See Storing CDRs in a Local File (page 1-19), Configuring the Gz/Rf Offline Charging Interface (page 6-7), and Sending Charging Records over the Ga Interface (page 5-15).</td><td>Added two new configuration parameters under the offline-charging interface:

<code>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <<i>interface-name</i>> [pgw|sgw|ggsn|wag|epdg] num-charge-condition-change [0..50]</code>

■ <code>num-charge-condition-change</code> – Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0.

<i>Note:</i> <i>A trigger is only counted towards the <code>num-charge-condition-change</code> if it is enabled under the <code>event-trigger-template</code>.</i>

<code>an3000-mcm-slot7cpu1(config)# service-construct interface offline-charging <<i>interface-name</i>> [pgw|ggsn] qos-change-trigger [pcrf-initiated|access-nw-initiated|any]</code>

■ <code>qos-change-trigger</code> – Specifies when to activate the qos-change-trigger.
– <code>pcrf-initiated</code> – Activate the qos-change-trigger for a PCRF-initiated QoS change only.
– <code>access-nw-initiated</code> – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only.
– <code>any</code> – (default) Activate the qos-change-trigger for any type of QoS change.</td></tr>
</table>

Feature	Description
Added support for populating the UWAN User Location Information field in local, GTPP and Gz/Rf PGW CDRs when the UE Local IP Address and the UE UDP Port IE are received from the ePDG See Chapter 5, Ga Interface, Chapter 6, Gz/Rf Interface, and Chapter 2, Local CDRs.	When the UE Local IP Address and the UE UDP Port are received from the ePDG, the MCC populates the UWAN User Location Information field in local (ASN.1 and CSV), GTPP and Gz/Rf PGW CDRs. This CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default `presentFlag` setting is `no`, indicating the data field is not present in CDRs. **Top-level CDR field** `an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtpp [interim|stop] <template-name> [uwanuserlocationinfo|uwanuserlocationinfoudpsourceport|uwanuserlocationinfouelocalipaddress] presentFlag [yes|no|conditional]` **Within the Change of Charging Condition Field (List of Traffic Data Volumes) and the Change of Service Condition field (List of Service Data)** `an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs gtpp [interim|stop] <template-name> [changeofcharconditionuwanuserlocationinfouelocalipaddress |changeofcharconditionuwanuserlocationinfoudpsourceport|changeofserviceconditionuwanuserlocationinfouelocalipaddress |changeofserviceconditionuwanuserlocationinfoudpsourceport] presentFlag [yes|no|conditional]`  This CDR field is configurable in local CSV Start, Interim and Stop data record templates. The default `presentFlag` setting is `no`, indicating the data field is not present in CDRs.  `an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw charging-local [start|interim|stop|event] <template-name> uwanuserlocationinfo presentFlag [yes|no|conditional]`  The UWAN-User-Location-Information AVP is configurable in ACR data record templates. The default `presentFlag` setting is `no`, indicating the field is not sent.  `an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw ofcs diameter [start|interim|stop|event] <template-name> uwanuserlocationinfo presentFlag [yes|no|conditional]`

Feature	Description
Added support for the CAMEL Charging Information CDR field and added 3GPP 32.298 version 10.0 as an option for formatting local ASN.1 and GTPP CDRs See Storing CDRs in a Local File (page 1-19), Sending Charging Records over the Ga Interface (page 5-15), and CAMEL Charging Information (page 3-6).	This feature adds support for the CAMEL Charging Information field in PGW/GGSN local ASN.1 and GTPP CDRs. The CAMEL Charging Information CDR field is configurable in local ASN.1 and GTPP Interim and Stop data record templates. The default `presentFlag` setting is `no`, indicating the data field is not present in CDRs. ***Note:*** *The CAMEL Charging Information CDR field is only applicable for 3G sessions.* `an3000-mcm-slot7cpu1(config)#` **`service-construct data-record-template [pgw|ggsn] ofcs gtpp [interim|stop] <template-name> camelinfo presentFlag [yes|no|conditional]`** This feature also adds version 10 as an option when configuring the 3GPP 32.298 version used to format local ASN.1 and GTPP CDR fields in Interim and Stop data records. To format local ASN.1 CDRs: `an3000-mcm-slot7cpu1(config)#` **`service-construct interface offline-charging <interface-name> [sgw|pgw|ggsn] local asn1-cdr-version [10.0|9.0|8.4|6.0]`** ***Note:*** *This parameter is only visible when the `encode-format` is set to `asn1` or `both`.* To format GTPP CDRs: `an3000-mcm-slot7cpu1(config)#` **`service-construct interface aaa-interface-group gtpp node-admin <node-name> version [10.0|9.0|8.4]`** **Limitation** When the `version` parameter is set to `9.0` for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to `no` if it is not needed. When the `version` parameter is set to `10.0` for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to `yes` (default).

Feature	Description
Added consent-id to Secure HTTP Identify Protocol (SHIP) services See the *Affirmed Networks Value Added Services Guide* and Consent Information (page 4-36).	This feature enables closer coordination between Content Providers (CP) and end users who have given permission for that coordination. For example, if a CP requests to share subscriber information, this feature will verify if the user has consented to share their information with a given CP. If consent was granted, the data is shared; if not, access to subscriber information is rejected. This feature receives data from the PCRF. A new string named 'ConsentID' is added to the PCRF. That string is relayed to the Proxy for processing. If this feature is invoked and the content provider's configured consent-id matches a VSA string in the metadata for that user, then MCC processes the request. Otherwise, MCC returns an error code. This feature adds a new content provider and content status to data records and also adds the SERVER PLUGIN FAILURES NO CONTENT statistic: `an3000-mcm-slot7cpu1# show services http-server statistics cluster-level server-plugin` This feature adds the following new show subscriber pdn-session parameter: `an3000-mcm-slot7cpu1# show subscriber pdn-session 239616` `Session VSA ConsentID <Consent ID String>` This feature adds the following new show subscriber proxy-session parameter: `an3000-mcm-slot7cpu1# show subscriber proxy-session detail 239616` `Session VSA - ConsentID         <Consent ID String>` This feature enables all proxy metadata across MCC to store up to 1022 bytes of data. Configure the consent-id as follows: `an3000-mcm-slot7cpu1(config)# services http-server ship content-provider <CP-name>` `consent-id     Consent ID for the content provider - string` This feature filters SHIP requests as follows: ■ If the configured content provider consent-id is '`*`' (wildcard) (default), then the SHIP request is processed, regardless of the user strings supplied. The SHIP request could still fail due to other errors. ■ If the configured content provider consent-id is supplied and matches a VSA string provided for the user (not case sensitive), then the SHIP request is processed. The SHIP request could still fail due to other errors. ■ If the configured content provider consent-id is supplied but does not match a VSA string supplied for the user, then the SHIP request fails with the failure return code (1004). In the HTTP Transaction data records (HTDRs), added the Consent Information HTDR, which contains Secure HTTP Identify Protocol (SHIP) consent-specific information.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	**Times New Roman bold font**	Use **configure_ha** to configure... Enter **y**. To continue, press **Enter**.
Selecting a menu item	**Menu** => **Option**	Select **Security** => **Group Management**.
Variable parameters for which you supply values	***<courier bold italic blue font>***	**configure_dr convert slave --masterhost *<host IP>***
User input	**Courier bold blue font**	**cd /opt/Affirmed/NMS/bin**
Screen messages, system output, sample source code	`Courier regular blue font`	`Checking for License...`
Required keywords and arguments in square brackets	**[ ]**	**cluster [id] node [id] reboot**
Optional keywords and arguments in curly brackets	**{ }**	**{*cdr-file-name*}**
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	|	`Is this correct? [y|n]`
Keywords representing a single element in angle brackets	**< >**	***<host name>***
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and pter titles, file path names, new terms, and emphasized text	*Times New Roman italic font*	See the *Acuitas User's Guide*. *Workflow* is an application running on the system.

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*

CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the latest version of Adobe Reader ®. To download the latest version of the Adobe Reader software from the Adobe web site, go to http://get.adobe.com/reader/

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24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Chapter 1

Charging Data Records

This section contains the following information about *Affirmed Networks Mobile Content Cloud (MCC)* Charging Data Records (CDRs).

CDR Overview

A *Charging Data Record* (CDR), also referred to as a Call Data Record or Call Detail Record, is a formatted collection of information about a chargeable event for use in billing and accounting. For example, when a mobile phone user activates a Packet Data Protocol (PDP) context, the network element generates chargeable events, such as the time of session set-up, the duration of the session, and the amount of data transferred (and so on). Offline (post-paid) charging has four main components:

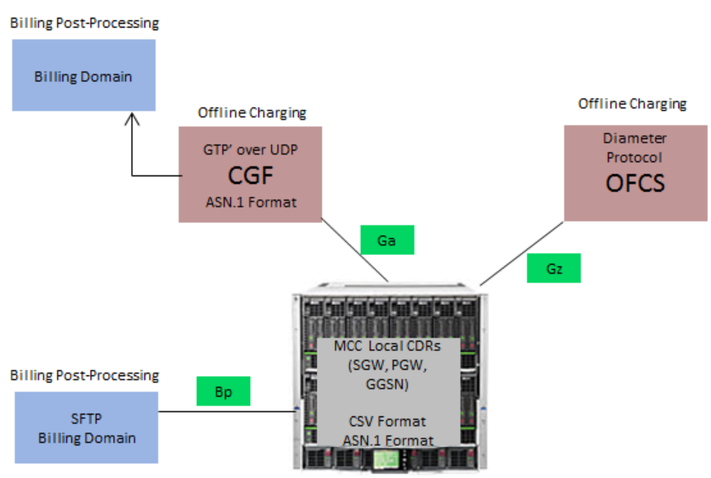
- *Charging Trigger Function* (CTF) – Generates charging event information and forwards it to the Charging Data Function.
- *Charging Data Function* (CDF) – Uses the event information to generate CDRs and transfers them to the Charging Gateway Function.
- *Charging Gateway Function* (CGF) – Uses the CDRs to generate CDR files and transfers CDRs to the Billing Domain for post-processing.
- *Billing Domain* – CDR post-processing for subscriber billing and accounting.

Using the internal charging gateway functionality, Affirmed Networks Mobile Content Cloud (MCC) devices can store CDRs locally in both *comma-separated values* (CSV) and *Abstract Syntax Notation 1* (ASN.1) format. You can configure the system to forward CDRs to the following external devices:

- Charging Gateway Function (CGF) via the Ga interface using GTP'
- Offline Charging System (OFCS) via the Gz/Rf interface using Diameter
- Billing Domain via the Bp interface using Secure File Transfer Protocol (SFTP)

Figure 1-1 illustrates the relevant CDR interfaces and external devices.

Figure 1-1 CDR Interfaces and External Devices



Offline Charging Support for CUPS

The MCC Control and User Plane Separation (CUPS) architecture supports the following offline charging interfaces used to transfer Charging Data Records (CDRs) to an Offline Charging System (OFCS):

- Local Interface – Stores local CDRs in text files (CSV/ASN.1 format).
- Ga Interface – Transfers CDRs using GTP Prime (GTPP).
- Gz/Rf Interface – Transfers CDRs using Diameter.

These offline charging interfaces are configured on the PGW-C. There is no change to the interface configuration. See the *Affirmed Networks Gateway Administration Guide* for more information on the MCC CUPS architecture.

The PGW-C sends charging information over PFCP to the PGW-U via the Sx interface using vendor-specific information elements (IEs). The PGW-U reports traffic usage information to the PGW-C via the Sx interface. The PGW-C then uses the traffic usage information to generate CDRs and transfers them via the offline charging interface. All existing event triggers, including time and volume, are supported using vendor-specific IEs.

Local CDRs

Local CDRs are maintained in text files and stored on the MCM in */public/logs/datarecord.lnk.* The amount of time in which they are stored and the size of the file is configurable. When the maximum size is reached, the system creates a new CDR file. The system generates separate CDRs in order for each subscriber to be billed for a subset or for an entire set of charges for a chargeable event. Multiple CDRs may be generated for a single chargeable event, for example, because of its long duration or because more than one party should be charged. Local CDRs are classified as follows:

- SGW CDRs (page 2-2)
- PGW CDRs (page 2-4)
- GGSN CDRs (page 2-7)
- WAG CDRs (page 2-12)
- ePDG CDRs (page 2-14)
- GTP Proxy CDRs (page 2-15)
- C-SGN CDRs (page 2-16)

Note: *Trusted WLAN Access Gateway (TWAG or WAG) and evolved Packet Data Gateway (ePDG) local CDRs are only stored in CSV format.*

Local CDR File Formats

You can configure MCC gateways to store local CDRs in either CSV format, ASN.1 format or both CSV and ASN.1 formats. For information on saving CSV and ASN.1 encoded CDR files to the local disk, see Storing CDRs in a Local File (page 1-19).

CSV Format

The system stores local CDRs in a comma-separated values (CSV) format. CSV files store tabular data (numbers and text) in plain-text form and consist of any number of records, separated by line breaks. Each record consists of fields, separated by characters or strings, most commonly a literal comma or tab. Typically, all records have an identical sequence of fields. Each individual field is configured in hex, decimal, or string format.

ASN.1 Format

Typically, Abstract Syntax Notation 1 (ASN.1) encoded CDR files are delivered over the Ga interface and are formatted per the ASN.1 definitions specified in 3GPP TS 32.298, in Tag-Length-Value (TLV) format. ASN.1 is a standard and flexible notation that describes rules and structures for representing, encoding, transmitting, and decoding data in telecommunications and computer networking. ASN.1 provides a consistent mechanism to record, transmit, and read data across diverse systems.

ASN.1 defines the abstract syntax of information but does not restrict the way the information is encoded. Various ASN.1 encoding rules provide the transfer syntax (a concrete representation) of the data values whose abstract syntax is described in ASN.1. The system uses Basic Encoding Rules (BER) to facilitate the exchange of structured data to external servers.

Note: *For a complete ASN.1 definition of the General Packet Radio Service (GPRS) records based on the ASN.1 definition in 3GPP TS 32.298, see* *Appendix B, ASN.1 Definition*.

Local CDR File Naming Convention

To facilitate CDR, HTDR, and event log file retrieval and processing, the MCC automatically appends a sequence number to the local CSV filename to indicate which file it is. The MCC uses the following file naming convention for local CSV CDR, HTDR, and event log files:

```
<filename-prefix><YYYYMMDDHHMMSS>_<123>.csv
```

where `filename-prefix` indicates the prefix of the filename in which to store CDRs, `YYYYMMDDHHMMSS` represents the timestamp and `123` represents the file sequence number. For example, `pgwcdr20161114163821_123.csv`.

The sequence number runs from 1 to 2147483647, so that at one record per second, it would take a little over 68 years to rollover. The sequence number resets to 1 after a system reboot, MCM switchover or system failure. Depending on the frequency of these events, CDR records could potentially have the same sequence number.

You can configure the width of the file sequence number using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<file-type> <filename-prefix> sequence-field-width <value>
```

`sequence-field-width` – Defines the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following:

```
<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv
```

Customizing the File Naming Convention for Local ASN.1 CDRs

Use the following command to customize the file naming convention for local CDRs (ASN.1 only). This feature only applies to the datarecord profile.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
datarecord <filename-prefix> file-name-format [default|custom1]
```

- **default** – The MCC uses the default file naming convention for local CDR files: `<filename-prefix><YYYYMMDDHHMMSS>_<123>.csv`
- **custom1** – The MCC uses the following custom file naming convention for local CDR files (ASN.1 only):

  `<filename-prefix>_MM_DD_YYYY_hh_mm_ss_<#records>_file<123>.u`

  where

  – `filename-prefix` – Indicates the prefix of the filename in which to store CDRs.

  – `MM_DD_YYYY_hh_mm_ss` – Represents the timestamp.

  – `#records` – Represents the number of records stored in the file, starting with 0.

  – `123` – Represents the file sequence number. The number of leading zeros prepended to the file sequence number depends on the value configured for the `sequence-field-width`. For example, if the `sequence-field-width` is set to 6, the sequence number will start with 000001.

Note: *In the rare case of an uncontrolled MCM switchover caused by a crash in the DataRecordAgentMCM process or an MCM kernel crash, the currently open CDR files will remain with a record count of 0, indicating that the number of records could not be determined when the file was closed.*

CDR Transfer Mode

Local CDRs can be transfered to a billing server in either push or pull mode.

Push Mode – The system sends (pushes) CDR files to a preconfigured destination using SFTP. The upload frequency is configurable on the MCC.

Pull Mode – Once the closing conditions are reached for CDR files, the system exports the files to the */public/logs/datarecord.lnk/export* directory according to the export rules and user permissions configured for the file management profile. A billing server then retrieves (pulls) the stored CDRs using SFTP from the */public/logs/datarecord.lnk/export* directory.

Once the billing server has successfully retrieved (pulled) the stored data record files, the files remain in the */public/logs/datarecord.lnk/export* directory and are cleaned up based on the purge rules configured for the file management profile. Alternatively, the export user (user configured with read/write access to the files in the */export* directory) can delete the files after a successful SFTP file transfer.

Data record export rules and user permissions are configurable using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
datarecord <filename-prefix> export-rule
[exportrule-default|exportrule-none] export-user <username>
```

- **export-rule** – Specifies the rules/actions to apply when exporting the file to the */public/logs/datarecord.lnk/export* directory for SFTP access.
  - **exportrule-default** – The system exports files to the */public/logs/datarecord.lnk/export* directory according to the default format.

    Uses the file naming convention:
    `<filename-prefix><YYYYMMDDHHMMSS>_<123>.extension`

    Applies purge rules for this file profile to all files in the directory.
  - **exportrule-none** – Specifies that there is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.

- **export-user** – Specifies the user that has read/write access to the files in the */public/logs/datarecord.lnk/export* directory. To set the default SFTP directory for the export user, issue the following command:

  ```
  an3000-mcm-slot7cpu1# user-access change-sftp-directory
  ```

  The system prompts:

  ```
  Enter sftp directory:
  ```

  Type **/public/logs/datarecord.lnk/export** to set the default SFTP directory.

Supported Record Types and Triggers

Local CDRs contain the following record types:

- *Start* – Starts an accounting session.
- *Interim* – Updates an accounting session.
- *Stop* – Ends an accounting session.
- *Event* – Indicates a one-time accounting event.

***Note:** ASN.1 supports partial and full record types based on network events. If the cause is coded as a normal or abnormal release, a full record is generated; otherwise a partial record is generated.*

Events that Trigger an Interim (Partial) Record

The following events trigger closure and sending of an interim (partial) CDR:

- When the time limit is reached for a bearer (`interim-update-interval`)
- When the volume limit per CDR is reached for a bearer (`data-volume-limit`)
- When the number of service containers added to a CDR reaches the configured limit (`num-service-containers`)
- When the event recurrence limit is reached, which specifies the maximum number of containers added to a CDR per closure event (See Offline Charging Event Trigger Template (page 1-11))
- When the time limit is reached for storing a closed service container (`cdr-close-time`)
- When there is a change in the QoS of the IP-CAN bearer (`qos-change-close-trigger`)
- When the context is transferred to a new SGW/SGSN (*applies to the GGSN/PGW*) or when the Mobility Management Entity (MME) is changed (*applies to the SGW*) (`serving-node-change-close-trigger`)
- When there is a change to the serving node's PLMN ID (`serving-node-plmn-change-close-trigger`) *Applies to PGW and GGSN only*
- When there is a change to the UE's time zone (`ue-time-zone-change-close-trigger`)
- When the Radio Access Technology (RAT) is changed within the same PGW (`rat-type-change-close-trigger`)

- When the user location information is changed (`user-location-change-close-trigger`) *Applies to SGW and PGW only*
- When there is a change to the user Closed Subscriber Group (CSG) information (`user-csg-change-close-trigger`) *Applies to SGW and PGW only*
- When there is a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) (`change-in-up-to-ue-close-trigger`) *Applies to the SGW only*

Note: *For information on configuring these values, see* *Storing CDRs in a Local File (page 1-19)*.

Events that Trigger a Stop (Final) Record

A CDR is closed as the Stop (Final) record of a subscriber session for the following events:

Gateway	Event
PGW-CDR Closure	■ Detach request received from UE. ■ Delete Session Request received from SGW. ■ Re-Auth-Request (RAR) received from PCRF requesting bearer deletion. ■ PGW-initiated session termination (expiry of idle or maximum timer duration). ■ Manual subscriber clearing (`subscriber clear` command). ■ Abnormal releases such as path failures.
SGW-CDR Closure	■ Detach request received from UE. ■ Delete Bearer Request received from PGW. ■ Delete Session Request received from MME. ■ SGW-initiated Delete Session Request (expiry of idle or maximum timer duration). ■ Manual subscriber clearing (`subscriber clear` command). ■ Abnormal releases such as path failures.
GGSN-CDR Closure	■ Delete PDP Context Request received from SGSN. ■ RAR received from PCRF requesting PDP context deletion. ■ GGSN-initiated PDP Context Deactivation (expiry of idle or maximum timer duration). ■ Manual subscriber clearing (`subscriber clear` command). ■ Abnormal releases such as path failures.

Gateway	Event
WAG-CDR Closure	■ Dynamic Host Configuration Protocol (DHCP) release received from UE. ■ DHCP lease timer expires. ■ WAG-initiated session termination (expiry of idle or maximum timer duration). ■ Manual subscriber clearing (`subscriber clear` command). ■ Abnormal releases such as path failures.
ePDG-CDR Closure	■ Detach request received from UE. ■ HSS-initiated Delete Request. ■ Delete Bearer Request received from PGW. ■ ePDG-initiated session termination (expiry of idle or maximum timer duration). ■ Manual subscriber clearing (`subscriber clear` command). ■ Abnormal releases such as path failures.

Note: *When the* `subscriber clear` *CLI command is executed on a gateway device, CDRs for the subscriber sessions are closed with the Management Intervention (20) cause code.*

Offline Charging Event Trigger Template

The offline charging event trigger template defines the events that trigger the closure and sending of an interim (partial) CDR on the local, Gz/Rf, and Ga interfaces. When CDR closure events happen, the MCC stores the events in service data containers (SDCs) until another event occurs to transfer the stored service containers in an interim (partial) CDR. Service containers are maintained separately for each offline charging interface (local, Ga, Gz/Rf).

For offline CDRs (Gz/Rf and Ga), Operators configure an event recurrence limit, which specifies the maximum number of containers added to a CDR per closure event. Upon reaching this limit, the MCC closes the interim CDR and transfers it to the offline server. The maximum service container limit per CDR closure event is 20.

If a single CDR closure event occurs with multiple changes (for example, an SGSN change with CGI change, RAI change, and QoS Change), MCC creates only one service container listing all of the change conditions. Using service containers to group offline CDR closure events reduces the number of Interim CDRs sent to offline charging servers. Furthermore, when generating CDRs, the MCC creates a separate service container for each service ID, so associating multiple PCC rules (Service Rules) with the same service ID (SDF ID) reduces the number of service containers created in each CDR.

Reporting the Cause for Record Closing in ASN.1 PGW CDRs

When multiple offline charging event triggers are configured with the same recurrence value in the event trigger template and the events occur simultaneously, the trigger with the highest priority is used as the reason for the CDR closure in the Cause for Record Closing field in PGW ASN.1 CDRs (local and offline).

The following order of priority is used to fill the Cause for Record Closing field in ASN.1 PGW CDRs when event triggers, configured with the same recurrence value, occur simultaneously:

1. rat-type-change-close-trigger
2. serving-node-change-close-trigger
3. ue-time-zone-change-close-trigger
4. serving-node-plmn-change-close-trigger
5. qos-change-close-trigger
6. user-location-change-close-trigger
7. user-csg-change-close-trigger

For example, when the recurrence value for the QoS change trigger and user location change trigger are both set to 1 in the event trigger template, the QoS change trigger takes higher priority and is used as the reason for the CDR closure when both events occur simultaneously.

Data Record Templates

Operators can use data record templates to include or exclude CDR fields in the CDR file and, in the case of CSV CDRs, to modify the position of fields.

Data Record Templates for CDRs in CSV Format

Local CDR fields in CSV format are presented in default order. Each complex field (such as ServiceDataContainer and TrafficDataVolume) is encoded in single quotes. If a complex field has an embedded complex field, it is encoded in two single quotes. The third level complex field is encoded in four single quotes, and so on. A field without a value set is identified by two consecutive commas (,,). If the same complex field is repeated, these complex fields are separated by a semicolon (;).

Each CDR field in CSV format contains the following six configurable parameters:

- *Present Flag* – Indicates whether to include the data field in the CDR. Options include:
  - Conditional – Indicates the data field is present in CDRs when available.
  - Yes – Indicates the data field is always present in CDRs.
  - No – Indicates the data field is not present in CDRs.
- *Position Index* – Location of the data field in the CDR. A pre-defined order exists for each data field if positionIndex is not configured.
- *Format Type* – The format for the data field. If not configured, the format type uses the default.
- *Length* – The fixed length of the data field. The length defaults to zero indicating the data field is written out as it is.
- *Alignment* – The alignment of the data field. If not configured, the alignment defaults to left aligned.
- *Padding* – The character used for padding when the field has a fixed length.

Note: *For information on the SGW/PGW/GGSN/WAG/ePDG CDR fields with respect to 3GPP TS 32.251 V9.9.0 and 3GPP TS 32.298 V10.0.0, see Chapter 3, CDR Descriptions.*

Data Record Templates for CDRs in ASN.1 Format

Operators can configure Interim (partial) and Stop (final) data record templates to include or exclude ASN.1 fields in local and GTPP (Ga interface) CDRs. Operators can customize these data record templates to ensure the generated Interim and Stop data records conform to the ASN.1 definitions specified in the desired 3GPP TS 32.298 release. By default, all CDR fields in Interim and Stop data record templates are set to conditional, indicating that the data field is filled in CDRs when available.

Alternatively, Operators can specify a 3GPP 32.298 version used to automatically format local ASN.1 and GTPP CDR fields in Interim and Stop data records.

Formatting Local ASN.1 CDRs

Use the following CLI command to specify a 3GPP 32.298 version used to format local ASN.1 CDRs.

Note: *This parameter is only visible when the* `encode-format` *for the local offline-charging interface is set to* `asn1` *or* `both`*.*

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> pgw local asn1-cdr-version
[10.0|9.0|8.4|6.0]
```

- `6.0` – Fields generated in local Interim and Stop data records are partially compliant to the ASN.1 syntax specified in 3GPP 32.298 version 6.0. The GPRSCallEventRecord value is set to 28 and the CallEventRecordType is set to 70, as per 3GPP 32.298 version 6.0.
- `8.4` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 8.4.
- `9.0` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 9.0.
- `10.0` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 10.0.

Formatting GTPP CDRs

Use the following CLI command to specify a 3GPP 32.298 version used to format GTPP CDRs.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group gtpp node-admin <service-profile> version
[10.0|9.0|8.4]
```

- `10.0` – Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 10.0.
- `9.0` – (default) Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 9.0.
- `8.4` – Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 8.4.

Note the following:

- If using a 3GPP 32.298 version beyond 10.0 to format local CDR fields, select 10.0 and then configure the Interim and Stop data record templates to conform to the ASN.1 definitions specified in the desired 3GPP TS 32.298 release.
- The 3GPP 32.298 version configured for local CDR fields and the version configured for GTPP CDR fields do not need to match.
- If a version is not configured for local CDR fields, local Interim and Stop data records will be compliant to the 3GPP 32.298 version configured for GTPP CDR fields.
- If a version is not configured for either local or GTPP CDR fields, local and GTPP Interim and Stop data records will be compliant to 3GPP 32.298 version 9.0.
- When the version parameter is set to `9.0` for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to `no` if it is not needed. When the `version` parameter is set to `10.0` for local ASN.1/GTPP CDRs, the Low Priority Indicator field in local ASN.1/GTPP Interim and Stop data record templates should be set to `yes` (default).

If you format CDR fields as per the ASN.1 syntax definition in 3GPP TS 32.298 Version 8.4, the following CDR fields are omitted from the Interim and Stop data records regardless of their configured setting in the Interim and Stop data record templates.

- PGW PLMN Identifier
- Start Time
- Stop Time
- Served MNNAI
- Served 3GPP 2 MEID
- Cause For Record Closing

- PDN Connection ID
- User CSG Information
  - CSG ID
  - CSG Access Mode
  - CSG Membership Indication
- IMSI Unauthenticated Flag
  - ECGI Change
  - TAI Change
  - User Location Change
- List of Service Data
  - EPC QoS Information

> ***Note:*** *For 3GPP TS 32.298 Version 8.4, the List of Service Data containers include the QoS Information Negotiated field in Octet String format. For 3GPP TS 32.298 Version 9.0 and beyond, the List of Service Data containers include EPC QoS Information field in Sequence format.*

CDR Storage

The CDR storage module is responsible for storing CDRs when the remote server is unavailable or congested. When the remote server becomes available, the pending CDRs are sent (pushed) to the billing server. This CDR storage feature is supported on all offline charging interfaces (Gz/Rf and Ga) and is enabled by default. The system reserves 20% of the total disk partition size to store CDRs when the remote server is unavailable.

For Gz/Rf SGW/PGW CDRs, Operators can configure the MCC to store CDRs in a predefined file and format when the remote server is unavailable. Once servers become available, an OFCS pulls the stored SGW/PGW CDRs using SFTP from the predefined file path. For more information, see Storing Gz/Rf CDRs to a File in the Event of a Failure (page 6-22).

> ***Note:*** *For information on storing CDRs to volatile storage for v-MCM node types, see the Affirmed Networks System Administration Guide.*

Co-Located SGW CDRs

The MCC enables the SGW and PGW to operate as stand-alone services on separate devices or as combined services running on the same device. If the SGW subscriber session is using a PGW operating as a combined SGW/PGW system, by default, only PGW CDRs are generated. However, you can configure the MCC to generate SGW CDRs for SGW subscriber sessions with co-located PGW sessions. For more information, see Generating SGW CDRs for Co-Located SGW/PGW (page 1-39).

Offline Charging

The offline charging feature enables Operators to configure the offline charging interface to an *Offline Charging System* (OFCS). Offline charging enables Operators to charge their customers, in real-time, based on service usage.

An OFCS is used for offline charging and receives charging events from the *Policy and Charging Enforcement Function* (PCEF). The PCEF generates CDRs that can be transferred to the billing system. The SGW, PGW, and GGSN log charging events to CDRs when CDRs are enabled. They can also send the charging events to offline charging servers in real-time using streaming protocols such as Diameter on the Gz/Rf interface and GPRS Tunneling Protocol prime (GTP') on the Ga interface.

- Ga interface – Transports ASN.1 encoded CDR files from the gateways (SGW/PGW/GGSN) to the CGF. The interface uses a GTP-based protocol (called GTP', GTP prime or GTPP), with modifications to support CDRs. For more information, see Chapter 5, Ga Interface.
- Gz/Rf interface – Transports CDR files in CSV format from the gateways (SGW/GGSN/PGW) to an OFCS. For more information, see Chapter 6, Gz/Rf Interface.

An MCC device is designed to perform offline charging on tiered subscriber billing access plans. MCC charging service interacts with the Operator's offline charging back-office systems and PCRF systems to recognize and bill for tiered usage on a non-real time basis. For example, the system is able to recognize three tiered billing plans and the associated overage of the plan and use that data to produce call detail records that can be used by the Operator to compile a monthly bill. The system is able to produce periodic CDRs using a 15-minute interval and transmit those CDRs electronically to the Operator's mediation system using the Operator's administrative network.

CDR Reporting

There are two CLI parameters that govern how CDRs are generated.

- CDR Reporting Level – Controls the granularity of the CDR itself. Applies to the PGW, SGW and GGSN.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> cdr-report-level
[bearer|pdn-session]
```

  - **`bearer`** – (default) Each bearer will have its own set of CDRs. When set to `bearer`, the rating group `reporting-level` must be set to `serviceRule`.
  - **`pdn-session`** – Only one set of CDRs is generated for the entire session, irrespective of the number of bearers. When set to `pdn-session`, the rating group `reporting-level` must be set to either `ratingGroup` or `ratingGroupAndQci`. *Only applies to Gz/Rf PGW and SGW CDRs.*

- Container Reporting Level – Controls the granularity of the service data container (SDC) inside the CDR. Applies to the PGW only.

  For local and Ga (GTPP) CDRs, the MCC only generates SDCs at the service-rule level.

  For Gz/Rf CDRs, the MCC generates SDCs at the rating-group level, the service-rule level, or the rating-group and QoS-Class-Identifier (QCI) level, depending on the `reporting-level` configuration of a particular rating group.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group-name> reporting-level [serviceRule|ratingGroup|
ratingGroupAndQci]
```

  - **`serviceRule`** – (default) Adds one SDC per policy rule in CDRs. When set to `serviceRule`, the offline-charging `cdr-report-level` must be set to `bearer`.
  - **`ratingGroup`** – Adds one SDC per rating group in CDRs. *Only applies to Gz/Rf PGW CDRs.*
  - **`ratingGroupAndQci`** – Adds one SDC per rating group and QCI combination. For example, when two bearers share the same rating group, the MCC generates two SDCs because there are two unique rating group and QCI combinations. *Only applies to Gz/Rf PGW CDRs.*

Note: *To generate SDCs at the rating group or rating group and QCI level, ensure that the `cdr-report-level` is set to `pdn-session`. For more information, see* *Generating Session-Level Gz/Rf CDRs (page 6-30).*

Table 1-1 shows the configuration combinations supported on the PGW.

Table 1-1 PGW CDR Reporting Configuration Combinations

Reporting Combination	Ga Interface (GTPP)	Gz/Rf Interface (Diameter)	Local Interface (CSV/ASN.1)
Bearer CDR + Service Rule SDC	✓	✓	✓
PDN Session CDR + Rating Group SDC	☒	✓	☒
PDN Session CDR + Rating Group and QCI SDC	☒	✓	☒

Note the following restrictions:

- All offline interfaces associated with a session need to have the same CDR reporting level + container reporting level.
- All rating groups associated with a session need to be configured with the same container reporting level.
- For SGW CDRs, usage data is captured in the Traffic Data Volume container.

Storing CDRs in a Local File

This section describes how to store CDRs in a local file. CDR files can be transfered to a billing server in either push or pull mode. For more information, see CDR Transfer Mode (page 1-7).

Note: *For local CDRs, the MCC only generates SDCs at the service-rule level. For more information, see* *CDR Reporting (page 1-17)*.

Note: *When configuring the local offline charging interface on the GTP Proxy, use the PGW object.*

Use the following sequence to configure the system to store CDRs locally.

1 Create a local file in which to store CDRs and configure the file management profile.

a Use the following CLI command to create a local CDR file and configure the file management profile to transfer CDR files using *push* mode.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management
profile datarecord <filename-prefix>
```

where **<filename-prefix>** specifies the prefix of the filename in which to store CDRs.

The following table describes the configurable parameters.

Parameter	Description
compress-enabled	Specifies whether this file should be compressed (zipped) when it is closed. Compressed data record files are stored in */public/logs/datarecord.lnk*. ■ **enabled** ■ **disabled**

Parameter	Description
compress-format	If compress is enabled, specify the format for compressing files: ■ **gzip** – (default) The data records are written to a file. The file is gzipped after it is closed. ■ **raw** – The data records are written in zlib raw format. The zlib raw format provides better performance compared to the gzip format because the records are compressed on the source before sending them to the agent for writing. For information on how to uncompress data records written in zlib raw format, see Uncompressing Data Records in zlib raw Format (page 1-38) ***Note:*** *Affirmed Networks recommends using the zlib raw format if you expect the system to generate data records at a rate higher than 10K data records per second.*
dest-port	Specifies the port of the file SFTP server.
dest-url	Specifies the URL/IP Address for the file SFTP server. If you specify a URL, ensure the DNS server IP address is configured for the management-context. an3000-mcm-slot7cpu1(config)# **network-context management-context dns server-address <*ip-address*>**
file-name-format	*Applies to local ASN.1 CDRs only.* Specifies the file naming convention to use for local ASN.1 CDRs. See Customizing the File Naming Convention for Local ASN.1 CDRs (page 1-6) for more information. ■ **default** – The MCC uses the default file naming convention for local CDR files: <filename-prefix><YYYYMMDDHHMMSS>_<123>.csv ■ **custom1** – The MCC uses the following custom file naming convention for local ASN.1 CDR files: <filename-prefix>_MM_DD_YYYY_hh_mm_ss_<#records>_file<123>.u
file-rotate-empty	Specifies whether empty CDR files are rotated. ■ **enabled** – Empty CDR files are rotated based on the configured file-rotate-frequency interval. ■ **disabled** – (default) Empty CDR/HTDR files are not rotated.
file-rotate-frequency	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (file-rotate-size-bytes) or max number of records (file-rotate-num-records). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
file-rotate-num-records	Specifies the number of records at which to rotate the files.

Parameter	Description
file-rotate-size-bytes	Specifies the size (in bytes) at which files should be rotated.
file-transfer-enabled	Specifies whether to enable or disable file transfer. ■ **enabled** ■ **disabled**
passwd	Specifies the password to log into the file SFTP server.
purge-enabled	Specifies whether to enable or disable purge/cleanup for this file. ■ **enabled** ■ **disabled**
purge-file-older-than	Specifies the frequency at which files should be purged.
purge-low-thr	Specifies the percentage utilization at which to stop transferring/purging files.
purge-thr	Specifies the percentage utilization at which to transfer/purge files.
sequence-field-width	Specifies the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following: `<filename><YYYYMMDDHHMMSS>_<0000123>.csv`
upload-dir-path	Specifies the path on the SFTP server to which to upload the files.
upload-frequency	Specifies the frequency to upload files.
upload-offset-secs	Specifies the offset at which to upload files after upload-frequency. This value (in seconds) is added to the time at which the file should be uploaded.
username	Specifies the username to log into the file SFTP server.

b Use the following CLI command to create a local CDR file and configure the file management profile to transfer CDR files using *pull* mode:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management
profile datarecord <filename-prefix>
```

where **<filename-prefix>** specifies the prefix of the filename in which to store CDRs.

The following table describes the configurable parameters.

Parameter	Description
compress-enabled	Specifies whether this file should be compressed (zipped) when it is closed. Compressed data record files are stored in */public/logs/datarecord.lnk*. ■ **enabled** ■ **disabled**
compress-format	If compress is enabled, specify the format for compressing files: ■ **gzip** – (default) The data records are written to a file. The file is gzipped after it is closed. ■ **raw** – The data records are written in zlib raw format. The zlib raw format provides better performance compared to the gzip format because the records are compressed on the source before sending them to the agent for writing. For information on how to uncompress data records written in zlib raw format, see Uncompressing Data Records in zlib raw Format (page 1-38) ***Note:*** *Affirmed Networks recommends using the zlib raw format if you expect the system to generate data records at a rate higher than 10K data records per second.*
export-rule	Specifies the rules/actions to apply when exporting the file to the */public/logs/datarecord.lnk/export* directory for SFTP access. For more information, see CDR Transfer Mode (page 1-7). ■ **exportrule-default** – The system exports files to the */public/logs/datarecord.lnk/export* directory and applies purge rules for this file profile to all files in the directory. ■ **exportrule-gz-custom1** – (*Applies to Gz/Rf CDRs only)*. For more information, see Storing Gz/Rf CDRs to a File in the Event of a Failure (page 6-22). ■ **exportrule-none** – There is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.
export-user	Specifies the user that has read/write access to the files in the */public/logs/datarecord.lnk/export* directory. For information on setting the default SFTP directory for the export user, see CDR Transfer Mode (page 1-7).

Parameter	Description
file-name-format	*Applies to local ASN.1 CDRs only.* Specifies the file naming convention to use for local ASN.1 CDRs. See Customizing the File Naming Convention for Local ASN.1 CDRs (page 1-6) for more information. ■ **default** – The MCC uses the default file naming convention for local CDR files: `<filename-prefix><YYYYMMDDHHMMSS>_<123>.csv` ■ **custom1** – The MCC uses the following custom file naming convention for local ASN.1 CDR files: `<filename-prefix>_MM_DD_YYYY_hh_mm_ss_<#records>_file<123>.u`
file-rotate-empty	Specifies whether empty CDR files are rotated. ■ **enabled** – Empty CDR files are rotated based on the configured `file-rotate-frequency` interval. ■ **disabled** – (default) Empty CDR files are not rotated.
file-rotate-frequency	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (`file-rotate-size-bytes`) or max number of records (`file-rotate-num-records`). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
file-rotate-num-records	Specifies the number of records at which to rotate the files.
file-rotate-size-bytes	Specifies the size (in bytes) at which files should be rotated.
purge-enabled	Specifies whether to enable or disable purge/cleanup for this file. ■ **enabled** ■ **disabled**
purge-file-older-than	Specifies the frequency at which files should be purged.
purge-low-thr	Specifies the percentage utilization at which to stop transferring/purging files.
purge-thr	Specifies the percentage utilization at which to transfer/purge files.
sequence-field-width	Specifies the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following: `<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv`

2 Configure the data record templates to select which CDR fields to include in the CDR file. For more information on data record templates, see Data Record Templates (page 1-12).

a To configure data record templates for generating CDRs in CSV format:

```
an3000-mcm-slot7cpu1(config)# service-construct
data-record-template [pgw|sgw|ggsn|wag|epdg] charging-local
[start|stop|interim|event] <template-name> <field>
```

where

Parameter	Description
`[pgw|sgw|ggsn|wag|epdg]`	Specifies the gateway type for this template.
`[start|stop|interim|event]`	Specifies the record type.
`<template-name>`	Specifies the name of the data record template.
`<field>`	Specifies the name of the CDR field you are modifying.

The following table describes the configurable parameters.

Parameter	Description
`alignment`	The alignment of the data field. If not configured, the alignment defaults to left aligned.
`formattype`	The format for the data field. There is a default format for each data field, if not configured.
`length`	The fixed length of the data field. The default value is zero indicating the data field is written out as it is.
`padding`	The character used for padding when the field has a fixed length.
`positionIndex`	The location of the data field in the CDR. There is predefined order for each data field if positionIndex is not configured.
`presentFlag`	Indicates whether to include the data field in the CDR. Options include: ■ **`conditional`** – Indicates the data field is present in CDRs when available. ■ **`yes`** – Indicates the data field is always present in CDRs. ■ **`no`** – Indicates the data field is not present in CDRs.

b To configure Interim (partial) and Stop (final) data record templates for generating CDRs in ASN.1 format:

```
an3000-mcm-slot7cpu1(config)# service-construct
data-record-template [sgw|pgw|ggsn] ofcs gtpp [interim|stop]
<template-name> <fields> presentFlag [yes|no|conditional]
```

where

Parameter	Description
`[pgw|sgw|ggsn]`	Specifies the gateway type for this template.
`[interim|stop]`	Specifies the record type.
`<template-name>`	Specifies the name of the data record template.
`<fields>`	Specifies the name of the CDR field you are modifying.
`presentFlag`	Indicates whether to include the data field in the CDR. Options include: ■ `conditional` – Indicates the data field is present in CDRs when available. ■ `yes` – Indicates the data field is always present in CDRs. ■ `no` – Indicates the data field is not present in CDRs. ***Note:*** *You can also use the* `presentFlag` *parameter to categorize local ASN.1 CDRs as either invalid or error if a specified set of fields are not present in the CDR. See Categorizing Local ASN.1 CDRs as Invalid or Error (page 1-45).*

3 Configure the event trigger template to specify the events that trigger an interim update on the offline charging interface and to trigger the closure of an interim CDR.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
```

The following table describes the configurable parameters.

Parameter	Description
`cgi-sai-change-close-trigger`	Specifies whether a change in the Cell Global Identification-Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close.
`change-in-up-to-ue-close-trigger`	Applies to offline interfaces only. Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close. ***Note:*** *Applies to the SGW only.*

Parameter	Description
`ecgi-change-close-trigger`	Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close.
`qos-change-close-trigger`	Specifies whether a QoS Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of QoS Change events to trigger a CDR close.
`rat-type-change-close-trigger`	Specifies whether a RAT Type Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAT Type Change events to trigger a CDR close.
`routing-area-change-close-trigger`	Specifies whether a change in the Routing Area Identity (RAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAI change events to trigger a CDR close.
`serving-node-change-close-trigger`	Specifies whether a Serving Node Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node Change events to trigger a CDR close. ***Note:*** *Applies to the GGSN/PGW when the context is transferred to a new SGW/SGSN or to the SGW when the Mobility Management Entity (MME) is changed.*
`serving-node-plmn-change-close-trigger`	Specifies whether a Serving Node PLMN Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node PLMN Change events to trigger a CDR close. ***Note:*** *Applies to PGW and GGSN only.*
`time-limit-close-trigger`	Applies to offline interfaces only. If interim updates are enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.
`tracking-area-change-close-trigger`	Specifies whether a change in the Tracking Area Identity (TAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of TAI change events to trigger a CDR close.

Parameter	Description
ue-time-zone-change-close-trigger	Specifies whether a UE Time Zone Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UE Time Zone Change events to trigger a CDR close.
user-csg-change-close-trigger	Specifies whether a CSG change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CSG change events to trigger a CDR close.
user-location-change-close-trigger	Specifies whether a User Location Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of User Location Change events to trigger a CDR close.
volume-limit-close-trigger	Applies to offline interfaces only. If Data Volume Limit is enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.

4 Configure the local interface for offline charging.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> admin-state [enabled|disabled]
[pgw|sgw|ggsn|wag|epdg]
```

where **<*interface-name*>** is the name of the local interface for offline charging, **admin-state** is the administrative state of the interface, and **[pgw|sgw|ggsn|wag|epdg]** specifies the gateway type.

The following table describes the configurable parameters.

Parameter	Description
apn-reporting	*Applies to the PGW only*. Specifies the Access Point Names (APNs) to be reported over the local offline-charging interface. ■ **original** — Initial APN ■ **mapped** — Final / current APN ■ **prior-mapped** — APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN) The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. If the apn-reporting value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the apn-reporting value configured for the Control Profile. For more information, see Configuring APN Reporting at the Interface Level (page 1-47). ***Note:*** *The interface-level apn-reporting parameter does not support the custom option.*
cdr-close-time	If interim-updates are enabled on the interface, specifies the number of seconds for storing closed service containers before transferring service containers in an interim CDR to the offline server. The default is 0 (disabled).
data-volume-limit	Specifies the maximum number of kilobytes that the gateway maintains across all service containers (for a bearer) before generating a new CDR. The volume usage counts are reset when a CDR is closed and sent. The default is 0 (disabled).
encode-nai-in-asn1-msisdn	*Applies to SGW/PGW/GGSN only*. Specifies whether to include/exclude the Nature of Address Indicator (NAI) in the Served Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) field when generating local ASN.1 or GTPP CDRs. By default, this parameter is enabled, indicating the NAI is included in the Served MSISDN CDR field. ■ **enabled** (default) ■ **disabled**
event-trigger-template	Specifies the event trigger template name.
interim-update-interval	Specifies the time interval, in seconds, after which a new service container or CDR is generated. The timer resets when a CDR is sent. The default is 0 (disabled). When this time interval expires, the MCC reports the current usage statistics (up to when the timer expired) in the CDR.

Parameter	Description
`interim-updates`	Specifies whether to enable or disable interim updates to the offline charging server. ■ `enabled` ■ `disabled`
`num-charge-condition-change`	Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0. ***Note:*** *A trigger is only counted towards the* `num-charge-condition-change` *if it is enabled under the* `event-trigger-template`.
`num-service-containers`	If `interim-updates` are enabled on the interface, specifies the maximum number of service containers that the gateway maintains per CDR at the bearer-level. Upon reaching this limit, the gateway transfers the service containers in an interim CDR to the offline server.
`qos-change-trigger`	Specifies when to activate the qos-change-trigger. ■ `pcrf-initiated` – Activate the qos-change-trigger for a PCRF-initiated QoS change only. ■ `access-nw-initiated` – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only. ■ `any` – (default) Activate the qos-change-trigger for any type of QoS change.
`suppress-no-interim-zero-traffic-stop-cdr`	Specifies whether to suppress zero-volume Stop CDRs for short-lived sessions for which there is no Interim CDR generated. *Applies to SGW/PGW ASN.1 and GTPP CDRs only*. ■ `enabled` – The MCC suppresses Stop CDRs for short-lived sessions with zero-volume service data containers (SDCs) for which there is no Interim CDR generated. ■ `disabled` – (default) The MCC generates Stop CDRs for short-lived sessions even if there is no traffic data to report (zero-volume SDCs).
`suppress-zero-traffic-containers`	Specifies whether to suppress the creation of service containers in PGW/GGSN CDRs for service rules with no traffic. *Applies to PGW/GGSN CDRs (CSV/ASN.1 format) only.* ■ `enabled` – When generating CDRs, the MCC does not create service containers for service rules with no traffic. ■ `disabled` – (default) When generating CDRs, the MCC creates service containers for service rules with no traffic. ***Note:*** *All service rules configured as event triggers, such as QoS change or ULI change, will have at least one service container to indicate the event trigger, even when the MCC is configured to suppress the creation of service containers for service rules with no traffic.*

Parameter	Description
unique-cdr-nodeid	Specifies whether the Node ID value in SGW/PGW CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. This feature applies to all offline-charging interfaces (local, Ga, Gz/Rf). ■ **Enabled** – The value for the Node ID field in CDRs consists of the configured gateway service name and the Service Type ID. For example, if the configured gateway service name is `MyPGW` and the Service Type ID is `3`, then the Node ID = `MyPGW_3`. For all CDR types (local, GTPP and Gz/Rf), the Local Sequence Number field will be incremental for all 32 bits. The Local Sequence Number field will not contain the Service Type ID. ■ **Disabled** – (default) The value for the Node ID field in CDRs is the gateway service name, configured using **zone <*name*> gateway [pgw|sgw] <*gw-name*>**. For example, if the configured gateway service name is `MyPGW`, then the Node ID = `MyPGW`. For local and GTPP CDRs, the Local Sequence Number field will consist of the incremental counter (20 bits) + the Service Type ID (12 bits). For Gz/Rf CDRs, the Local Sequence Number field will be incremental for all 32 bits. ***Note:*** *After performing a Geographic Redundancy switchover, it is possible that the Task Service Type ID is different and in that case the Node ID value will change in the CDRs for the same subscriber.*

5 (Optional) Specify the SGW mode for generating SGW CDRs.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> sgw cdr-mode
[stand-alone|roam-in|co-located|imsi-range]
```

The following table describes the configurable parameters.

Parameter	Description
`offline-charging`	Specifies the name of the local interface for offline charging.
`cdr-mode`	Specifies the SGW mode for generating SGW CDRs. ■ `stand-alone` – (default) SGW CDRs are generated only for stand-alone SGW sessions (home or roaming). Stand-alone SGW sessions have PGW sessions anchored on a device other than the MCC. ■ `roam-in` – SGW CDRs are generated only for a subset of stand-alone SGW sessions that are roaming-in (the subscriber's PLMN is part of the roaming PLMN list configured on the SGW). *Applies to Gz/Rf and GTPP SGW CDRs only. Does not apply to SGW CDRs generated on the local offline charging interface.* ■ `co-located` – SGW CDRs are generated for stand-alone SGW sessions and co-located SGW sessions (home or roaming). Co-located sessions have both SGW and PGW services located on the same MCC. ■ `imsi-range` – SGW CDRs are generated based on a list of configured IMSI ranges. For more information, see Generating SGW CDRs Based on IMSI ranges (page 1-40).

6 (Optional) Specify the User-Name format used on offline-charging interfaces if not present in the PCO.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> user-name-type
[apn|default-username|imsi|imsi-nai-no-prefix|imsi-nai-
prefix-0|imsi-plus-apn|imsi-plus-apnnionly|msisdn|msisdn-plus-apn|msis
dn-plus-apnnionly|none]
```

The following table describes the configurable parameters.

Parameter	Description
offline-charging	Specifies the name of the local interface for offline charging.
user-name-type	Specifies the User-Name format used on offline-charging interfaces if not present in the Protocol Configuration Options (PCO). This setting overrides the `user-name-type` present at the APN level. However, if this `user-name-type` is set to `none`, the APN level `user-name-type` is used. ■ **apn** ■ **default-username** ■ **imsi** ■ **imsi-nai-no-prefix** ■ **imsi-nai-prefix-0** ■ **imsi-plus-apn** ■ **imsi-plus-apnnionly** ■ **msisdn** ■ **msisdn-plus-apn** ■ **msisdn-plus-apnnionly** ■ **none**

7 Configure the parameters for streaming CDRs locally and add the data record templates to the local offline charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn|wag|epdg] local
```

where `<interface-name>` is the name of the local interface for offline charging and `[pgw|sgw|ggsn|wag|epdg]` specifies the gateway type.

The following table describes the configurable parameters.

Parameter	Description
`asn1-cdr-version`	Specifies a 3GPP 32.298 version used to format ASN.1 local CDR fields in Interim and Stop data records. ■ `6.0` – Fields generated in local Interim and Stop data records are partially compliant to the ASN.1 syntax specified in 3GPP 32.298 version 6.0. The GPRSCallEventRecord value is set to 28 and the CallEventRecordType is set to 70, as per 3GPP 32.298 version 6.0. ■ `8.4` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 8.4. ■ `9.0` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 9.0. ■ `10.0` – Fields generated in local Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 10.0. ***Note:*** *For more information, see Data Record Templates for CDRs in ASN.1 Format (page 1-13).* ***Note:*** *This parameter is only visible when the `encode-format` is set to `asn1` or `both`.*
`cdr-file-name`	Specifies the file name for streaming CDR records locally.
`cdr-file-type`	Specifies the file type for streaming CDR records locally.
`encode-format`	Specifies the format in which encoded CDRs are saved to the local disk. ■ `asn1` ■ `csv` ■ `both` ***Note:*** *WAG and ePDG CDRs are only stored in CSV format.*
`custom-format`	Specifies the format for GGSN local CDRs. ■ `3gpp` ■ `custom`
`start-template`	Specifies the Start data record template name for CDRs in CSV format.

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>stop-template</code></td><td>Specifies the Stop data record template name for CDRs in CSV format.</td></tr>
<tr><td><code>interim-template</code></td><td>Specifies the Interim data record template name for CDRs in CSV format.</td></tr>
<tr><td><code>event-template</code></td><td>Specifies the Event data record template name for CDRs in CSV format.</td></tr>
<tr><td><code>stop-template-asn1</code></td><td>Specifies the Stop (final) data record template name for ASN.1 encoded CDRs.

Note: The ASN.1 Interim and Stop templates are visible only when the <code>encode-format</code> is set to either <code>asn1</code> or <code>both</code>.</td></tr>
<tr><td><code>interim-template-asn1</code></td><td>Specifies the Interim (partial) data record template name for ASN.1 encoded CDRs.

Note: The ASN.1 Interim and Stop templates are visible only when the <code>encode-format</code> is set to either <code>asn1</code> or <code>both</code>.</td></tr>
<tr><td><code>report-epcqos-in-kbps</code></td><td>Specifies whether the EPC QoS Information fields in local (CSV/ASN.1) and GTPP CDRs should report values in Kilobits-per-second (Kbps).
<code>enabled</code> – The EPC QoS information fields in CDRs are filled as follows:
Sets the current (non-extended) bit rates in Kbps.
Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1.
<code>disabled</code> – (default) The EPC QoS information fields in CDRs are filled as follows:
Sets the current (non-extended) bit rates in bps. For bandwidth values greater than 2^32-1, sets the value to 2^32-1.
Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1.
Note: See Configuring EPC QoS Information Reporting for 5G NSA (page 1-51) for more information.</td></tr>
</table>

Parameter	Description
`report-time-usage-on-zero-traffic`	Specifies whether the Time First Usage, Time Last Usage, and Time Usage CDR subfields should be set in Service Data Containers even if there is no data to be reported. *Applies to PGW local ASN.1 and GTPP CDRs only.* ■ `enabled` – If there is no traffic data to report, the Time First Usage and the Time Last Usage subfields are set to the Time of Container Report and the Time Usage subfield is set to 0 in Stop and Interim PGW local ASN.1 and GTPP CDRs. ■ `disabled` – (default) If there is no traffic data to report, the Time First Usage, Time Last Usage, and Time Usage subfields are not included in Stop and Interim PGW local ASN.1 and GTPP CDRs. ***Note:*** *These CDR subfields are contained within the List of Service Data field, which is known as Change of Service Condition in ASN.1 format.*
`ie-customization`	Allows you to customize certain fields in ASN.1 CDRs and to specify whether to include the record sequence number in ASN.1 CDRs. *Applies to local ASN.1 and GTPP CDRs only*. See Customizing ASN.1 CDR Fields (page 1-42).
`error-cdr-file-name`	*Applies to local ASN.1 CDRs only.* Specifies the file name for streaming ASN.1 CDRs locally when the CDR is categorized as error. See Categorizing Local ASN.1 CDRs as Invalid or Error (page 1-45) for more information.
`invalid-cdr-file-name`	*Applies to local ASN.1 CDRs only.* Specifies the file name for streaming ASN.1 CDRs locally when the CDR is categorized as invalid. See Categorizing Local ASN.1 CDRs as Invalid or Error (page 1-45) for more information.

8 Perform the following steps to complete the CDR configuration:

a For the PGW, GGSN, TWAG and ePDG, bind the offline charging interface to a workflow control profile and ensure the workflow control profile is bound to a subscriber analyzer.

b For the SGW, configure the local offline charging interface directly on the SGW object using the following command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
default-offline-interface local-interface <interface-name>
```

Note: *Enter a question mark (?) after* `local-interface` *to display a list of available local offline interfaces.*

The workflow control profile and SGW support a maximum of two offline charging interfaces, one for local CDRs and one for streaming to the OFCS (Diameter or GTPP). For information on the workflow control profile, see the *Affirmed Networks Gateway Administration Guide.*

Note: *If the Control Profile with which an offline rating group is associated does not have a* `default-offline-interface` *configured, the MCC still allows the PDN session to be established and allows offline charging traffic. If the Control Profile has a local CDR interface configured, the MCC will generate CDRs and store them in a local file.*

Optional CDR Configuration Tasks

This section describes optional CDR configuration tasks.

Uncompressing Data Records in zlib raw Format

If you select `raw` as the compress-format for data records, you must use the Affirmed Networks uncompress utility to uncompress the data records. Affirmed Networks recommends copying the data records to a local server/directory to which you have write permissions. You must be running Linux 2.6.22 or higher to use this utility.

The utility is located on the MCC as follows:

```
/tos/<release_number>/opt/thoroughbred/bin/affirmedUnzip.
```

Copy this utility to the location on your data records server where the compressed data records are being SFTPed. You can run this utility offline. For example:

```
maint@an3000-mcm-slot7cpu1:~# affirmedUnzip

linux: affirmedUnzip <file | directory> <file | directory> ...
```

This utility uncompresses files using the ZLIB RAW option (files with .zraw extension). For example:

```
-rw-r--r-- 1 maint maint 10486201 Mar 13 12:42
bob20150313123001_132.csv.zraw
```

When a directory is entered, the utility uncompresses all `*.zraw` files in that directory. Use a wildcard in the file name to uncompress multiple files matching a certain file name. For example:

```
linux:~# affirmedUnzip
/var/log/datarecord/test/testcdr20150311162909_2244.csv.zraw

Uncompressing testcdr20150311162909_2244.csv.zraw
..................... SUCCEEDED

.(6.0.0.0-458.REL)maint@an3000-mcm-slot7cpu1:~#
```

Generating SGW CDRs for Co-Located SGW/PGW

MCC enables the SGW and PGW to operate as standalone services on separate devices or as combined services running on the same device. If the SGW subscriber session is using a PGW operating as a combined SGW/PGW system, only PGW CDRs are generated by default. However, you can configure the MCC to generate SGW CDRs for SGW subscriber sessions with co-located PGW sessions.

Co-located sessions have both SGW and PGW services located on the same MCC. If this functionality is enabled, SGW CDRs are generated as per the offline interface configuration on the SGW for co-located sessions. If disabled, the CDRs are generated only for standalone SGW sessions. Standalone SGW sessions have PGW sessions anchored on a device other than the MCC.

Use the following CLI command to enable the MCC to generate co-located SGW CDRs.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> sgw cdr-mode co-located
```

Note: *To generate SGW CDRs for co-located SGW/PGW sessions based on a list of configured IMSI ranges, see* *Generating SGW CDRs Based on IMSI ranges (page 1-40)**.*

Generating SGW CDRs Based on IMSI ranges

This feature adds the ability to generate SGW CDRs based on a list of configured IMSI ranges. This feature applies to all offline-charging interfaces (local, Ga, Gz) and both standalone and co-located SGW sessions.

Note: *Standalone SGW sessions have PGW sessions anchored on a device other than the MCC. Co-located sessions have both SGW and PGW services located on the same MCC.*

When the SGW receives a Create Session Request and the SGW `cdr-mode` is configured as `imsi-range`, the SGW tries to match the subscriber's IMSI with any of the IMSI ranges within the list. If an entry is found, the SGW generates CDRs for that session. If an entry is not found, the SGW does not generate CDRs for the session.

To generate SGW CDRs based on a list of configured IMSI ranges, all offline-charging interfaces configured on an MCC SGW must have the `cdr-mode` configured as `imsi-range` and must be configured with the same IMSI range list.

If an IMSI for a session, which was not generating SGW CDRs initially, is added to the IMSI list, SGW CDRs are generated for subsequent sessions from the same IMSI. However, sessions that exist before an IMSI range is added to a list will not generate SGW CDRs.

Set parameters for this feature using the following commands:

1. Configure the IMSI range.

   ```
   an3000-mcm-slot7cpu1(config)# service-construct imsi-range
   <imsi-range-name> start <imsi-start-value> end <imsi-end-value>
   ```

2. Configure the list of IMSI ranges.

   ```
   an3000-mcm-slot7cpu1(config)# service-construct imsi-range-list
   <imsi-list-name> imsiRangeList <imsi-range-names>
   ```

3. Enable the MCC to generate SGW CDRs based on a list of configured IMSI ranges.

   ```
   an3000-mcm-slot7cpu1(config)# service-construct interface
   offline-charging <interface-name> sgw cdr-mode imsi-range
   ```

4. Associate the IMSI range list with the offline-charging interface.

   ```
   an3000-mcm-slot7cpu1(config)# service-construct interface
   offline-charging <interface-name> sgw offline-charging-imsi-list
   <imsi-list-name>
   ```

Note: *The `offline-charging-imsi-list` parameter is only visible when `cdr-mode` is set to `imsi-range`.*

---

5 Ensure the local and offline charging interfaces are associated with the SGW object.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
default-offline-interface [local-interface|offline-interface]
<interface-name>
```

Viewing Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface level.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

```
sdc-total-uplink-bytes/sdc-total-downlink-bytes

sdc-total-uplink-packets/sdc-total-downlink-packets

total-uplink-bytes/total-downlink-bytes

total-uplink-packets/total-downlink-packets
```

Customizing ASN.1 CDR Fields

Customizing the PDP/PDN Type CDR Field

Use the following command to customize the PDP/PDN Type ASN.1 CDR field.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] [local|gtpp]
ie-customization pdp-pdn-type pdp-pdn-type-prefix <value>
populate-pdn-type-in-pdp-type [enabled|disabled]
```

where

Parameter	Description
pdp-pdn-type-prefix	*Applies to local ASN.1 and GTPP CDRs only.* Specifies the prefix used to fill in the spare bits of the PDP/PDN Type field in ASN.1 CDRs. The default prefix for the PDP/PDN Type field is 0xF, as specified by 3GPP.
populate-pdn-type-in-pdp-type	*Applies to local ASN.1 and GTPP CDRs only.* Specifies whether the PDP/PDN Type field in ASN.1 CDRs is populated with 3G or 4G PDP types. ■ **enabled** – The PDP/PDN Type field is populated with 4G PDP types. When enabled, the following PDN type values are defined: – IPv4 – 0x01 – IPv6 – 0x02 – IPv4/IPv6 – 0x03 ■ **disabled** – (default) The PDP/PDN Type field is populated with 3G PDP types. See PDP/PDN Type (page 3-27).

Customizing the Encoding Format for the APN NI CDR Field

Use the following command to specify the ASN.1 encoding format for the Access Point Name Network Identifier (APN NI) ASN.1 CDR field.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] [local|gtpp]
ie-customization apn-ni format-per-3gpp [enabled|disbaled]
```

where

Parameter	Description
apn-ni format-per-3gpp	*Applies to local ASN.1 and GTPP CDRs only.* Specifies the ASN.1 encoding format for the Access Point Name Network Identifier (APN NI) ASN.1 CDR field. ■ **enabled** – The APN NI ASN.1 CDR field is encoded as defined in 3GPP 23.003. The APN NI may contain multiple labels and each label is coded as a one octet length field followed by that number of octets coded as 8-bit ASCII characters. ■ **disabled** (default) – The APN NI ASN.1 CDR field is encoded with all the labels in ASCII characters and each label is separated by a dot (.).

Customizing the Record Sequence Number in Local ASN.1 and GTPP CDRs

Use the following command to customize the record sequence number in local ASN.1 and GTPP CDRs.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] [local|gtpp]
ie-customization record-sequence-number
include-only-if-interim-was-sent [enabled|disabled]
start-record-sequence-number-from-zero [enabled|disabled]
```

where

Parameter	Description
include-only-if-interim-was-sent	*Applies to local ASN.1 and GTPP CDRs only.* Specifies whether to include the record sequence number in ASN.1 CDRs. If enabled, the record sequence number will only be included in local ASN.1 and GTPP Stop CDRs if at least one Interim CDR was sent prior to the Stop CDR. ■ **enabled** ■ **disabled** (default)

Parameter	Description
start-record-sequence-number-from-zero	*Applies to local ASN.1 and GTPP CDRs only.* Specifies whether the record sequence number in local ASN.1 and GTPP CDRs should start with 0. ■ **enabled** – The record sequence number in local ASN.1/GTPP CDRs (whichever is generated first) will start with 0. ■ **disabled** – The record sequence number in local ASN.1/GTPP CDRs (whichever is generated first) will start with 1.

Customizing the IMEISV Within the Served IMEISV CDR field

Use the following command to customize the IMEISV sent within the Served IMEISV CDR field on the Ga and local (ASN.1) interfaces.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw] [local|gtpp]
ie-customization served-imeisv allow-zero-value [enabled|disabled]
```

where

Parameter	Description
ie-customization served-imeisv allow-zero-value	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the Served IMEISV CDR field. *Applies to local ASN.1 and GTPP SGW/PGW CDRs only.* ■ **enabled** – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the Served IMEISV CDR field on the Ga and local (ASN.1) interfaces. ■ **disabled** – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the Served IMEISV CDR field on the Ga and local (ASN.1) interfaces.

Categorizing Local ASN.1 CDRs as Invalid or Error

Use the following steps to categorize local ASN.1 CDRs as either invalid or error if a specified set of fields are not present in the CDR. Using the `presentFlag` setting under GTPP Interim and Stop CDR data record templates, Operators can categorize fields as `mandatoryElseError` or `mandatoryElseInvalid`. If a field, categorized with either of these settings, is missing from the CDR, the system categorizes the CDR accordingly and stores it in a predefined file under the *datarecord* directory.

1 Create two local files; one to store local ASN.1 CDRs categorized as invalid and another to store local ASN.1 CDRs categorized as error.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
datarecord <filename-prefix>
```

2 Use the following command to set the `presentFlag` setting for a GTPP Interim and Stop CDR field.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|sgw|ggsn] ofcs gtpp [interim|stop] <template-name> <field>
presentFlag [mandatoryElseError|mandatoryElseInvalid]
```

3 Use the following CLI command to specify the file names on the local offline-charging interface for streaming ASN.1 CDRs locally when the CDR is categorized as invalid or error.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] local
[error-cdr-file-name|invalid-cdr-file-name] <filename>
```

Note: *3GPP mandatory fields, which are not configurable in data record templates, are categorized as error if the field is missing from the CDR.*

Customizing the APN Reported on External Interfaces

Operators can customize the Access Point Name (APN) reported on external interfaces. The customized APN is used in the Called-Station-ID AVP and sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- Credit-Control-Request (CCR) on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the customized APN-reporting value is filled in the Access Point Name Network Identifier CDR field.

Set parameters for this feature using the following commands:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
apn-reporting custom custom-reporting-apn-name <apn-name>
```

To enable this feature, set the `apn-reporting parameter` to `custom`. The value set in the `custom-reporting-apn-name` parameter is used to fill the Called-Station-Id value in messages to external interfaces.

Note: *The `custom-reporting-apn-name` parameter is visible only if `apn-reporting` is set to `custom`.*

Use the following command to display the reported APN name (`reporting-apn-name`) sent as the Called-Station-Id AVP on external interfaces.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id>
```

Configuring APN Reporting at the Interface Level

You can configure apn-reporting at the interface level to allow for different Access Point Names (APNs) to be reported over different external interfaces. The `apn-reporting` value configured at the interface level is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the `apn-reporting` value configured at the interface level is filled in the Access Point Name Network Identifier CDR field.

The `apn-reporting` value configured at the interface level overrides the `apn-reporting` value configured for the Control-Profile. If the `apn-reporting` value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the `apn-reporting` value configured for the Control Profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <name> pgw apn-reporting
[original|mapped|prior-mapped]
```

Note: *The interface-level* `apn-reporting` *parameter does not support the* `custom` *option and does not apply to the S6b interface.*

Note: *This feature applies to the PGW only.*

Excluding the Service ID in Service Data Containers

Each service rule is configured with a service ID (`service-data-flow-id`), which is used in CDRs to identify the service rule. The MCC creates a separate service data container (SDC) for each service ID when generating CDRs. Use the following CLI command to specify whether to include or exclude the service ID in SDCs.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule
<rule-name> include-service-data-flow-id-in-offline-sdc
[enabled|disabled]
```

- **`enabled`** – (default) When generating CDRs, the MCC includes the service ID (service data flow ID) in SDCs.
- **`disabled`** – When generating CDRs, the MCC excludes the service ID (service data flow ID) in SDCs.

The following guidelines apply when configuring the MCC to exclude the service ID in SDCs:

- Applies to static and predefined service rules only, which are pre-configured on the MCC.
- Applies to Gz/Rf, Ga (GTPP), and local (CSV/ASN.1) CDRs.
- The offline-charging `cdr-report-level` must be set to `bearer`.
- The rating group `reporting-level` must be set to `serviceRule`.
- Configure the same service ID (`service-data-flow-id`) for all service rules under each rating group that does not need the service ID reported in SDCs. Only the rating group will be included in the SDC along with the usage information for all service rules associated with that rating group. However, the service ID will not be included in the SDC.
- Ensure the service ID (`service-data-flow-id`) for service rules configured with the `include-service-data-flow-id-in-offline-sdc` parameter set to `disabled` does not overlap with another service rule which has the `include-service-data-flow-id-in-offline-sdc` parameter set to `enabled`.
- Suppress the Service-Identifier AVP under the Multiple-Services-Credit-Control AVP in Gy interface data record templates.

Transferring CDR Files Through North/South Interface

You can use a separate network-context loopback-ip to enable file-management export users to access CDR files without having to connect through the standard management-context and mgmt-ip address.

You can enable and disable the network-context to act as an SFTP login port. Connections are made through the loopback-ip address on the north-south subnet using port 2221, for example:

```
sftp -P 2221 admin@<loopback-ip>
```

Note: *Some export transfers may experience lower performance rates using this port.*

The infrastructure file-management rule requires that both an `export-rule` and `export-user` be defined for export users to access the network-context address through SFTP.

Note: *Depending on the loopback-ip address you choose for the export-ip network context, you may need to create a default route on the client machine to forward traffic intended for the loopback-ip to the network-context's interface-ip address. The system will recognize this address as the correct route.*

Issue the following commands:

1 Create a loopback-ip on the network context:

```
an3000-mcm-slot7cpu1(config)# network-context <network-context-name>
loopback-ip <loopback-ip-name> ip-address <ip-address>
```

2 Associate the loopback-ip with the export object:

```
an3000-mcm-slot7cpu1(config)# infrastructure export-ip mgmtip
network-context <network-context-name> loopback-ip <loopback-ip-name>
```

3 Configure the export rule to apply when exporting the file to the */export* directory.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-rule
[exportrule-default|exportrule-gy-custom1|exportrule-gz-custom1|export
rule-none]
```

Where:

- `exportrule-default` – The system exports files to the */export* directory and applies purge rules for this file profile to all files in the directory.
  - Gz/Rf CDRs are exported to */public/logs/datarecord.lnk/gz/export*
  - Gy CCR-Terminate records are exported to */public/logs/datarecord.lnk/gy/export*
  - Local CDRs are exported to */public/logs/datarecord.lnk/export*
- `exportrule-gy-custom1` – The system exports files to the */public/logs/datarecord.lnk/gy/export/AP_primary* directory according to the custom1 format configured for CCR-T records. See the *Affirmed Networks Gy Interface Specification* for more information on this option.
- `exportrule-gz-custom1` – The system exports files to the */public/logs/datarecord.lnk/gz/export* directory according to the custom1 format configured for Gz/Rf SGW and PGW CDRs. See Storing Gz/Rf CDRs to a File in the Event of a Failure (page 6-22) for more information on this option.
- `exportrule-none` – Specifies that there is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.

4 Specify the export user that has read/write access to the files in the */export* directory.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<type> <filename-prefix> export-user <string>
```

Show Command

```
an3000-mcm-slot7cpu1# show running-config infrastructure export-ip
mgmtip network-context GN loopback-ip fmExportLoopback
```

Configuring EPC QoS Information Reporting for 5G NSA

The following local (ASN.1/CSV) and GTPP QoS information CDR subfields were added to support the higher throughput rates required for the 5G non-standalone (NSA) solution, as defined in 3GPP 32.298 version 15.2.0.

- Extended Max Requested BW UL
- Extended Max Requested BW DL
- Extended GBR UL
- Extended GBR DL
- Extended APN AMBR UL
- Extended APN AMBR DL

These EPC QoS Information CDR subfields are configurable in local ASN.1 and GTPP (Ga interface) Interim and Stop data record templates. The default `presentFlag` setting is `no`, indicating the data field is not present in CDRs. For example:

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[sgw|pgw] ofcs gtpp [interim|stop] <template-name>
changeofcharconditionepcqosinformationextendedapnambrdl presentFlag
[yes|no|conditional
```

The EPC QoS Information CDR subfields appear as the following in local ASN.1 and GTPP Interim and Stop data record templates:

- `changeofcharconditionepcqosinformationextendedmrbul`
- `changeofcharconditionepcqosinformationextendedmrbdl`
- `changeofcharconditionepcqosinformationextendedgbrul`
- `changeofcharconditionepcqosinformationextendedgbrdl`
- `changeofcharconditionepcqosinformationextendedapnambrul`
- `changeofcharconditionepcqosinformationextendedapnambrdl`

- `changeofserviceconditionqosinformationnegextendedmrbul`
- `changeofserviceconditionqosinformationnegextendedmrbdl`
- `changeofserviceconditionqosinformationnegextendedgbrul`
- `changeofserviceconditionqosinformationnegextendedgbrdl`
- `changeofserviceconditionqosinformationnegextendedapnambrul`
- `changeofserviceconditionqosinformationnegextendedapnambrdl`

Note: *The EPC QoS Information subfields are not configurable in local CSV data record templates and therefore are present in CDRs within the QoS Information field.*

Setting the EPC QoS Information Reporting Value

Use the following command to specify whether the EPC QoS Information fields in local (CSV/ASN.1) and GTPP CDRs should report values in Kilobits-per-second (Kbps).

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw] [local|gtpp]
report-epcqos-in-kbps [enabled|disabled]
```

- **`enabled`** – The EPC QoS information fields in CDRs are filled as follows:
  - Sets the current (non-extended) bit rates in Kbps.
  - Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1.
- **`disabled`** – (default) The EPC QoS information fields in CDRs are filled as follows:
  - Sets the current (non-extended) bit rates in bps. For bandwidth values greater than 2^32-1, sets the value to 2^32-1.
  - Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1.

Configuring the Node ID in CDRs

This feature adds a new configuration option used to configure the Node ID value under the gateway service, which will be used as the Node ID value in local (ASN.1 and CSV), GTPP and Gz/Rf SGW/PGW/GGSN CDRs. If this parameter is not configured, the gateway service name is used as the value for the Node ID field in SGW/PGW/GGSN CDRs.

```
an3000-mcm-slot7cpu1(config)# zone default gateway [sgw|pgw|ggsn]
<gateway-service-name> node-id <node-id-string>
```

Note: *The* `node-id` *parameter under the gateway service should not be configured or changed mid-session. If it is, the node-id in subsequent CDRs will contain the newly configured value.*

There is an existing configuration parameter under the offline-charging interface that allows you to specify whether the Node ID value in CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [sgw|pgw|ggsn] unique-cdr-nodeid
[enabled|disabled]
```

Table 1-2 summarizes the MCC behavior when using the `node-id` parameter under the gateway service and the `unique-cdr-nodeid` parameter under the offline-charging interface together.

Table 1-2 Configuring the `node-id` and `unique-cdr-nodeid` parameters together

zone default gateway [sgw|pgw|ggsn] <*gateway-service-name*> node-id <*node-id-string*>	**service-construct interface offline-charging <*offline-charging-name*> [pgw|sgw|ggsn] unique-cdr-nodeid [enabled|disabled]**	**Value of Node ID in CDR**
`<node-id-string>` configured	`enabled`	`<node-id-string>_Service-Type-ID`
`<node-id-string>` not configured	`enabled`	`<gateway-service-name>_Service-Type-ID`
`<node-id-string>` not configured	`disabled`	`<gateway-service-name>`
`<node-id-string>` configured	`disabled`	`<node-id-string>`

Chapter 2

Local CDRs

This section lists the local CDR fields and file formats supported by the system for the CSV format. For a list of ASN.1 supported CDR fields, see ASN.1 Supported Fields (page 5-41). For a detailed description of these CDR fields, see Chapter 3, CDR Descriptions.

- SGW CDRs
- PGW CDRs
- GGSN CDRs
- WAG CDRs
- ePDG CDRs
- GTP Proxy CDRs
- C-SGN CDRs

SGW CDRs

Table 2-1 lists the field, name, format and default position for the SGW CDRs. For a detailed description, see Chapter 3, CDR Descriptions.

Table 2-1 SGW Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
acctrecordtype	Accounting Record Type	Decimal	1	1	1	1
recordtype	Record Type	Decimal	2	2	2	2
servedimsi	Served IMSI	String	3	3	3	3
imsiunauthenticatedflag	IMSI Unauthenticated Flag	Decimal	4	4	4	4
servedimeisv	Served IMEISV	String	5	5	5	5
sgwaddress	SGW Address	Dotted IP address/Hex	6	6	6	6
chargingid	Charging ID	Decimal	7	7	7	7
pdnconnectionid	PDN Connection ID	Decimal	8	8	8	8
servingnodeaddr	Serving Node Address	Dotted IP address	9	9	9	9
sgwchange	SGW Change	Decimal	10	-	-	-
pgwplmnid	PGW PLMN ID	String	11	10	10	10
apnid	Access Point Name Network Identifier	String	12	11	11	11
pdptype	PDP/PDN Type	Decimal	13	12	12	12
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	14	13	13	13
dynamicaddressflag	Dynamic Address Flag	Decimal	15	14	14	14
trafficdatavolume	List of Traffic Data Volumes	Contents are filled according to the Diameter Traffic-Data-Volume AVP definition	-	15	15	15
recordopeningtime	Record Opening Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	16	16	16
mstimezone	MS Time Zone	String	16	17	17	17
duration	Duration	Decimal	-	29	18	-
causeforrecclosing	Cause for Record Closing	Decimal	-	32	19	-
diagnostics	Diagnostics	Decimal	-	-	20	-
nodeid	Node ID	String	17	18	21	18

Table 2-1 SGW Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
localsequencenumber	Local Sequence Number	Decimal	18	19	22	19
apnselectionmode	APN Selection Mode	Decimal	19	20	23	20
servedmsisdn	Served MSISDN	String	20	21	24	21
userlocationinfo	User Location Information	Default (Binary-Coded Decimal (BCD) encoding)	21	22	25	22
usercsginfo	User CSG Information	Default (BCD encoding)	22	23	26	23
chargingcharacteristics	Charging Characteristics	Hex	23	24	27	24
chargingselectionmode	Charging Selection Mode	Decimal	24	25	28	25
pgwaddress	PGW Address	Dotted IP address/Hex	25	26	29	26
servnodeplmnid	Serving Node PLMN Identifier	Default (BCD encoding)	26	27	30	27
rattype	RAT Type	Decimal	27	28	31	28
stoptime	Stop Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	-	32	-
starttime	Start Time	time-secs-YYYY-MM-DDTH H:MM:SS	28	30	33	-
servingnodeipaddr	Serving Node IP Address	Dotted IP address/Hex	29	31	34	29
sgwaddressv6	SGW Address (IPv6)	Dotted IP address/Hex	30	33	35	30
recordsequencenumber	Record Sequence Number	Decimal	31	34	36	31
servedpdpaddrv6	Served PDP Address (IPv6)	Dotted IP address/Hex	32	35	37	32
pgwaddressv6	PGW Address (IPv6)	Dotted IP address/Hex	33	36	38	33
lowpriorityindicator	Low Priority Indicator	Decimal	34	-	-	-
servingplmnratecontrol	Serving PLMN Rate Control	Default (BCD encoding)	35	38	40	-
cpciotepsoptimisationindicator	CP CIoT EPS Optimisation Indicator	Decimal	36	37	39	-
unipducponlyflag	UNI PDU CP Only Flag	Decimal	37	39	41	-
pdppdntypeextension	PDP/PDN Type Extension	Decimal	38	40	42	-
secondaryratusagereports	List of RAN Secondary RAT Usage Reports	See TS 3GPP 32.298 Version 15.2.0 for encoding details.	-	41	43	-

PGW CDRs

Table 2-2 lists the field, name, format, and default position for the PGW CDRs. For a detailed description, see Chapter 3, CDR Descriptions.

Table 2-2 PGW Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
acctrecordtype	Accounting Record Type	Decimal	1	1	1	1
recordtype	Record Type	Decimal	2	2	2	2
servedimsi	Served IMSI	String	3	3	3	3
imsiunauthenticatedflag	IMSI Unauthenticated Flag	Decimal	4	4	4	4
servedimeisv	Served IMEISV	String	5	5	5	5
served3gpp2meid	Served 3GPP 2 MEID (Mobile Equipment Identifier)	String	6	6	6	6
servedmnnai	Served MNNAI (Served Mobile Node identifier in Network Access Identifier format (based on IMSI), if available)	String	7	7	7	7
pgwaddress	PGW Address	Dotted IP address/Hex	8	8	8	8
chargingid	Charging ID	Decimal	9	9	9	9
pdnconnectionid	PDN Connection ID	Decimal	10	10	10	10
servingnodeaddr	Serving Node Address	Dotted IP address	11	11	11	11
pgwplmnid	PGW PLMN ID	String	12	12	12	12
apnid	Access Point Name Network Identifier	String	13	13	13	13
pdptype	PDP/PDN Type	Decimal	14	14	14	14
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	15	15	15	15
dynamicaddressflag	Dynamic Address Flag	Decimal	16	16	16	16
servicedata	List of Service Data ***Note:*** *Filled if service data flow (SDF) charging is configured. See the Affirmed Networks Gateway Administration Guide for more information on configuring charging components, such as rating groups, billing plans and charging instances.*	Contents are filled according to the Diameter Service-Data-Container AVP definition	-	17	17	17

Table 2-2 PGW Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
trafficdatavolume	List of Traffic Data Volumes	Contents are filled according to the Diameter Traffic-Data-Volume AVP definition	-	18	18	18
recordopeningtime	Record Opening Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	19	19	19
mstimezone	MS Time Zone	String	17	20	20	20
duration	Duration	Decimal	-	33	21	-
causeforrecclosing	Cause for Record Closing	Decimal	-	38	22	-
diagnostics	Diagnostics	Decimal	-	-	23	-
nodeid	Node ID	String	18	21	24	21
localsequencenumber	Local Sequence Number	Decimal	19	22	25	22
apnselectionmode	APN Selection Mode	Decimal	20	23	26	23
servedmsisdn	Served MSISDN	String	21	24	27	24
userlocationinfo	User Location Information	Default (BCD encoding)	22	25	28	25
usercsginfo	User CSG Information	Default (BCD encoding)	23	26	29	26
gpp2userlocationinfo	3GPP2 User Location Information	Default (BCD encoding)	24	27	30	27
chargingcharacteristics	Charging Characteristics	Hex	25	28	31	28
chargingselectionmode	Charging Selection Mode	Decimal	26	29	32	29
servnodeplmnid	Serving Node PLMN Identifier	Default (BCD encoding)	27	30	33	30
psfurnishcharging	PS Furnish Charging Information (Online Charging Session Information)	Default (BCD encoding)	28	31	34	31
rattype	RAT Type	Decimal	29	32	35	32
stoptime	Stop Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	-	36	-
starttime	Start Time	time-secs-YYYY-MM-DDTH H:MM:SS	30	34	37	-
mvnoresellerid	MVNO Reseller ID	UTF8String	31	35	38	33
mvnosubclassid	MVNO Sub Class ID	UTF8String	32	36	39	34
servingnodeipaddr	Serving Node IP Address	Dotted IP address/Hex	33	37	40	35
recordsequencenumber	Record Sequence Number	Decimal	34	39	41	36

Table 2-2 PGW Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
pgwaddressv6	PGW Address (IPv6)	Dotted IP address/Hex	35	40	42	37
servedpdpaddrext	Served PDP Address Extension	Dotted IP address/Hex	-	41	43	38
servedpdpaddrv6	Served PDP Address (IPv6)	Dotted IP address/Hex	36	42	44	39
lowpriorityindicator	Low Priority Indicator	Decimal	37	-	-	-
sgiptptunnelingmethod	SGi PtP Tunnelling Method	Decimal	38	43	45	-
unipducponlyflag	UNI PDU CP Only Flag	Decimal	39	44	46	-
servingplmnratecontrol	Serving PLMN Rate Control	Default (BCD encoding)	40	45	47	-
apnratecontrol	APN Rate Control	Default (BCD encoding)	41	46	48	-
pdppdntypeextension	PDP/PDN Type Extension	Decimal	42	47	49	-
scsasaddress	SCS/AS Address	Default (BCD encoding)	43	48	50	-
secondaryratusagereports	List of RAN Secondary RAT Usage Reports	See TS 3GPP 32.298 Version 15.2.0 for encoding details.	-	49	51	-
uwanuserlocationinfo	UWAN User Location Information	Default (BCD encoding)	44	50	52	40

GGSN CDRs

GGSN CDRs support the following formats for generating CDR files:

- Standard 3GPP Format
- Custom1 Format

Standard 3GPP Format

Table 2-3 lists the field, name, format, and default position for 3GPP GGSN CDRs.

Table 2-3 GGSN Local CDR Fields and Formats (3GPP)

Field	Name	Format	Start	Interim	Stop	Event
acctrecordtype	Accounting Record Type	Decimal	1	1	1	1
recordtype	Record Type	Decimal	2	2	2	2
ntwkinitiatedpdp	Network Initiated PDP	Decimal	3	3	3	3
servedimsi	Served IMSI	String	4	4	4	4
servedimeisv	Served IMEISV	String	5	5	5	5
ggsnaddress	GGSN Address (GGSN Address Used)	Dotted IP address/Hex	6	6	6	6
chargingid	Charging ID	Decimal	7	7	7	7
servingnodeaddr	Serving Node Address	Dotted IP address	8	8	8	8
apnid	Access Point Name Network Identifier	String	9	9	9	9
pdptype	PDP/PDN Type	Decimal	10	10	10	10
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	11	11	11	11
dynamicaddressflag	Dynamic Address Flag	Decimal	12	12	12	12
trafficdatavolume	List of Traffic Data Volumes	Contents are filled according to the Diameter Traffic-Data-Volume AVP definition	-	13	13	13
recordopeningtime	Record Opening Time	time-secs-YYYY-MM-DDTH H:MM:SS	23	14	14	14
mstimezone	MS Time Zone	String	13	15	15	15
nodeid	Node ID	String	14	16	16	16
localsequencenumber	Local Sequence Number	Decimal	15	17	17	17

Table 2-3 GGSN Local CDR Fields and Formats (3GPP)

Field	Name	Format	Start	Interim	Stop	Event
apnselectionmode	APN Selection Mode	Decimal	16	18	18	18
servedmsisdn	Served MSISDN	String	17	19	19	19
userlocationinfo	User Location Information	Default (BCD encoding)	18	20	20	20
chargingcharacteristics	Charging Characteristics	Hex	19	21	21	21
chargingselectionmode	Charging Selection Mode	Decimal	20	22	22	22
servnodeplmnid	Serving Node PLMN Identifier	Default (BCD encoding)	21	23	23	23
rattype	RAT Type	Decimal	22	24	24	24
starttime	Start Time	time-secs-YYYY-MM-DDTH H:MM:SS	24	-	-	-
duration	Duration	Decimal	38	38	38	38
causeforclosing	Cause for Record Closing	Decimal	39	39	39	39
psfurnishcharging	PS Furnish Charging Information (Online Charging Session Information)	Default (BCD encoding)	46	46	46	46
diagnostics	Diagnostics	Decimal	-	-	47	-
stoptime	Stop Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	-	48	-
servicedata	List of Service Data ***Note:*** *Filled if service data flow (SDF) charging is configured. See the Affirmed Networks Gateway Administration Guide for more information on configuring charging components, such as rating groups, billing plans and charging instances.*	Contents are filled according to the Diameter Service-Data-Container AVP definition	-	48	50	48
mvnoresellerid	MVNO Reseller ID	UTF8String	48	49	52	49
mvnosubclassid	MVNO Sub Class ID	UTF8String	49	50	53	50
servingnodeipaddr	Serving Node IP Address	Dotted IP address/Hex	50	51	54	51
usercsginfo	User CSG Information	Default (BCD encoding)	51	52	55	52
ggsnaddressv6	GGSN Address (IPv6)	Dotted IP address/Hex	52	53	56	53
servedpdpaddrext	Served PDP Address Extension	Dotted IP address/Hex	53	54	57	54

Custom1 Format

Table 2-4 lists the CDR fields available to the custom1 format and lists the default position for the CDR fields as they appear in data record templates. However, the field positions are customizable. See Table 2-5 for an example on organizing and structuring CDR fields according to the custom1 format.

Table 2-4 GGSN Custom1 CDR Fields and Default Positions

Field	Name	Format	Size (in Bytes)	Start	Interim	Stop	Event
systemtype	System Type	Decimal	1	25	25	25	25
potentialduplicate	Potential Duplicate	Decimal	1	26	26	26	26
imschargingid	IMS Charging ID	String	50	27	27	27	27
chargingclass	Charging Class	Decimal	8	28	28	28	28
serviceid	Service ID	Decimal	10	29	29	29	29
uplink	Uplink	Decimal	12	30	30	30	30
downlink	Downlink	Decimal	12	31	31	31	31
uplinkhits	Uplink Hits	Decimal	12	32	32	32	32
downlinkhits	Downlink Hits	Decimal	12	33	33	33	33
contentuplink	Content Uplink	Decimal	12	34	34	34	34
contentdownlink	Content Downlink	Decimal	12	35	35	35	35
changecondition	Change Condition	Decimal	3	36	36	36	36
changetime	Change Time	time-secs-YYYYMMDDHHMMSS	14	37	37	37	37
storedclosingcause	Stored Closing Cause	Decimal	3	40	40	40	40
qosnegociatedprofile	QoS Negotiated Profile	String	28	41	41	41	41
uliglt	ULI GLT (Geographic Location Type)	Hex	2	42	42	42	42
ulimncmcc	ULI MNC-MCC (Geographic Location Area)	Default (BCD encoding)	6	43	43	43	43
ulilac	ULI LAC (Location Area Code)	Hex	4	44	44	44	44
ulisac	ULI SAC (Call Identity (CI) or Service Area Code (SAC))	Hex	4	45	45	45	45
pdptypeorg	PDP Type Organization	String	1	47	47	49	47

Table 2-4 GGSN Custom1 CDR Fields and Default Positions

Field	Name	Format	Size (in Bytes)	Start	Interim	Stop	Event
QCI	QCI (QoS Class Identifier)	Decimal	1	-	-	51	-

GGSN CDR Custom1 Format (Example)

Table 2-5 provides an example for organizing and structuring CDR fields according to the custom1 format.

Table 2-5 GGSN CDR Custom1 Format (Example)

Field	Name	Format	Position ID	Size (in Bytes)
recordtype	Record Type	Decimal	1	3
systemtype	System Type	Decimal	2	1
potentialduplicate	Potential Duplicate	Decimal	3	1
localsequencenumber	Local Sequence Number	Decimal	4	12
recordopeningtime	Record Opening Time	time-secs-YYYYMMDDH HMMSS	5	14
servedimsi	Served IMSI	String	6	16
servedimeisv	Served IMEISV	String	7	16
servedmsisdn	Served MSISDN	String	8	18
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	9	32
ggsnaddress	GGSN Address	Dotted IP address/Hex	10	32
servingnodeipaddr	Serving Node IP Address	Dotted IP address/Hex	11	32
imschargingid	IMS Charging ID	String	12	50
chargingid	Charging ID	Decimal	13	10
chargingclass	Charging Class	Decimal	14	8
apnid	Access Point Name Network Identifier	String	15	101
pdptypeorg	PDP Type Organization	String	16	1
pdptype	PDP/PDN Type	Decimal	17	2
servnodeplmnid	Serving Node PLMN Identifier	Default (BCD encoding)	18	6
rattype	RAT Type	Decimal	19	3

Table 2-5 GGSN CDR Custom1 Format (Example)

Field	Name	Format	Position ID	Size (in Bytes)
serviceid	Service ID	Decimal	20	10
chargingcharacteristics	Charging Characteristics	Hex	21	2
nodeid	Node ID	String	22	20
downlink	Downlink	Decimal	23	12
uplink	Uplink	Decimal	24	12
contentuplink	Content Uplink	Decimal	25	12
contentdownlink	Content Downlink	Decimal	26	12
changecondition	Change Condition	Decimal	27	3
causeforrecclosing	Cause for Record Closing	Decimal	28	3
storedclosingcause	Stored Closing Cause	Decimal	29	3
changetime	Change Time	time-secs-YYYYMMDDHHMMSS	30	14
duration	Duration	Decimal	31	12
uplinkhits	Uplink Hits	Decimal	32	12
downlinkhits	Downlink Hits	Decimal	33	12
qosnegociatedprofile	QoS Negotiated Profile	String	34	28
uliglt	ULI GLT (Geographic Location Type)	Hex	35	2
ulimncmcc	ULI MNC-MCC (Geographic Location Area)	Default (BCD encoding)	36	6
ulilac	ULI LAC (Location Area Code)	Hex	37	4
ulisac	ULI SAC (Call Identity (CI) or Service Area Code (SAC))	Hex	38	4
psfurnishcharging	PS Furnish Charging Information (Online Charging Session Information)	Default (BCD encoding)	39	59

WAG CDRs

Table 2-6 lists the field, name, format, and default position for the WAG CDRs. For a detailed description, see Chapter 3, CDR Descriptions.

Note: *WAG local CDRs are only stored in CSV format.*

Table 2-6 WAG Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
acctrecordtype	Accounting Record Type	Decimal	1	1	1	1
recordtype	Record Type	Decimal	2	2	2	2
servedimsi	Served IMSI	String	3	3	3	3
servedmnnai	Served MNNAI	String	4	4	4	4
servedmac	Served MAC	String	5	5	5	5
wagname	WAG Name	String	6	6	6	6
apaddr	Access Point Address	Dotted IP address/Hex	7	7	7	7
aaaname	AAA Name	String	8	8	8	8
chargingid	Charging ID	Decimal	9	9	9	9
pdnconnectionid	PDN Connection ID	Decimal	10	10	10	10
apnid	Access Point Name Network Identifier	String	11	11	11	11
pdptype	PDP/PDN Type	Decimal	12	12	12	12
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	13	13	13	13
localsequencenumber	Local Sequence Number	Decimal	14	17	20	17
servicedata	List of Service Data ***Note:*** *Filled if service data flow (SDF) charging is configured. When configuring SDF charging on the TWAG, ensure the charging-instance charging characteristic is set to 0 (wildcard). See the Affirmed Networks Gateway Administration Guide for more information on configuring charging components, such as rating groups, billing plans and charging instances.*	Contents are filled according to the Diameter Service-Data-Container AVP definition	-	14	14	14

Table 2-6 WAG Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
trafficdatavolume	List of Traffic Data Volumes	Contents are filled according to Diameter Traffic-Data-Volume AVP definition	-	15	15	15
recordopeningtime	Record Opening Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	16	16	16
duration	Duration	Decimal	-	21	17	-
causeforrecclosing	Cause for Record Closing	Decimal	-	-	18	-
diagnostics	Diagnostics	Decimal	-	-	19	-
starttime	Start Time	time-secs-YYYY-MM-DDTH H:MM:SS	15	18	22	-
stoptime	Stop Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	-	21	-
servedmsisdn	Served MSISDN	String	16	19	23	18
servedpdpaddrext	Served PDP Address Extension	Dotted IP address/Hex	17	20	24	19
servedimeisv	Served IMEISV	String	18	22	25	20

ePDG CDRs

Table 2-7 lists the field, name, format, and default position for the MCC ePDG CDRs.

***Note:** ePDG local CDRs are only stored in CSV format.*

Table 2-7 ePDG Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
acctrecordtype	Accounting Record Type	Decimal	1	1	1	1
recordtype	Record Type	Decimal	2	2	2	2
servedimsi	Served IMSI	String	3	3	3	3
servedmnnai	Served MNNAI	String	4	4	4	4
epdgname	ePDG Name	String	5	5	5	5
epdgfqdn	ePDG FQDN	String	6	6	6	6
aaaname	AAA Name	String	7	7	7	7
chargingid	Charging ID	Decimal	8	8	8	8
pdnconnectionid	PDN Connection ID	Decimal	9	9	9	9
apnid	Access Point Name Network Identifier	String	10	10	10	10
pdptype	PDP/PDN Type	Decimal	11	11	11	11
servedpdpaddr	Served PDP Address	Dotted IP address/Hex	12	12	12	12
localsequencenumber	Local Sequence Number	Decimal	13	16	19	16
servicedata	List of Service Data ***Note:** Filled if service data flow (SDF) charging is configured. See the Affirmed Networks Gateway Administration Guide for more information on configuring charging components, such as rating groups, billing plans and charging instances.*	Contents are filled according to Diameter Service-Data-Container AVP definition	-	13	13	13
trafficdatavolume	List of Traffic Data Volumes	Contents are filled according to Diameter Traffic-Data-Volume AVP definition	-	14	14	14

Table 2-7 ePDG Local CDR Fields and Formats

Field	Name	Format	Start	Interim	Stop	Event
recordopeningtime	Record Opening Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	15	15	15
duration	Duration	Decimal	-	19	16	-
causeforrecclosing	Cause for Record Closing	Decimal	-	-	17	-
diagnostics	Diagnostics	Decimal	-	-	18	-
starttime	Start Time	time-secs-YYYY-MM-DDTH H:MM:SS	14	17	21	-
servedpdpaddrext	Served PDP Address Extension	Dotted IP address	15	18	22	17
stoptime	Stop Time	time-secs-YYYY-MM-DDTH H:MM:SS	-	-	20	-
ikeremoteipaddr	IKE Remote IP Address	Dotted IP address	16	20	23	-
ikeremoteport	IKE Remote Port	Decimal	17	21	24	-
ikelocalipaddr	IKE Local IP Address	Dotted IP address	18	22	25	-
ikelocalport	IKE Local Port	Decimal	19	23	26	-
servedimeisv	Served IMEISV	String	20	24	27	18

GTP Proxy CDRs

The GTP Proxy generates local PGW CDRs for both PGW and GGSN GTP Proxy sessions and supports all of the PGW CDR fields listed in Table 2-2. When used for a GTP Proxy session, the following PGW CDR fields are filled as follows:

- PGW Address – Contains the GTP Proxy IPv4 address
- PGW Address (IPv6) – Contains the GTP Proxy IPv6 address (if available)
- Charging ID – Contains the Charging ID generated by the GTP Proxy
- Serving Node Address – Contains the SGW/SGSN IP address

For more information on the GTP Proxy, see the *Affirmed Networks GTP Proxy Administration Guide.*

C-SGN CDRs

The C-SGN generates the following NB-IoT-related SGW and PGW CDRs. For a description of each of these CDR fields, see CDR Descriptions (page 3-1). For a complete list of local SGW and PGW CDRs, see SGW CDRs (page 2-2) and PGW CDRs (page 2-4).

Table 2-8 C-SGN NB-IoT-related SGW and PGW CDRs

C-SGN PGW CDRs	C-SGN SGW CDRs
APN Rate Control	CP CIoT EPS Optimisation Indicator
Low Priority Indicator	Low Priority Indicator
PDP/PDN Type Extension	PDP/PDN Type Extension
SCS/AS Address	Serving PLMN Rate Control
Serving PLMN Rate Control	UNI PDU CP Only Flag
SGi PtP Tunnelling Method	3GPP-RAT-Type
UNI PDU CP Only Flag	List of Traffic Data Volumes (Change of Charging Condition) – CP CIoT EPS Optimisation Indicator – RAT Type – Serving PLMN Rate Control
3GPP-RAT-Type	
List of Service Data (Change of Service Condition) – APN Rate Control – RAT Type – Serving PLMN Rate Control	

Chapter 3

CDR Descriptions

This section describes the following CDR fields in alphabetical order and contains a description, length, format, and example for each CDR.

Note: *For more information on ASN.1 encoded CDR files, see Chapter 5, Ga Interface.*

Note: *Refer to the 3GPP specifications and RFCs listed in References and Specifications (page 2-xxv) for more details.*

Note: *WAG and ePDG local CDRs are only stored in CSV format.*

CDR Descriptions

This section describes the following CDRs based on 3GPP TS 32.298.

Table 3-1 CDR Descriptions

Table 3-1 CDR Descriptions

AAA Name

Description	The name of the Authentication, Authorization, and Accounting (AAA) server. *Applicable only to WAG and ePDG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	AAAGROUP1	N/A

Access Point Address

Description	The data plane IP addresses (IPv4 or IPv6) of the remote access point (AP). *Applicable only to WAG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String (dotted IP address) or Hex	N/A
Example	2001470886520443	N/A

Access Point Name Network Identifier

Description	Logical name of the connected access point to the external packet data network (Network Identifier part of Access Point Name). To report the Network-ID (NI) without the Operator-ID (OI) in the Access Point Name Network Identifier CDR field, the `apn-matching-type` must be set to `ni-only` and there should also be an APN configured that matches with the NI portion of the APN name reported in the CDR. Configure the `apn-matching-type` using the following CLI command: `an3000-mcm-slot7cpu1(config)#` **`zone default gateway profile gateway-common profile-1 apn-matching-type`** You can specify the ASN.1 encoding format for the Access Point Name Network Identifier (APN NI) ASN.1 CDR field. See Customizing ASN.1 CDR Fields (page 1-42) for more information.	
	CSV	**ASN.1 Code (87)**
Length		1-63 Bytes
Format	String	IA5string
Example	consumer.att	870b73746172656e742e636f6d

Accounting Record Type

Description	Identifies the type of accounting record.	
	CSV	**ASN.1 Code (80)**
Length		N/A
Format	Decimal	N/A
Value Range	■ 1 = EVENT ■ 2 = START ■ 3 = INTERIM ■ 4 = STOP	ASN.1 supports partial and full record types based on network events. If the cause is coded as a normal or abnormal release, a full record is generated; otherwise a partial record is generated.
Example	1	N/A

APN Rate Control

Description	Contains the APN Rate Control as specified in 3GPP TS 29.128 [244], which is used during the record for the PDN connection to the SCEF. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0*	
	CSV	**ASN.1 Code** **bf43 (PGW)**
Length		Variable
Format	Default (BCD encoding)	Sequence: ■ APN Rate Control Uplink – Additional Exception Reports – Rate Control Time Unit – Rate Control Max Rate ■ APN Rate Control Downlink – Rate Control Time Unit – Rate Control Max Rate – Rate Control Max Message Size
Example	1, 2, 1000	bf430ba009800101810102820164

APN Selection Mode

Description	An index indicating how the Access Point Name (APN) was selected.	
	CSV	**ASN.1 Code (95)**
Length		1 Byte
Format	Decimal	Enumerated
Value Range	■ 0 = MS or network provided APN, subscription verified ■ 1 = MS provided APN, subscription not verified ■ 2 = Network provided APN, subscription not verified	■ 0 = MS or network provided APN, subscription verified ■ 1 = MS provided APN, subscription not verified ■ 2 = Network provided APN, subscription not verified
Example	1	950101

Application Service Provider Identity

Description	Identifies the application service provider that is delivering a service to an end user. Subfield within the List of Service Data field (List of Service Data (page 3-17)).	
	CSV	**ASN.1 Code (9a)**
Length		Variable
Format	UTF8String	Octet string
Example	sppidpcrf	737070696470637266

CAMEL Charging Information

Description	Contains the Customized Application for Mobile Enhanced Logic (CAMEL) information. This field is present if CAMEL Charging Information is received by the GGSN in the GTP Create PDP Context Request. ***Note:*** *The CAMEL Charging Information CDR field is only applicable for 3G sessions.*	
	CSV	**ASN.1 Code (9f21)**
Length		Variable
Format	N/A	Octet string
Example	N/A	9f210a31323334353637383930

Cause for Record Closing

Description	Indicates the reason for the Charging Data Record (CDR) closing. ASN.1 supports partial and full record types based on network events. If the cause is coded as a normal or abnormal release, a full record is generated; otherwise a partial record is generated. When event triggers, configured with the same recurrence value in the event trigger template, occur simultaneously, the following order of priority is used to fill the Cause for Record Closing field in ASN.1 PGW CDRs: **1** rat-type-change-close-trigger **2** serving-node-change-close-trigger **3** ue-time-zone-change-close-trigger **4** serving-node-plmn-change-close-trigger **5** qos-change-close-trigger **6** user-location-change-close-trigger **7** user-csg-change-close-trigger

	CSV	**ASN.1 Code (8f)**
Length		1 Byte
Format	Decimal	Integer
Value Range	■ 0 = Normal Release – IP-CAN bearer release or detach. ■ 1 = Abnormal Release – Any other abnormal release. ■ 2 = QoS Change ■ 3 = Volume Limit – Configured volume threshold has been exceeded. ■ 4 = Time Limit – Configured interval has been reached. ■ 5 = Serving Node Change – Serving Node address list overflow. ■ 6 = SGSN PLMN ID Change – Change of PLMN-ID. ■ 7 = User Location Change ■ 8 = RAT Change – Change of radio interface from EUTRAN<–>GSM<–> UMTS, and so on. ■ 9 = MS Time Zone Change – MS changes time zone. ■ 10 = Tariff Time Change ■ 11 = Service Idled Out ■ 12 = Service Specific Unit Limit ■ 13 = Max Change Condition – Maximum number of changes in charging conditions.	■ 0 = Normal Release – IP-CAN bearer release or detach. ■ 4 = Abnormal Release – Any other abnormal release. ■ 5 = CAMEL Init Call Release ■ 16 = Volume Limit – Configured volume threshold has been exceeded. ■ 17 = Time Limit – Configured interval has been reached. ■ 18 = Serving Node Change – Serving Node address list overflow. ■ 19 = Max Change Condition – Maximum number of changes in charging conditions. ■ 20 = Management Intervention – Request due to O&M reasons. ■ 21 = Intra SGSN Intersystem Change ■ 22 = RAT Change – Change of radio interface from EUTRAN<–>GSM<–> UMTS, and so on. ■ 23 = MS Time Zone Change – MS changes time zone. ■ 24 = SGSN PLMN ID Change – Change of PLMN-ID. ■ 25 = SGW Change

Value Range *Continued*	■ 14 = CGI SAI Change ■ 15 = RAI Change ■ 16 = ECGI Change ■ 17 = TAI Change ■ 18 = Service Data Volume Limit ■ 19 = Service Data Time Limit ■ 20 = Management Intervention – Request due to O&M reasons. ■ 21 = Service Stop ■ 22 = No Condition ■ 25 = SGW Change ■ 36 = Change in UP to UE	■ 52 = Unauthorized Requesting Network ■ 53 = Unauthorized LCS Client ■ 54 = Position Method Failure ■ 58 = Unknown or Unreachable LCS Client ■ 59 = List of Downstream Node Change
Example	1	8F0114

Charging Characteristics

Description	The charging characteristics used by the Serving GPRS Support Node (SGSN). ***Note:*** *When charging characteristics are not provided by GTP or AAA, the default charging characteristics configured under the workflow data profile are used to select a billing plan. In this case, the configured charging characteristics are also included in CDRs.*	
	CSV	**ASN.1 Code (97)**
Length		2 Bytes
Format	Hex	Octet string
Example	08	97020008

Charging ID

Description	IP CAN bearer charging identifier used to identify this IP CAN bearer in different records created by process control networks (PCNs). ***Note:*** *When used for a GTP Proxy session, this field contains the charging ID generated by the GTP Proxy. The GTP Proxy ignores/discards the charging ID received from the PGW/GGSN.*	
	CSV	**ASN.1 Code (85)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	0583157843	850422c24853

Charging Selection Mode

Description	Indicates the charging characteristic type that the gateway applies to the CDR.	
	CSV	**ASN.1 Code (98)**
Length		1 Byte
Format	Decimal	Enumerated
Value Range	■ 0 = Serving Node Supplied ■ 1 = Subscription Specific ■ 2 = APN Specific ■ 3 = Home Default ■ 4 = Roaming Default ■ 5 = Visiting Default	■ 0 = Serving Node Supplied ■ 1 = Subscription Specific ■ 2 = APN Specific ■ 3 = Home Default ■ 4 = Roaming Default ■ 5 = Visiting Default
Example	1	980100

CP CIoT EPS Optimisation Indicator

Description	Contains the indication on whether Control Plane CIoT EPS optimisation is used by the PDN connection during data transfer with the UE (for example, Control Plane NAS PDU via S11-U between SGW and MME) or not (for example, User Plane via S1-U between SGW and eNB). *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code 9f3b (SGW)**
Length		1 Byte
Format	Decimal	Boolean
Value Range	■ 0 = Not Apply ■ 1 = Apply	■ False ■ True
Example	1	9f3b0100

Diagnostics

Description	Provides more details for the release of the connection, such as the Non-Stratum-Access (NAS) cause code.	
	CSV	**ASN.1 Code (b0)**
Length		1-5 Bytes
Format	Decimal	Choice: ■ gsm0408Cause ■ network specific cause
Example	Not Filled	b032a330060c2b06010401090a30010202658 10100a21d301b81091410311213282d040082 0122a30680040a40816c8403020114

The following vendor-specific ASN.1 CDR field is part of the Diagnostics field:

Network Specific Cause

Network Specific Cause ::= Sequence of Management Extensions

Field Name	Description	Format	ASN.1 Code
Identifier	OBJECT IDENTIFIER for diagnostics	OCTET STRING	0x6
Significance	BOOLEAN DEFAULT FALSE	BOOLEAN	0x81
Information	SEQUENCE of CauseInformation	OCTET STRING	0xA2
CauseInformation	SEQUENCE		0x30
msgTimeStamp	Message Time Stamp	TimeStamp	0x81
msgType	messageType	INTEGER	0x82
msgSourceIp	Source IP Address	IP Address	0x83
msgCause	CauseCode	OCTET STRING	0x84

Duration

Description	Indicates the length of time of the record, in seconds. The value is reset for each new partial CDR.	
	CSV	**ASN.1 Code (8e)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	159	8e019f

Dynamic Address Flag

Description	Indicates whether the served Policy Decision Point/Public Data Network (PDP/PDN) address is dynamic, which is allocated during IP CAN bearer activation, initial attach (E-UTRAN or over S2x) and user equipment (UE) requested PDN connectivity. This field is missing if the address is static.	
	CSV	**ASN.1 Code (8b)**
Length		1 Byte
Format	Decimal	Boolean
Value Range	■ 0 = Static ■ 1 = Dynamic	■ False ■ True
Example	1	8b0101

Dynamic Address Flag Extension

Description	This field indicates that the IPv4 address has been dynamically allocated for a particular IP CAN bearer (PDN connection) of PDN type IPv4v6, and the dynamic IPv6 prefix is indicated in the Dynamic Address Flag. This field is missing if the IPv4 address is static. Dynamic address allocation might be relevant for charging (for example, as one resource offered and possibly owned by the network operator).	
	CSV	**ASN.1 Code (8b)**
Length		1 Byte
Format	N/A	Boolean
Value Range	N/A	■ False ■ True
Example	N/A	8b0101

ePDG FQDN

Description	Contains the Fully Qualified Domain Name of the Evolved Packet Data Gateway (ePDG). *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	epdg.epc.mnc12.mcc456.pub.3gppnetwork.org	N/A

ePDG Name

Description	Contains the service name of the ePDG. *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	ePDG2	N/A

GGSN Address

Description	The control plane IP address of the PGW or GGSN used.	
	CSV	**ASN.1 Code (a4)**
Length		6 or 18 Bytes (v4 or v6)
Format	String (dotted IP address) or Hex	Octet string
Example	10.64.64.8 or 0A404008	a40680040A404008

GGSN Address (IPv6)

Description	Indicates the serving GGSN or PGW IPv6 Address for the Control Plane.	
	CSV	**ASN.1 Code**
Length	N/A	N/A
Format	Dotted IP address or Hex	N/A
Example	2001:470:8865:4081:81::2d	N/A

3GPP2 User Location Information

Description	Contains the 3GPP2 User Location Information as described in TS 29.212 [220].	
	CSV	**ASN.1 Code**
Length		N/A
Format	Default (BCD encoding)	N/A
Example		N/A

IKE Local IP Address

Description	Indicates the destination IP address of the Internet Key Exchange (IKE) tunnel. *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length	N/A	N/A
Format	Dotted IP address	N/A
Example	10.64.133.26	N/A

IKE Local Port

Description	Indicates the destination port of the IKE tunnel. *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length	N/A	N/A
Format	Decimal	N/A
Example	500	N/A

IKE Remote IP Address

Description	Indicates the source IP address of the IKE tunnel. *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length	N/A	N/A
Format	Dotted IP address	N/A
Example	10.71.32.142	N/A

IKE Remote Port

Description	Indicates the destination port of the IKE tunnel. *Applicable only to ePDG.*	
	CSV	**ASN.1 Code**
Length	N/A	N/A
Format	Decimal	N/A
Example	4500	N/A

IMS Signaling Context

Description	Indicates whether the IP-CAN bearer is used for IMS signaling. It is only present if the IP-CAN bearer is an IMS signaling bearer.	
	CSV	**ASN.1 Code**
Length	N/A	1
Format	N/A	Integer
Example	N/A	N/A

IMSI Unauthenticated Flag

Description	Indicates whether the provided served International Mobile Subscriber Identity (IMSI) is authenticated (emergency bearer service situation). ***Note:*** *The IMSI Unauthenticated Flag ASN.1 CDR field is not applicable to 3GPP 32.298 Version 8.4.*	
	CSV	**ASN.1 Code**
Length		1
Format	Decimal	Integer
Value Range	■ 0 = Authenticated ■ 1 = Unauthenticated	N/A
Example	0	N/A

List of RAN Secondary RAT Usage Reports

Description	This list applies to SGW and PGW CDRs, and includes one or more containers reported from the RAN for a secondary RAT. *Applies to 5G NSA.*	
	CSV	**ASN.1 Code** **bf40 (SGW)** **bf49 (PGW)**
Length	N/A	Variable
Format	■ Data Volume Uplink ■ Data Volume Downlink ■ RAN Start Time ■ RAN End Time ■ Secondary RAT Type See TS 3GPP 32.298 Version 15.2.0 for encoding details.	Sequence: ■ Data Volume Uplink ■ Data Volume Downlink ■ RAN Start Time ■ RAN End Time ■ Secondary RAT Type
Example	'250,100,1526580685,1526580745,1'	bf498187301f81011482011e83091805171406 452d04008409180517140745 2d0400850101

List of Service Data

Description	Provides a list of changes in charging conditions for all service data flows within this IP CAN bearer categorized per rating group. Each change is time stamped. Charging conditions are used to categorize traffic volume, elapsed time and number of events per tariff period. This field also lists initial and subsequently changed Quality of Service (QoS) and corresponding data values. ***Note:*** *In ASN.1 format, this field is known as Change of Service Condition.* ***Note:*** *Filled if service data flow charging is configured. See the Affirmed Networks Gateway Administration Guide for more information on configuring charging components, such as the rating group, billing plan, and charging instance.*	
	CSV	**ASN.1 Code (bf22)**
Length		Variable
Format	Complex field encoded in single quotes. Each embedded complex field (such as qosinfo) is encoded in two single quotes. Third level complex fields are encoded in four single quotes, and so on. A field without a value set is identified by two consecutive commas (,,). The subfields are in fixed order. See List of Service Data Subfields (CSV) (page 3-17).	Sequence of Change of Service Condition. See Change of Service Condition Subfields (ASN.1) (page 3-20).
Example	‘,WDP_UC04_MP4_5,1489478,13283,20041960,20886,,,160000,,160000,,,,,,’	bf224e304c810164840107880500080000000a9108101098601008703012 4f888030493e0aa068004ac1e17618c0202d08d0202d08e091403011002072b0000910164940d1844f100233d44f00201605901

List of Service Data Subfields (CSV)

Table 3-2 List of Service Data Subfields (CSV)

Field	Name	Format	Position
afchargingid	AF Charging ID	Unsigned32	1
chargingrulebasename	Charging Rule Base Name	String	2
acctinputoctets	Account Input Octets (Downlink)	Unsigned64	3
acctinputpackets	Account Input Packets (Downlink)	Unsigned64	4
acctoutputoctets	Account Output Octets (Uplink)	Unsigned64	5
acctoutputpackets	Account Output Packets (Uplink)	Unsigned64	6
localsequencenumber	Local Sequence Number	Unsigned32	7

Table 3-2 List of Service Data Subfields (CSV)

Field	Name	Format	Position
qosinfo	QoS Information – QoS Class ID – Uplink MBR – Downlink MBR – Uplink GBR – Downlink GBR – Bearer ID – Allocation Retention Priority – APN Uplink AMBR – APN Downlink AMBR – Extended Max Requested BW UL – Extended Max Requested BW DL – Extended GBR UL – Extended GBR DL – Extended APN AMBR UL – Extended APN AMBR DL		8
ratinggroup	Rating Group Quota ID	Unsigned32	9
changetime	Change Time	Unsigned32	10
serviceid	Service Data Flow ID	Unsigned32	11
serviceinfo	Service Specific Information – Service Specific Data – Service Specific Type		12
sgsnaddress	SGSN Address	Unsigned32	13
servingnodeaddr	Serving Node Address (Replaces SGSN Address field)	Bytes	14
timefirstusage	Time First Usage	Unsigned32	15
timelastusage	Time Last Usage	Unsigned32	16
duration	Duration	Unsigned32	17
changecondition	Change Condition	Unsigned32	18
userlocationinfo	3GPP User Location Information	Bytes	19
timeofreport	Time of Report	Unsigned32	20

Table 3-2 List of Service Data Subfields (CSV)

Field	Name	Format	Position
qoSInformationNegRel84	QoS Information Negotiated (Only applicable for 3GPP TS 32.298 version 8.4)	String	21
servingnodeipaddr	Serving Node IP Address (Replaces Serving Node Address field)	Bytes	22
newlocalsequencenumber	New Local Sequence Number	Unsigned32	23
cumulativeacctinputoctets	Cumulative Account Input Octets	Unsigned64	24
cumulativeacctoutputoctets	Cumulative Account Output Octets	Unsigned64	25
sponsorid	Sponsor Identity	UTF8String	26
appserviceproviderid	Application Service Provider Identity	UTF8String	27
afrecordinformation	AF-Record-Information – AF Charging Identifier – Flows		28
psfurnishcharging	PS Furnish Charging Information	Default (BCD)	29
failurehandlingcontinue	Failure Handling Continue	Boolean	30
gyeventchangecondition	Gy Event Change Condition	Unsigned32	31
gpprattype	3GPP-RAT-Type	Integer	32
servingplmnratecontrol	Serving PLMN Rate Control	Default (BCD)	33
apnratecontrol	APN Rate Control	Default (BCD)	34
uwanuserlocationinfo	Uwan-User-Location-Info	Default (BCD)	35

Change of Service Condition Subfields (ASN.1)

Note: *In ASN.1 format, the List of Service Data field is known as Change of Service Condition.*

Table 3-3 Change of Service Condition Subfields (ASN.1)

Field Name
Rating Group
Charging Rule Base Name
Local Sequence Number
Time First Usage
Time Last Usage
Time Usage (Duration)
Change Condition
QoS Information Negotiated (Only applicable for 3GPP TS 32.298 version 8.4 in Octet String format. See Data Record Templates for CDRs in ASN.1 Format (page 1-13) for more information) – APN Aggregate Max Bitrate DL – APN Aggregate Max Bitrate UL – ARP – Guaranteed Bitrate DL – Guaranteed Bitrate UL – Max Requested Bandwith DL – Max Requested Bandwith UL – QCI

Table 3-3 Change of Service Condition Subfields (ASN.1)

Field Name
EPC QoS Information (Applicable for 3GPP TS 32.298 version 9.0 and beyond in Sequence format. See Data Record Templates for CDRs in ASN.1 Format (page 1-13) for more information) – QCI – Max Requested Bandwith UL – Max Requested Bandwith DL – Guaranteed Bitrate UL – Guaranteed Bitrate DL – ARP – APN Aggregate Max Bitrate DL – APN Aggregate Max Bitrate UL – Extended Max Requested BW UL – Extended Max Requested BW DL – Extended GBR UL – Extended GBR DL – Extended APN AMBR UL – Extended APN AMBR DL
Serving Node Address (SGSN Address)
Data Volume FBC Uplink (Account Output Octets)
Data Volume FBC Downlink (Account Input Octets)
Time of Report
Failure Handling Continue
Service ID
PS Furnish Charging Information
AF Record Information – AF Charging Identifier – Flows
User Location Information
Sponsor Identity
Application Service Provider Identity
3GPP RAT Type
Serving PLMN Rate Control
APN Rate Control
UWAN User Location Info

List of Traffic Data Volumes

	CSV	ASN.1 Code
Description	Provides a list of changes in charging conditions for this QCI/ARP pair. Each change is time stamped. Charging conditions are used to categorize traffic volume per tariff period. This field also lists initial and subsequently changed QoS and corresponding data values. ***Note:*** *In ASN.1 format, this field is known as Change of Charging Condition.*	
Length		Variable
Format	Complex field encoded in single quotes. Each embedded complex field (such as qosinfo) is encoded in two single quotes. Third level complex fields are encoded in four single quotes, and so on. A field without a value set is identified by two consecutive commas (,,). The subfields are in fixed order. See List of Traffic Data Volumes Subfields (CSV) (page 3-22).	Sequence of Change of Charging Condition. See Change of Charging Condition Subfields (ASN.1) (page 3-24).
Example	'''2,0,0,0,0,05,,12000,12000'',20147730,22045, 1698843,14746,,,'	ac26302483010084010085010086 00a917810109820100830100840100850100870221c0880221c0

List of Traffic Data Volumes Subfields (CSV)

Table 3-4 List of Traffic Data Volumes Subfields (CSV)

Field	Name	Format	Position
qosinfo	QoS Information – QoS Class ID – Uplink MBR – Downlink MBR – Uplink GBR – Downlink GBR – Bearer ID – Allocation Retention Priority – APN Uplink AMBR – APN Downlink AMBR – Extended Max Requested BW UL – Extended Max Requested BW DL – Extended GBR UL – Extended GBR DL – Extended APN AMBR UL – Extended APN AMBR DL		1

Table 3-4 List of Traffic Data Volumes Subfields (CSV)

Field	Name	Format	Position
acctinputoctets	Account Input Octets (Downlink)	Unsigned64	2
acctinputpackets	Account Input Packets (Downlink)	Unsigned64	3
acctoutputoctets	Account Output Octets (Uplink)	Unsigned64	4
acctoutputpackets	Account Output Packets (Uplink)	Unsigned64	5
changecondition	Change Condition	Unsigned32	6
changetime	Change Time	Unsigned32	7
userlocationinfo	3GPP User Location Information	Bytes	8
qoSInformationNegRel84	QoS Information Negotiated (Only applicable for 3GPP TS 32.298 version 8.4)	String	9
gpprattype	3GPP RAT Type	Integer	10
cpciotepsoptimisationindicator	CP CIoT-EPS Optimisation Indicator	Decimal	11
servingplmnratecontrol	Serving PLMN Rate Control	Default (BCD)	12
uwanuserlocationinfo	Uwan-User-Location-Info	Default (BCD)	13

Change of Charging Condition Subfields (ASN.1)

Note: *In ASN.1 format, the List of Traffic Data Volumes field is known as Change of Charging Condition.*

Table 3-5 Change of Charging Condition Subfields (ASN.1)

Field Name
QoS Information Requested
QoS Information Negotiated (Only applicable for 3GPP TS 32.298 version 8.4 in Octet String format) – APN Aggregate Max Bitrate DL – APN Aggregate Max Bitrate UL – ARP – Guaranteed Bitrate DL – Guaranteed Bitrate UL – Max Requested Bandwidth DL – Max Requested Bandwidth UL – QCI
Data Volume GPRS Uplink (Output Octets)
Data Volume GPRS Downlink (Input Octets)
Change Condition
Change Time
User Location Information
EPC QoS Information (Applicable for 3GPP TS 32.298 version 9.0 and beyond in Sequence format) – QCI – Max Requested Bandwidth UL – Max Requested Bandwidth DL – Guaranteed Bitrate UL – Guaranteed Bitrate DL – ARP – APN Aggregate Max Bitrate DL – APN Aggregate Max Bitrate UL – Extended Max Requested BW UL – Extended Max Requested BW DL – Extended GBR UL – Extended GBR DL – Extended APN AMBR UL – Extended APN AMBR DL

Table 3-5 Change of Charging Condition Subfields (ASN.1)

Field Name
3GPP RAT Type
CP CIoT-EPS Optimisation Indicator
Serving PLMN Rate Control
UWAN User Location Info

Local Sequence Number

Description	Indicates the consecutive record number created by this node. The number starts from zero and is allocated sequentially among all CDR types.	
	CSV	**ASN.1 Code (94)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	7340033	940101

Low Priority Indicator

Description	Indicates if the PDN connection has a low priority, for example, for Machine Type Communication. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code 9f2e (PGW) 9f2c (SGW)**
Length		1 Byte
Format	Decimal	Integer
Example	1	9f2e00bf (PGW) 9f2c00bf (SGW)

MS Time Zone

Description	Indicates the time zone in which the mobile station (MS) is currently located as defined in 3GPP TS 29.060.	
	CSV	**ASN.1 Code (9f1f)**
Length		2 Bytes
Format	String	Octet String
Example	4000	9f1f020101

Network Initiated PDP

Description	Provides a flag indicating whether the PDP is initiated from the network.	
	CSV	**ASN.1 Code (9f1f)**
Length		Not Filled
Format	Decimal	Not Filled
Example	0	Not Filled

Node ID

Description	Contains the service name of the gateway device. For more information, see Configuring the Node ID in CDRs (page 1-53).	
	CSV	**ASN.1 Code (92)**
Length		1-20 Bytes
Format	String	IA5string
Example	fichgg01	92086669636867673031

PDN Connection ID

Description	Indicates the PDN connection (IP-CAN session) identifier used to identify different records belonging to the same PDN connection.	
	CSV	**ASN.1 Code (9f29)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	139460617	9F2904070000d9

PDP/PDN Type

Description	Indicates the served PDP/PDN type. ***Note:*** *To populate the PDP/PDN Type field in ASN.1 CDRs with 4G PDP types, see Customizing ASN.1 CDR Fields (page 1-42).*	
	CSV	**ASN.1 Code (88)**
Length		2 Bytes
Format	Decimal	Octet string
Value Range	■ 0 = IPv4 ■ 1 = PPP ■ 2 = IPv6 ■ 3 = IPv4IPv6	■ PDP Type Organization ■ PDP/PDN Type Number ***Note:*** *See TS 3GPP 29.060 [215] for encoding details.*
Example	0	8802f121

PDP/PDN Type Extension

Description	This integer is a 1:1 copy of the PDP type value as defined in TS 29.061 [215]. Used for Non-IP PDP type. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code 9f44 (PGW) 9f3e (SGW)**
Length		1 Byte
Format	Decimal	Integer
Example	4 (Non-IP)	9f440104 (PGW) 9f3e0104 (SGW)

PGW Address

Description	Indicates the serving PGW IP Address for the Control Plane. If both an IPv4 and IPv6 address for the PGW is available, the PGW includes the IPv4 address in this field and the IPv6 address in the PGW Address (IPv6) field. ***Note:*** *When used for a GTP Proxy session, this field contains the GTP Proxy IPv4 address.*	
	CSV	**ASN.1 Code (a4)**
Length		6 or 18 Bytes (v4 or v6)
Format	String (dotted IP address) or Hex	Octet string
Example	10.64.64.8 or 0A404008	a40680040a404008

PGW Address (IPv6)

Description	Indicates the serving PGW IPv6 Address for the Control Plane. ***Note:*** *When used for a GTP Proxy session, this field contains the GTP Proxy IPv6 address (if available).*	
	CSV	**ASN.1 Code (bf32)**
Length		18 Bytes (IPv6)
Format	Dotted IP address or Hex	Octet string
Example	2001:470:8865:4081:81::2d	bf32128110200104708865408000800000000 0006c

PGW PLMN ID

Description	Indicates the PGW Public Land Mobile Network (PLMN) Identifier (Mobile Country Code/Mobile Network Code), configured using the following command: `an3000-mcm-slot7cpu1(config)# service-construct plmnid-list home-list plmnid <plmnid>` **PGW CDRs** When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the `home-plmnid-list`, the configured PLMN ID is used in the PGW CDR. When multiple PLMN ID values are configured in the `home-plmnid-list`, the first PLMN ID from the list to match the PLMN ID received in the subscriber's IMSI is used in the CDR. If an entry is not found in the list, the first entry in the `home-plmnid-list` is used in the PGW CDR. **SGW CDRs** The PLMN ID (MCC + MNC), received in the subscriber's IMSI, is matched with the `home-plmnid-list` and if not found, it is matched with the `roaming-plmnid-list`. If an entry is found, the configured PLMN ID is used in the SGW CDR. If the IMSI MCC/MNC is not found in these lists, the PGW PLNM ID length is set to 5 (MCC 3 digits, MNC 2 digits) and the IMSI MCC/MNC is used in the SGW CDR.	
	CSV	**ASN.1 Code (9f25)**
Length		3 Bytes
Format	String	Octet string
Example	111222	9f2503130089

PS Furnish Charging Information

Description	Contains online charging session information.	
	CSV	**ASN.1 Code (bc)**
Length		Variable
Format	Default (BCD encoding)	Sequence
Example	Not Filled	bc10810e4672656546726f6d617444617461

RAT Type

Description	Indicates the Radio Access Technology (RAT) type currently used by the Mobile Station, when available. The field is provided by the SGSN/MME and transferred. This RAT type is defined in TS 29.060 [204] for GTP, in TS 29.274 [210] for eGTP, and in TS 29.275 [211] for PMIP. ***Note:*** *For the C-SGN, this field indicates which RAT type is currently serving the UE as defined in TS 29.061 [205]. (NB-IoT).*	
	CSV	**ASN.1 Code (9e)**
Length		1 Byte
Format	Decimal	Integer
Value Range	■ 0 = <reserved> ■ 1 = UTRAN ■ 2 = GERAN ■ 3 = WLAN ■ 4 = GAN ■ 5 = HSPA Evolution ■ 6 = EUTRAN ■ 8= EUTRAN-NB-IoT ■ 7 - 255 = <spare>	■ 0 = <reserved> ■ 1 = UTRAN ■ 2 = GERAN ■ 3 = WLAN ■ 4 = GAN ■ 5 = HSPA Evolution ■ 6 = EUTRAN ■ 8 = EUTRAN-NB-IoT ■ 7 - 255 = <spare>
Example	1	9e0102

Record Extensions

Enables network operators and/or manufacturers to add their own extensions to the standard ASN.1 record definitions.

The following vendor-specific ASN.1 CDR fields are part of the Record Extension field:

Extension Information

Field Name	Description	Format	ASN.1 Code
recordExtensions	Management Extensions	Set of Management Extensions	0xB3
Extension Information	SEQUENCE		0x30
extensionType		INTEGER	0x80
length		INTEGER	0x81
serviceList	SEQUENCE of ServiceEvent		0xA2
transparentVSA	The TMO-Transparent-Data VSA received from the AAA server is written transparently into the Record Extension.	OCTET STRING	0x86

MVNO Information

Field Name	Description	Format	ASN.1 Code
recordExtensions	Management Extensions	Set of Management Extensions	0xB3
ManagementExtension	Sequence of MvnoInformation		0x30
Identifier	OBJECT IDENTIFIER for MVNO Information	OBJECT IDENTIFIER	0x6
Significance	BOOLEAN DEFAULT FALSE	BOOLEAN	0x81
Information	SEQUENCE of MvnoInformation	OCTET STRING	0xA2
MvnoInformation	SEQUENCE		0x30
ResellerID	MVNO Reseller Identifier	OCTET STRING	0x80
SubClassID	MVNO Sub class Identifier	OCTET STRING	0x81

RAN Cause Codes

Field Name	Description	Format	ASN.1 Code
recordExtensions	Management Extensions	Set of Management Extensions	0xB3
ManagementExtension	Sequence of RAN Cause Code		0x30
Identifier	OBJECT IDENTIFIER for RAN Cause Code	OCTET STRING	0x6
Significance	BOOLEAN DEFAULT FALSE	BOOLEAN	0x81
Information	SEQUENCE of CauseInformation	OCTET STRING	0xA2
CauseInformation	SEQUENCE		0x30
msgTimeStamp	Message Time Stamp	TimeStamp	0x81
msgType	messageType	INTEGER	0x82
msgSourceIp	Source IP Address	IP Address	0x83
msgCause	CauseCode	OCTET STRING	0x84

Record Opening Time

Description	Time stamp when IP CAN bearer is activated or record opening time on subsequent partial records.	
	CSV	**ASN.1 Code (8d)**
Length		9 Bytes
Format	Available format types: ■ time-secs-YYYY-MM-DDTHH:MM:SS – (default) The system expects to receive the timestamp in seconds since Epoch time. ■ time-secs-YYYYMMDDHHMMSS ■ time-nsecs-YYYY-MM-DDTHH:MM:SS.ZONE – The system expects to receive the timestamp in nanoseconds since Epoch time.	BCD encoded octet string
Example	■ 2011-09-22T12:23:57-0500 (default format) ■ 20110922122357 ■ 2011-09-22T12:23:57-0500	8d091311190836242d0500

Record Sequence Number

Description	Contains a running sequence number used to link partial records generated by the SGSN/SGW/PGW for a particular session (characterized with the same Charging ID and PGW address pair).	
	CSV	**ASN.1 Code (91)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	3	910106

Record Type

Description	Indicates the node function.	
	CSV	**ASN.1 Code (80)**
Length		1 Byte
Format	Decimal	Integer
Value Range	■ 8 = SGW ■ 9 = PGW ■ 10 = GGSN ■ 226 = Custom ■ 227 = WAG ■ 229 = ePDG	■ ASN.1 (3GPP TS 32.298, V9 and V10) – 84 = SGW – 85 = PGW – 85 = GGSN ■ ASN.1 (3GPP TS 32.298, V6) – 19 = GGSN
Example	9	800155

SCS/AS Address

Description	Indicates the address of the Service Capability Server/Application Server (SCS/AS). *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code** **bf47 (PGW)**
Length		Variable
Format	Default (BCD encoding)	Set: ■ SCS Address ■ SCS Realm
Example	10.37.7.61 : 00013d07250a	bf4715a10680040a35017c820b31302e35332e0010312e313234

Served 3GPP 2 MEID

Description	Mobile Equipment Identifier (MEID) of the served party's terminal equipment for 3GPP 2 access.	
	CSV	**ASN.1 Code**
Length		Not Filled
Format	String	Not Filled
Example	Not Filled	Not Filled

Served IMEISV

Description	Contains the International Mobile Equipment Identity (IMEI) or IMEI software version (IMEISV) of the equipment served, if available. It is used to identify the user in case the served International Mobile Subscriber Identity (IMSI) is not present during emergency bearer service.	
	CSV	**ASN.1 Code (9d)**
Length		8 Bytes
Format	String	BCD encoded octet string
Example	3517540100359201	9d0899990000000051f0

Served IMSI

Description	Contains the International Mobile Subscriber Identity (IMSI) of the served party.	
	CSV	**ASN.1 Code (83)**
Length		3-8 Bytes
Format	String	BCD encoded octet string
Example	208101825411443	8308440100000001042f8

Served MAC

Description	Contains the mobile node Medium Access Control (MAC) address. *Applicable only to WAG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	aa-bb-29-00-00-08	N/A

Served MNNAI

Description	Indicates the Served Mobile Node (MN) identifier in Network Access Identifier (NAI) format (based on IMSI), if available. This information is used during EAP-AKA authentication of the UE. 0<IMSI>@nai.epc.mnc<MNC>.mcc<MCC>.3gppnetwork.org Example: 0234150999999999@nai.epc.mnc015.mcc234.3gppnetwork.org where: ■ IMSI = 234150999999999 ■ MCC = 234 ■ MNC = 15	
	CSV	**ASN.1 Code (bf24)**
Length		Variable
Format	String	Set
Example	0456124000000000@nai.epc.mnc123.mcc456.pub.3gppnetwork.org	bf243a80010381353031373536373534373839373839393406e61692e6570632e6d6e633637352e6d63633137352e336770706e6574776f726b2e6f7267

Served MSISDN

Description	The field tracks the primary Mobile Station (MS) ISDN number (MSISDN) of the subscriber.	
	CSV	**ASN.1 Code (96)**
Length		1-9 Bytes
Format	String	BCD encoded octet string The MSISDN is encoded in international format according to the numbering plan defined in the ITU-T E.164 standard. It contains the *nature of address indicator*, which indicates whether the number is an international number: GTPv2: First octet (nature of address octet) encoded as 0x00 if the address digits <=10 else 0x91 GTPv1: The digits are passed through as they are.
Example	000000033609182185	9691063170969959f3

Served PDP Address

Description	This field contains the PDP address of the served IMSI. This is a network layer address (IPv4 or IPv6). The address for each PDP type is allocated either temporarily or permanently (see Dynamic Address Flag). This parameter is present except when both the PDP type is PPP and dynamic PDP address assignment is used.	
	CSV	**ASN.1 Code (a9)**
Length		N/A
Format	String (dotted IP address) or Hex	N/A
Example	10.64.64.8 or 0A404008	N/A

Served PDP Address (IPv6)

Description	This field contains the PDP connection IPv6 address when the SGW/PGW supports dual stack (configured with both IPv4 and IPv6 addresses). *Applicable only to SGW and PGW.*	
	CSV	**ASN.1 Code (a9)**
Length		N/A
Format	String (dotted IP address) or Hex	N/A
Example	2001:470:8865:2d01::2	N/A

Served PDP/PDN Address

Description	This field contains the IP address for the PDN connection (PDP context, IP-CAN bearer). This is a network layer address (IPv4 or IPv6). The address for each Bearer type is allocated either temporarily or permanently (see Dynamic Address Flag). This parameter is present except when both the Bearer type is PPP and dynamic address assignment is used.	
	CSV	**ASN.1 Code (a9)**
Length	N/A	8 or 20 Bytes
Format	N/A	Octet string
Example	N/A	a908a00680040A404008

Served PDP Address Extension

Description	This field contains the IPv6 address for the PDN connection (PDP context, IP-CAN bearer) when dual-stack IPv4/IPv6 is used and the IPv4 prefix is included in Served PDP Address or Served PDP/PDN Address. *Applicable only to GGSN, WAG, and ePDG.* ***Note:*** *For PGW CDR CSV records, the Served PDP Address Extension field is deprecated in favor of the Served PDP Address (IPv6) field. The default* `presentFlag` *setting for this field in PGW CDR records is "*`no`*". In PGW Start records, this field has a default position of 10000 to ensure it appears at the end of the CDR CSV file.*	
	CSV	**ASN.1 Code (a9)**
Length		8 Bytes
Format	String (dotted IP address) or Hex	Octet string
Example	2001:470:8865:2d01::2	a914a012811020010470886566c010000000000000003

Serving Node Address

Description	The current serving node address used during this record. ***Note:*** *The Serving Node Address field is deprecated in favor of the Serving Node IP Address field.*	
	CSV	**ASN.1 Code (a6)**
Length		N/A
Format	Dotted IP Address	N/A
Example	10.64.64.8	N/A

Serving Node IP Address

Description	List of serving node control plane IP addresses (for example, SGSN, MME) used during this record. When used for a GTP Proxy session, this field contains the SGW/SGSN IP address.	
	CSV	**ASN.1 Code (a6)**
Length		6 or 18 Bytes
Format	String (dotted IP address) or Hex	Octet string
Example	10.64.64.8 or 0A404008	a60680040A404008

Serving Node Address (IPv6)

Description	These fields contain one or several control plane IPv6 addresses of SGSN, MME, ePDG, HSGW, WAG or SGW, which have been connected during the record.	
	CSV	**ASN.1 Code (bf31)**
Length	N/A	18 Bytes
Format	N/A	Octet string
Example	N/A	bf31128110200104708865408100810000000 0006c

Serving Node PLMN Identifier

Description	A serving node PLMN Identifier (Mobile Country Code and Mobile Network Code) used during this record, if available.	
	CSV	**ASN.1 Code (9b)**
Length		3 Bytes
Format	Default (BCD encoding)	Octet string
Example	02F801	9b03130181

Serving Node Type

Description	The IP address of the current SGSN.	
	CSV	**ASN.1 Code (bf23)**
Length	N/A	Variable
Format	N/A	Sequence of serving node type
Example	N/A	bf23030a0102

Serving PLMN Rate Control

Description	Contains the Serving PLMN Rate Control as specified in TS 23.401 [245], which is used during the record for the PDN connection to the PGW. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code** **bf42 (PGW)** **bf3d (SGW)**
Length		Variable
Format	Default (BCD encoding) ■ Uplink Rate Limit – Maximum of 10 NAS PDUs in uplink direction per 6 minute interval ■ Downlink Rate Limit – Maximum of 20 NAS PDUs in downlink direction per 6 minute interval	Sequence: ■ Serving PLMN Downlink Rate Control Value ■ Serving PLMN Uplink Rate Control Value
Example	10, 20	bf4208800226ac810226b5 (PGW) bf3d08800226ac810226b5 (SGW)

SGi PtP Tunnelling Method

Description	Indicates whether SGi PtP tunnelling method is based on UDP/IP or other methods for a non-IP PDN type PDN connection. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code** **9f40 (PGW)**
Length		1 Byte
Format	Decimal	Enumerated
Value Range	■ 0 = UDP IP Based ■ 1 = Others	■ 0 = UDP IP Based ■ 1 = Others
Example	1	9f400100

SGSN Address

Description	The IP address of the current SGSN.	
	CSV	**ASN.1 Code (aa)**
Length		6 or 18 Bytes
Format	String (dotted IP address) or Hex	Octet string
Example	10.64.64.8 or 0A404008	aa0680040A404008

SGW Address

Description	The control plane IP address of the SGW used.	
	CSV	**ASN.1 Code (a4)**
Length		6 or 18 Bytes
Format	String (dotted IP address) or Hex	Octet string
Example	10.64.64.8 or 0A404008	a40680040a20230a

SGW Address (IPv6)

Description	Indicates the serving SGW IPv6 Address for the Control Plane.	
	CSV	**ASN.1 Code (bf31)**
Length	N/A	18 Bytes (IPv6)
Format	Dotted IP address or Hex	Octet string
Example	2001:470:8865:4080:80::2c	bf311281102001047088654081008100000000006c

SGW Change

Description	This field is present if this is the first record after an SGW change.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Value Range	■ 0 = Not-changed ■ 1 = Changed	For ASN.1 encoding, refer to Cause for Record Closing (page 3-7).
Example	1	N/A

Sponsor Identity

Description	Identifies the sponsor willing to pay for the Operator's charge for connectivity. Subfield within the List of Service Data field (see List of Service Data).	
	CSV	**ASN.1 Code(99)**
Length		Variable
Format	UTF8String	Octet string
Example	sponsorid	73706F6E736F726964

Start Time

Description	The timestamp indicating when the User IP-CAN session starts. Available in the CDR for the first bearer in an IP-CAN session.	
	CSV	**ASN.1 Code (9f26)**
Length		9 Bytes
Format	Available format types: ■ time-secs-YYYY-MM-DDTHH:MM:SS – (default) The system expects to receive the timestamp in seconds since Epoch time. ■ time-secs-YYYYMMDDHHMMSS ■ time-nsecs-YYYY-MM-DDTHH:MM:SS.ZONE – The system expects to receive the timestamp in nanoseconds since Epoch time.	Octet string
Example	■ 2011-09-22T12:23:57-0500 (default format) ■ 20110922122357 ■ 2011-09-22T12:23:57-0500	9f2609110922122357 2b0000

Stop Time

Description	The timestamp indicating when the User IP-CAN session terminates. Available in the CDR for the last bearer in an IP-CAN session.	
	CSV	**ASN.1 Code (9f27)**
Length		9 Bytes
Format	Available format types: ■ time-secs-YYYY-MM-DDTHH:MM:SS – (default) The system expects to receive the timestamp in seconds since Epoch time. ■ time-secs-YYYYMMDDHHMMSS ■ time-nsecs-YYYY-MM-DDTHH:MM:SS.ZONE – The system expects to receive the timestamp in nanoseconds since Epoch time.	Octet string
Example	■ 2011-09-22T12:23:57-0500 (default format) ■ 20110922122357 ■ 2011-09-22T12:23:57-0500	9f27091109221223572b0000

UNI PDU CP Only Flag

Description	Indicates whether this PDN connection is applied with “Control Plane Only flag”. For example, transfer using Control Plane NAS PDUs only when Control Plane CIoT EPS Optimisation is enabled. This field is missing if both the user plane and control plane UNI for PDU transfer (for example, S1-U and S11-U from SGW) are allowed when Control Plane CIoT EPS Optimisation is enabled. *Applicable to the C-SGN as defined in 3GPP TS 32.298 V13.7.0.*	
	CSV	**ASN.1 Code** **9f41 (PGW)** **9f3c (SGW)**
Length		1 Byte
Format	Decimal	Boolean
Value Range	■ 0 = UNI-PDU-both-UP-CP ■ 1 = UNI-PDU-CP-Only	■ False ■ True
Example	1	PGW – 9f410101 SGW – 9f3c01ff

User CSG Information

Description	The Closed Subscriber Group (CSG) information of the UE, if available, including CSG ID, access mode and CSG membership indication.	
	CSV	**ASN.1 Code (bf2b)**
Length		1-4 Bytes
Format	Default (BCD encoding)	Sequence: ■ CSG ID ■ Access Mode ■ CSG Membership Indication – For ASN.1 encoding, the CSG Membership Indication is only encoded if its value is 1 indicating that the subscriber is a CSG member.
Example	22773,0,1	bf2b0b8004000059098101008200

User Location Information

Description	The User Location Information of the MS as defined in 3GPP TS 29.060 [203] for GPRS, and in 3GPP TS 29.274 [210] for EPC, if available. ***Note:*** *When a CDR is closed, the top-level ULI value is filled with the same ULI used to open the CDR, even if the ULI value is updated. This information is maintained at the bearer level, so when the next CDR is opened, the CDR will use the most recent, updated ULI value.*	
	CSV	**ASN.1 Code (9f20)**
Length		5-13 Bytes
Format	Default (BCD encoding)	Octet String
Example	0102f801c1c17983	9f2006081254631234

UWAN User Location Information

Description	Contains the UE location in an Untrusted Wireless Access Network (UWAN), which includes the UE local IP address and optionally UDP source port number (if NAT is detected).	
	CSV	**ASN.1 Code(bf3e)**
Length		
Format	Default (BCD encoding)	Sequence: ■ UE Local IP Address ■ UDP Source Port
Example	00010a406f20,6536	bf3e18a0128110fe80000000000000020c29fff e8436dd81021af4

WAG Name

Description	Contains the service name of the Wireless Access Gateway (WAG). *Applicable only to WAG.*	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	WAG2	N/A

Custom1 CDR Descriptions (GGSN)

This section describes the GGSN CDRs specific to custom1.

Change Condition

	CSV	ASN.1 Code (88)
Description	Indicates the change in charging condition causing: ■ Sending of data record from the node ■ Volume counts container closing for an IP-CAN bearer ■ Service data container closing (for ASN.1 this is referred to as Cause for Record Closing (page 3-7))	
Length		5 Bytes
Format	Decimal	Bit String
Value Range	■ 0 = Normal Release ■ 1 = Abnormal Release ■ 2 = Qos Change ■ 3 = Volume Limit ■ 4 = Time Limit ■ 5 = Serving Node Change ■ 8 = RAT Change	■ 0 = QOS Change ■ 1 = SGSN Change ■ 2 = SGSN PLMN ID Change ■ 3 = Tariff Time Switch ■ 4 = PDP Context Release ■ 5 = RAT Change ■ 8 = Configuration Change ■ 9 = Service Stop ■ 25 = Time Limit ■ 26 = Volume Limit ■ 31 = User Location Change – CGI-SAI Change – RAI Change – ECGI Change – TAI Change
Example	2	88050008000000

Change Time

	CSV	ASN.1 Code
Description	The time in UTC format when the volume counts associated with the IP-CAN bearer or the service data container are closed and reported due to a change in charging conditions.	
Length		N/A
Format	time-secs-YYYYMMDDHHMMSS	N/A
Example	20110922122357	N/A

Charging Class

Description	Indicates a unique identifier for the charging class used to bill the user for using one or more flows or one or more URLs. Filled with the rating group ID.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	1	N/A

Content Downlink

Description	The number of downlink bytes transmitted for the user for a given service rule or rating group.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	10000	N/A

Content Uplink

Description	The number of uplink bytes transmitted for the user for a given service rule or rating group.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	20000	N/A

Downlink

Description	The number of downlink layer 3 bytes transmitted for the user for a given service.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	0	N/A

Downlink Hits

Description	The number of downlink packets transmitted for the user for a given service.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example		N/A

IMS Charging ID

Description	Indicates the IMS Charging Identifier (ICID) as generated by the International Mobile Subscriber (IMS) node for the Session Initiation Protocol (SIP) session. The ICID is 0 for non-IMS calls.	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	0	N/A

PDP Type Organization

Description	Indicates the served PDP type organization.	
	CSV	**ASN.1 Code (88)**
Length		2 Bytes
Format	String	Octet string
Example	1	8802F101

Potential Duplicate

Description	Indicates a potentially duplicated CDR packet.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	0	N/A

QCI

Description	Indicates the QoS Class Identifier (QCI) for the default bearer.	
	CSV	**ASN.1 Code**
Length	1 byte	N/A
Format	Decimal	N/A
Example	7	N/A

QoS Negotiated Profile

Description	Indicates the negotiated QoS attributes.	
	CSV	**ASN.1 Code**
Length		N/A
Format	String	N/A
Example	0000017396D3FE7483FFFF004A00	N/A

Service ID

Description	Provides a unique identifier for the service (service rule ID).	
	CSV	**ASN.1 Code (91)**
Length		1-5 Bytes
Format	Decimal	Integer
Example	119	910164

Stored Closing Cause

Description	Indicates a reason for the CDR closure. For value mapping, see Cause for Record Closing (page 3-7).	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	13	N/A

System Type

Description	Indicates the GRPS release.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Value Range	■ 1 = GPRS release 2 ■ 2 = 3G release 1 ■ 3 = GPRS release 3 ■ 4 = 3G release 2 ■ 5 = ICD release 1.1 ■ 9 = NG NSN GGSN	N/A
Example	5	N/A

ULI GLT

Description	Indicates the geographic location type.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Hex	N/A
Example	1	N/A

ULI MNC-MCC

Description	Indicates the geographic location area, composed of the MNC and MCC.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Default (BCD encoding)	N/A
Example	02F801	N/A

ULI LAC

Description	Indicates the location area code.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Hex	N/A
Example	5208	N/A

ULI SAC

Description	Indicates the call identity (CI) or service area code (SAC).	
	CSV	**ASN.1 Code**
Length		N/A
Format	Hex	N/A
Example	478B	N/A

Uplink

Description	Indicates the number of uplink bytes transmitted for the user for a given service (identified by a service ID and a charging class).	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	0	N/A

Uplink Hits

Description	Indicates the number of uplink layer 3 packets transmitted for the user for a given service.	
	CSV	**ASN.1 Code**
Length		N/A
Format	Decimal	N/A
Example	8	N/A

Vendor-Specific CDR Descriptions

This section describes the supported Vendor-Specific CDRs.

MVNO Reseller ID

Description	The Mobile Virtual Network Operator (MVNO) name or identifier. *Applicable only to PGW.* ***Note:*** *This field is included in the Record Extensions field of the ASN.1 CDR record.*	
	CSV	**ASN.1 Code**
Length	Variable	N/A
Format	UTF8String	N/A
Example	TRACFONE	N/A

MVNO Sub Class ID

Description	The Mobile Virtual Network Operator (MVNO) sub-name or sub-identifier. *Applicable only to PGW.* ***Note:*** *This field is included in the Record Extensions field of the ASN.1 CDR record.*	
	CSV	**ASN.1 Code**
Length	Variable	N/A
Format	UTF8String	N/A
Example	CLASS1ID	N/A

Chapter 4

HTTP Transaction Data Records

This section contains the following information about *Affirmed Networks Mobile Content Cloud (MCC)* HTTP transaction data records (HTDRs).

HTDR Overview

HTDRs provide information about each HTTP transaction that traverses the MCC Multi Protocol Proxy. The data is represented in comma-separated values (CSV) format and written as a string of CSV values.

To facilitate HTDR file retrieval and processing, the MCC automatically appends a sequence number to the local CSV filename to indicate which file it is. The MCC uses the following file naming convention for local HTDR files:

```
<filename-prefix><YYYYMMDDHHMMSS>_<123>.csv
```

where `filename-prefix` indicates the prefix of the filename in which to store HTDRs, `YYYYMMDDHHMMSS` represents the timestamp and `123` represents the file sequence number. For example, `htdr20161114163821_123.csv`.

The sequence number runs from 1 to 2147483647, so that at one record per second, it would take a little over 68 years to rollover. The sequence number resets to 1 after a system reboot, MCM switchover or system failure. Depending on the frequency of these events, CDR records could potentially have the same sequence number.

You can configure the width of the file sequence number using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
<file-type> <filename-prefix> sequence-field-width <value>
```

`sequence-field-width` – Defines the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following:

```
<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv
```

Note: *See the Affirmed Networks Value Added Services Guide for information on the CLI commands used to configure and monitor Multi Protocol Proxy.*

Local HTDRs

This section lists the supported local HTTP transaction data record (HTDR) fields and data types. Table 4-1 lists the default position, field, name and data type for each top-level HTDR field. Each top-level HTDR field contains a set of subfields.

Table 4-1 Top-Level HTTP Transaction Data Record Fields

Position	Field	Name	Data Type
1	time-information	Time Information	Subfield
2	subscriber-information	Subscriber Information	Subfield
3	connection-information	Client Connection Information	Subfield
4	request-information	HTTP Request Information	Subfield
5	response-information	HTTP Response Information	Subfield
6	compression-information	Compression Information	Subfield
7	web-image-optimization-information	Web-Image Optimization Information	Subfield
8	video-information	Video Information	Subfield
9	minification-information	Minification Information	Subfield
10	preempt-dns-information	Preempt DNS Information	Subfield
11	cache-information	Cache Information	Subfield
12	tcp-splicing-information	TCP Splicing Information	Subfield
13	manifest-information	Manifest Information	Subfield
14	tcp-client-connection-information	TCP Client Connection Information	Subfield
15	filter-information	Filter Information	Subfield
16	tcp-server-connection-information	TCP Server Connection Information	Subfield
17	entitlement-information	Entitlement Information	subfield
18	udp-proxy-information	UDP Proxy Information	subfield
19	dpi-information	DPI Information	subfield
20	consent-information	Consent Information	subfield

Time Information

Table 4-2 lists the default position, name, data type and description for each subfield contained in the Time Information HTDR.

Table 4-2 Time Information Subfields

Position	Name	Data Type	Description
1	Transaction Timestamp	Integer64	The timestamp indicating when the HTTP transaction starts, in Epoch time (seconds).
2	Request Latency	Unsigned64	Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP request to the server. (Starts counting from the moment the HTTP proxy receives the first byte of data from the client to when it sends the last byte of data to the server.)
3	Response Latency	Unsigned64	Indicates the amount of time, in milliseconds, it takes the HTTP proxy to forward an HTTP response to the client. (Starts counting from the moment the HTTP proxy receives the first byte of data from the server to when it sends the last byte of data to the client.)
4	Transaction Duration	Unsigned64	Indicates the total amount of time of the transaction, in milliseconds.
5	Transaction Time Sub-Seconds	Unsigned64	The millisecond portion of the timestamp, indicating when the HTTP transaction starts. ***Note:*** *Field 1 and Field 5 together provide the date and time of the transaction to the millisecond level.*
6	Transaction ID	Unsigned Integer	Indicates the transaction identity.
7	Recording Entity	String	Indicates the name of the proxy process.

Subscriber Information

Table 4-3 lists the default position, name, data type and description for each subfield contained in the Subscriber Information HTDR.

Table 4-3 Subscriber Information Subfields

Position	Name	Data Type	Description
1	IMSI	Unsigned64	Indicates the International Mobile Subscriber Identity (IMSI) of the User Equipment (UE) that originated the request.
2	IMEI	Unsigned64	Indicates the International Mobile Equipment Identity (IMEI) of the UE that originated the request.
3	MSISDN	Unsigned64	Indicates the primary Mobile Station (MS) ISDN number (MSISDN) of the subscriber.

Table 4-3 Subscriber Information Subfields

Position	Name	Data Type	Description
4	Session ID	Unsigned Integer	Identifies the Mobile Content Cloud’s session ID for the subscriber's PDN connection.
5	Serving NW MCC	Unsigned Integer	Indicates the Mobile Country Code (MCC) of the serving network.
6	Serving NW MNC	Unsigned Integer	Indicates the Mobile Network Code (MNC) of the serving network.
7	Home PLMN ID Length	Unsigned Integer	Indicates the length of the home Public Land Mobile Network (PLMN) Identifier.
8	RAT Type	Unsigned Integer	Indicates the Radio Access Technology (RAT) type currently used by the Mobile Station, when available. The following values are defined: ■ 0 = Reserved ■ 1 = UTRAN ■ 2 = GERAN ■ 3 = WLAN ■ 4 = GAN ■ 5 = HSPA-Evolution ■ 6 = EUTRAN ■ 101 = IEEE 802.16e ■ 102 = 3GPP2-eHRPD ■ 103 = 3GPP2-HRPD ■ 104 = 3GPP2-1xRTT ■ 105 = 3GPP-EPS
9	Roaming Status	Boolean	Indicates whether the subscriber is roaming. The following values are defined: ■ 0 = False ■ 1 = True
10	SGW IP	Hexadecimal Bytes	Indicates the IP address of the SGW used (can be an IPv4 or IPv6 address).
11	APN (Mapped)	String	Indicates the logical name of the mapped access point to the external packet data network.
12	APN (Received)	String	Indicates the logical name of the connected access point to the external packet data network.

Table 4-3 Subscriber Information Subfields

Position	Name	Data Type	Description
13	Device Type	Enumerated	Indicates the type of device. The following values are defined: ■ 0 = Unknown ■ 1 = Handheld ■ 2 = FeaturePhone ■ 3 = SmartPhone ■ 4 = Modem ■ 5 = Dongle ■ 6 = Tablet ■ 7 = Vehicle ■ 8 = ConnectedDevice ■ 9 = Router ■ 10 = Module ■ 11 = eBook ■ 12 = FemotoAp
14	Device Make	String	Indicates the make (brand) of the device.
15	Device Model	String	Indicates the model of the device.
16	Device Screen Width	Unsigned Integer	Indicates the screen width of the device.
17	Device Screen Height	Unsigned Integer	Indicates the screen height of the device.
18	C Plane session ID	Unsigned Integer	Indicates the session identifier on C-Plane.
19	Proxy Instance Name	String	Indicates the name of the HTTP proxy instance.
20	Session Operator Specific 1	String	Indicates Operator-specific data.
21	User Location Info	Bytes	Indicates user location information.

Client Connection Information

Table 4-4 lists the default position, name, data type and description for each subfield contained in the Client Connection Information HTDR.

Table 4-4 Client Connection Information Subfields

Position	Name	Data Type	Description
1	Gi Network Context	String	Indicates the network context of the Gi interface.
2	Source IP Address	Hexadecimal Bytes	Indicates the source IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).
3	Source Port	Unsigned Integer	Indicates the source port of the IP packets received from the UE.
4	Destination IP Address	Hexadecimal Bytes	Indicates the destination IP address of the IP packets issued by the UE (can be an IPv4 or IPv6 address).
5	Destination Port	Unsigned Integer	Indicates the destination port of the IP packets received from the UE.
6	IP-Transparency	Boolean	Indicates whether IP-transparency is enabled or disabled. With IP-transparency, HTTP requests forwarded by the HTTP proxy to origin servers contain the UE's original IP address as the source TCP/IP address in the IP packet. The following values are defined: ■ 0 = Disabled ■ 1 = Enabled

HTTP Request Information

Table 4-5 lists the default position, name, data type and description for each subfield contained in the HTTP Request Information HTDR.

Table 4-5 HTTP Request Information Subfields

Position	Name	Data Type	Description
1	HTTP Request Method	Enumerated	Indicates the reason for issuing the HTTP request. The following values are defined: ■ 0 = Unknown ■ 1 = OPTIONS ■ 2 = GET ■ 3 = HEAD ■ 4 = POST ■ 5 = PUT ■ 6 = DELETE ■ 7 = TRACE ■ 8 = CONNECT ■ 9 = MKCOL ■ 10 = COPY ■ 11 = MOVE ■ 12 = PROPFIND ■ 13 = PROPATCH ■ 14 = LOCK ■ 15 = UNLOCK ■ 16 = PATCH
2	Client Request URL (Original)	String (Percent-Encoded)	Indicates the URL that the client (UE) used to make the request.
3	Server Request URL	String (Percent-Encoded)	Indicates the URL sent to the server. ***Note:*** *This field is filled only if the HTTP proxy updates the URL sent to the server.*
4	Client Side Protocol	Enumerated	Indicates the protocol used by the client to communicate with the server. The following values are defined: ■ 0 = TCP ■ 1 = TLS ■ 2 = WAP 1.x Connectionless ■ 3 = WAP 1.x Connection Oriented ■ 4 = WAP 1.x Connectionless with Security ■ 5 = WAP 1.x Connection Oriented with Security

Table 4-5 HTTP Request Information Subfields

Position	Name	Data Type	Description
5	Server Side Protocol	Enumerated	Indicates the protocol used by the server to communicate with the client. The following values are defined: ■ 0 = TCP ■ 1 = TLS ■ 2 = WAP 1.x Connectionless ■ 3 = WAP 1.x Connection Oriented ■ 4 = WAP 1.x Connectionless with Security ■ 5 = WAP 1.x Connection Oriented with Security
6	User Agent	String (Percent-Encoded)	Identifies the client software originating the request as received by the HTTP proxy.
7	Content-Type	String (Percent-Encoded)	Indicates the kind of content contained in the HTTP request. Only applicable for the HTTP POST request method. The following values are defined: ■ 0 = Not Applicable ■ 1 = video/x-flv ■ 2 = video/mp4 ■ 3 = video/webm ■ 4 = video/other ■ 5 = app/octetstream ■ 6 = app/javascript ■ 7 = text/javascript ■ 8 = text/plain ■ 9 = text/html ■ 10 = text/css ■ 11 = image/bmp ■ 12 = image/gif ■ 13 = image/jpeg ■ 14 = image/png
8	Content-Length	Unsigned64	Indicates the amount of data (in bytes) in the Content-Length Header of the HTTP request from the client.
9	Client Request Header Size	Unsigned64	Indicates the size of the headers in the HTTP request from the client, in bytes.
10	Client Request Body Size	Unsigned64	Indicates the size of the body in the HTTP request from the client, in bytes.
11	Server Request Header Size	Unsigned64	Indicates the size of the Header in the HTTP request sent from the HTTP proxy to the server, in bytes.
12	Server Request Body Size	Unsigned64	Indicates the size of the body in the HTTP request sent from the HTTP proxy to the sever, in bytes.

Table 4-5 HTTP Request Information Subfields

Position	Name	Data Type	Description
13	Range Request	Boolean	Indicates whether the HTTP request is a Range Request. Range Requests are requests for a subset of a larger file based on one or more ranges. The following values are defined: ■ 0 = False ■ 1 = True
14	Request Successful	Boolean	Indicates whether the HTTP request is successful. The following values are defined: ■ 0 = False ■ 1 = True
15	Chunked Request	Boolean	Indicates whether the HTTP request is a chunked request. The following values are defined: ■ 0 = False ■ 1 = True
16	Explicit Proxy Request	Boolean	Indicates whether the HTTP request is an explicit proxy request. With explicit proxy, the client's mobile equipment is configured to send requests directly to the HTTP proxy. The following values are defined: ■ 0 = False ■ 1 = True
17	DNS Resolution Applied	Boolean	Indicates whether Domain Name System (DNS) resolution was used. The following values are defined: ■ 0 = False ■ 1 = True
18	Proxy Local IP Address	Hexadecimal Bytes	Indicates the HTTP proxy local loopback address used (can be an IPv4 or IPv6 address).
19	Proxy Local IP Port	Unsigned Integer	Indicates the HTTP proxy local port.
20	Origin Server IP Address	Hexadecimal Bytes	Indicates the IP address of the Origin Server (can be an IPv4 or IPv6 address).
21	Origin Server Port	Unsigned Integer	Indicates the Origin Server port.
22	BEF Applied	Boolean	Indicates whether Back-End Forwarding (BEF) was used. The following values are defined: ■ 0 = False ■ 1 = True

Table 4-5 HTTP Request Information Subfields

Position	Name	Data Type	Description
23	Proxy Network Context	String	Indicates the network context of the HTTP proxy. ***Note:*** *The value for this field might differ from the value in the Gi Network Context subfield (Client Connection Information) if the server-connection network context field provisioned in the HTTP Proxy Instance is different from the incoming network context.*
24	Referer URL	String (Percent-Encoded)	Indicates the URL of the original domain which originated the particular request.
25	Applied Policy Name	String (Percent-Encoded)	Indicates the name of the optimization policy that was matched and applied to the request.
26	Client Range Start	Unsigned64	Indicates the start of an HTTP Range Request from the client. Range Requests are requests for a subset of a larger file based on one or more ranges.
27	Client Range End	Unsigned64	Indicates the end of an HTTP Range Request from the client.
28	Failure Reason	Enumerated	Indicates the HTTP SVC error code. The following values are defined: 0 = No Error – init 1 = Client Close - Incomplete Request 2 = Client Close - Incomplete Response 3 = Client Close - Pipeline Drop 4 = Request Overdue 5 = Invalid Request - Non-HTTP 6 = Invalid Request - Too Large 7 = Invalid Request - Line Parse Failure 8 = Invalid Request - Headers Parse Failure 9 = Invalid Request - Header Bad Attribute 10 = Invalid Request - Header Duplicate Attribute 11 = Invalid Request - Bad Hostname 12 = Invalid Request - No Host Found 13 = Request Process Failure - Unsupported 14 = Request Process Failure - Expect 15 = Request Process Failure - Body Read 16 = Request Process Failure - Request Compile Error 17 = Request Process Failure - Server Init 18 = Server Close - Incomplete Request 19 = Server Close - Incomplete Response 20 = Invalid Response - Header Too Large

Table 4-5 HTTP Request Information Subfields

Position	Name	Data Type	Description
			21 = Invalid Response - Status Parse Failure 22 = Invalid Response - Headers Parse Failure 23 = Response Process Failure 24 = Cached Response Read Failure 25 = Cached Response Process Failure 26 = Filter Response Failure 27 = Precondition Failure 28 = Client Close Timeout 29 = Server Close Timeout 30 = Maintenance Connection Close 31 = LDAP Reject 32 = MMS Maximum Message Size Exceeded 33 = MMS Maximum Recipients Exceeded 34 = CFI Failure 35 = Connect Request Error 36 = Filter Reject 37 = AOC Reject 38 = Service Unavailable 39 = Service Resource Unavailable 40 = Redirected 41 = Blocked 42 = Plugin Server 43 = Host IP Same As Proxy Local IP 44 = Client Handoff For Bypass Failed 45 = Cache Incorrect Status 46 = Request Bypass 47 = No Response From Server 48 = Content Insertion Server 49 = Unsupported Method 50 = Request Connect To Local Address 51 = Out of Buffer

HTTP Response Information

Table 4-6 lists the default position, name, data type and description for each subfield contained in the HTTP Response Information HTDR.

Note: *The HTTP Response Information subfields are not populated for TCP-spliced transactions.*

Table 4-6 HTTP Response Information Subfields

Position	Name	Data Type	Description
1	HTTP Response Status	Unsigned Integer	Indicates the HTTP response status code.
2	Server Content-Type (if applicable)	String (Percent-Encoded)	Indicates the type of content contained in the HTTP response sent from the server (if applicable).
3	Client Content-Type (if updated)	String (Percent-Encoded)	Indicates the type of content contained in the HTTP response sent to the client (UE). ***Note:*** *This field is filled only if the HTTP proxy updates the content-type sent to the client.*
4	Server Content-Length (if present)	Unsigned64	Indicates the amount of data (in bytes) in the body of the HTTP response from the server (if present).
5	Client Content-Length (if updated)	Unsigned64	Indicates the amount of data (in bytes) in the body of the HTTP response sent to the client, if updated by the HTTP proxy. ***Note:*** *This field is filled only if the HTTP proxy updates the content-length sent to the client.*
6	Server Response Header Size	Unsigned64	Indicates the size of the headers in the HTTP response from the server, in bytes.
7	Server Response Body Size	Unsigned64	Indicates the size of the body in the HTTP response from the server, in bytes.
8	Client Response Header Size	Unsigned64	Indicates the size of the headers in the HTTP response sent to the client, in bytes.
9	Client Response Body Size	Unsigned64	Indicates the size of the body in the HTTP response sent to the client, in bytes.
10	Chunked Response	Boolean	Indicates whether the HTTP response is chunked. The following values are defined: ■ 0 = False ■ 1 = True

Table 4-6 HTTP Response Information Subfields

Position	Name	Data Type	Description
11	Response Served from Cache	Boolean	Indicates whether the HTTP proxy served the response from the cache. The following values are defined: ■ 0 = False ■ 1 = True
12	Response Cached	Boolean	Indicates whether the HTTP proxy cached the response. The following values are defined: ■ 0 = False ■ 1 = True
13	Detected Content-Type (If updated)	Enumerated	Indicates the type of content detected in the HTTP response sent to the client (UE). The following values are defined: ■ 1 = video/x-flv ■ 2 = video/mp4 ■ 3 = video/webm ■ 4 = video/other ■ 5 = application/octet-stream ■ 6 = application /javascript ■ 7 = text/javascript ■ 8 = text/plain ■ 9 = text/html ■ 10 = text/css ■ 11 = image/bmp ■ 12 = image /gif ■ 13 = image /jpeg ■ 14 = image/png ■ 15 = video/dash ■ 16 = video/hls ■ 17 = video/hds ■ 18 = application/xml ***Note:*** *This field is filled only if the HTTP proxy updates the detected content-type sent to the client.*
14	Applied Policy Name	String (Percent-Encoded)	Indicates the name of the optimization policy that was matched and applied to the request.
15	Server Range Start	Unsigned64	Indicates the start of an HTTP Range Request to the server from the MCC. Range Requests are requests for a subset of a larger file based on one or more ranges.
16	Server Range End	Unsigned64	Indicates the end of an HTTP Range Request to the server from the MCC.

Compression Information

Table 4-7 lists the default position, name, data type and description for each subfield contained in the Compression Information HTDR.

Table 4-7 Compression Information Subfields

Position	Name	Data Type	Description
1	Compression Method (if any)	Enumerated	Indicates the method used for compressing the HTTP data, if compression is applied. The following values are defined: ■ 0 = Identity ■ 1 = gzip ■ 2 = Deflate ■ 3 = Last
2	Compressed by Server	Boolean	Indicates whether the origin server compressed the HTTP response. The following values are defined: ■ 0 = False ■ 1 = True
3	Compressed by Proxy	Boolean	Indicates whether the HTTP proxy compressed the HTTP response sent to the client. The following values are defined: ■ 0 = False ■ 1 = True
4	Compression Ratio (if any)	Unsigned Integer	Indicates the percentage of bandwidth saved, if compression is applied.
5	Bytes Optimized Out	Unsigned Integer	Indicates the number of bytes saved, if compression is applied.

Web-Image Optimization Information

Table 4-8 lists the default position, name, data type and description for each subfield contained in the Web-Image Optimization Information HTDR.

Table 4-8 Web-Image Optimization Information Subfields

Position	Name	Data Type	Description
1	Server Image Type	String	Indicates the type of image sent from the server.
2	Client Image Type	String	Indicates the type of image sent to the client.
3	Server Resolution	String	Indicates the resolution of the image sent from the server, displayed as width x height.
4	Client Resolution	String	Indicates the resolution of the image sent to the client, displayed as width x height.
5	Optimization Ratio (if any)	Unsigned Integer	Indicates the percentage of bandwidth saved, if Web optimization is applied.
6	Bytes Optimized Out	Unsigned Integer	Indicates the number of bytes saved, if Web optimization is applied.

Video Information

Table 4-9 lists the default position, name, data type and description for each subfield contained in the Video Information HTDR.

> ***Note:*** *Video Information subfields 8 through 23 apply to online transcoding only.*

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
1	Served Container Type	Enumerated	Indicates the type of container served to the client. The following values are defined: ■ 1 = video/x-flv ■ 2 = video/mp4 ■ 3 = video/webm ■ 4 = other video types (not supported for transcoding) ■ 15 = video/dash ■ 16 = video/hls ■ 17 = video/hds
2	Online Transcoding Applied	Boolean	*Not supported in version 8.2 and later.* Indicates whether online transcoding was used. Online transcoding is used to compress video content in real-time when video is playing back from a user's device. The following values are defined: ■ 0 = False ■ 1 = True
3	Served Adapted Video (offline-transcoding)	Boolean	*Not supported in version 8.2 and later.* Indicates whether the HTTP proxy served a transcoded video from the cache. The following values are defined: ■ 0 = False ■ 1 = True
4	Video Paced	Boolean	Indicates whether the video was paced. The following values are defined: ■ 0 = False ■ 1 = True
5	Quality of Experience (if adapted)	Unsigned Integer	*Not supported in version 8.2 and later.* Indicates the Quality of Experience (QoE) grade level (0-5).
6	Paced Video Bit Rate	Unsigned Integer	Indicates the bit rate of the video, in bits per second (bps).

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
7	Bytes Optimized Out	Unsigned Integer	*Not supported in version 8.2 and later.* Indicates the number of bytes saved, if video adaptation is used.
8	Source Media Width	Integer	*Not supported in version 8.2 and later.* Indicates the width of the source media.
9	Target Media Width	Integer	*Not supported in version 8.2 and later.* Indicates the width of the target media.
10	Source Media Height	Integer	*Not supported in version 8.2 and later.* Indicates the height of the source media.
11	Target Media Height	Integer	*Not supported in version 8.2 and later.* Indicates the height of the target media.
12	Source Media Bit Rate	Integer	*Not supported in version 8.2 and later.* Indicates the bit rate of the source media, in bps.
13	Target Media Bit Rate	Integer	*Not supported in version 8.2 and later.* Indicates the bit rate of the target media, in bps.
14	Source Media Frame Rate	Double (floating point)	*Not supported in version 8.2 and later.* Indicates the frame rate of the source media, in frames per second (fps).
15	Target Media Frame Rate	Double (floating point)	*Not supported in version 8.2 and later.* Indicates the frame rate of the target media, in fps.
16	Source Media Audio Sampling Rate	Integer	*Not supported in version 8.2 and later.* Indicates the audio sampling rate of the source media, in Hertz (Hz).
17	Target Media Audio Sampling Rate	Integer	*Not supported in version 8.2 and later.* Indicates the audio sampling rate of the target media, in Hz.
18	Source Media Video Codec	String	*Not supported in version 8.2 and later.* Indicates the video codec name of the source media, such as H265 or VP9.
19	Target Media Video Codec	String	*Not supported in version 8.2 and later.* Indicates the video codec name of the target media, such as H265 or VP9.
20	Source Media Audio Codec	String	*Not supported in version 8.2 and later.* Indicates the audio codec name of the source media, such as Advanced Audio Coding (AAC) or Vorbis.
21	Target Media Audio Codec	String	*Not supported in version 8.2 and later.* Indicates the audio codec name of the target media, such as Advanced Audio Coding (AAC) or Vorbis.

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
22	Source Media Audio Channels	Integer	*Not supported in version 8.2 and later.* Indicates the audio channel used for the source media. The following values are defined: ■ 1 = Monaural ■ 2 = Stereo
23	Target Media Audio Channels	Integer	*Not supported in version 8.2 and later.* Indicates the audio channel used for the target media. The following values are defined: ■ 1 = Monaural ■ 2 = Stereo

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
24	Video Bypass Cause	Enumerated	*Not supported in version 8.2 and later.* Indicates the reason that the file was bypassed. The following values are defined: ■ 0 = Bypass Success ■ 1 = Source Format Not Supported ■ 2 = Source Media Format Not Streamable ■ 3 = Source Media Format Error ■ 4 = Above Maximum Source Object Size ■ 5 = Below Minimum Source Object Size ■ 6 = Above Maximum Source Duration ■ 7 = Below Minimum Source Duration ■ 8 = Above Maximum Source Bitrate ■ 9 = Above Maximum Source Framerate ■ 10 = Operation Canceled ■ 11 = In Error State ■ 12 = Above Maximum Picture Dimensions ■ 13 = No Source Audio Stream ■ 14 = No Source Video Stream ■ 15 = Unknown Source Media Format ■ 16 = Could Not Find Source Media Stream Info ■ 17 = Unsupported Source Audio Codec ■ 18 = Unsupported Source Video Codec ■ 19 = Moov Atom Missing ■ 20 = Bad Moov Atom ■ 21 = Below Bitrate Savings ■ 22 = MAP Table Entry Indicates Bypass ■ 23 = No MAP Table Entry Match ■ 24 = Operating Mode Invalid ■ 25 = Maximum Quality Grade List Truncated ■ 26 = Maximum Quality Grade Out of Range ■ 27 = No Profile Database or MAP ■ 28 = Invalid Parameter (Speed Level) ■ 29 = Invalid Parameter (Maximum Object Size) ■ 30 = Invalid Parameter (Minimum Object Size)

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
24	Video Bypass Cause (Continued)	Enumerated	■ 31 = Invalid Parameter (Maximum Duration) ■ 32 = Invalid Parameter (Minimum Duration) ■ 33 = Invalid Parameter (Maximum Bitrate) ■ 34 = Invalid Parameter (Maximum Framerate) ■ 35 = Invalid Parameter (Bitrate Savings) ■ 36 = Supported Container Type Not In Database ■ 37 = No Quality Grades Supported in Database ■ 38 = Maximum Horizontal Pixels Not in Database ■ 39 = Error Reading Property in Database ■ 40 = Property Value Syntax Error in Database ■ 41 = Property Value Range Error in Database ■ 42 = Maximum Vertical Pixels Not in Database ■ 43 = Match Criteria Range Error In MAP Table ■ 44 = Xcode Parameter Range Error in MAP Table ■ 45 = Target Format Not Supported ■ 46 = Target Format Parameters Not Valid ■ 47 = Target Transcoding Parameters Not Valid ■ 48 = Unsupported Target Audio Codec ■ 49 = Unsupported Target Video Codec ■ 50 = Source Media Decoding Failure ■ 51 = Target Media Transcoding Failure ■ 52 = Internal Xcode Error ■ 53 = Video Optimization Disabled ■ 54 = Video Optimization Not Required ■ 55 = Optimization Overload ■ 56 = No QoE ■ 57 = Internal Failure ■ 58 = Media Not Applicable
25	Video Pacing Bytes Saved	Unsigned64	Indicates the number of bytes saved from video pacing.
26	ABR Rate Limiting Applied	Boolean	Indicates whether rate limiting was applied. The following values are defined: ■ 0 = False ■ 1 = True

Table 4-9 Video Information Subfields

Position	Name	Data Type	Description
27	ABR Rate Limiting Bypass Reason	Enumerated	Indicates the reason for bypassing rate limiting. The following values are defined: ■ 1 = ABR_SUBSCRIBER_OPTOUT: Subscriber Opt-out ■ 2 = ABR_NO_MATCHING_PROFILE: No Matching ABR Profile ■ 3 = ABR_VIDEO_PACING_APPLIED: Video Pacing Applied ■ 4 = ABR_SUBSCRIBER_OPTIN_REQ_NOT_MET: Subscriber Opt-in Requirement Not Met ■ 5 = ABR_CANCEL_TOOSHORT: Segment too short, rate limiting canceled ■ 6 = ABR_CP_MANIFEST_NO_MAX_RATE: Manifest contains no maximum rate ■ 7 = ABR_CP_MANIFEST_INVALID_MAX_RATE: Manifest contains invalid maximum rate ■ 8 = ABR_SAVINGS_ESTIMATION
28	ABR Rate Limiting Bitrate	Integer	Indicates the actual bit rate of the video segment, in bps.
29	ABR Session ID	Unsigned64	Indicates the unique session ID for the ABR session.
30	ABR Bytes Saved	Unsigned64	Indicates the total bytes saved by enforcing ABR.
31	Applied Policy Name	String (Percent-Encoded)	Indicates the name of the optimization policy that was matched and applied to the request.

Minification Information

Table 4-10 lists the default position, name, data type and description for each subfield contained in the Minification Information HTDR.

Table 4-10 Minification Information Subfields

Position	Name	Data Type	Description
1	Minify Type	Enumerated	Indicates the type of data minified. The following values are defined: ■ 7 = text/javascript ■ 9 = text/html ■ 10 = text/css
2	Bytes Optimized Out	Unsigned Integer	Indicates the number of bytes saved, if minification is used.

Preempt DNS Information

Table 4-11 lists the default position, name, data type and description for each subfield contained in the Preempt Domain Name system (DNS) Information HTDR.

Table 4-11 Preempt DNS Information Subfields

Position	Name	Data Type	Description
1	Applied to the Response	Boolean	Indicates whether the HTTP proxy applied preemptive DNS to the HTTP response. Preemptive DNS replaces URLs in the HTTP response with the IP address of one of the Affirmed Networks Gi local loopback addresses and configured DNS Token before sending the response to the client. The following values are defined: ■ 0 = False ■ 1 = True
2	Number of URLs Adapted	Unsigned Integer	Indicates the number of URLs in the Web page for which preemptive DNS was applied.

Cache Information

Table 4-12 lists the default position, name, data type and description for each subfield contained in the Cache Information HTDR.

Table 4-12 Cache Information Subfields

Position	Name	Data Type	Description		
1	Age of the Cached Request	Signed64	Indicates the amount of time, in seconds, that the fetched item has been in the cache.		
2	Max-Age in the Request Header	Integer	If there is a max-age in the HTTP request header, this field indicates its value, in seconds.		
3	Max-Age in the Response Header	Integer	If there is a max-age in the HTTP response header, this field indicates its value, in seconds.		
4	Cache Control Flags in the Request	Bitmap Integer value	Indicates the different cache-control flags present in the HTTP request header. If multiple cache control flags are present, the bit mask values are added together and the sum is presented as a decimal value in the HTDR. For example, a value of 65538 (0x10002) is equal to CACHE_CONTROL_MAX_AGE + CACHE_CONTROL_NO_ CACHE. The following values are defined:		
				Bit Mask Value (Hex)	**Decimal Value**
			CACHE_CONTROL_NO_HIDDEN	0x00000001	1
			CACHE_CONTROL_NO_CACHE	0x00000002	2
			CACHE_CONTROL_PUBLIC	0x00000004	4
			CACHE_CONTROL_PRIVATE	0x00000008	8
			CACHE_CONTROL_NO_STORE	0x00000010	16
			CACHE_CONTROL_NO_TRANSFORM	0x00000020	32
			CACHE_CONTROL_MUST_REVALIDATE	0x00000040	64
			CACHE_CONTROL_PROXY_REVALIDATE	0x00000080	128
			CACHE_CONTROL_ONLY_IF_CACHED	0x00000100	256
			CACHE_CONTROL_VARY	0x00000200	512
			CACHE_CONTROL_COOKIE	0x00000400	1024
			CACHE_CONTROL_MISMATCH	0x00000800	2048
			CACHE_CONTROL_NO_PRAGMA	0x00001000	4096
			CACHE_CONTROL_ALWAYS	0x00002000	8192
			CACHE_CONTROL_MAX_AGE	0x00010000	65536
			CACHE_CONTROL_S_MAXAGE	0x00020000	131072
			CACHE_CONTROL_MIN_FRESH	0x00040000	262144
			CACHE_CONTROL_MAX_STALE	0x00080000	524288
			CACHE_CONTROL_CHUNKED	0x00100000	1048576

Table 4-12 Cache Information Subfields

Position	Name	Data Type	Description		
5	Cache Control Flags in the Response	Bitmap Integer Value	Indicates the different cache-control flags present in the HTTP response header. If multiple cache control flags are present, the bit mask values are added together and the sum is presented as a decimal value in the HTDR. For example, a value of 65538 (0x10002) is equal to CACHE_CONTROL_MAX_AGE + CACHE_CONTROL_NO_ CACHE. The following values are defined:		
				Bit Mask Value (Hex)	**Decimal Value**
			CACHE_CONTROL_NO_HIDDEN	0x00000001	1
			CACHE_CONTROL_NO_CACHE	0x00000002	2
			CACHE_CONTROL_PUBLIC	0x00000004	4
			CACHE_CONTROL_PRIVATE	0x00000008	8
			CACHE_CONTROL_NO_STORE	0x00000010	16
			CACHE_CONTROL_NO_TRANSFORM	0x00000020	32
			CACHE_CONTROL_MUST_REVALIDATE	0x00000040	64
			CACHE_CONTROL_PROXY_REVALIDATE	0x00000080	128
			CACHE_CONTROL_ONLY_IF_CACHED	0x00000100	256
			CACHE_CONTROL_VARY	0x00000200	512
			CACHE_CONTROL_COOKIE	0x00000400	1024
			CACHE_CONTROL_MISMATCH	0x00000800	2048
			CACHE_CONTROL_NO_PRAGMA	0x00001000	4096
			CACHE_CONTROL_ALWAYS	0x00002000	8192
			CACHE_CONTROL_MAX_AGE	0x00010000	65536
			CACHE_CONTROL_S_MAXAGE	0x00020000	131072
			CACHE_CONTROL_MIN_FRESH	0x00040000	262144
			CACHE_CONTROL_MAX_STALE	0x00080000	524288
			CACHE_CONTROL_CHUNKED	0x00100000	1048576

Table 4-12 Cache Information Subfields

Position	Name	Data Type	Description
6	Cache-Bypass Cause When Storing Data	Enumerated	Indicates the reason for bypassing the cache when attempting to store the data. The following values are defined: ■ 1 = CACHE_BYPASS_INTERNAL_ERROR ■ 2 = CACHE_BYPASS_UPDATING ■ 3 = CACHE_BYPASS_METHOD_NOT_CACHEABLE ■ 4 = CACHE_BYPASS_SERVICE_RULE_BYPASS ■ 5 = CACHE_BYPASS_RESPONSE_CODE_NOT_CACHEABLE ■ 6 = CACHE_BYPASS_CACHE_CONTROL ■ 7 = CACHE_BYPASS_RESPONSE_CHUNKED ■ 8 = CACHE_BYPASS_CACHE_DISABLED ■ 9=CACHE_BYPASS_SERVICE_RULE_DONT_CACHE_POLICY ■ 10 = CACHE_BYPASS_MAX_AGE_ZERO ■ 11 = CACHE_BYPASS_MULTIPART_REQUEST ■ 12 = CACHE_BYPASS_BEF_NO_CACHE ■ 13 = CACHE_BYPASS_FE_PROXY ■ 14=CACHE_BYPASS_CONTENT_OPTIMIZATION_MISMATCH ■ 15 = CACHE_BYPASS_STALE_ENTRY ■ 16 = CACHE_BYPASS_HEAD_INVALIDATE ■ 17 = CACHE_BYPASS_RESOURCE_UNAVAILABLE
7	Cache-Bypass Cause When Retrieving Data	Enumerated	Indicates the reason for bypassing the cache when retrieving data. The following values are defined: ■ 1 = CACHE_BYPASS_INTERNAL_ERROR ■ 2 = CACHE_BYPASS_UPDATING ■ 3 = CACHE_BYPASS_METHOD_NOT_CACHEABLE ■ 4 = CACHE_BYPASS_SERVICE_RULE_BYPASS ■ 5 = CACHE_BYPASS_RESPONSE_CODE_NOT_CACHEABLE ■ 6 = CACHE_BYPASS_CACHE_CONTROL ■ 7 = CACHE_BYPASS_RESPONSE_CHUNKED ■ 8 = CACHE_BYPASS_CACHE_DISABLED ■ 9=CACHE_BYPASS_SERVICE_RULE_DONT_CACHE_POLICY ■ 10 = CACHE_BYPASS_MAX_AGE_ZERO ■ 11 = CACHE_BYPASS_MULTIPART_REQUEST ■ 12 = CACHE_BYPASS_BEF_NO_CACHE ■ 13 = CACHE_BYPASS_FE_PROXY ■ 14=CACHE_BYPASS_CONTENT_OPTIMIZATION_MISMATCH ■ 15 = CACHE_BYPASS_STALE_ENTRY ■ 16 = CACHE_BYPASS_HEAD_INVALIDATE ■ 17 = CACHE_BYPASS_RESOURCE_UNAVAILABLE

Table 4-12 Cache Information Subfields

Position	Name	Data Type	Description
8	Is the Cache Entry Stale	Boolean	Indicates whether the HTTP response in the cache is stale based on the Expires header field in the HTTP response. If this field is true, the Cache Bypass Cause When Retrieving subfield is set to CACHE_BYPASS_STALE_ENTRY. The following values are defined: ■ 0 = False ■ 1 = True

TCP Splicing Information

Table 4-13 lists the default position, name, data type and description for each subfield contained in the TCP Splicing Information HTDR.

Table 4-13 TCP Splicing Information Subfields

Position	Name	Data Type	Description
1	Upstream Bytes Received	Unsigned64	Indicates the number of bytes received from the client.
2	Upstream Bytes Sent	Unsigned64	Indicates the number of bytes sent to the server.
3	Downstream Bytes Received	Unsigned64	Indicates the number of bytes received from the server.
4	Downstream Bytes Sent	Unsigned64	Indicates the number of bytes sent to the client.
5	Status	Enumerated	Indicates the status of the request. The following values are defined: ■ 0 = Init ■ 1 = Idle Timeout ■ 2 = Client Close ■ 3 = Server Close ■ 4 = Client Error Event ■ 5 = Server Error Event ■ 6 = Client RX Error ■ 7 = Client TX Error ■ 8 = Server RX Error ■ 9 = Server TX Error ■ 10 = Idle Force ■ 11 = No Buffer ■ 12 = No Data ■ 13 = Connect Fail ■ 14 = Hand off ■ 15 = Client Error ■ 16 = Server Error ■ 17 = DNS Request Fail ■ 18 = DNS Response Fail ■ 19 = Reject ■ 20 = TFD Reject
6	Local IP	Hexadecimal Bytes	Indicates the local IP address.
7	Local Port	Unsigned Integer	Indicates the local port number.

Table 4-13 TCP Splicing Information Subfields

Position	Name	Data Type	Description
8	Server IP	Hexadecimal Bytes	Indicates the server IP address.
9	Server Port	Unsigned Integer	Indicates the server port number.
10	Proxy NC	String	Indicates the network context of the proxy.
11	Protocol	Enumerated	Indicates the type of protocol used to communicate. The following values are defined: ■ 0 = Invalid ■ 1 = HTTP 1.x TLS ■ 2 = HTTP 2.0 ■ 3 = HTTP 2.0 TLS ■ 4 = Web Socket ■ 5 = HTTP ■ 6 = Others ■ 7 = WebRTC ■ 8 = FacebookZero
12	TLS SNI	String	Indicates the Transport Layer Security (TLS) Server Name Indication (SNI).
13	TLS Server Common Name	String	Indicates the TLS server common name.
14	Rate Limiting Applied	Boolean	Indicates whether rate limiting was applied. The following values are defined: ■ 0 = False ■ 1 = True
15	Rate Limiting Bitrate	Unsigned64	Indicates the rate-limiting bit rate of the video segment, in bps.
16	Rate Limiting Bypass Cancel Reason	Enumerated	Indicates the reason for bypassing rate limiting. The following values are defined: ■ 0 = Cancel_TooShort: Bytes to send is less than the minimum rate limiting bitrate ■ 1 = Bypass_NoAbr: Subscriber Opt-out ■ 2 = Bypass_Disable: Rate limiting admin-state is disabled ■ 3 = Bypass_opt_in_NoAbr: Opt-in requirements are not met ■ 4 = Bypass_No_MaxRate_in_CP_Manifest ■ 5 = Bypass_Invalid_MaxRate_in_CP_Manifest ■ 6 = Savings Estimation
17	Rate Limiting Burst Size	Unsigned64	Indicates the rate limiting burst size, in bytes.

Table 4-13 TCP Splicing Information Subfields

Position	Name	Data Type	Description
18	Negotiated ALPN	String	Indicates the Application Layer Protocol Negotiation (ALPN) between the client and server. The following values are defined: ■ HTTP/1.1 with the identification sequence http/1.1 ■ HTTP/2 over TLS with the identification sequences h2 or h2-8 ■ HTTP/2 over TCP with the identification sequence h2c ■ WebRTC with the identification sequence webrtc ■ Empty when no protocol indication exists in the client or server hello
19	TCP Rate-Limited Session ID	Unsigned64	Indicates the unique session ID for rate-limited connections in a TCP rate-limited session. Reported for all rate-limited connections in a TCP rate-limited session
20	TCP Rate-Limited Session Duration	Unsigned64	Indicates the TCP rate-limited session duration, in milliseconds. Reported only when the last rate-limited connection in that TCP rate-limited session has ended.
21	TCP Rate-Limited Session Bitrate	Unsigned64	Indicates the bit rate of the TCP rate-limited session, in bps. Reported only when the last rate-limited connection in that TCP rate-limited session has ended.
22	Applied Policy Name	String (Percent-Encoded)	Indicates the name of the optimization policy that was matched and applied to the request.
23	TCP RL Bytes Saved	Unsigned64	Indicates the TCP rate-limited bytes saved.

Manifest Information

Table 4-14 lists the default position, name, data type and description for each subfield contained in the Manifest Information HTDR.

Table 4-14 Manifest Information Subfields

Position	Name	Data Type	Description
1	Manifest Matched	Boolean	Indicates whether the Content Provider (CP) manifest matches with the URL. The following values are defined: ■ 0 = False ■ 1 = True
2	Manifest Max Rate	Unsigned64	Indicates the maximum rate received from the manifest, in kilobits per second (Kbps).

TCP Client Connection Information

Table 4-15 lists the default position, name, data type and description for each subfield contained in the TCP Client Connection Information HTDR.

Table 4-15 TCP Client Connection Information Subfields

Position	Name	Data Type	Description
1	Minimum Round Trip Time	Unsigned64	Indicates the minimum round trip time, in microseconds.
2	Average Round Trip Time	Unsigned64	Indicates the average round trip time, in microseconds.
3	Maximum Round Trip Time	Unsigned64	Indicates the maximum round trip time, in microseconds.
4	Minimum Downstream Maximum Segment Size (MSS)	Unsigned64	Indicates the minimum downstream MSS, in bytes.
5	Average Downstream Maximum Segment Size (MSS)	Unsigned64	Indicates the average downstream MSS, in bytes.
6	Maximum Downstream Maximum Segment Size (MSS)	Unsigned64	Indicates the maximum downstream MSS, in bytes.
7	Minimum Downstream Congestion Window	Unsigned64	Indicates the minimum downstream congestion window.
8	Average Downstream Congestion Window	Unsigned64	Indicates the average downstream congestion window.
9	Maximum Downstream Congestion Window	Unsigned64	Indicates the maximum downstream congestion window.

Table 4-15 TCP Client Connection Information Subfields

Position	Name	Data Type	Description
10	Minimum Downstream Throughput	Unsigned64	Indicates the minimum downstream throughput, in kilobits per second (Kbps).
11	Average Downstream Throughput	Unsigned64	Indicates the average downstream throughput, in Kbps.
12	Maximum Downstream Throughput	Unsigned64	Indicates the maximum downstream throughput, in Kbps.
13	Downstream Segments Retransmitted	Unsigned64	Indicates the number of downstream segments retransmitted.
14	Estimated Downstream Segments	Unsigned64	Indicates the number of estimated downstream segments.
15	TCP Client Connection Measurement Bypass Reason	Enumerated	Indicates the reason for bypassing the TCP client connection measurement. The following values are defined: ■ 0 = Bypass_ConnTooShort ■ 1 = Bypass_BrandNewConn ■ 2 = Bypass_MissedSampInterval ■ 3 = Bypass_BEFConn
16	Congestion Algorithm	String	Indicates the TCP congestion algorithm.
17	Congestion Level None Time	Unsigned64	Indicates how long the client connection stayed in the none congestion level, in milliseconds.
18	Congestion Level Mild Time	Unsigned64	Indicates how long the client connection stayed in the mild congestion level, in milliseconds.
19	Congestion Level Moderate Time	Unsigned64	Indicates how long the client connection stayed in the moderate congestion level, in milliseconds.
20	Congestion Level Severe Time	Unsigned64	Indicates how long the client connection stayed in the severe congestion level, in milliseconds.

Filter Information

Table 4-16 lists the default position, name, data type and description for each subfield contained in the Filter Information HTDR.

Table 4-16 Filter Information Subfields

Position	Name	Data Type	Description
1	Filter Name	String	Indicates the name of the filter used to block or allow HTTP/HTTPS traffic. The URL received from the UE is compared with the filter database.
2	Filter Type	Enumerated	Indicates the type of filter. ■ 0 = global whitelist ■ 1 = global blacklist ■ 2 = global service-rule ■ 3 = instance phishing ■ 4 = instance IWF ■ 5 = instance service-rule ■ 6 = instance category ■ 7 = instance online category ■ 8 = instance malware category ■ 9 = instance filter-hash-list
3	Action Applied	Enumerated	Indicates the action applied to the HTTP/HTTPS traffic when matched. ■ 0 = BLOCK ■ 1 = REDIRECT ■ 2 = TERMINATE ■ 3 = ALLOW ■ 4 = NONE
4	Filter Instance Name	String	Indicates the name of the filter instance configured on the MCC.
5	Online Category	Enumerated	Indicates the online category if categorizing matched content. ■ 1 = url ■ 2 = malware ■ 3 = phishing ■ 4 = iwf

TCP Server Connection Information

Table 4-17 lists the default position, name, data type and description for each subfield contained in the TCP Server Connection Information HTDR.

Table 4-17 TCP Server Connection Information Subfields

Position	Name	Data Type	Description
1	Round Trip Time (micro-seconds)	Unsigned64	Indicates the round trip time (RTT), in microseconds, for the server connections. These are connections initiated by the proxy to the origin server.
2	Estimated Downstream Segments Retransmitted	Unsigned64	Indicates the number of estimated downstream segments retransmitted for the server connections. These are connections initiated by the proxy to the origin server.
3	Estimated Downstream Segments Out of Order	Unsigned64	Indicates the number of estimated downstream segments out of order for the server connections. These are connections initiated by the proxy to the origin server.
4	Congestion Algorithm	String	Indicates the TCP congestion algorithm.

Entitlement Information

Table 4-18 lists the default position, name, data type and description for each subfield contained in the Entitlement Information HTDR.

Table 4-18 Entitlement Information Subfields

Position	Name	Data Type	Description
1	Entitlements	String	Indicates the tethering entitlement values.

UDP Proxy Information

Table 4-19 lists the default position, name, data type and description for each subfield contained in the UDP Proxy Information HTDR.

Table 4-19 UDP Proxy Information Subfields

Position	Name	Data Type	Description
1	Upstream Bytes received	Unsigned64	Indicates the number of upstream bytes received.
2	Upstream bytes sent	Unsigned64	Indicates the number of upstream bytes sent.
3	Downstream bytes received	Unsigned64	Indicates the number of downstream bytes received.
4	Downstream bytes sent	Unsigned64	Indicates the number of downstream bytes sent.

Table 4-19 UDP Proxy Information Subfields

Position	Name	Data Type	Description
5	Local IP Address	Hexadecimal Bytes	Indicates the local IP address.
6	Local IP Port	Unsigned64	Indicates the local IP port.
7	Server IP Address	Hexadecimal Bytes	Indicates the server IP address.
8	Server IP Port	Unsigned64	Indicates the server IP port.
9	proxyNC	String	Identifies the Network Context (NC) selected by the proxy.
10	QUIC SNI	String	Identifies the Server Name Indication (SNI) sent over the Quick UDP Internet Connection (QUIC) Client Hello that indicates the server to which the client intends to connect.
11	Quic Version	String	Identifies the version of the QUIC protocol selected by the client.
12	Quic Connection ID	Unsigned64	Identifies the QUIC connection between the client and the server.
13	Rate Limiting Applied	Boolean	Indicates whether rate limiting was applied. The following values are defined: ■ 0 = False ■ 1 = True
14	Rate Limiting Bitrate	Unsigned64	Indicates the rate-limiting bit rate of the video segment, in bps.
15	Rate Limiting Bypass Reason	Enumerated	Indicates the reason for bypassing rate limiting. The following values are defined: ■ 0 = Flow was not QUIC ■ 1 = Rate-Limiting Disabled ■ 2 = No Matching Policy ■ 3 = No Profile ■ 4 = Protocol Error ■ 5 = Internal Error ■ 6 = Rate Limiting Opted Out by Subscriber ■ 7 = Rate Limiting Subscriber Opt-in Requirement Not Met ■ 8 = Savings Estimation
16	Rate Limiting Burst Size	Unsigned64	Indicates the rate limiting burst size, in bytes.
17	Applied Policy Name	String (Percent-Encoded)	Indicates the name of the optimization policy that was matched and applied to the request.
18	UDP Rate Limiting Bytes Saved	Unsigned64	Indicates the byte savings for rate-limited QUIC video traffic.

Table 4-19 UDP Proxy Information Subfields

Position	Name	Data Type	Description
19	UDP RL Session ID	Unsigned64	Indicates the unique session ID for rate-limited flow in a UDP rate-limited session.
20	UDP RL Session Duration	Unsigned64	Indicates the UDP rate-limited session duration, in milliseconds. Reported only when the last rate-limited flow in that UDP rate-limited session has ended.
21	UDP RL Session Bitrate	Unsigned64	Indicates the bit rate of the UDP rate-limited session, in bps. Reported only when the last rate-limited flow in that UDP rate-limited session has ended.
22	Bytes Dropped Due to Buffer Limiting	Unsigned64	Indicates the number of bytes dropped by rate limiting.
23	Failure Reason	Enumerated	Indicates the UDP proxy failure reason. The following values are defined: ■ 0 = No Failure ■ 1 = Server Connection Init Failure ■ 2 = Out of Socket ■ 3 = No Connection Found ■ 4 = No Packets ■ 5 = Other Reasons

DPI Information

Table 4-20 lists the default position, name, data type and description for each subfield contained in the Deep Packet Inspection (DPI) Information HTDR.

Table 4-20 DPI Information Subfields

Position	Name	Data Type	Description
1	Application	Bytes	Indicates the application detected on a particular flow/connection.

Consent Information

Table 4-21 lists the default position, name, data type and description for each subfield contained in the Consent Information HTDR, which contains Secure HTTP Identify Protocol (SHIP) consent-specific information.

Table 4-21 SHIP Information Subfields

Position	Name	Data Type	Description
1	Content Provider	String	Indicates the Content Provider.

Table 4-21 SHIP Information Subfields

Position	Name	Data Type	Description
2	AppID	Unsigned64	Not used.
3	Consent Status	Enumerated	Indicates the consent status. ■ 0 = Consent Not Available ■ 1 = Consent Not Matched ■ 2 = Consent Matched
4	SHIP Status	Enumerated	Indicates the SHIP status. ■ 0 = SHIP success ■ 1 = Content Provider Not Found ■ 2 = Info Not Available ■ 3 = Consent Validation Failed ■ 4 = SHIP Internal Server Error

Storing HTDRs in a Local File

This section describes how to store HTDRs in a local file. HTDR files can be transfered to a billing server in either push or pull mode. For more information, see CDR Transfer Mode (page 1-7).

Note: *For a list of the supported HTDR fields, see Local HTDRs (page 4-3).*

Use the following sequence to configure the system to store HTDRs locally.

1 Create a local file in which to store HTDRs and configure the file management profile.

a Use the following CLI command to create an HTDR file and configure the file management profile to transfer HTDR files using *push* mode.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management
profile datarecord <filename-prefix>
```

where `<filename-prefix>` specifies the prefix of the filename in which to store HTDRs.

The following table describes the configurable parameters.

Parameter	Description
`compress-enabled`	Specifies whether this file should be compressed (zipped) when it is closed. Compressed data record files are stored in */public/logs/datarecord.lnk*. ■ `enabled` ■ `disabled`
`compress-format`	If compress is enabled, specify the format for compressing files: ■ `gzip` – (default) The data records are written to a file. The file is gzipped after it is closed. ■ `raw` – The data records are written in zlib raw format. The zlib raw format provides better performance compared to the gzip format because the records are compressed on the source before sending them to the agent for writing. For information on how to uncompress data records written in zlib raw format, see Uncompressing Data Records in zlib raw Format (page 1-38) ***Note:*** *Affirmed Networks recommends using the zlib raw format if you expect the system to generate data records at a rate higher than 10K data records per second.*
`dest-port`	Specifies the port of the file SFTP server.

Parameter	Description
dest-url	Specifies the URL/IP Address for the file SFTP server. If you specify a URL, ensure the DNS server IP address is configured for the management-context. `an3000-mcm-slot7cpu1(config)#` **`network-context management-context dns server-address <ip-address>`**
file-rotate-empty	Specifies whether empty HTDR files are rotated. ■ **enabled** – Empty HTDR files are rotated based on the configured `file-rotate-frequency` interval. ■ **disabled** – (default) Empty HTDR files are not rotated.
file-rotate-frequency	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (`file-rotate-size-bytes`) or max number of records (`file-rotate-num-records`). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
file-rotate-num-records	Specifies the number of records at which to rotate the files.
file-rotate-size-bytes	Specifies the size (in bytes) at which files should be rotated.
file-transfer-enabled	Specifies whether to enable or disable file transfer. ■ **enabled** ■ **disabled**
passwd	Specifies the password to log into the file SFTP server.
purge-enabled	Specifies whether to enable or disable purge/cleanup for this file. ■ **enabled** ■ **disabled**
purge-file-older-than	Specifies the frequency at which files should be purged.
purge-low-thr	Specifies the percentage utilization at which to stop transferring/purging files.
purge-thr	Specifies the percentage utilization at which to transfer/purge files.
sequence-field-width	Specifies the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following: `<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv`

Parameter	Description
`upload-dir-path`	Specifies the path on the SFTP server to which to upload the files.
`upload-frequency`	Specifies the frequency to upload files.
`upload-offset-secs`	Specifies the offset at which to upload files after upload-frequency. This value (in seconds) is added to the time at which the file should be uploaded.
`username`	Specifies the username to log into the file SFTP server.

b Use the following CLI command to create an HTDR file and configure the file management profile to transfer HTDR files using *pull* mode:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management
profile datarecord <filename-prefix>
```

where **`<filename-prefix>`** specifies the prefix of the filename in which to store HTDRs.

The following table describes the configurable parameters.

Parameter	Description
`compress-enabled`	Specifies whether this file should be compressed (zipped) when it is closed. Compressed data record files are stored in */public/logs/datarecord.lnk*. ■ `enabled` ■ `disabled`
`compress-format`	If compress is enabled, specify the format for compressing files: ■ `gzip` – (default) The data records are written to a file. The file is gzipped after it is closed. ■ `raw` – The data records are written in zlib raw format. The zlib raw format provides better performance compared to the gzip format because the records are compressed on the source before sending them to the agent for writing. For information on how to uncompress data records written in zlib raw format, see Uncompressing Data Records in zlib raw Format (page 1-38) ***Note:*** *Affirmed Networks recommends using the zlib raw format if you expect the system to generate data records at a rate higher than 10K data records per second.*

Parameter	Description
export-rule	Specifies the rules/actions to apply when exporting the file to the */public/logs/datarecord.lnk/export* directory for SFTP access. For more information, see CDR Transfer Mode (page 1-7). ■ **exportrule-default** – The system exports files to the */public/logs/datarecord.lnk/export* directory and applies purge rules for this file profile to all files in the directory. ■ **exportrule-none** – There is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.
export-user	Specifies the user that has read/write access to the files in the */public/logs/datarecord.lnk/export* directory. For information on setting the default SFTP directory for the export user, see CDR Transfer Mode (page 1-7).
file-rotate-empty	Specifies whether empty HTDR files are rotated. ■ **enabled** – Empty HTDR files are rotated based on the configured `file-rotate-frequency` interval. ■ **disabled** – (default) Empty HTDR files are not rotated.
file-rotate-frequency	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (`file-rotate-size-bytes`) or max number of records (`file-rotate-num-records`). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
file-rotate-num-records	Specifies the number of records at which to rotate the files.
file-rotate-size-bytes	Specifies the size (in bytes) at which files should be rotated.
purge-enabled	Specifies whether to enable or disable purge/cleanup for this file. ■ **enabled** ■ **disabled**
purge-file-older-than	Specifies the frequency at which files should be purged.
purge-low-thr	Specifies the percentage utilization at which to stop transferring/purging files.
purge-thr	Specifies the percentage utilization at which to transfer/purge files.

Parameter	Description
`sequence-field-width`	Specifies the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following: `<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv`

2 Configure the HTTP transaction data record template to select which top-level HTDR fields to include in the HTDR file and to modify the position of top-level HTDR fields.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
multi-protocol-proxy http-transaction <template-name> <field>
```

where

Parameter	Description
`<template-name>`	Specifies the name of the HTTP transaction data record template.
`<field>`	Specifies the name of the top-level HTDR field you are modifying.

Each top-level HTDR field in CSV format contains the following six configurable parameters:

Parameter	Description
`alignment`	The alignment of the data field. If not configured, the alignment defaults to left aligned.
`format`	The format for the data field. There is a default format for each data field, if not configured.
`length`	The fixed length of the data field. The default value is zero indicating the data field is written out as it is.
`padding`	The character used for padding when the field has a fixed length.
`position`	The location of the data field within the HTDR file. If not configured, there is a predefined order for each data field.
`present`	Indicates whether to include the data field in the HTDR file. By default, this value is set to `conditional` for all data fields, indicating that the data field is filled in the HTDR file when available.

3 Configure the HTTP Transaction Record Profile.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy
http-transaction-record-profile <name> file-name <filename> file-type
datarecord encode-format csv transaction-template <template-name>
record-url-length <value>
```

The following table describes the configurable parameters.

Parameter	Description
`http-transaction-record-profile`	Specify the name of the HTTP transaction record profile.
`file-name`	Specify the file name for streaming HTDR records locally.
`file-type`	Specify `datarecord` as the file type for streaming HTDR records locally.
`encode-format`	Specify `csv` as the format in which encoded HTDRs are saved to the local disk.
`transaction-template`	Specify the HTTP transaction data record template name.
`record-url-length`	Specify the length to which the URL field should be truncated in the record. The range is 10-65535.

4 Specify the HTTP Transaction Record Profile in the Multi Protocol Proxy Instance tied to the Workflow Data Profile for it to take effect.

```
an3000-mcm-slot7cpu1(config)# services multi-protocol-proxy instance
<name> http-transaction-record-profile <name>
```

Note: *The Multi Protocol Proxy instance must already exist. See the Affirmed Networks Value Added Services Guide for information on configuring a Multi Protocol Proxy instance.*

Chapter 5

Ga Interface

This section contains the following information used to control and monitor the Ga interface:

Note: *WAG and ePDG CDRs are only stored in CSV format.*

Note: *For a complete ASN.1 definition of the General Packet Radio Service (GPRS) records based on the ASN.1 definition in 3GPP TS 32.298, see* *Appendix B, ASN.1 Definition**.*

Interface Overview

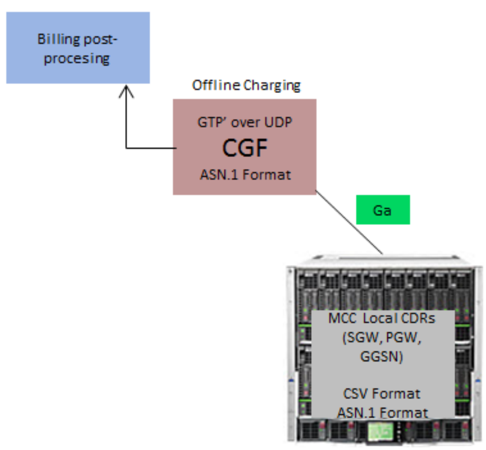
The Ga interface transports Abstract Syntax Notation 1 (ASN.1) encoded CDR files from the gateway agents (SGW/PGW/GGSN) to the *Charging Gateway Function* (CGF). The CGF is a device on the GSM GPRS or UMTS network that collects and maintains CDRs for subscriber contexts.

The Ga interface uses the GTP Prime protocol (GTP' or GTPP), with modifications, to transport CDRs. GTP' is required when the CGF resides outside the CDR generating nodes. In most cases, this means the CGF collects and processes CDRs from many individual network elements, such as the GGSNs, and sends them to a centralized computer which then delivers the charging data to the network Operator's billing center. GTP' path communications are conducted using UDP/TCP over IP.

Note: *For a complete ASN.1 definition of the General Packet Radio Service (GPRS) records based on the ASN.1 definition in 3GPP TS 32.298, see* *Appendix B, ASN.1 Definition*.

Figure 5-1 illustrates the framework for CDR transport over the Ga interface.

Figure 5-1 CDR Transport Over Ga Interface

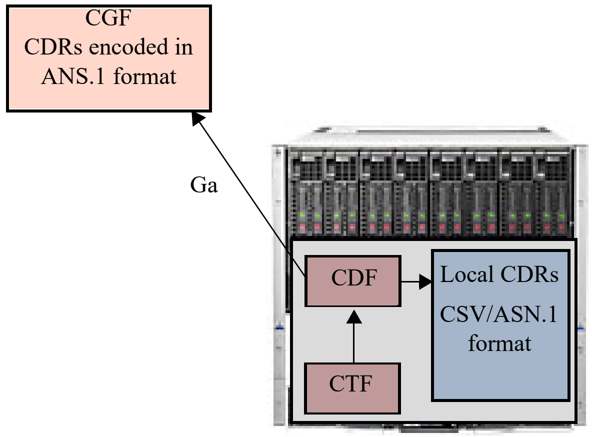


Within the gateways (SGW/PGW/GGSN), the following two network elements work to generate CDRs and send them to the CGF for further processing.

- *Charging Trigger Function* (CTF) – Generates charging event information based on service usage and forwards this information to the CDF.
- *Charging Data Function* (CDF) – Generates CDRs using the event information received from the CTF, stores them locally on the system and also transfers the CDRs to the CGF.

Figure 5-2 illustrates these two network elements.

Figure 5-2 CTF and CDF Framework



Data Record Templates

Operators can use data record templates to include or exclude CDR fields in the CDR file. For more information, see Data Record Templates for CDRs in ASN.1 Format (page 1-13).

GTP’

The system uses the GTP’ protocol to transport ASN.1 encoded CDRs from the gateways (SGW/PGW/GGSN) to the peer CGF via the Ga interface. The MCC complies with 3GPP 32.295 as follows:

- Uses the UDP as the path protocol.
- Maintains the link towards the peer CGF.
- Encapsulates the encoded charging records from the gateways into a GTP’ Data Transfer Request.

GTP’ performs the following functions:

- Transfers CDRs between the CDF and the CGF.
- Redirects CDRs to another CGF.
- Detects communication failures between the communicating peers, using Echo messaging.
- Advertises to peers about its CDR transfer capability (for example, after a period of service downtime).
- Prevents duplicate CDRs that may arise during redundancy operations.

  If configured, the CDR duplication prevention function may also be carried out by marking potentially duplicated CDR packets and delegating the final duplicate deletion task to a CGF or the Billing Domain (instead of handling the possible duplicates solely by GTP' messaging).

GSN-GSN communication occurs using GTP’ over UDP (compliant with STD 0006[404]) over IP as follows:

Ports for signaling the request messages:

- The UDP Destination Port may be the server port number 3386 which has been reserved for GTP’.
- Alternatively another port can be used, which has been configured by O&M, except Port Number 2123, which is used by GTPv2-C.
- The UDP Source Port is a locally allocated port number at the sending network element.

Note: *The UDP Source Port number should be different from the UDP Source Port number allocated to GTPv2 as defined in 3GPP TS 29.274.*

Ports for signaling the response messages:

- The UDP Destination Port value is the value of the Source Port of the corresponding request.
- The UDP Source Port is the value from the Destination Port of the corresponding request message.

Ga Message Flows

This section provides examples of message flows for transferring CDRs to the CGF over the Ga interface:

- CDR Transfer Over the Ga Interface (page 5-7)
- Failover to Secondary CGF (page 5-8)
- Forced Failover to Secondary CGF (page 5-10)
- Preventing Duplicate CDRs During CGF Failover (page 5-12)

Note: *When the GTPP server is not reachable, the PGW/SGW generates GTPP Interim CDRs for all CDR events, including volume/time, and buffers them in memory and disk. Once the GTPP server link is restored, the MCC plays back all Interim CDRs buffered to the server. To reduce the load on the server after a GTPP failover, the PGW/SGW sends up to 128 Interim CDRs from heap or 64 Interim CDRs from disk. The GTPP process buffers a total of 7000 transactions.*

CDR Transfer Over the Ga Interface

Figure 5-3 demonstrates the message flow for a typical CDR transfer where CDRs are successfully transfered between the CDF and CGF over the Ga interface.

Figure 5-3 Typical CDR Transfer Over the Ga Interface

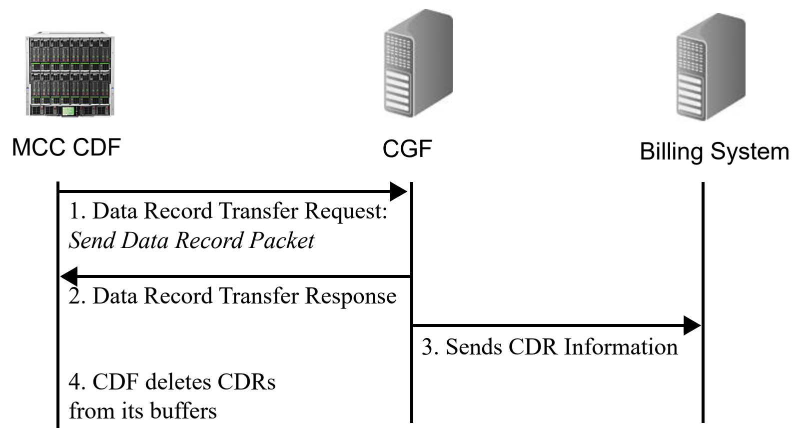


Table 5-1 Typical CDR Transfer Over the GA Interface

Step	Description
1.	The CDF sends a Data Record Transfer Request with the message Send Data Record Packet to the CGF.
2.	The CGF sends a Data Record Transfer Response to the CDF acknowledging the successful receipt of the Data Record Transfer message(s).
3.	The CGF sends the CDR information to the Billing system for post-processing.
4.	The CDF deletes all successfully sent CDRs from its buffer.

Failover to Secondary CGF

Figure 5-4 demonstrates the message flow during a failover from the primary to the secondary CGF.

Figure 5-4 Failover to Secondary CGF

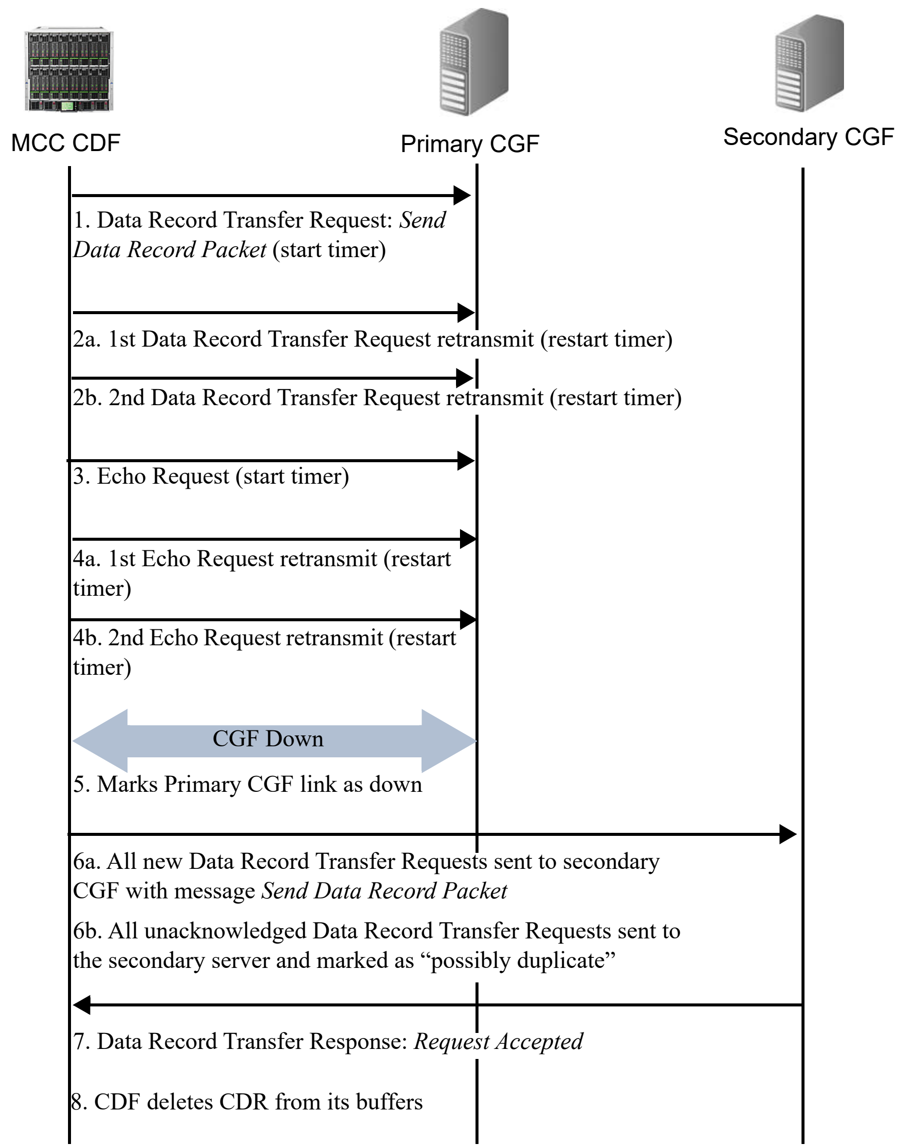


Table 5-2 Failover to Secondary CGF

Step	Description
1.	The CDF sends a Data Record Transfer Request with the message *Send Data Record Packet* to the primary CGF.

Table 5-2 Failover to Secondary CGF

Step	Description
2.	If the primary CGF does not respond to the Data Record Transfer Request within one second, the CDF retransmits the request two more times, restarting the timer with each retransmission.
3.	After three consecutive Data Record Transfer Requests with no response, the CDF sends an Echo Request to determine if the primary CGF is responsive.
4.	If the primary CGF does not respond to the Echo Request within one second, the CDF retransmits the request two more times, restarting the timer with each retransmission.
5.	After three consecutive Echo Requests with no response, the CDF marks the primary CGF as down.
6.	The CDF redirects all Data Record Transfer Requests to the secondary CGF. ■ 6a – All new Data Record Transfer Requests are sent with the message Send Data Record Packet informing the CGF to process this Request normally. For more information, see Figure 5-3 on page 5-7. ■ 6b – All unacknowledged Data Record Transfer Requests (all Requests sent to the primary server with no response) are sent with the message Send possibly duplicated Data Record Packet. This message informs the CGF that this CDR packet could be a duplicate. CDRs flagged as “possibly duplicate” are handled differently. For more information, see Figure 5-6 on page 5-12.
7.	The secondary CGF sends a Data Record Transfer Response to the CDF with the message *Request Accepted*.
8.	The CDF deletes the CDR from its buffer.

Forced Failover to Secondary CGF

Figure 5-5 demonstrates the message flow during a forced failover from the primary to the secondary CGF. A forced failover occurs when the primary CGF is removed from service for maintenance, for example.

Figure 5-5 Forced Failover to Secondary CGF

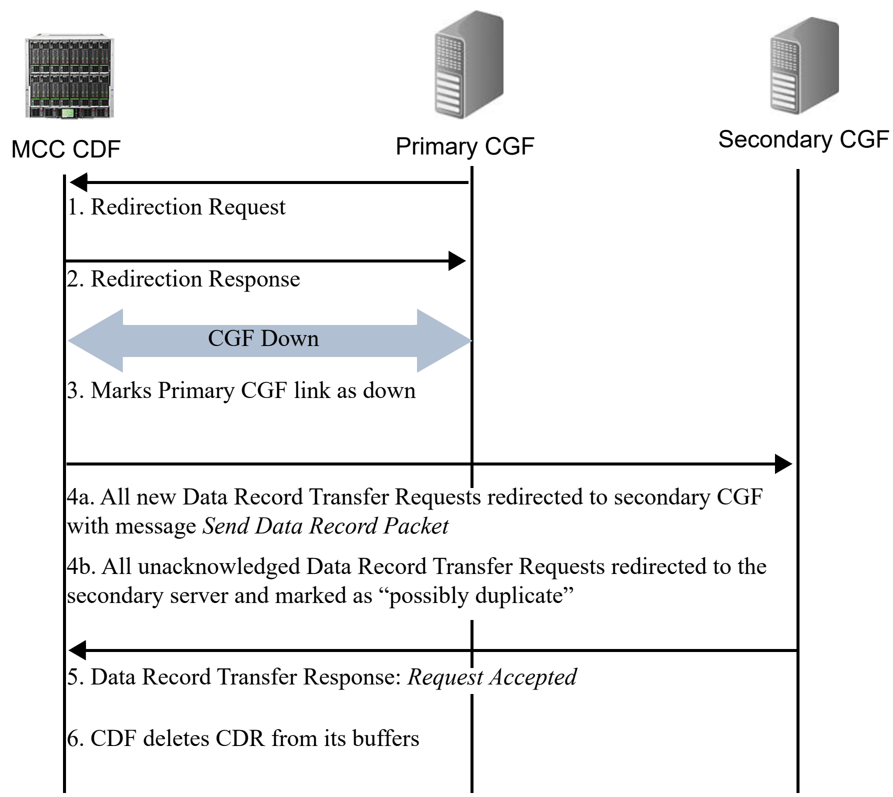


Table 5-3 Forced Failover to Secondary CGF

Step	Description
1.	The primary CGF sends a Redirection Request to the CDF informing the CDF to redirect CDR packets to the secondary CGF.
2.	The CDF sends a Redirection Response to the primary CGF acknowledging the successful receipt of the Redirection Request.
3.	The CDF marks the primary CGF as down.

Table 5-3 Forced Failover to Secondary CGF

Step	Description
4.	The CDF redirects all Data Record Transfer Requests to the secondary CGF. ■ 4a – All new Data Record Transfer Requests are sent with the message Send Data Record Packet informing the CGF to process this request normally. For more information, see Figure 5-3 on page 5-7. ■ 4b – All unacknowledged Data Record Transfer Requests (all Requests sent to the primary server with no response) are sent with the message Send possibly duplicated Data Record Packet. This message informs the CGF that this CDR packet could be a duplicate. CDRs flagged as "possibly duplicate" are handled differently. For more information, see Figure 5-6 on page 5-12.
5.	The secondary CGF sends a Data Record Transfer Response to the CDF with the message Request Accepted.
6.	The CDF deletes the CDR from its buffer.

Preventing Duplicate CDRs During CGF Failover

Figure 5-6 demonstrates how duplicate CDRs are prevented during CGF failover.

Figure 5-6 Preventing Duplicate CDRs During CGF Failover

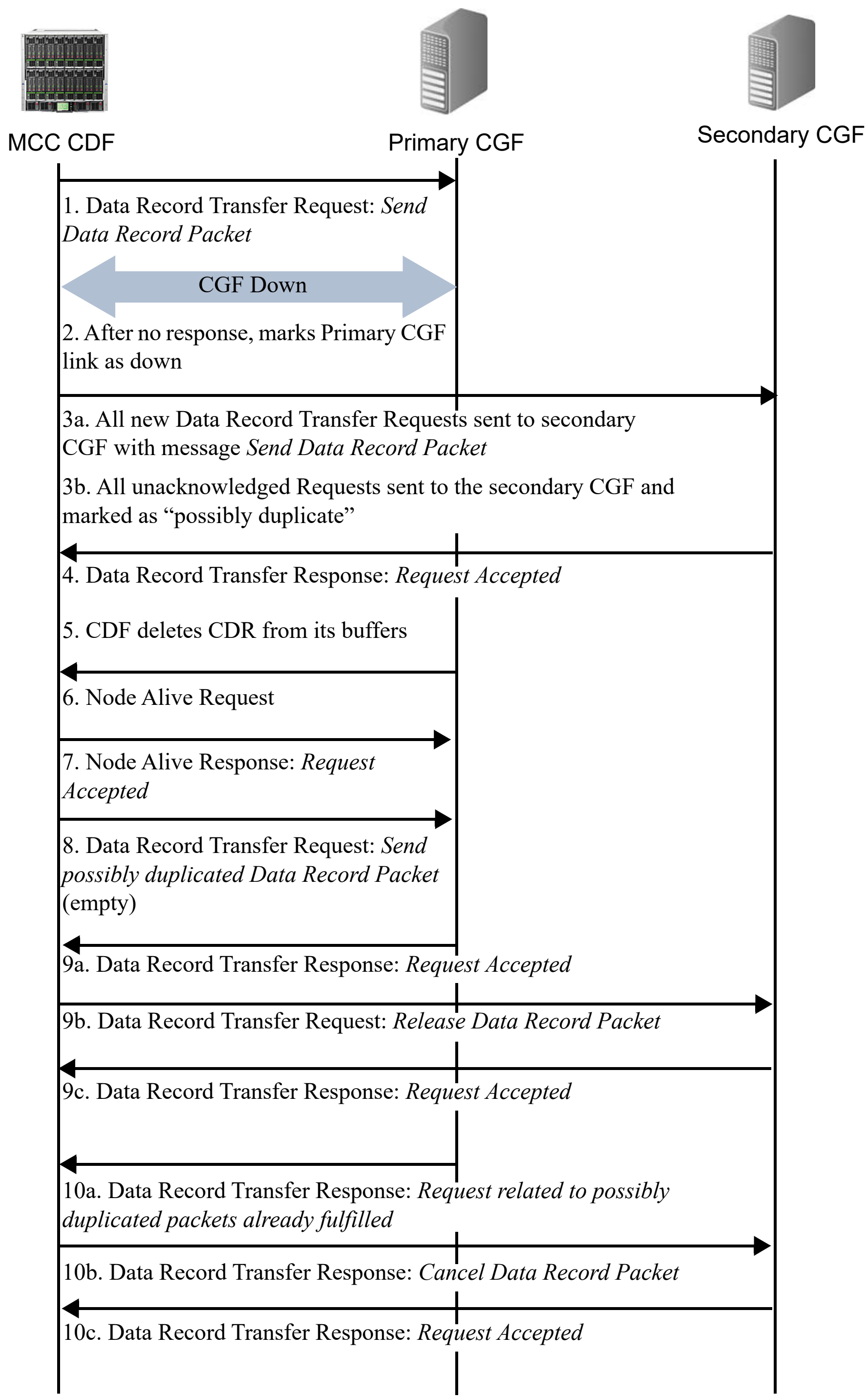


Table 5-4 Preventing Duplicate CDRs During CGF Failover

Step	Description
1.	The CDF sends a Data Record Transfer Request with the message Send Data Record Packet to the primary CGF. For detailed information on the message flow during a CGF failover, see Figure 5-4 on page 5-8.
2.	The CDF determines that the primary CGF is unavailable and marks it as down.
3.	The CDF redirects all Data Record Transfer Requests to the secondary CGF. ■ 3a – All new Data Record Transfer Requests are sent with the message Send Data Record Packet informing the CGF to process this request normally. For more information, see Figure 5-3 on page 5-7. ■ 3b – All unacknowledged Data Record Transfer Requests (all Requests sent to the primary server with no response) are sent with the message Send possibly duplicated Data Record Packet. This message informs the secondary CGF that this CDR packet could be a duplicate. To avoid duplicating CDRs, the secondary CGF stores the flagged CDR packet in its buffer in a special queue for "possibly duplicate" CDRs. These CDR packets require a second confirmation from the CDF before they are sent to the billing system for post-processing.
4.	The secondary CGF sends a Data Record Transfer Response to the CDF with the message *Request Accepted.*
5.	The CDF deletes the CDR from its buffer.
6.	If the primary CGF becomes available again, it sends the CDF a Node Alive Request.
7.	The CDF responds with a Node Alive Response: *Request Accepted* acknowledging the successful receipt of the Node Alive Request.
8.	To determine whether the primary CGF has already sent the flagged CDR packets to the billing system, the CDF sends the primary CGF an empty Data Record Transfer Request, consisting only of sequence numbers, with the message *Send possibly duplicated Data Record Packet*.
9.	If the primary CGF *has not sent* the CDR message to the billing system, the following occurs: ■ 9a – The primary CGF responds with a Data Record Transfer Response with the message Request Accepted. ■ 9b – The CDF sends a Data Record Transfer Request to the secondary CGF with the message Release Data Record Packet. This message informs the secondary CGF to remove the CDR packet from the "possibly duplicated" queue and send the CDR packet to the Billing system. ■ 9c – The secondary CGF sends the CDR to the billing system for post-processing and responds to the CDF with a Data Record Transfer Response: Request Accepted.

Table 5-4 Preventing Duplicate CDRs During CGF Failover

Step	Description
10.	If the primary CGF has sent the CDR message to the billing system, the following occurs: ■ 10a – The primary CGF responds with a Data Record Transfer Response with the message Request related to possibly duplicated packets already fulfilled informing the CDF that it has already sent the flagged CDR packet. ■ 10b – The CDF sends a Data Record Transfer Request to the secondary CGF with the message Cancel Data Record Packet. This message informs the secondary CGF to remove the CDR packet from the “possibly duplicated” queue and discard the packet. ■ 10c – The secondary CGF deletes the CDR packet(s) from its buffer and responds to the CDF with a Data Record Transfer Response: Request Accepted.

Sending Charging Records over the Ga Interface

This section describes the steps for configuring the GTP'-based Ga interface on an MCC gateway. For more information on the CLI commands in this section, see the *Affirmed Networks CLI Reference Guide*. For information on how Ga (GTPP) CDRs are reported, see CDR Reporting (page 1-17).

> ***Note:*** *When configuring the Ga interface on the GTP Proxy, use the PGW object.*

Use the following sequence to configure the Ga offline charging interface.

1 Create a network context for the Ga interface and configure the Ga interface loopback IP address.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address>
```

where

Parameter	Description
`network-context`	Specifies the name of the network context.
`loopback-ip`	Specifies the name of the loopback address associated with this network context.
`ip-address`	Specifies a loopback IP address to use for a service endpoint (Ga interface) on the network context. Can be IPv4/IPv6.

2 Configure the GTP' server group on the gateway (SGW/PGW/GGSN).

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group gtpp node-admin <server-group>
```

where ***<server-group>*** specifies the name of the GTP' server group on the gateway (SGW, PGW, GGSN).

> ***Note:*** *For information on configuring multiple server groups on the Ga interface, see* *Configuring Multiple Server Groups on the Ga Interface (page 5-31).*

The following table describes the configurable parameters.

Parameter	Description
`admin-state`	Specifies the administrative state of the GTP' server group. ■ `enabled` ■ `disabled`
`data-record-response-as-echo-success`	The MCC performs a failover from the primary to the secondary Charging Gateway Function (CGF) when an Echo Response is not received during the Echo Response timeout period, even if a Data Record Transfer Response is received from the primary CGF during the same period. When this feature is enabled, the MCC considers a Data Record Transfer Response received during the Echo Response timeout period to be a successful Echo Response, in order to maintain the primary active link from the PGW to the CGF over the Ga interface. Modifications to this parameter are applied to both new and existing peers. ■ `enabled` ■ `disabled` (default)
`dscp-value`	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this server group. For more information, see Configuring DSCP Values for Control Plane Traffic (page 5-30)
`ga-record-version`	Specifies the Data Record Format Version sent in the data record packet. The `ga-record-version` encodes the GTP' message header with the release version and application identifier as specified. The default version for this parameter is 3GPP TS 32.298 10.8.0. ■ `application-identifier` – Identifies the CDR application. For example, the Application Identifier has a value of 1 (decimal) if used for charging purposes. ■ `release-identifier` – Indicates the 3GPP TS 32.298 release used to encode the CDR. The Release Identifier value corresponds to the first digit of the version number of the 3GPP TS 32.298 document, as shown on the cover sheet. Values include: 8, 9, 10, 11, and 12. ■ `version-identifier` – Identifies the version of the 3GPP TS 32.298 document used to encode the CDR. The Version Identifier value corresponds to the second digit in the version number of the 3GPP TS 32.298 document, as shown on the cover sheet, plus 1. For example, the Version Identifier for version 10.8.0 is 9. ***Note:*** *The system does not automatically generate the fields in Interim and Stop data records according to the 3GPP 32.298 release identified in this parameter. This parameter is only used to configure the GTP' message header sent in the data record packet. Use the* `version` *parameter to automatically format CDR fields in Interim and Stop data records according to a specific 3GPP 32.298 release.*

Parameter	Description
`geo-redundant-group`	Specifies the Geographic Redundancy Group name. The Geographic Redundancy Group must already exist. For information on configuring a Geographic Redundancy Group, see the *Acuitas User's Guide*.
`local-port`	Specifies the server group local port number. For example, 3386.
`loopback-ip`	Specifies the name of the primary loopback IP address (associated with this network context) used for the Ga endpoint. Can be IPv4 or IPv6.
`network-context`	Specifies the name of the network context on which the Ga endpoint is anchored.
`pacer-transaction-window-size`	Specifies the maximum number of CDR transactions allowed in the queue per GTPP server group across all gateway tasks. This parameter is only visible in the CLI when set to a non-zero value. Use the no form of this command to disable this parameter. Command: **`no pacer-transaction-window-size`** If this parameter is not configured, the MCC uses the value configured for the `pacer-max-transactions-per-interval` parameter. When neither the `pacer-transaction-window-size` nor the `pacer-max-transaction-per-interval` is configured, the default value of 128 transactions per task will be set. If a GTPP server group is not shared by both the SGW and PGW, it is recommended that you set this parameter to twice the expected amount. If the number of outstanding transactions exceeds the value set for this parameter, the MCC will declare the GTPP server group congested and no other transaction will be initiated until the congestion is cleared. Unsent CDR messages will be buffered both in memory and on disk and retransmitted once the congestion state clears.
`send-potentially-duplicate`	Specifies whether to allow the duplicate record detection mechanism on a server group. See Configuring Multiple Server Groups on the Ga Interface (page 5-31) for more information. ■ **`true`** (default) ■ **`false`**

Parameter	Description
`version`	Specifies the 3GPP 32.298 version used to format GTPP CDR fields in Interim and Stop data records. ■ `10.0` – Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 10.0. ■ `9.0` – (default) Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 9.0. ■ `8.4` – Fields generated in Interim and Stop data records are compliant to the ASN.1 syntax specified in 3GPP 32.298 version 8.4. ***Note:*** *Configure the* `ga-record-version` *parameter to ensure the GTP' message header in the data record packet matches the 3GPP TS 32.298 release used to format CDRs.* ***Note:*** *For more information, see* *Data Record Templates for CDRs in ASN.1 Format (page 1-13).*

3 Configure the primary GTP' peer with which the gateway (SGW, PGW, GGSN) communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group gtpp node-admin <service-group> peer
<peer-host-name>
```

where ***<service-group>*** specifies the name of the GTP' server group on the gateway (SGW, PGW, GGSN) and ***<peer-host-name>*** specifies the host name of the GTP' peer.

The following table describes the configurable parameters.

Parameter	Description
`admin-state`	Specifies the administrative state of the GTP' peer. When enabled, the system attempts to establish peer connection. ■ `enabled` (default) ■ `disabled`
`data-record-max-retries`	Specifies the maximum number of transmit attempts for sending data records to the GTP' peer. The default is 3.
`data-record-response-timeout`	Specifies the number of seconds to wait before the system retransmits the data records to the GTP' peer. The default is 1.
`echo-max-retries`	Specifies the maximum number of Echo-Request attempts to the GTP' peer. The default is 3.
`echo-response-timeout`	Specifies the number of seconds to wait before the system retransmits the Echo-Request to the GTP' peer. The default is 1.
`ip-address`	Specifies the IP address of the GTP' peer. Can be IPv4/IPv6.

Parameter	Description
pacer-max-transactions-per-interval	Specifies the maximum number of Data Record Transfer Requests sent to a GTPP peer per time interval. This parameter is only visible in the CLI when set to a non-zero value. Use the no form of this command to disable this parameter. Command: **no pacer-max-transactions-per-interval** When neither the `pacer-transaction-window-size` nor the `pacer-max-transaction-per-interval` is configured, the default value of 128 transactions per task will be set.
pacer-time-interval	Specifies the CDR transaction time interval, in milliseconds, after which the pacer resets its outgoing transaction count. ***Note:*** *This parameter is only visible in the CLI when the* `pacer-max-transaction-per-interval` *parameter is set to a non-zero value.*
periodic-health-check-timeout	Specifies the number of seconds between periodic health check messages from the gateway agent (SGW, PGW, GGSN) to the GTP' peer. The default is 60.
port	Specifies the signaling port number.
role	Specifies the primary or secondary GTP' peer. ■ **primary** (default) ■ **secondary**
transport	Specifies the peer transport protocol.

4 (Optional) Configure a secondary GTP' peer with which the gateway (SGW, PGW, GGSN) communicates if the primary goes down.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group gtpp node-admin <service-profile> peer
<peer-host-name>
```

5 Configure the data record templates to filter CDR fields in Interim and Stop data records.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|sgw|ggsn] ofcs gtpp [interim|stop] <template-name> <fields>
presentFlag [yes|no|conditional]
```

where

Parameter	Description
[pgw|sgw|ggsn]	Specifies the gateway type for this template.
[interim|stop]	Specifies the record type.
<template-name>	Specifies the name of the data record template.

Parameter	Description
<fields>	Specifies the name of the CDR field you are modifying.
presentFlag	Indicates whether to include the data field in the CDR. Options include: ■ **conditional** – (default) Indicates the data field is present in CDRs when available. ■ **yes** – Indicates the data field is always present in CDRs. ■ **no** – Indicates the data field is not present in CDRs.

To view a list of configurable CDR fields in Interim and Stop data record templates, use the following CLI command:

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|sgw|ggsn] ofcs gtpp [interim|stop] <template-name> ?
```

Note: *The optional CDR fields listed in Interim and Stop data record templates are either top-level CDR fields or nested CDR fields. For example,* `changeofserviceconditionchargingrulebasename` *is a nested CDR field for which Charging Rulebase Name is a subfield of the List of Service Data (change of service condition) field.*

6 Configure the event trigger template to specify the events that trigger an interim update on the offline charging interface and to trigger the closure of an Interim CDR.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
```

The following table describes the configurable parameters.

Parameter	Description
cgi-sai-change-close-trigger	Specifies whether a change in the Cell Global Identification-Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close.
change-in-up-to-ue-close-trigger	Applies to offline interfaces only. Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close. ***Note:*** *Applies to the SGW only.*

Parameter	Description
`ecgi-change-close-trigger`	Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close.
`qos-change-close-trigger`	Specifies whether a QoS Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of QoS Change events to trigger a CDR close.
`rat-type-change-close-trigger`	Specifies whether a RAT Type Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAT Type Change events to trigger a CDR close.
`routing-area-change-close-trigger`	Specifies whether a change in the Routing Area Identity (RAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAI change events to trigger a CDR close.
`serving-node-change-close-trigger`	Specifies whether a Serving Node Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node Change events to trigger a CDR close. ***Note:*** *Applies to the GGSN/PGW when the context is transferred to a new SGW/SGSN or to the SGW when the Mobility Management Entity (MME) is changed.*
`serving-node-plmn-change-close-trigger`	Specifies whether a Serving Node PLMN Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node PLMN Change events to trigger a CDR close. ***Note:*** *Applies to PGW and GGSN only.*
`time-limit-close-trigger`	Applies to offline interfaces only. If interim updates are enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.
`tracking-area-change-close-trigger`	Specifies whether a change in the Tracking Area Identity (TAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of TAI change events to trigger a CDR close.

Parameter	Description
ue-time-zone-change-close-trigger	Specifies whether a UE Time Zone Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UE Time Zone Change events to trigger a CDR close.
user-csg-change-close-trigger	Specifies whether a CSG change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CSG change events to trigger a CDR close.
user-location-change-close-trigger	Specifies whether a User Location Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of User Location Change events to trigger a CDR close.
volume-limit-close-trigger	Applies to offline interfaces only. If Data Volume Limit is enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.

7 Configure the Ga offline charging interface for streaming CDRs to offline charging servers (GTP' peers).

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> admin-state [enabled|disabled]
[pgw|sgw|ggsn]
```

where **<*interface-name*>** is the name of the Ga offline charging interface, **admin-state** is the administrative state of the interface, and **[pgw|sgw|ggsn]** specifies the gateway type for the offline charging interface.

The following table describes the configurable parameters.

Parameter	Description
`apn-reporting`	*Applies to the PGW only*. Specifies the Access Point Names (APNs) to be reported over the Ga interface. ■ **`original`** — Initial APN ■ **`mapped`** — Final / current APN ■ **`prior-mapped`** — APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN) The `apn-reporting` value configured at the interface level overrides the `apn-reporting` value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. If the `apn-reporting` value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the `apn-reporting` value configured for the Control Profile. For more information, see Configuring APN Reporting at the Interface Level (page 1-47). ***Note:*** *The interface-level `apn-reporting` parameter does not support the `custom` option.*
`cdr-close-time`	If `interim-updates` are enabled on the interface, specifies the maximum number of seconds for storing closed service containers before transferring service containers in an Interim CDR to the offline charging server.
`data-volume-limit`	Specifies the maximum number of kilobytes that the gateway maintains across all service containers (for a bearer) before generating a new CDR. The volume usage counts are reset when a CDR is closed and sent. When the GTPP server is not reachable, volume/time triggered Interim CDRs are buffered in memory and disk. Once the GTPP server link is restored, the MCC plays back all Interim CDRs buffered to the server.
`encode-nai-in-asn1-msisdn`	*Applies to SGW/PGW/GGSN only*. Specifies whether to include/exclude the Nature of Address Indicator (NAI) in the Served Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) field when generating local ASN.1 or GTPP CDRs. By default, this parameter is `enabled`, indicating the NAI is included in the Served MSISDN CDR field. ■ **`enabled`** (default) ■ **`disabled`**
`event-trigger-template`	Specifies the event trigger template name.

Parameter	Description
`interim-update-interval`	Specifies the time interval, in seconds, after which a new service container or CDR is generated. The timer resets when a CDR is sent. The default is 0 (disabled). When this time interval expires, the MCC reports the current usage statistics (up to when the timer expired) in the CDR.
`interim-updates`	Specifies whether to enable or disable interim updates to the offline charging server. ■ `enabled` ■ `disabled`
`num-charge-condition-change`	Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0. ***Note:*** *A trigger is only counted towards the* `num-charge-condition-change` *if it is enabled under the* `event-trigger-template`.
`num-service-containers`	If `interim-updates` are enabled on the interface, specifies the maximum number of service containers that the gateway maintains per CDR at the bearer-level. Upon reaching this limit, the gateway transfers the service containers in an Interim CDR to the offline server.
`qos-change-trigger`	Specifies when to activate the qos-change-trigger. ■ `pcrf-initiated` – Activate the qos-change-trigger for a PCRF-initiated QoS change only. ■ `access-nw-initiated` – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only. ■ `any` – (default) Activate the qos-change-trigger for any type of QoS change.
`suppress-no-interim-zero-traffic-stop-cdr`	Specifies whether to suppress zero-volume Stop CDRs for short-lived sessions for which there is no Interim CDR generated. *Applies to SGW/PGW ASN.1 and GTPP CDRs only*. ■ `enabled` – The MCC suppresses Stop CDRs for short-lived sessions with zero-volume service data containers (SDCs) for which there is no Interim CDR generated. ■ `disabled` – (default) The MCC generates Stop CDRs for short-lived sessions even if there is no traffic data to report (zero-volume SDCs).

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td>suppress-zero-traffic-service-containers</td><td>Specifies whether to suppress the creation of service containers in PGW/GGSN CDRs for service rules with no traffic. Applies to PGW/GGSN CDRs (CSV/ASN.1 format) only.
■ enabled – When generating CDRs, the MCC does not create service containers for service rules with no traffic.
■ disabled – (default) When generating CDRs, the MCC creates service containers for service rules with no traffic.
Note: All service rules configured as event triggers, such as QoS change or ULI change, will have at least one service container even when the MCC is configured to suppress the creation of service containers for service rules with no traffic. This is to indicate the event trigger.</td></tr>
<tr><td>unique-cdr-nodeid</td><td>Specifies whether the Node ID value in SGW/PGW CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. This feature applies to all offline-charging interfaces (local, Ga, Gz/Rf).
■ Enabled – The value for the Node ID field in CDRs consists of the configured gateway service name and the Service Type ID.
For example, if the configured gateway service name is MyPGW and the Service Type ID is 3, then the Node ID = MyPGW_3. For all CDR types (local, GTPP and Gz/Rf), the Local Sequence Number field will be incremental for all 32 bits. The Local Sequence Number field will not contain the Service Type ID.
■ Disabled – (default) The value for the Node ID field in CDRs is the gateway service name, configured using zone <<i>name</i>> gateway [pgw|sgw] <<i>gw-name</i>>.
For example, if the configured gateway service name is MyPGW, then the Node ID = MyPGW. For local and GTPP CDRs, the Local Sequence Number field will consist of the incremental counter (20 bits) + the Service Type ID (12 bits). For Gz/Rf CDRs, the Local Sequence Number field will be incremental for all 32 bits.
<i>Note: After performing a Geographic Redundancy switchover, it is possible that the Task Service Type ID is different and in that case the Node ID value will change in the CDRs for the same subscriber.</i></td></tr>
</table>

8 (Optional) Specify the SGW mode for generating SGW CDRs.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> sgw cdr-mode
[stand-alone|roam-in|co-located|imsi-range]
```

The following table describes the configurable parameters.

Parameter	Description
offline-charging	Specifies the name of the Ga offline charging interface.

Parameter	Description
`cdr-mode`	Specifies the SGW mode for generating SGW CDRs. ■ `stand-alone` – (default) SGW CDRs are generated only for stand-alone SGW sessions (home or roaming). Stand-alone SGW sessions have PGW sessions anchored on a device other than the MCC. ■ `roam-in` – SGW CDRs are generated only for a subset of stand-alone SGW sessions that are roaming-in (the subscriber's PLMN is part of the roaming PLMN list configured on the SGW). *Applies to Gz/Rf and GTPP SGW CDRs only. Does not apply to SGW CDRs generated on the local offline charging interface.* ■ `co-located` – SGW CDRs are generated for stand-alone SGW sessions and co-located SGW sessions (home or roaming). Co-located sessions have both SGW and PGW services located on the same MCC. ■ `imsi-range` – SGW CDRs are generated based on a list of configured IMSI ranges. For more information, see Generating SGW CDRs Based on IMSI ranges (page 1-40).

9 (Optional) Specify the User-Name format used on offline-charging interfaces if not present in the PCO.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> user-name-type
[apn|default-username|imsi|imsi-nai-no-prefix|imsi-nai-
prefix-0|imsi-plus-apn|imsi-plus-apnnionly|msisdn|msisdn-plus-apn|msis
dn-plus-apnnionly|none]
```

The following table describes the configurable parameters.

Parameter	Description
`offline-charging`	Specifies the name of the Ga offline charging interface.

Parameter	Description
`user-name-type`	Specifies the User-Name format used on offline-charging interfaces if not present in the Protocol Configuration Options (PCO). This setting overrides the `user-name-type` present at the APN level. However, if this `user-name-type` is set to `none`, the APN level `user-name-type` is used. ■ **`apn`** ■ **`default-username`** ■ **`imsi`** ■ **`imsi-nai-no-prefix`** ■ **`imsi-nai-prefix-0`** ■ **`imsi-plus-apn`** ■ **`imsi-plus-apnnionly`** ■ **`msisdn`** ■ **`msisdn-plus-apn`** ■ **`msisdn-plus-apnnionly`** ■ **`none`**

10 Associate the GTP' server group and data record templates with the Ga offline charging interface and configure parameters for customizing certain ASN.1 CDR fields.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] gtpp
```

where ***`<interface-name>`*** is the name of the Ga offline charging interface and **`[pgw|sgw|ggsn]`** specifies the gateway type.

The following table describes the configurable parameters.

Parameter	Description
`server-group`	Specifies the name of the GTP' service profile on the gateway (PGW, SGW, GGSN).
`stop-template`	Specifies the Stop data record template name for ASN.1 encoded CDRs.
`interim-template`	Specifies the Interim data record template name for ASN.1 encoded CDRs.
`ie-customization`	Allows you to customize certain fields in ASN.1 CDRs and to specify whether to include the record sequence number in ASN.1 CDRs. *Applies to local ASN.1 and GTPP CDRs only.* See Customizing ASN.1 CDR Fields (page 1-42).

Parameter	Description
`report-epcqos-in-kbps`	Specifies whether the EPC QoS Information fields in local (CSV/ASN.1) and GTPP CDRs should report values in Kilobits-per-second (Kbps). ■ `enabled` – The EPC QoS information fields in CDRs are filled as follows: – Sets the current (non-extended) bit rates in Kbps. – Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1. ■ `disabled` – (default) The EPC QoS information fields in CDRs are filled as follows: – Sets the current (non-extended) bit rates in bps. For bandwidth values greater than 2^32-1, sets the value to 2^32-1. – Sets the extended bit rates in Kbps if the current bit rates are greater than 2^32-1. ***Note:*** *See* *Configuring EPC QoS Information Reporting for 5G NSA (page 1-51)* *for more information.*
`report-time-usage-on-zero-traffic`	Specifies whether the Time First Usage, Time Last Usage, and Time Usage CDR subfields should be set in Service Data Containers even if there is no data to be reported. *Applies to PGW local ASN.1 and GTPP CDRs only.* ■ `enabled` – If there is no traffic data to report, the Time First Usage and the Time Last Usage subfields are set to the Time of Container Report and the Time Usage subfield is set to 0 in Stop and Interim PGW local ASN.1 and GTPP CDRs. ■ `disabled` – (default) If there is no traffic data to report, the Time First Usage, Time Last Usage, and Time Usage subfields are not included in Stop and Interim PGW local ASN.1 and GTPP CDRs. ***Note:*** *These CDR subfields are contained within the List of Service Data field, which is known as Change of Service Condition in ASN.1 format.*

11 Perform the following steps to complete the Ga interface configuration:

a For the PGW/GGSN, bind the offline charging interface to a workflow control profile and ensure the workflow control profile is bound to a subscriber analyzer.

b For the SGW, configure the Ga offline charging interface directly on the SGW object using the following command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
default-offline-interface offline-interface <interface-name>
```

Note: *Enter a question mark (?) after the* `offline-interface` *parameter to display a list of available offline interfaces based on interface type (diameter or GTPP).*

The workflow control profile and SGW support a maximum of two offline charging interfaces, one for local CDRs and one for streaming to the OFCS (Diameter or GTPP). For information on these configuration steps, see the *Affirmed Networks Gateway Administration Guide.*

Note: *If the Control Profile with which an offline rating group is associated does not have a* `default-offline-interface` *configured, the MCC still allows the PDN session to be established and allows offline charging traffic. If the Control Profile has a local CDR interface configured, the MCC will generate CDRs and store them in a local file.*

Optional Ga Interface Configuration Tasks

This section describes optional Ga interface configuration tasks.

- Configuring DSCP Values for Control Plane Traffic (page 5-30)
- Configuring Multiple Server Groups on the Ga Interface (page 5-31)
- Synchronizing Ga and Gy Charging Data (page 5-33)

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on Diameter interfaces using Differentiated Services Code Point (DSCP).

GTPP interface (Ga) – DSCP value configured at the GTPP server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group gtpp node-admin <name> dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note: *The DSCP value configured at the Diameter server group level overrides the network-context DSCP value.*

Configuring Multiple Server Groups on the Ga Interface

Operators can associate multiple GTPP server groups with the Ga offline charging interface, with each server group consisting of one primary and one secondary peer. A single GTPP peer can act as a primary peer in one server group and as a secondary peer in another server group.

When the SGW/PGW sends a Data Record Transfer request on the Ga interface to the Charging Gateway Function (CGF), the MCC uses a round-robin approach to select from among the active server groups in the configured list. Once a server group is selected, the MCC continues to send all CDRs for that session to the selected server group. This feature also adds a new configuration option to disable the duplicate record detection mechanism on a server group.

Table 5-5 shows an example of the configuration used to configure multiple GTPP server groups (GroupA/GroupB/GroupC). Once the GTPP server groups are configured, use the following command to associate these GTPP server groups with the Ga interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging GTPPCHRG_PGW pgw gtpp server-group
[GroupA|GroupB|GroupC]
```

Table 5-5 Multiple Server Groups on the Ga Interface (Example)

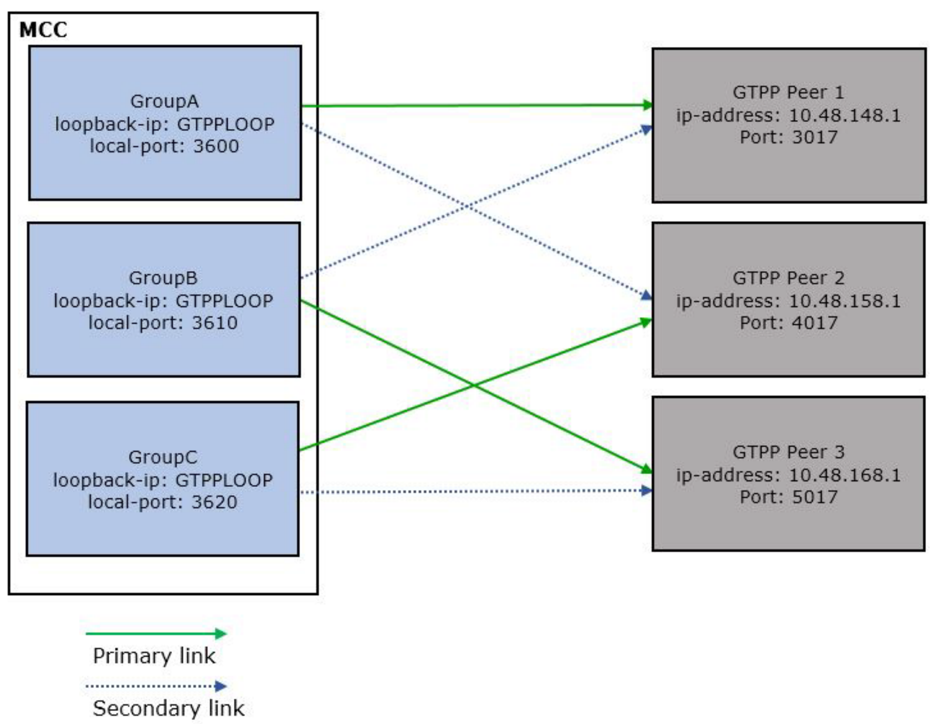
GTPP Server Group A Configuration			
an3000-mcm-slot7cpu1(config)# **service-construct interface aaa-interface-group gtpp node-admin GroupA local-port 3600 loopback-ip GTPPLOOP send-potentially-duplicate false**			
Group A Peers			
peer	**GTPPpeerP**	**peer**	**GTPPpeerS**
ip-address	**10.48.148.1**	**ip-address**	**10.48.158.1**
port	**3017**	**port**	**4017**
role	**primary**	**role**	**secondary**
GTPP Server Group B Configuration			
an3000-mcm-slot7cpu1(config)# **service-construct interface aaa-interface-group gtpp node-admin GroupB local-port 3610 loopback-ip GTPPLOOP send-potentially-duplicate false**			
Group B Peers			
peer	**GTPPpeerP**	**peer**	**GTPPpeerS**
ip-address	**10.48.168.1**	**ip-address**	**10.48.148.1**
port	**5017**	**port**	**3017**

Table 5-5 Multiple Server Groups on the Ga Interface (Example)

role	**primary**	**role**	**secondary**
GTPP Server Group C Configuration			
an3000-mcm-slot7cpu1(config)# **service-construct interface aaa-interface-group gtpp node-admin GroupC local-port 3620 loopback-ip GTPPLOOP send-potentially-duplicate false**			
Group C Peers			
peer	**GTPPpeerP**	**peer**	**GTPPpeerS**
ip-address	**10.48.158.1**	**ip-address**	**10.48.168.1**
port	**4017**	**port**	**5017**
role	**primary**	**role**	**secondary**

The configuration example presented in Figure 5-7 has three GTPP server groups, each consisting of one primary and one secondary peer. There are three GTPP peers in total, with each peer acting as a primary peer in one server group and as a secondary peer in another server group. Figure 5-7 demonstrates the network setup for this sample configuration.

Figure 5-7 GTPP Server Group and Peer Network Setup



Synchronizing Ga and Gy Charging Data

The Ga (OFCS) and Gy (OCS) interfaces are both used to report subscriber data usage for billing and accounting purposes. To ensure the reported subscriber data usage between the two interfaces is synchronized, Operators can configure a Gy event trigger template that defines the Gy events that trigger the closure and sending of an Interim CDR on the Ga (GTPP) interface. The MCC generates bearer-level CDRs with service-rule level service data containers (SDCs) for Gy reporting events.

When a configured Gy reporting event occurs, the MCC stores the event in a service-rule level SDC until a configured threshold is reached, such as the Gy event recurrence count, which specifies the maximum number of containers added to a CDR per closure event or the maximum container limit, which specifies the maximum number of service containers added to a CDR. Upon reaching the configured threshold, the MCC sends the stored SDC in an Interim CDR over the Ga interface.

Note the following when configuring bearer-level CDR reporting for Gy events:

- For session-level CDRs with Gy event support, all offline interim triggers, such as Volume/Time/RAN triggers, generate SDCs for both offline and online rating groups. Gy event triggers generate SDCs only for online rating groups.
- For bearer-level CDRs with Gy event support, all offline interim triggers, such as Volume/Time/RAN triggers, generate SDCs only for offline service rules (service rules associated with offline rating groups). Gy event triggers generate SDCs only for online service rules (service rules associated with online rating groups).
- If all the rating groups are online and bearer-level CDRs are enabled for Gy events, then offline interim triggers, such as Time/RAN triggers, will generate CDRs with no SDCs because there are no offline service rules. In this case, Affirmed Networks recommends disabling these offline events or creating a dummy service rule to be included in CDRs for offline events.
- If the PS-Furnish-Charging-Information (FCI) field is received from the OCS at the command level in CCA messages, the MCC will populate the command-level FCI field in bearer-level CDRs. If the FCI field is received from the OCS at the MSCC level in CCA messages, the MCC will generate service-rule level SDCs for Gy events. In this case, the FCI field is included within the Change of Service Condition field in GTPP PGW/GGSN CDRs. If no FCI field is present in the CCA at the command or MSCC level, the Gy transaction timeout expires, or no response is received, the MCC will generate bearer-level CDRs and service-rule level SDCs without an FCI value.
- The Cause For Record Closing field in GTPP/local ASN.1 CDRs is set to:
  - 16 = Volume Limit – Configured volume threshold has been exceeded.
- Bearer-level CDR reporting for Gy events only supports GTPP/local ASN.1 CDRs.

Use the following steps to configure offline charging reporting at the bearer-level for Gy events.

Note: *To use the Gy event trigger template to synchronize Ga and Gy interface charging data, ensure all online rating groups have the* `reporting-level` *set to* `serviceRule`.

---

1 Ensure bearer-level CDR reporting is configured on the Ga interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> cdr-report-level bearer
```

2 Enable offline charging service data container creation based on Gy reporting events on the Gy online charging interface. The Gy online charging interface must already exist. For more information on configuring the Gy interface, see the *Affirmed Networks Gy Interface Specification*.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> offline-cdr-for-gy-event enabled
```

3 Configure the Gy event trigger template to specify the Gy events that trigger the closure and sending of an Interim CDR on the Ga interface.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template gy-event-trigger-template <template-name>
<events> change-condition-value <value> recurrence-count <value>
```

The following table describes the configurable parameters.

Parameter	Description
`gy-event-trigger-template`	Specifies the name of the Gy event trigger template.
`<events>`	Specifies the name of the Gy event. ■ `final-close-trigger` ■ `forced-reauthorisation-close-trigger` ■ `quota-exhausted-close-trigger` ■ `quota-holding-time-close-trigger` ■ `rating-condition-change-close-trigger` ■ `tariff-time-change-close-trigger` ■ `validity-time-close-trigger` ■ `volume-quota-threshold-close-trigger`

Parameter	Description
change-condition-value	Specifies the change-condition value to use in the service-rule level SDC created for this Gy event. The change-condition-value range is 0-31 and by default, matches the ASN.1 change-condition values defined in 3GPP 32.298. If set to within the 0-31 range, the Gy event change-condition value in the SDC is set to the ASN.1 change-condition value defined in 3GPP 32.298. If set to any value beyond 31, the SDC will have “Volume Limit” as the change condition.
recurrence-count	Specifies the maximum number of service data containers added to a CDR per closure event. Upon reaching this limit, offline charging CDRs are closed and transfered to the OFCS. Value 0 disables the trigger, value 1 - 20 enables the trigger.

4 Associate the Gy event trigger template with the Ga offline charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|ggsn] gy-event-trigger-template
<template-name>
```

Monitoring

This section describes how to monitor the operational status of the CGF server via the Ga interface and how to display statistical information for CGF server communication messages. For more information on statistics, see *Acuitas Performance Statistics*.

GTPP Node Status

Use the following CLI command to display the operational status of the GTPP node.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group gtpp node-status <node-name>
```

The system displays output similar to the following:

```
adminstatus                enabled
operstatus                 outofservice
operational-reason         "Peer Down"
host-port                  3386
host-ip-address            10.7.98.181
dropped-packets            0
duplicate-packets          0
sequence-over-run-packets  0
```

Peer Status

Use the following CLI command to display the operational status of the CGF server.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group gtpp peer-status <node-name> <peer-host-name>
```

The system displays output similar to the following:

```
ip-address                       10.48.148.1
port                             3027
transport                        udp
role                             primary
adminstatus                      enabled
operstatus                       down
dataRecordRetryCount             3
dataRecordRespTimeout            1
dataRecordResponseAsEchoSuccess  disabled
echoRetryCount                   3
echoRespTimer                    1
periodicHealthCheckTimer         60
```

Peer-cmd-Stats

Use the following CLI command to display the number of protocol messages sent per command-code between the system and the specified CGF server.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group gtpp peer-cmd-stats <node-name> <peer-host-name>
```

The system displays output similar to the following:

```
Echo-Req-Transmitted                        8097
Echo-Resp-Recvd                             0
Echo-Req-Recvd                              0
Echo-Resp-Transmitted                       0
Node-Alive-Req-Transmitted                  4
Node-Alive-Resp-Rcvd                        0
Node-Alive-Req-Rcvd                         0
Node-Alive-Resp-Transmitted                 0
Data-Record-Req-Transmitted                 0
Data-Record-Resp-Rcvd                       0
Redirection-Resp-Rcvd                       0
Redirection-Resp-Transmitted                0
Potential-Duplicate-Rec-Transmitted         0
Potential-Duplicate-EmptyRec-Transmitted 0
Cancel-Req-Transmitted                      0
Release-Req-Transmitted                     0
Potential-Duplicate-Record-Rcvd             0
Request-Accepted-Rcvd                       0
Request-Fullfilled-Rcvd                     0
```

Peer Stats

Use the following CLI command to display the number of inbound/outbound requests/responses, protocol errors and failures between the system and the specified CGF server.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group gtpp peer-stats <node-name> <peer-host-name>
```

The system displays output similar to the following:

```
num-protocol-error          0
num-permanent-failure       0
num-routing-failure         0
num-pacer-no-credit-failure 0
num-out-request             490
num-out-response            0
num-in-request              0
num-in-response             0
num-link-bounces            0
pacer-high-water-mark       0
```

GTPP Peer Performance Statistics

Use the following CLI command to display GTPP peer performance statistics. For more information on performance statistics, see *Acuitas Performance Statistics.*

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group gtpp statistics cluster-level peer node-name
```

The system displays output similar to the following:

```
num-echo-req-tx                                  497
num-echo-req-rx                                  0
num-echo-resp-tx                                 0
num-echo-resp-rx                                 0
num-node-alive-req-tx                            3
num-node-alive-req-rx                            0
num-node-alive-resp-tx                           0
num-node-alive-resp-rx                           0
num-redirection-req-rx                           0
num-redirection-resp-tx                          0
num-data-record-transfer-req-tx                  0
num-data-record-transfer-resp-rx                 0
num-potential-duplicate-data-record-rx           0
num-potential-duplicate-empty-data-record-tx     0
num-cancel-request-tx                            0
```

```
num-release-request-tx                                0
num-req-fulfilled-rx                                  0
num-req-accepted-rx                                   0
Discontinuity Timestamp                               1560845356
num-link-declared-as-up-based-on-data-record-response 0
num-first-data-record-req-transmitted                 0
```

Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface level.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

```
sdc-total-uplink-bytes/sdc-total-downlink-bytes

sdc-total-uplink-packets/sdc-total-downlink-packets

total-uplink-bytes/total-downlink-bytes

total-uplink-packets/total-downlink-packets
```

Alarms

A major-level trap similar to the following occurs when a GTP' peer goes down. The alarm clears after the connection is restored.

```
anGtppPeerDown
Description:        The Transport connection to Gtpp peer is down.
Trap ID:            93
Status:             Current
Severity:           Major
Auto Clearing:      No
Potential Impact:   This might result in Ga Interface not able to send
                    the Call Data Records to Billing system via the
                    CGF| mediation function.
Corrective Action:  Check the routing/connectivity towards the GTP
                    peer. The alarm clears after the connection is
                    restored. Once you have restored the IP
                    connectivity make sure the Gtpp Peer is up and
                    routes are correctly configured and GTPP
                    configuration is correct.
```

Message Definitions

Table 5-6 lists the supported messages for the Ga interface.

Table 5-6 Ga Interface Messages

Command	Command Code	Reference
Echo Request	1	3GPP TS 32.295 V9.0.0
Echo Response	2	3GPP TS 32.295 V9.0.0
Version Not Supported	3	3GPP TS 32.295 V9.0.0
Node Alive Request	4	3GPP TS 32.295 V9.0.0
Node Alive Response	5	3GPP TS 32.295 V9.0.0
Redirection Request	6	3GPP TS 32.295 V9.0.0
Redirection Response	7	3GPP TS 32.295 V9.0.0
Data Record Transfer Request	240	3GPP TS 32.295 V9.0.0
Data Record Transfer Response	241	3GPP TS 32.295 V9.0.0

ASN.1 Supported Fields

Table 5-7 lists all the ASN.1 fields supported on the system for GTP'. Fields excluded from Table 5-7 are not supported and the system discards unsupported fields and does not send them.

Note: *ASN.1 specification descriptions take precedence over other descriptions in case of discrepancies.*

Note: *For more information on the supported ASN.1 CDR fields, see Chapter 3, CDR Descriptions.*

Table 5-7 ASN.1 Supported Fields

Field	ASN.1 Code	Format	Size (in bytes)
Record Type	80	Integer	1
Served IMSI	83	BCD Encoded Octet String	3-8
PGW Address	a4	Octet String	6 or 18 (depending on v4 or v6 address)
SGW Address	a4	Octet String	6 or 18 (depending on v4 or v6 address)
Charging ID	85	Integer	1-5
Serving Node Address	a6	Octet String	6 or 18
Access Point Name Network Identifier	87	IA5String	1-63
PDP/PDN Type	88	Octet String	2
Served PDP/PDN Address	a9	Octet String	8 or 20
Dynamic Address Flag	8b	Boolean	1
Record Opening Time	8d	BCD Encoded Octet String	9
Duration	8e	Integer	1-5
Cause for Record Closing	8f	Integer	1
Diagnostics	b0	Choice	1-5
Record Sequence Number	91	Integer	1-5

Table 5-7 ASN.1 Supported Fields

Field	ASN.1 Code	Format	Size (in bytes)
Node ID	92	IA5String	1-20
Record Extensions Extension Information MVNO Information RAN Cause Codes	b3	Management Extensions	Variable
Local Sequence Number	94	Integer	1-5
APN Selection Mode	95	Enumerated	1
Served MSISDN	96	BCD Encoded Octet String	1-9
Charging Characteristics	97	Octet String	2
Charging Selection Mode	98	Enumerated	1
IMS Signaling Context	N/A	Integer	1
Serving Node PLMN Identifier	9b	Octet String	3
Served IMEISV	9d	BCD Encoded Octet String	8
RAT Type	9e	Integer	1
MS Time Zone	9f1f	Octet String	2
User Location Information	9f20	Octet String	5-13
UWAN User Location Information	bf3e	Sequence	Variable
CAMEL Charging Information	9f21	Octet String	Variable
List of Service Data ***Note:*** *In ASN.1 format, this field is known as Change of Service Condition.*	bf22	Sequence	Variable
Rating Group	81	Integer	1-5
Charging Rulebase Name	82	IA5String	1-16
Local Sequence Number	84	Integer	1-5
Time of First Usage	85	BCD Encoded Octet String	9
Time of Last Usage	86	BCD Encoded Octet String	9

Table 5-7 ASN.1 Supported Fields

Field	ASN.1 Code	Format	Size (in bytes)
Time Usage (Duration)	87	Integer	1-5
Change Condition	88	Bit String	5
QoS Information Negotiated	a9	Sequence (or Octet String depending on the 3GPP 32.298 version specified) See Data Record Templates (page 1-12) for more information.	Variable
Serving Node Address	aa	Octet String	6 or 18
Data Volume FBC Uplink	8c	Integer	1-5
Data Volume FBC Downlink	8d	Integer	1-5
Time of Report	8e	BCD Encoded Octet String	9
Failure Handling Continue	90	Boolean	1
Service ID	91	Integer	1-5
User Location Information (LOSD)	94	Octet String	6-13
Sponsor Identity	99	Octet String	Variable
Application Service Provider Identity	9a	Octet String	Variable
AF Record Information	b3	Sequence	Variable
PS Furnish Charging Information	bc	Sequence	Variable
3GPP RAT Type	9e	Integer	1
Serving PLMN Rate Control	bf23	Sequence	Variable
APN Rate Control	bf43	Sequence	Variable
UWAN User Location Info		Sequence	Variable
List of Traffic Data Volumes ***Note:*** *In ASN.1 format, this field is known as Change of Charging Condition.*	ac	Sequence	Variable

Table 5-7 ASN.1 Supported Fields

Field	ASN.1 Code	Format	Size (in bytes)
Serving Node Type	bf23	Sequence of serving node type	Variable
Served MNNAI	bf24	Set	Variable
PGW PLMN ID	9f25	Octet String	3
Start Time	9f26	Octet String	9
Stop Time	9f27	Octet String	9
PDN Connection ID	9f29	Integer	1-5
IMSI Unauthenticated Flag ***Note:*** *The IMSI Unauthenticated Flag ASN.1 CDR field is not applicable to 3GPP 32.298 Version 8.4.*	N/A	Integer	1
User CSG Information	bf2b	Sequence of CSG Information	1-4
Served PDP Address Extension	bf2d	Octet String	8
Dynamic Address Flag Extension	9f2f	Boolean	1
SGW Address (IPv6)	bf30	Octet String	18
Serving Node Address (IPv6)	bf31	Octet String	18
PGW Address (IPv6)	bf32	Octet String	18
PS Furnish Charging Information	bc	Sequence	Variable
APN Rate Control	bf43	Sequence	Variable
Low Priority Indicator	9f2e (PGW) 9f2c (SGW)	Integer	1
PDP/PDN Type Extension	9f44 (PGW) 9f3e (SGW)	Integer	1
SCS/AS Address	bf47	Set	Variable
Serving PLMN Rate Control	bf42 (PGW) bf3d (SGW)	Sequence	Variable
SGi PtP Tunnelling Method	9f40	Enumerated	1

Table 5-7 ASN.1 Supported Fields

Field	ASN.1 Code	Format	Size (in bytes)
UNI PDU CP Only Flag	9f41 (PGW) 9f3c (SGW)	Boolean	1
CP CIoT EPS Optimisation Indicator	9f3b (SGW)	Boolean	1
List of RAN Secondary RAT Usage Reports	bf40 (SGW) bf49 (PGW)	Sequence	Variable

Chapter 6

Gz/Rf Interface

This section contains the following information used to control and monitor the Diameter-based Gz/Rf interface:

See Appendix A, Diameter AVP Descriptions for a description of the Attribute Value Pairs (AVPs) supported on the Gz/Rf interface.

Note: *See the Affirmed Networks CLI Reference Guide for more information on the CLI commands used to configure and monitor the Gz/Rf interface.*

Interface Overview

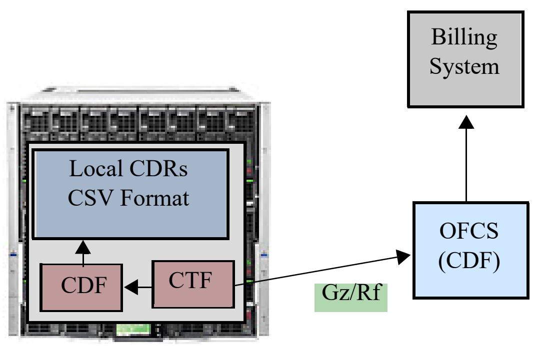
The Gz/Rf Interface is a Diameter-based interface used to transport charging information in comma-separated values *(*CSV) format from the gateways (SGW/GGSN/PGW) to an *Offline Charging System* (OFCS). Offline charging is a process where charging information for a subscriber session is collected and converted into call detail records (CDRs), which are then transferred to the Operator's postpaid billing system for processing.

MCC gateways can send the charging events to offline charging servers in real-time, providing Operators with real-time accounting information within the current subscriber session.

The *Charging Trigger Function* (CTF), within the MCC gateway generates charging event information based on service usage and forwards this information to the *Charging Data Function* (CDF) in the OFCS. The CDF generates CDRs using the event information received from the CTF and forwards the CDRs to the Operator's billing system for post-processing.

Figure 6-1 illustrates the framework for transporting charging information over the Gz/Rf interface.

Figure 6-1 Transporting Charging Information over the Gz/Rf Interface



This section describes the Diameter Gz/Rf interface implementation in the Affirmed Networks Mobile Content Cloud (MCC) in accordance with 3GPP TS 32.251, 3GPP TS 32.298 and 3GPP TS 32.299.

Accounting Request Message Types

The MCC implementation of the Gz/Rf interface expects the OFCS to be a stateless accounting state machine as specified in RFC 3588. A stateless accounting state machine can receive accounting information in any order and at any time.

MCC gateways (SGW, PGW, GGSN) report offline charging events to the OFCS using Diameter Accounting-Request (ACR) messages. The charging events can either be one-time events or session-based. Table 6-1 lists the four types of ACR records used to send accounting data.

Table 6-1 ACR Records

ACR Records	Description
START	Starts an accounting session.
INTERIM	Updates an accounting session.
STOP	Ends an accounting session.
EVENT	Indicates a one-time accounting event.

The START, INTERIM and STOP records are used to record charging information for the duration of a session. The EVENT record is used to indicate a configured charging event occurred or to indicate that a session failed to establish.

Data Record Templates

Operators can use data record templates to include or exclude certain AVPs in ACR messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|sgw|ggsn] ofcs diameter [start|stop|interim|event]
<template-name> <fields> presentFlag [yes|no|conditional]
```

where

Parameter	Description
[pgw|sgw|ggsn]	Specifies the gateway type for this template.
[start|stop|interim|event]	Specifies the record type.
<template-name>	Specifies the name of the data record template.
<fields>	Specifies the name of the AVP field you are modifying.
presentFlag	Indicates whether to include the AVP in the ACR message. Options include: ■ **conditional** – Indicates the AVP field is included in ACR messages if it is available. ■ **yes** – Indicates the AVP field is included in ACR messages. ■ **no** – Indicates the AVP field is not included in ACR messages. ***Note:*** *To suppress an AVP in ACR messages that is embedded within a grouped AVP, set the* `presentFlag` *setting for the top-level AVP to* `yes` *and the* `presentFlag` *setting for the embedded AVP to* `no`.

Operators can suppress the Subscription-ID AVP in ACR messages based on the Subscription-Id-Type (IMSI/MSISDN) using ACR data record templates. In the ACR data record templates, the Subscription-ID options appear as `subscriptionidimsi` and `subscriptionidmsisdn`.

Use the following CLI command to list the configurable AVPs in the ACR data record templates.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|sgw|ggsn] ofcs diameter [start|stop|interim|event]
<template-name> ?
```

Diameter Routing Agent

A Diameter Routing Agent (DRA) provides routing capabilities between a PGW and an OFCS when multiple OFCSs are deployed in a Diameter realm. By default, the PGW sends the statically configured Destination-Host and Destination-Realm AVPs in Diameter request messages to the OFCS. With this feature, you can control whether the PGW sends the statically configured Destination-Host and Destination-Realm on the Gz/Rf interface in the initial ACR message. The initial ACR message is the first ACR sent to the peer after the bearer creation or after the previous or backup peer stops responding.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
send-destination-host-avp [true|false]
```

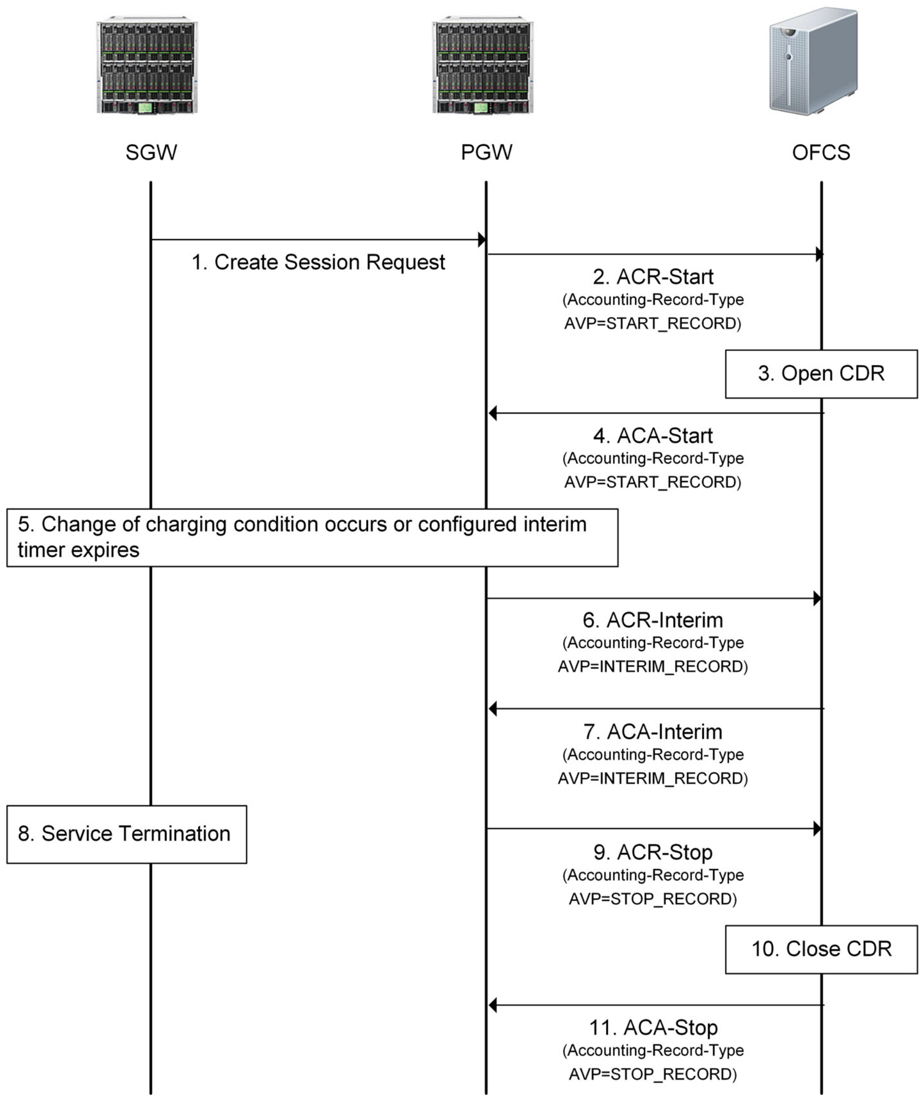
- **`true`** – (default) The PGW sends the statically configured Destination-Host and Destination-Realm in the initial ACR message to the OFCS via the Gz/Rf interface.

- **`false`** – The PGW does not send the statically configured Destination-Host and Destination-Realm in the initial ACR message to the OFCS via the Gz/Rf interface. The PGW uses the values from the Origin-Host/Origin-Realm AVPs received in the Accounting-Answer (ACA) response to fill the Destination-Host and Destination-Realm AVPs in subsequent ACR messages.

Message Flow

Figure 6-2 demonstrates the basic message flow to perform session-based charging over the Gz/Rf offline interface.

Figure 6-2 Session-based Charging over the Gz/Rf Interface



Configuring the Gz/Rf Offline Charging Interface

This section describes the steps for configuring the Diameter-based Gz/Rf interface on an MCC gateway.

Note: *A maximum of 30 service data containers (SDCs) can be included in Gz/Rf CDRs and any additional SDCs are dropped.*

Use the following sequence to configure the Gz/Rf offline charging interface:

1 Create a network context for the Gz/Rf interface and configure the Gz/Rf interface loopback IP address.

```
an3000-mcm-slot7cpu1(config)# network-context <name> loopback-ip
<loopback-name> ip-address <ip-address>
```

where

Parameter	Description
network-context	Specifies the name of the network context.
loopback-ip	Specifies the name of the loopback address associated with this network context.
ip-address	Specifies a loopback IP address to use for a service endpoint (Gz/Rf interface) on the network context. Can be IPv4/IPv6.

2 Configure the local diameter node.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
```

where ***<node-name>*** specifies the name of the Diameter service profile on the gateway (PGW, SGW, GGSN).

The following table describes the configurable parameters.

Parameter	Description
cea-timeout	Specifies the wait time (in seconds) for receiving a Capability-Exchange-Answer (CEA).
connection-attempt-count	Specifies the maximum number of connection retry attempts from the PGW to the peer.
connection-attempt-timeout	Specifies the Tc timer, in seconds, as described in RFC 6733. Specifies how frequently to send transport connection attempts to a peer with whom no active transport connection exists.

Parameter	Description
`dscp-value`	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this Diameter server group. For more information, see Configuring DSCP Values for Control Plane Traffic (page 6-21)
`dwr-retry-count`	Specifies the maximum number of DWR attempts to the diameter peer before declaring the peer down.
`dwr-retry-timer`	Specifies the number of seconds the diameter node waits for a DWA after retransmitting a DWR once the first dwr-timer expires. ***Note:*** *The `dwr-retry-timer` is typically set at a shorter interval than the `dwr-timer`.*
`dwr-timer`	Specifies the number of seconds the diameter node waits for a Device-Watchdog-Answer (DWA) after sending a Device-Watchdog-Request (DWR) to the diameter peer. ***Note:*** *When the `dwr-timer` expires, the diameter node retransmits the DWR to the diameter peer and the `dwr-retry-timer` is started.*
`geo-redundant-group`	Specifies the Geographic Redundancy Group name. The Geographic Redundancy Group must already exist. For information on configuring a Geographic Redundancy Group, see the *Acuitas User's Guide*.
`link-type`	Specifies whether the link used to monitor neighboring network elements is a shared link among the primary and secondary NE's or a private (local) link: ■ `private` ■ `shared`
`loopback-ip`	Specifies the name of the primary loopback IP address (associated with this network context) used for the Gz/Rf endpoint. Can be IPv4 or IPv6.
`net-health-report-enabled`	Specifies whether to monitor network health towards the AAA interface: ■ `true` ■ `false`
`network-context`	Specifies the name of the network context on which the Gz/Rf endpoint is anchored. Restricts the inbound Diameter messages by accepting only inbound messages on the IP interfaces in the interface group.
`node-role`	Specifies the local node role: ■ `client` ■ `server`

Parameter	Description
origin-host	Specifies the local hostname in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the diameter request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin host.*
origin-realm	Specifies the Local Realm Name. The value is used in the Origin-Realm AVP in the diameter request message. ***Note:*** *You can configure multiple Diameter server groups with the same origin realm.*
port	Specifies the TCP listening port number used when `node-role` is set to `server`. The default is 3868. ***Note:*** *This is the TCP listening port, not the Gz source port used when `node-role` is set to `client`. The Gz source port is dynamically assigned and is not configurable.*
product-name	Specifies the diameter product name.
send-destination-host-avp	Specifies whether the PGW sends the statically configured Destination-Host and Destination-Realm on the Gz/Rf interface in the initial ACR message. The initial ACR message is the first ACR sent to the peer after the bearer creation or after the previous or backup peer stops responding. ■ **true** – (default) The PGW sends the statically configured Destination-Host and Destination-Realm in the initial ACR message to the OFCS via the Gz/Rf interface. ■ **false** – The PGW does not send the statically configured Destination-Host and Destination-Realm in the initial ACR message to the OFCS via the Gz/Rf interface. The PGW uses the values from the Origin-Host/Origin-Realm AVPs received in the ACA response to fill the Destination-Host and Destination-Realm AVPs in subsequent ACR messages.
transport	Specifies the transport type. ■ **tcp** (default) ■ **sctp** – To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gz/Rf interface, you must configure an SCTP profile and associate the profile with the local Diameter node. See Configuring an SCTP Profile (page 6-36).
vendor-id	Specifies the diameter vendor ID.
sctp-profile	Specifies the name of the SCTP profile. The SCTP profile must already exist. See Configuring an SCTP Profile (page 6-36). ***Note:*** *The `sctp-profile` parameter is only visible when the local Diameter node transport type is set to SCTP.*

Parameter	Description
secondary-loopback-ip	Specifies a secondary loopback IP address for the local Diameter node to support SCTP multi-homing. See Configuring an SCTP Profile (page 6-36) for more information. ***Note:*** *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*

3 Configure the diameter peer with which the gateway (PGW, SGW, GGSN) communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name>
```

where **<*node-name*>** specifies the name of the Diameter service profile on the gateway (PGW, SGW, GGSN) and **<*peer-host-name*>** specifies the peer host name. The peer host name is used in the Destination-Host AVP to send diameter request messages.

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administrative state of the diameter peer. When enabled, the system attempts to establish peer connection. ■ **enabled** (default) ■ **disabled**
backup-host-name	Identifies the secondary peer used during a failover from the primary to the secondary peer. ***Note:*** *This parameter is only allowed for the secondary peer and must be configured if failover is supported.*
peer-identifier	Specifies the external identifier for this peer.
port	Specifies the signaling port number.
primary-ip-address	Specifies the IP address of the peer (can be an IPv4 or IPv6 address).
realm	Specifies the peer realm name. Realm is used in the Destination-Realm AVP in diameter requests.
role	Specifies the type of peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ **Primary** – (default) Active peers used for diameter sessions. ■ **Secondary** – Used for diameter sessions, only when the primary peers are down. ***Note:*** *If failover to a secondary peer is supported, you must enter the host name of the secondary server in the* `backup-host-name` *parameter.*

Parameter	Description
secondary-loopback-ip	Specifies a secondary loopback IP address for the Diameter peer to support SCTP multi-homing. See Configuring an SCTP Profile (page 6-36) for more information. ***Note:*** *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*
weight	Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-Update/CCR-Terminate) for existing sessions. *Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.* See Weighted Round-Robin Distribution to Multiple Diameter Peers (page 6-38) for more information.

4 Configure the data record templates to filter optional AVPs in the ACR messages.

`an3000-mcm-slot7cpu1(config)#` **service-construct data-record-template [pgw|sgw|ggsn] ofcs diameter [start|stop|interim|event] *<template-name>* *<fields>* presentFlag [yes|no|conditional]**

where

Parameter	Description
[pgw|sgw|ggsn]	Specifies the gateway type for this template.
[start|stop|interim|event]	Specifies the record type.
<template-name>	Specifies the name of the data record template.
<fields>	Specifies the name of the AVP field you are modifying.
presentFlag	Indicates whether to include the AVP in the ACR message. Options include: ■ **conditional** – Indicates the AVP field is included in ACR messages if it is available. ■ **yes** – Indicates the AVP field is included in ACR messages. ■ **no** – Indicates the AVP field is not included in ACR messages. ***Note:*** *To suppress an AVP in ACR messages that is embedded within a grouped AVP, set the* `presentFlag` *setting for the top-level AVP to* `yes` *and the* `presentFlag` *setting for the embedded AVP to* `no`*.*

5 Configure the event trigger template to specify the events that trigger an interim update on the offline charging interface and to trigger the closure of an interim CDR.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
```

The following table describes the configurable parameters.

Parameter	Description
`cgi-sai-change-close-trigger`	Specifies whether a change in the Cell Global Identification-Service Area Identifier (CGI-SAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CGI-SAI change events to trigger a CDR close.
`change-in-up-to-ue-close-trigger`	Applies to offline interfaces only. Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close. ***Note:*** *Applies to the SGW only.*
`ecgi-change-close-trigger`	Specifies whether a change in the E-UTRAN Cell Global Identifier (ECGI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of ECGI change events to trigger a CDR close.
`qos-change-close-trigger`	Specifies whether a QoS Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of QoS Change events to trigger a CDR close.
`rat-type-change-close-trigger`	Specifies whether a RAT Type Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAT Type Change events to trigger a CDR close.
`routing-area-change-close-trigger`	Specifies whether a change in the Routing Area Identity (RAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of RAI change events to trigger a CDR close.
`serving-node-change-close-trigger`	Specifies whether a Serving Node Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node Change events to trigger a CDR close. ***Note:*** *Applies to the GGSN/PGW when the context is transferred to a new SGW/SGSN or to the SGW when the Mobility Management Entity (MME) is changed.*

Parameter	Description
serving-node-plmn-change-close-trigger	Specifies whether a Serving Node PLMN Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of Serving Node PLMN Change events to trigger a CDR close. ***Note:*** *Applies to PGW and GGSN only.*
time-limit-close-trigger	Applies to offline interfaces only. If interim updates are enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-20.
tracking-area-change-close-trigger	Specifies whether a change in the Tracking Area Identity (TAI) triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of TAI change events to trigger a CDR close.
ue-time-zone-change-close-trigger	Specifies whether a UE Time Zone Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UE Time Zone Change events to trigger a CDR close.
user-csg-change-close-trigger	Specifies whether a CSG change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of CSG change events to trigger a CDR close.
user-location-change-close-trigger	Specifies whether a User Location Change triggers an update. Value 0 disables the trigger, value 1 - 50 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of User Location Change events to trigger a CDR close.
volume-limit-close-trigger	Applies to offline interfaces only. If Data Volume Limit is enabled on the interface, specifies the number of events to trigger a CDR close. Enter a value between 1-50.

6 Configure the Gz/Rf offline charging interface for streaming CDRs to the OFCS.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> admin-state [enabled|disabled]
[pgw|sgw|ggsn]
```

where ***<interface-name>*** is the name of the Gz/Rf offline charging interface, **admin-state** is the administrative state of the interface, and **[pgw|sgw|ggsn]** specifies the gateway type for the offline charging interface.

The following table describes the configurable parameters.

Parameter	Description
apn-reporting	*Applies to the PGW only*. Specifies the Access Point Names (APNs) to be reported over the Gz interface. ■ **original** — Initial APN ■ **mapped** — Final / current APN ■ **prior-mapped** — APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN) The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. If the apn-reporting value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the apn-reporting value configured for the Control Profile. For more information, see Configuring APN Reporting at the Interface Level (page 1-47). ***Note:*** *The interface-level apn-reporting parameter does not support the custom option.*
cdr-close-time	If interim-updates are enabled on the interface, specifies the maximum number of seconds for storing closed service containers before transferring service containers in an interim CDR to the offline charging server.
data-volume-limit	Specifies the maximum number of kilobytes that the gateway maintains across all service containers (for a bearer) before generating a new CDR. The volume usage counts are reset when a CDR is closed and sent. ***Note:*** *When session-level Gz/Rf CDR reporting is enabled, the data-volume-limit specifies the maximum number of kilobytes maintained across all service containers*
event-trigger-template	Specifies the event trigger template name.
interim-update-interval	Specifies the time interval, in seconds, after which a new service container or CDR is generated. The timer resets when a CDR is sent. The default is 0 (disabled). When this time interval expires, the MCC reports the current usage statistics (up to when the timer expired) in the CDR.
interim-updates	Specifies whether to enable or disable interim updates to the offline charging server. ■ **enabled** ■ **disabled**

Parameter	Description
`num-charge-condition-change`	Specifies the maximum number of change in charging conditions (such as RAT change, QoS change, ULI change, etc.) that the gateway maintains per CDR. Upon reaching this limit, the gateway transfers the SDCs in an interim CDR to the offline server. The default is 0. ***Note:*** *A trigger is only counted towards the `num-charge-condition-change` if it is enabled under the `event-trigger-template`.*
`num-service-containers`	If `interim-updates` are enabled on the interface, specifies the maximum number of service containers that the gateway maintains per CDR at the bearer-level. Upon reaching this limit, the gateway transfers the service containers in an interim CDR to the offline server.
`qos-change-trigger`	Specifies when to activate the qos-change-trigger. ■ **`pcrf-initiated`** – Activate the qos-change-trigger for a PCRF-initiated QoS change only. ■ **`access-nw-initiated`** – Activate the qos-change-trigger for an access-network/UE-initiated QoS change only. ■ **`any`** – (default) Activate the qos-change-trigger for any type of QoS change.
`suppress-zero-traffic-service-containers`	Specifies whether to suppress the creation of service containers in PGW/GGSN CDRs for service rules with no traffic. *Applies to PGW/GGSN CDRs (CSV/ASN.1 format) only.* ■ **`enabled`** – When generating CDRs, the MCC does not create service containers for service rules with no traffic. ■ **`disabled`** – (default) When generating CDRs, the MCC creates service containers for service rules with no traffic. ***Note:*** *All service rules configured as event triggers, such as QoS change or ULI change, will have at least one service container even when the MCC is configured to suppress the creation of service containers for service rules with no traffic. This is to indicate the event trigger.*

Parameter	Description
unique-cdr-nodeid	Specifies whether the Node ID value in SGW/PGW CDRs reports the configured gateway service name along with the Service Type ID or just the configured gateway service name. This feature applies to all offline-charging interfaces (local, Ga, Gz/Rf). ■ **Enabled** – The value for the Node ID field in CDRs consists of the configured gateway service name and the Service Type ID. For example, if the configured gateway service name is MyPGW and the Service Type ID is 3, then the Node ID = MyPGW_3. For all CDR types (local, GTPP and Gz/Rf), the Local Sequence Number field will be incremental for all 32 bits. The Local Sequence Number field will not contain the Service Type ID. ■ **Disabled** – (default) The value for the Node ID field in CDRs is the gateway service name, configured using **zone <*name*> gateway [pgw|sgw] <*gw-name*>**. For example, if the configured gateway service name is MyPGW, then the Node ID = MyPGW. For local and GTPP CDRs, the Local Sequence Number field will consist of the incremental counter (20 bits) + the Service Type ID (12 bits). For Gz/Rf CDRs, the Local Sequence Number field will be incremental for all 32 bits. ***Note:*** *After performing a Geographic Redundancy switchover, it is possible that the Task Service Type ID is different and in that case the Node ID value will change in the CDRs for the same subscriber.*

7 (Optional) Specify the contents and format of certain AVPs sent via the Gz/Rf interface to the OFCS and the CDR reporting level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <name>
```

The following table describes the configurable parameters.

Parameter	Description
offline-charging	Specifies the name of the Gz/Rf interface for offline charging.
service-context-id	Specifies the Service-Context-Id AVP sent in ACR messages to the OFCS. The default Service-Context-Id is 9.32251@3gpp.org.
user-name-type	Specifies the User-Name AVP format used in ACR messages if not present in the Protocol Configuration Options (PCO). This setting overrides the `user-name-type` present at the APN level. However, if this `user-name-type` is set to `none`, the APN level `user-name-type` is used. ■ **apn** ■ **default-username** ■ **imsi** ■ **imsi-nai-no-prefix** ■ **imsi-nai-prefix-0** ■ **imsi-plus-apn** ■ **imsi-plus-apnnionly** ■ **msisdn** ■ **msisdn-plus-apn** ■ **msisdn-plus-apnnionly** ■ **none**
session-id-format	Specifies the format for the Session-Id AVP sent on the Gz/Rf interface. For more information, see Session-ID (page A-20). ■ **v0** – (default) The Session-ID AVP format represents the optional value for the session setup time in decimal format. For example: *example.com;1450803134;2148532240;8;***1450812272***;262229* where ***1450812272*** represents the session setup time in decimal format. ■ **v1** – The Session-ID AVP format represents the optional value for the session setup time in hex format. *This format applies to the Gz/Rf interface only.* For example: **01**-*example.com;1450803134;2148532240;8;***5679A370-***262229* where ***01*** is a prefix indicating that this Session-ID format is using version 1 (v1) and ***5679A370*** represents the session setup time in hex format. Note that the delimiter after the timestamp uses a dash (-) instead of a semi-colon (;).

Parameter	Description
`cdr-report-level`	Specifies whether to enable session-level Gz/Rf CDR reporting. See Generating Session-Level Gz/Rf CDRs (page 6-30) for more information. ■ **`pdn-session`** – Enables session-level Gz/Rf CDR reporting. Only one set of CDRs is generated for the entire session, irrespective of the number of bearers. When set to `pdn-session`, ensure that the local interface for offline charging is disabled and that all rating groups have the `reporting-level` set to either `ratingGroup` or `ratingGroupAndQc`i. *Only applies to Gz/Rf PGW and SGW CDRs.* ■ **`bearer`** (default) – Each bearer will have its own set of CDRs. When set to `bearer`, the rating group `reporting-level` must be set to `serviceRule`.

8 (Optional) Customize the IMEISV sent within the User-Equipment-Info AVP and 3GPP-IMEISV AVP on the Gz/Rf interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw] diameter avp-customization
[user-equipment-info-imeisv|3gpp-imeisv] allow-zero-value
[enabled|disabled]
```

The following table describes the configurable parameters.

Parameter	Description
`avp-customization user-equipment-info-imeisv allow-zero-value`	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the User-Equipment-Info AVP. *Applies to the SGW and PGW only.* ■ **`enabled`** – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the User-Equipment-Info AVP on the Gz/Rf interface. ■ **`disabled`** – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the User-Equipment-Info AVP on the Gz/Rf interface.
`avp-customization 3gpp-imeisv allow-zero-value`	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the 3GPP-IMEISV AVP. *Applies to the SGW and PGW only.* ■ **`enabled`** – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the 3GPP-IMEISV AVP on the Gz/Rf interface. ■ **`disabled`** – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the 3GPP-IMEISV AVP on the Gz/Rf interface.

9 (Optional) Specify the SGW mode for generating SGW CDRs and specify when to trigger the SGW Gz/Rf Start record in the current SGW Mode.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> sgw cdr-mode
[stand-alone|roam-in|co-located|imsi-range] cdr-start-trigger
[create-session-request|modify-bearer-request]
```

The following table describes the configurable parameters.

Parameter	Description
offline-charging	Specifies the name of the local interface for offline charging.
cdr-mode	Specifies the SGW mode for generating SGW CDRs. ■ **stand-alone** – (default) SGW CDRs are generated only for standalone SGW sessions (home or roaming). Standalone SGW sessions have PGW sessions anchored on a device other than the MCC. ■ **roam-in** – SGW CDRs are generated only for a subset of standalone SGW sessions that are roaming-in (the subscriber's PLMN is part of the roaming PLMN list configured on the SGW). *Applies to Gz/Rf and GTPP SGW CDRs only. Does not apply to SGW CDRs generated on the local offline charging interface.* ■ **co-located** – SGW CDRs are generated for standalone SGW sessions and co-located SGW sessions (home or roaming). Co-located sessions have both SGW and PGW services located on the same MCC. ■ **imsi-range** – SGW CDRs are generated based on a list of configured IMSI ranges. For more information, see Generating SGW CDRs Based on IMSI ranges (page 1-40).
cdr-start-trigger	Specifies when to trigger the SGW Gz/Rf Start record in the current SGW Mode. By default, the SGW Gz/Rf Start record is triggered with the Create-Session-Request message and the trigger can be changed to the Modify-Bearer-Request message. *Applies to Gz/Rf SGW CDRs only.* ■ **create-session-request** (default) ■ **modify-bearer-request**

10 Associate the Diameter server group and the data record templates with the Gz/Rf offline charging interface and configure the retransmit-count and retransmit-timeout.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn] diameter
```

The following table describes the configurable parameters.

Parameter	Description
server-group	Specifies the Service Profile for the selected streaming protocol.
start-template	Specifies the Start data record template name.

Parameter	Description
stop-template	Specifies the Stop data record template name.
interim-template	Specifies the Interim data record template name.
event-template	Specifies the Event data record template name.
retransmit-count	Specifies the maximum number of times the gateway transmits the ACR to the OFCS.
retransmit-timeout	Specifies the time interval, in milliseconds, before the system retransmits the ACR to the OFCS.

11 Perform the following steps to complete the offline charging interface configuration:

a For the PGW/GGSN, bind the offline charging interface to a workflow control profile and ensure the workflow control profile is bound to a subscriber analyzer.

b For the SGW, configure the offline charging interface directly on the SGW object using the following command:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <sgw-name>
default-offline-interface offline-interface <interface-name>
```

The workflow control profile and SGW support a maximum of two offline charging interfaces, one for local CDRs and one for streaming to the OFCS (diameter or GTP’). For information on these configuration steps, see the *Affirmed Networks Gateway Administration Guide.*

Note: *If the Control Profile with which an offline rating group is associated does not have a* `default-offline-interface` *configured, the MCC still allows the PDN session to be established and allows offline charging traffic. If the Control Profile has a local CDR interface configured, the MCC will generate CDRs and store them in a local file.*

Optional Gz/Rf Interface Configuration Tasks

This section describes optional Gz/Rf interface configuration tasks.

- Configuring DSCP Values for Control Plane Traffic (page 6-21)
- Storing Gz/Rf CDRs to a File in the Event of a Failure (page 6-22)
- Generating Session-Level Gz/Rf CDRs (page 6-30)
- Synchronizing Gz/Rf and Gy Charging Data (page 6-32)
- Modifying Default Diameter AVP Flags (page 6-34)
- Configuring an SCTP Profile (page 6-36)
- Weighted Round-Robin Distribution to Multiple Diameter Peers (page 6-38)

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on Diameter interfaces using Differentiated Services Code Point (DSCP).

Diameter interfaces (S6b, Gx, Gy, Gz/Rf) – DSCP value configured at the Diameter server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <name> dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>
control-dscp-value <value>
```

Note: *The DSCP value configured at the GTPP server group level overrides the network-context DSCP value.*

Storing Gz/Rf CDRs to a File in the Event of a Failure

This feature adds support for storing offline charging Gz/Rf SGW/PGW CDRs in a predefined file and format when the remote server is unavailable. Once servers become available, an OFCS retrieves (pulls) the stored CDRs using SFTP from the predefined file path.

Note: *This feature applies to Gz/Rf SGW and PGW CDRs only.*

If the Gz/Rf CDR file storage functionality is enabled and both the primary and secondary offline charging servers are unavailable, the MCC automatically stores the contents of the Gz/Rf ACR messages in a predefined file and format based on the configuration until the servers become available and retrieve (pull) the stored charging records. The amount of time in which they are stored and the size of the file is configurable. When the maximum size is reached, the system creates a new CDR file. File rotation is also configurable based on the number of records and the age of the file. If the file fails to write to disk, the system generates an alarm notification.

This section includes:

- Export Rules (page 6-23)
- File Naming Conventions (page 6-24)
- custom1 File Format (page 6-25)
- Configuring Gz/Rf CDR File Storage Functionality (page 6-27)

Export Rules

Once a file's closing conditions are reached, the file is moved to the */public/logs/datarecord.lnk/gz/export* directory according to the export rules and user permissions configured for the file management profile. The MCC supports several different export rules for Gz/Rf SGW/PGW CDRs configurable using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
gzrecord <filename-prefix> export-rule
```

- **`exportrule-default`** – The system exports files to the */public/logs/datarecord.lnk/gz/export* directory according to the default format.
  - Uses the file naming convention: `<filename-prefix><YYYYMMDDHHMMSS>_<123>.extension`
  - Applies purge rules for this file profile to all files in the directory.

- **`exportrule-gz-custom1`** – The system exports files to the */public/logs/datarecord.lnk/gz/export* directory according to the custom1 format configured for Gz/Rf CDRs.
  - Uses the file naming convention: `<filename-prefix>_-_<RC>.CTS<date>_-_<time>CTE[.<PI>][.<FE>]`
  - Updates the filename with the last modified timestamp when exporting the file to the */public/logs/datarecord.lnk/gz/export* directory.
  - Once the stored CDR files have been successfully pulled by the OFCS, the export user (user configured with read/write access to the files in the /export directory) can rename the file with a *.lock* extension and the purge rules configured for the file profile apply to the **.lock* files. Alternatively, the user can delete the files after a successful SFTP file transfer.

    If neither of these options are performed, files will remain in the */public/logs/datarecord.lnk/gz/export* directory and will only be purged when storage becomes full. They will not be cleaned up based on the purge rules configured for this file profile.

- **`exportrule-none`** – Specifies that there is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.

Note: *For more information on file naming conventions, see* *File Naming Conventions (page 6-24)**. For information on the custom1 file format, see* *custom1 File Format (page 6-25)**.*

File Naming Conventions

The MCC uses the following file naming conventions for storing Gz/Rf SGW/PGW CDRs if the Gz/Rf CDR file storage functionality is enabled.

- *Default* – The default file naming convention is used when the `export-rule` is set to `exportrule-default`.

  `<filename-prefix><YYYYMMDDHHMMSS>_<123>.extension`

  where `filename-prefix` indicates the prefix of the filename in which to store Gz/Rf CDRs, `YYYYMMDDHHMMSS` represents the timestamp and `123` represents the file sequence number.

  The sequence number runs from 1 to 2147483647, so that at one record per second, it would take a little over 68 years to rollover. The sequence number resets to 1 after a system reboot, MCM switchover or system failure. Depending on the frequency of these events, CDR records could potentially have the same sequence number.

- *custom1* – The custom1 file naming convention is used when the `export-rule` is set to `exportrule-gz-custom1`.

  `<filename-prefix>_-_<RC>.CTS<date>_-_<time>CTE[.<PI>][.<FE>]`

  where

  – `filename-prefix` – Indicates the prefix of the filename in which to store Gz/Rf CDRs.

  – `RC` – Indicates the file running counter value, which starts with 1 and is incremented by 1 for every file added. When it reaches 65535, it is reset to 1.

  – `CTS` – Creation Time-Stamp Start

  – `date` – Indicates the date (YYYYMMDD) when the file is closed.

  – `time` – Indicates the time in UTC format when the file is closed.

  – `CTE` – Creation Time-Stamp End

  – `PI` – Indicates the optional private information appended to the filename.

  – `FE` – Indicates the optional file extension appended to the filename.

Note: *For more information on export rules, see* *Export Rules (page 6-23).*

---

custom1 File Format

You can configure the encoding format for Gz/Rf SGW/PGW CDRs as part of the offline charging object.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [sgw|pgw] diameter
write-cdr-to-disk-on-failure enabled gz-cdr-encode-format custom1
```

Currently, the MCC only supports the custom1 encoding format. The Gz/Rf CDR custom1 file format is illustrated below. The custom1 file format does not impact the Diameter header and AVPs, only the File header and ACR header.

File Header
ACR Header
Diameter Header
AVP 1
AVP 2
ACR Header
Diameter Header
AVP 1
AVP 2

Table 6-2 illustrates the Gz/Rf CDR custom1 File header format. Note the following:

- If the ACR routing filter is not applicable, the *length of ACR routing filter* and *ACR routing filter* fields are not present in the File header.
- If the Private Extension is not applicable, the *Length of Private Extension* and *Private Extension* fields are not present in the File header.

Table 6-2 Gz/Rf CDR custom1 File Header Format

Bits								
Octets	8	7	6	5	4	3	2	1
1..4	File Length							
5..8	Header Length							
9	High Release Identifier			High Version Identifier				
10	Low Release Identifier			Low Version Identifier				
11..14	File opening timestamp							
15..18	Timestamp when last ACR was appended to file							
19..22	Number of ACRs in file							
23..26	File sequence number							
27	File closure trigger reason							
28..47	IP address of node that generated file							
48	Lost ACR indicator							
49..50	Length of ACR routing filter							
51..xy	ACR routing filter							
xy+1..xy+2	Length of Private Extension							
xy+3..n	Private Extension							

Table 6-3 illustrates the Gz/Rf CDR custom1 ACR header format.

Table 6-3 Gz/Rf CDR custom1 ACR Header Format

Bits								
Octets	8	7	6	5	4	3	2	1
1..2	ACR Length							
3	Release Identifier			Version Identifier				
4	Data Record Format			TS Number				

Configuring Gz/Rf CDR File Storage Functionality

Use the following sequence to configure the system to store Gz/Rf SGW/PGW CDRs in a file in the event of a failure.

1 Issue the following command to set the default SFTP directory for the export-user:

```
an3000-mcm-slot7cpu1# user-access change-sftp-directory
```

The system prompts:

```
Enter sftp directory:
```

2 Type **/public/logs/datarecord.lnk/gz/export** to set the default SFTP directory.

3 Create a local file in which to store Gz/Rf SGW/PGW CDRs in the event of a failure.

```
an3000-mcm-slot7cpu1(config)# infrastructure file-management profile
gzrecord <filename-prefix>
```

where **<filename-prefix>** specifies the prefix of the filename in which to store Gz/Rf CDRs.

The following table describes the configurable parameters.

Parameter	Description
compress-enabled	Specifies whether this file should be compressed (zipped) when it is closed. ■ **enabled** ■ **disabled**
compress-format	If compress is enabled, specify the format for compressing files: ■ **gzip** – (default) The data records are written to a file. The file is gzipped after it is closed.

Parameter	Description
export-rule	Specifies the rules/actions to apply when exporting the CDR file to the */public/logs/datarecord.lnk/gz/export* directory for SFTP access. ■ **exportrule-default** – The system exports files to the */public/logs/datarecord.lnk/gz/export* directory and applies purge rules for this file profile to all files in the directory. ■ **exportrule-gz-custom1** – The system exports files to the */public/logs/datarecord.lnk/gz/export* directory according to the custom1 format configured for Gz/Rf SGW/PGW CDRs. For more information, see Export Rules (page 6-23). ■ **exportrule-none** – Specifies that there is no export rule for this file profile. In this case, the MCC sends (pushes) the files to the remote SFTP server based on the upload parameters configured for the file profile.
export-user	Specifies the user that has read/write access to the files in the */public/logs/datarecord.lnk/gz/export* directory.
file-rotate-empty	Specifies whether empty CDR files are rotated. ■ **enabled** – Empty CDR files are rotated based on the configured `file-rotate-frequency` interval. ■ **disabled** – (default) Empty CDR/HTDR files are not rotated.
file-extension	Specifies the optional file-extension appended to the filename.
file-rotate-frequency	Specifies the frequency at which files should be rotated if the current file has not reached the maximum size (`file-rotate-size-bytes`) or max number of records (`file-rotate-num-records`). For example, MCC rotates the file on the hour if 1hour is selected, or at midnight, if 1day is selected, etc. Midnight is determined by the timezone and is not based on UTC.
file-rotate-num-records	Specifies the number of records at which to rotate the files.
file-rotate-size-bytes	Specifies the size (in bytes) at which files should be rotated.
private-information	Specifies optional private information appended to the filename.
purge-enabled	Specifies whether to enable or disable purge/cleanup for this file. ■ **enabled** ■ **disabled**
purge-file-older-than	Specifies the frequency at which files should be purged.
purge-low-thr	Specifies the percentage utilization at which to stop transferring/purging files.

Parameter	Description
purge-thr	Specifies the percentage utilization at which to transfer/purge files.
sequence-field-width	Specifies the number of leading zeros to prepend to the filename sequence number. Specify a value between 0 and 10. The default is 0. For example, if the file sequence number is `123` and the `sequence-field-width` is `7`, the file sequence number would appear as the following: `<filename-prefix><YYYYMMDDHHMMSS>_<0000123>.csv`

4 Configure the Gz/Rf CDR file storage functionality on the Gz/Rf offline charging interface.

an3000-mcm-slot7cpu1(config)# **service-construct interface offline-charging *<interface-name>* [sgw|pgw] diameter write-cdr-to-disk-on-failure [enabled|disabled] gz-cdr-encode-format custom1 gz-cdr-file-profile *<filename>***

The following table describes the configurable parameters.

Parameter	Description
write-cdr-to-disk-on-failure	Specifies whether to enable or disable the Gz/Rf CDR file storage functionality used to store CDRs when the remote server is unavailable. ■ **enabled** – When the remote server is unavailable, Gz/Rf SGW/PGW CDRs are stored in a predefined file and format based on the configuration. Once servers become available, an OFCS pulls the stored CDRs using SFTP from the predefined file path. ■ **disabled** (default) – When the remote server is unavailable, the system stores Gz/Rf SGW/PGW CDRs and sends (pushes) the pending CDRs to the billing server when the remote server becomes available. The predefined file and format does not apply in this case.
gz-cdr-encode-format	Specifies the format in which to encode the Gz/Rf SGW/PGW CDRs in the file. **custom1** – Gz/Rf CDRs are stored according to the custom1 format. See custom1 File Format (page 6-25) for more information. ***Note:*** *Visible in the CLI only when the* `write-cdr-to-disk-on-failure` *parameter is set to* `enabled`.
gz-cdr-file-profile	Specifies the filename used to store Gz/Rf SGW/PGW CDRs records in the event of a failure. ***Note:*** *Visible in the CLI only when the* `write-cdr-to-disk-on-failure` *parameter is set to* `enabled`.

Generating Session-Level Gz/Rf CDRs

For offline charging CDRs sent via the Gz/Rf interface to the OFCS, you can enable session-level Gz/Rf CDR reporting based on the default bearer. When session-level CDR reporting is enabled, the MCC sends the Start record when the default bearer is activated upon initial session setup. When a session is terminated and the default bearer is deactivated, the MCC generates the Stop record that includes all service containers stored within the default bearer context. When a dedicated bearer is deactivated, the MCC generates a rating-group level container and stores it within the default bearer context.

Use the following CLI command to enable session-level Gz/Rf CDR reporting.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> cdr-report-level pdn-session
```

***Note:** For SGW Gz/Rf CDRs, you can specify whether to trigger the Start record with the Create-Session-Request message or the Modify-Bearer-Request message when you configure the offline-charging interface. See Configuring the Gz/Rf Offline Charging Interface (page 6-7) for configuration information.*

When session-level Gz/Rf CDR reporting is enabled, ensure that the local interface for offline charging is disabled and that all Rating Groups have the `reporting-level` set to `ratingGroup`.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group <name>
reporting-level ratingGroup
```

When session-level Gz/Rf CDR reporting is enabled and the rating group `reporting-level` is set to `ratingGroup`, the service data containers within the CDR contain information corresponding to the entire rating group.

***Note:** For PGW Gz/Rf CDRs, you can generate SDCs at the rating group and QoS Class Identifier (QCI) level. For more information, see CDR Reporting (page 1-17).*

The events that trigger the closure and sending of an interim (partial) CDR for bearer-level CDRs are the same for session-level CDRs, with the exception of the following:

- Time and volume-based triggers are per session.
- The QoS change trigger is applicable for session-level QOS changes, such as an APN-AMBR change.

For a list of these event triggers, see Events that Trigger an Interim (Partial) Record (page 1-8).

Synchronizing Gz/Rf and Gy Charging Data

The Gz/Rf (OFCS) and Gy (OCS) interfaces are both used to report subscriber data usage for billing and accounting purposes. To ensure the reported subscriber data usage between the two interfaces is synchronized, Operators can configure a Gy event trigger template that defines the Gy events that trigger the closure and sending of an interim CDR on the Gz/Rf interface. The MCC generates session-level CDRs with rating-group level service data containers for Gy reporting events.

When a configured Gy reporting event occurs, the MCC stores the event in a service data container until a configured threshold is reached, such as the Gy event recurrence count, which specifies the maximum number of containers added to a CDR per closure event or the maximum container limit, which specifies the maximum number of service containers added to a CDR. Upon reaching the configured threshold, the MCC sends the stored service container in an interim CDR over the Gz/Rf interface.

Note: *To use the Gy event trigger template to synchronize Gz/Rf and Gy interface charging data, ensure all Rating Groups have the* `reporting-level` *set to* `ratingGroup` *and that session-level Gz/Rf CDR reporting is enabled. For information on enabling session-level CDR reporting on the Gz/Rf interface, see* *Generating Session-Level Gz/Rf CDRs (page 6-30)*.

Use the following sequence to configure offline charging reporting at the session-level for Gy events.

1 Enable offline charging service data container creation based on Gy reporting events on the Gy online charging interface. The Gy online charging interface must already exist. For more information on configuring the Gy interface, see the *Affirmed Networks Gateway Administration Guide*.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging <interface-name> offline-cdr-for-gy-event enabled
```

2 Configure the Gy event trigger template to specify the Gy events that trigger the closure and sending of an interim CDR on the Gz/Rf interface.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template gy-event-trigger-template <template-name>
<events> change-condition-value <value> recurrence-count <value>
```

The following table describes the configurable parameters.

Parameter	Description
`gy-event-trigger-template`	Specifies the name of the Gy event trigger template.

Parameter	Description
<events>	Specifies the name of the Gy event. ■ **final-close-trigger** ■ **forced-reauthorisation-close-trigger** ■ **quota-exhausted-close-trigger** ■ **quota-holding-time-close-trigger** ■ **rating-condition-change-close-trigger** ■ **tariff-time-change-close-trigger** ■ **validity-time-close-trigger** ■ **volume-quota-threshold-close-trigger**
change-condition-value	Specifies the change-condition value to use in the rating-group level service data container created for this Gy event. If set to `0`, the change-condition value in the service data container is set as defined in 3GPP 32.298.
recurrence-count	Specifies the maximum number of service data containers added to a CDR per closure event. Upon reaching this limit, offline charging CDRs are closed and transfered to the OFCS. Value 0 disables the trigger, value 1 - 20 enables the trigger.

3 Associate the Gy event trigger template with the Gz/Rf offline charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
offline-charging <interface-name> [pgw|sgw|ggsn]
gy-event-trigger-template <template-name>
```

Modifying Default Diameter AVP Flags

Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b). Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. An AVP can have one or both of these flags set.

- Mandatory “M” flag – Indicates that support for this AVP is required. If either the AVP or its value is not recognized, then the message is not processed.
- Vendor-specific “V” flag – Indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.

For example, if the RAT-Type AVP has the Mandatory flag and Vendor-Specific flag set by default, all Diameter messages must have this AVP, otherwise the message is not processed. Also, when this AVP is sent, the Vendor ID field will be populated with the value associated with the specific Vendor.

With this feature, if the Operator changes the RAT-Type AVP Flag setting so that only the Vendor-Specific flag is set for the Gx interface, then subsequent Gx messages may or may not have this AVP. However, the Vendor ID field will still be populated with the value associated with the specific Vendor when this AVP is sent.

Use the following CLI command to modify the default Diameter AVP flag values.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter avp-flags <entry-name> attribute-id
<attribute-id> vendor-id <vendor-id> [gx|gy|gz|swm|sta|s6b] flag-value
[none|mandatory|vendor-specific|mandatory-and-vendor-specific]
```

where

Parameter	Description
avp-flags	Specifies the configured name for the AVP flag entry. The entry name is a custom name used for the AVP flag entry and is not necessarily the name of the Diameter AVP.
attribute-id	Specifies the Diameter AVP attribute ID. ***Note:*** *The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.*
vendor-id	Specifies the Diameter AVP vendor ID. ***Note:*** *The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.*
[gx|gy|gz|swm|sta|s6b]	Specifies the Diameter interface.

Parameter	Description
`flag-value`	Specifies the flag value for the Diameter AVP. ■ If an AVP has a non-zero Vendor ID, the possible flag values are `vendor-specific` and `mandatory-and-vendor-specific`. ■ If an AVP has a Vendor ID of zero, the possible flag values are `none` and `mandatory`.

Configuring an SCTP Profile

To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gz/Rf interface, you must configure an SCTP profile and associate the profile with the local Diameter node.

You also have the option of configuring a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. SCTP multi-homing enables the MCC to switchover to an alternate path in the case of a failure on the primary path. The MCC will revert back to the primary path once the primary connection becomes available. The primary and secondary loopback IP addresses can be IPv4 or IPv6. MCC supports socket creation to be associated to a single network context. SCTP multi-homing will be supported in a single network context.

Note: *If using a mix of IPv4/IPv6, ensure the primary loopback IP address is IPv4 and the secondary is IPv6. SCTP Multi-homing will not work if the primary loopback IP address is IPv6 and the secondary is IPv4. For this reason, Affirmed Networks recommends not using a mix of IPv4/IPv6 addresses for SCTP Multi-homing.*

---

1 Create an SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct sctp-profile
<profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
`cookie-life`	Specifies the lifetime of the SCTP cookie, in 100 milliseconds.
`heartbeat-interval`	Specifies the periodic time interval (in milliseconds) at which the SCTP heart beat messages are sent. Value zero disables SCTP heart beats.
`init-retry`	Specifies the maximum number of attempts to retransmit the INIT message after which the attempted association is terminated.
`path-max-retransmit`	Specifies the maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive.
`max-in-streams`	Specifies the maximum number of incoming SCTP streams.
`max-out-streams`	Specifies the number of outgoing SCTP streams.
`path-mtu`	Specifies the maximum transmission unit (MTU), in bytes, for SCTP streams. Value 0 enables auto-discovery of the path MTU.

Parameter	Description
rto-initial	Specifies the initial retransmission timeout (RTO) value, in milliseconds. Subsequent SCTP RTO will be recomputed starting from this value.
rto-max	Specifies the maximum RTO value (in milliseconds) that any subsequent recomputed RTO may never exceed.
rto-min	Specifies the minimum RTO value (in milliseconds) that any subsequent recomputed RTO may never go below.
sack-period	Specifies the delay before sending the Selective Acknowledgment (SACK), in milliseconds. Value 0 disables the delay value.

2 Specify SCTP as the transport type for the local Diameter node and associate the SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> transport sctp
sctp-profile <profile-name>
```

3 (Optional) Configure a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. The secondary loopback IP address is only configurable for the SCTP transport type.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
secondary-loopback-ip <loopback-name>
```

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> secondary-loopback-ip <loopback-name>
```

Note: *The* `secondary-loopback-ip` *parameter is only visible when the local Diameter node transport type is set to SCTP.*

Weighted Round-Robin Distribution to Multiple Diameter Peers

Operators can configure the MCC to use a weighted round-robin algorithm to distribute initial session requests among the available Diameter peers within a Diameter server group based on the configured weight assigned to each peer. The Diameter peer with the highest weight (with 10 being the highest and 1 being the lowest) receives the most distribution of new session requests. The MCC only uses the configured weight for the initial peer selection (for example, CCR-Initial).

For example, if Peer A is set to 4 and Peer B is set 2, the MCC will distribute the initial session requests as ABABAA. In this example, the MCC alternates between both peers starting with Peer A because it has the highest weight.

When a Diameter peer is configured with a weight of 0, the MCC places the peer in "maintenance mode" and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-Update/CCR-Terminate) for existing sessions.

If the Diameter peer with the highest weight is unavailable, the request is sent to the peer with the next highest weight in the same Diameter server group. If multiple Diameter peers are assigned the highest weight (10), the MCC uses a round-robin approach to select from among those peers for new requests.

Operators can configure the MCC to use a weighted round-robin algorithm to select a Diameter peer from among a list of primary peers or from among a list of secondary peers, but not *across* primary and secondary peers.

Use the following command to assign a weight to a Diameter peer:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> weight <value>
```

- **weight** – Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in "maintenance mode" and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-Update/CCR-Terminate) for existing sessions. *Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.*

Monitoring

This section describes how to monitor the operational status of the OFCS via the Gz/Rf interface and how to display statistical information for OFCS communication messages. This section also describes the alarm notification that occurs when the TCP connection between the GGSN/PGW/SGW and the OFCS fails. For more information on statistics, see *Acuitas Performance Statistics.*

Node Status

Use the following CLI command to display the operational status of the local diameter node.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter node-status <diameter-node-name>
```

The system displays output similar to the following:

NAME	ORIGIN HOST	ORIGIN REALM	HOST IP ADDRESS	TRANSPORT	PORT	GEO REDUNDANT GROUP	STATUS
OFCS	client-affirmed.net	affirmed.net	10.48.59.2	tcp	3868		inservice

Peer Status

Use the following CLI command to display the operational status of the TCP connection between the system and the specified OFCS.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-status <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

NODE NAME	HOST NAME	REALM	IP ADDRESS	PORT	TRANS PORT	ROLE	BACKUP HOST NAME	STATUS
OFCS	ofcs-server.affirmed.com	affirmed.com	10.6.1.217	9100	tcp	primary		Okay

Peer-cmd-Stats

Use the following CLI command to display the number of request/response messages sent per command-code between the system and the specified OFCS.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-cmd-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

NODE NAME	HOST NAME	COMMAND CODE	REALM	NUM OUT REQ	NUM OUT RSP	NUM IN REQ	NUM IN RSP
OFCS	ofcs-server.affirmed.com	257	affirmed.com	3	0	0	3
OFCS	ofcs-server.affirmed.com	271	affirmed.com	3	0	0	3
OFCS	ofcs-server.affirmed.com	280	affirmed.com	618	0	0	618
OFCS	ofcs-server.affirmed.com	282	affirmed.com	0	1	1	0

Peer Stats

Use the following CLI command to display the number of inbound/outbound requests/responses, protocol errors and failures between the system and the specified OFCS.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

```
node name                  OFCS
host name                  ofcs-server.affirmed.com
realm                      affirmed.com
num-protocol-error         0
num-permanent-failure      0
num-routing-failure        0
num-out-request            626
num-out-response           1
num-in-request             1
num-in-response            626
```

Diameter Message-Level Statistics

Use the following CLI command to display statistics about diameter messages exchanged between the local diameter node and diameter peer via the Gz/Rf interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type GZ <diameter-node-name> <peer-host-name>
```

Note: *For more information on the Gz/Rf interface performance statistics, see Acuitas Performance Statistics.*

Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface level.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

- sdc-total-uplink-bytes/sdc-total-downlink-bytes
- sdc-total-uplink-packets/sdc-total-downlink-packets
- total-uplink-bytes/total-downlink-bytes
- total-uplink-packets/total-downlink-packets

Alarms

A major-level trap similar to the following occurs when the TCP connection between the system and the OFCS goes down. The alarm clears after connectivity is restored.

```
anDiameterPeerLinkDown
Description:        Transport connection to the diameter peer is down.
Trap ID:            89
Status:             Current
Severity:           Major
Auto Clearing:      No
Potential Impact:   The Diameter Links are Down towards the OFCS
                    server. This might result in Gz Interface not able
                    to send the Call Data Records to Billing system.
```

```
Corrective Action:  The alarm clears when the interface becomes
                    active. Check the routing/connectivity towards the
                    Diameter peer. The alarm clears after the
                    connectivity is restored. Once you have restored
                    the IP connectivity make sure the Diameter Peer is
                    up and routes are correctly configured and
                    Diameter configuration is correct.
```

Failure Handling

The system uses the watchdog request-based mechanism defined in RFC 3588 and RFC 3539 for failure detection. By exchanging Device-Watchdog-Request (DWR) and Device-Watchdog-Answer (DWA) messages between the SGW/PGW and OFCS, the system can detect OFCS or network-related failures and migrate sessions to a secondary OFCS, if available.

This section describes failure handling on the Gz/Rf interface and includes the following sections.

- Peer Selection (page 6-43)
- Failover Handling (page 6-44)

Peer Selection

The diameter node (SGW/PGW/GGSN) contains a list of diameter peers (OFCS) with which it communicates. Each peer must be assigned one of two roles; primary or secondary. Primary peers are active peers used for diameter sessions and secondary peers are used for diameter sessions only when the active peers are unavailable or the OFCS returns a temporary failure (result code 3xxx/4xxx).

You can configure a secondary peer to act as the backup server for a primary peer so that when a failover event occurs, the system automatically sends all new and pending requests to the backup server. If the backup server fails, the system automatically fails over to the primary server.

For new session requests, the system uses a round-robin approach to select peers from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers.

For an existing session, if the system determines that the server to which the initial ACR was sent is unavailable, the system sends subsequent messages to the backup server, if configured. The backup server may be a primary server if the existing session was originally initiated with a secondary server. If no backup server is configured for the server to which the initial ACR was sent, the system retransmits subsequent messages to the same server.

When the system detects that a peer is down, it removes the peer from the active primary or secondary peer list, depending on its role. When the peer comes back up (CER/CEA exchanges successfully complete), the system adds it back to the active peer list.

Failover Handling

Failover to an alternate OFCS can be triggered if one of the following events occurs:

- The OFCS returns a temporary failure (result code 3xxx/4xxx).
- The ACR timer (`retransmit-timeout`) expires. You can configure this value when you configure the offline charging interface. (see Configuring the Gz/Rf Offline Charging Interface (page 6-7)).
- The OFCS does not respond to DWRs.

You can configure a secondary peer to act as the backup server for a primary peer so that when a failover event occurs, the system automatically sends all new and pending requests to the backup server. If the backup server fails, the system automatically fails over to the primary server.

Note: *When configuring a secondary peer as a backup server, you must enter the host name of the secondary server in the* `backup-host-name` *parameter. See* *Configuring the Gz/Rf Offline Charging Interface (page 6-7)* *for more information.*

When a failover event occurs, the ACR message is retransmitted to the backup peer, if configured. When the primary OFCS recovers, the system sends new ACR messages to the primary OFCS. Existing accounting sessions that were redirected to the secondary OFCS, stay on the secondary server, even when the primary OFCS recovers.

If no backup peer is available, only the Start record is retransmitted to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers.

For all other record types (Interim/Stop/Event), if no backup server is configured, the system retransmits the ACR messages to the *same* peer for the configured number of retransmit counts.

Note the following:

- If a successful response is not received from the OFCS peer, the PGW will retry the ACR message every 30 seconds until a successful response is received even if the `retransmit-count` parameter is disabled (value 0).
- If the PGW receives two temporary failure responses (result code 3xxx/4xxx) from the OFCS, the PGW stops the current retransmit attempts even if the `retransmit-count` is higher than two. However, the PGW will retry the ACR message every 30 seconds until a successful response is received.

Failure Handling Message Flows

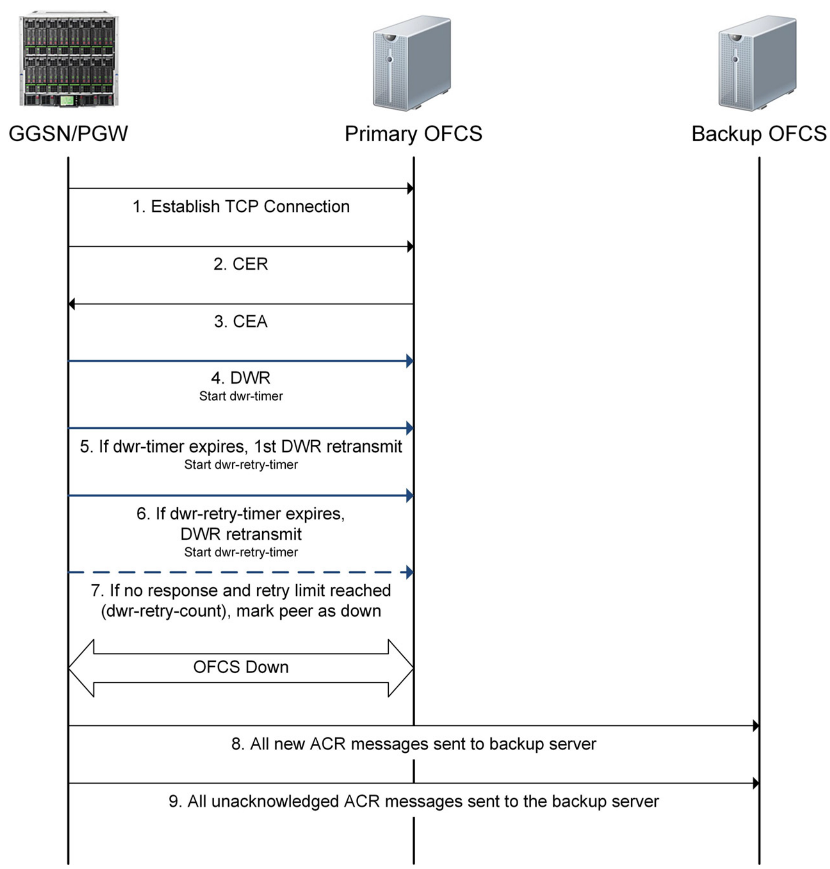
This section provides examples of the following message flows:

- Failover to Backup OFCS (Transport Failure) (page 6-46)
- Failover to Backup OFCS (Application Failure) (page 6-48)
- Failure Handling - No OFCS Available (page 6-50)

Failover to Backup OFCS (Transport Failure)

Figure 6-3 demonstrates the message flow during a failover from the primary to the backup OFCS when the watchdog request-based mechanism detects a failure. A failure at the transport level affects all sessions.

Figure 6-3 Failover to Backup OFCS (Transport Failure)



1 The GGSN/PGW establishes a TCP Connection with the OFCS.

2 The GGSN/PGW sends a Capabilities-Exchange-Request (CER) to the OFCS.

3 Assuming both diameter peers share the same capabilities, such as diameter applications and security mechanism, the OFCS returns with a successful Capabilities-Exchange-Answer (CEA). The TCP connection is successfully established and the peers can begin servicing accounting sessions.

4 Periodically, the GGSN/PGW sends DWRs to the OFCS to monitor the TCP connection.

5 If the OFCS does not respond to the first DWR with a Device-Watchdog-Answer (DWA) within the configured time limit (`dwr-timer`), the system retransmits the DWR and starts the DWR retry timer (`dwr-retry-timer`) with each retransmission.

Note: *The `dwr-retry-timer` is typically set at a shorter interval than the `dwr-timer`.*

---

6 If the OFCS does not respond within the time limit (`dwr-retry-timer`), the GGSN/PGW retransmits the DWR. This cycle continues until the GGSN/PGW receives a valid DWA response from the OFCS or the configured retry limit (`dwr-retry-count`) is reached.

7 If the configured retry limit (`dwr-retry-count`) is reached, the GGSN/PGW marks the primary OFCS as down.

8 The GGSN/PGW sends all new ACR messages to the secondary OFCS.

9 All unacknowledged ACR messages (all requests sent to the primary server with no response) are sent to the secondary OFCS.

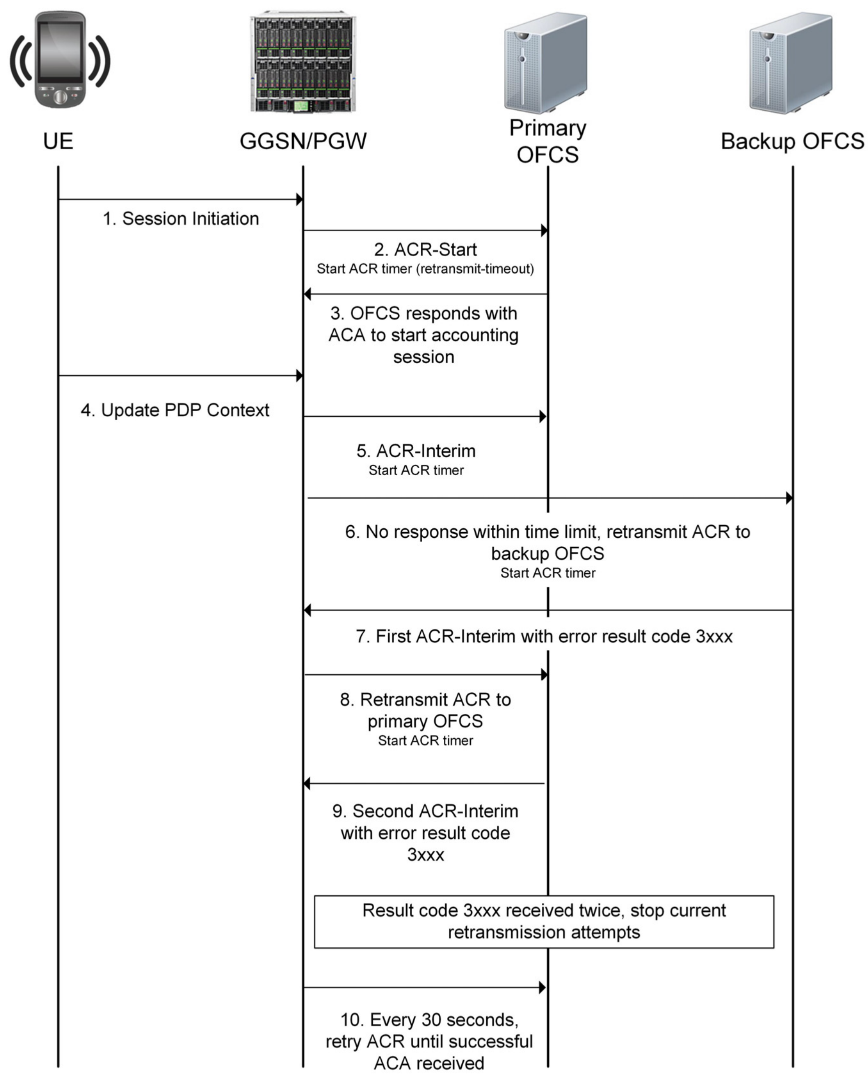
Note: *If there is no available secondary server and the failure occurs because the Watchdog mechanism expired, all messages are saved and sent when the OFCS server becomes available.*

Failover to Backup OFCS (Application Failure)

Figure 6-4 demonstrates the message flow during a failover from the primary to the backup OFCS when the ACR timer (`retransmit-timeout`) expires or the GGSN/PGW receives two ACA messages containing result code 3xxx during an existing session.

While a failure at the transport level affects all sessions, a failure at the application level only affects the session on which the ACR expiration occurs.

Figure 6-4 Failover to Backup OFCS (Application Failure)



1. The User Equipment (UE) initiates a session with the GGSN/PGW.
2. The GGSN/PGW sends an ACR message to the OFCS.
3. OFCS responds with an ACA message to start the accounting session.
4. During the session, an event occurs triggering an update.
5. The GGSN/PGW sends the primary OFCS an ACR-Interim message to update the accounting records and starts the ACR timer (`retransmit-timeout`) with each retransmission.
6. If the OFCS does not respond within the time limit (`retransmit-timeout`), the GGSN/PGW retransmits the ACR to the backup OFCS. This cycle continues until the GGSN/PGW receives a successful ACA response from the OFCS or the configured retransmit count is reached. If the retransmit count is reached, the GGSN/PGW retries the ACR-Interim message every 30 seconds until a successful ACA is received.
7. The backup OFCS responds with the first ACA message containing error result code 3xxx.
8. The GGSN/PGW retransmits the ACR-Interim to the primary OFCS.
9. Upon receiving the second ACA message containing error result code 3xxx, the GGSN/PGW stops the current retransmit attempts.
10. The GGSN/PGW retries the ACR-Interim message every 30 seconds until a successful ACA is received.

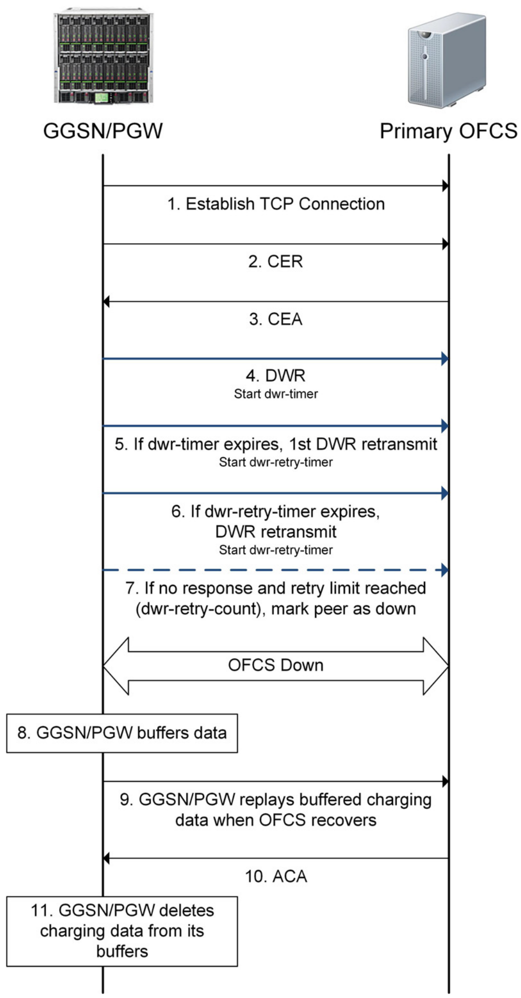
Failure Handling - No OFCS Available

If the connection to the OFCS fails and a secondary OFCS is not available, the system buffers charging data in memory and replays the buffered data in a staggered manner to the OFCS when it becomes available.

If the amount of buffered charging data exceeds a certain threshold, the system writes subsequent charging data to disk. When the OFCS recovers, the system replays the stored charging records (from both memory and disk) to the OFCS in the order in which they were stored. The system does not delete charging information from its buffers until an ACA message is received.

Figure 6-5 demonstrates the message flow when a failure occurs and no secondary OFCS is available.

Figure 6-5 Failure Handling (No Secondary OFCS)



1 The GGSN/PGW establishes a TCP Connection with the OFCS.

2 The GGSN/PGW sends a Capabilities-Exchange-Request (CER) to the OFCS.

3 Assuming both diameter peers share the same capabilities, such as diameter applications and security mechanisms, the OFCS returns with a successful Capabilities-Exchange-Answer (CEA). The TCP connection is successfully established and the peers can begin servicing accounting sessions.

4 Periodically, the GGSN/PGW sends DWRs to the OFCS to monitor the TCP connection.

5 If the OFCS does not respond to the first DWR with a Device-Watchdog-Answer (DWA) within the configured time limit (`dwr-timer`), the system retransmits the DWR and starts the DWR retry timer (`dwr-retry-timer`) with each retransmission.

Note: *The `dwr-retry-timer` is typically set at a shorter interval than the `dwr-timer`.*

---

6 If the OFCS does not respond within the time limit (`dwr-retry-timer`), the GGSN/PGW retransmits the DWR. This cycle continues until the GGSN/PGW receives a valid DWA response from the OFCS or the configured retry limit (`dwr-retry-count`) is reached.

7 If the configured retry limit (`dwr-retry-count`) is reached, the GGSN/PGW marks the primary OFCS as down.

8 The GGSN/PGW buffers the charging data in memory.

9 The GGSN/PGW replays the buffered charging data in a staggered manner to the OFCS when it becomes available.

10 The OFCS sends an ACA message to the GGSN/PGW acknowledging the received accounting request.

11 Upon receiving the ACA from the OFCS, the GGSN/PGW deletes the accounting data from its buffers.

Message Definitions

Table 6-4 describes the message definitions for the Gz/Rf interface.

Table 6-4 Gz/Rf Interface Message Definitions

Command	Acronym	Command Code	Reference
Capabilities-Exchange-Request (page 6-53)	CER	257	RFC 3588
Capabilities-Exchange-Answer (page 6-54)	CEA	257	RFC 3588
Device-Watchdog-Request (page 6-54)	DWR	280	RFC 3588
Device-Watchdog-Answer (page 6-55)	DWA	280	RFC 3588
Disconnect-Peer-Request (page 6-55)	DPR	282	RFC 3588
Disconnect-Peer-Answer (page 6-55)	DPA	282	RFC 3588
Accounting-Request (page 6-56)	ACR	271	RFC 3588
Accounting-Answer (page 6-63)	ACA	271	RFC 3588

Table 6-5 describes the special symbols in the AVP name for each message definition.

Table 6-5 AVP Name Symbols

Symbol	Description
`<>`	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
`{}`	AVP is mandatory.
`[]`	AVP is optional.
`*[]`	AVP is optional and is repeated.
`M`	If the Mandatory header bit is set, it indicates that support for this AVP is required. For further details, see RFC 3588.
`V`	If Vendor-Specific header bit is set, it indicates that the optional Vendor-ID field is present in the AVP header. For further details, see RFC 3588.
`M,V`	Indicates that this AVP has both the mandatory “M” flag and vendor-specific “V” flag set.

Note: *Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b). See* *Modifying Default Diameter AVP Flags (page 6-34)* *for more information.*

Capabilities-Exchange-Request

The Capabilities-Exchange-Request (CER), indicated by the Command-Code set to 257 and the Command Flags' 'R' bit set, is sent to exchange local capabilities. Upon detection of a transport failure, this message must not be sent to an alternate peer.

Message Format:

```
<CER> ::= < Diameter Header: 257, REQ >
       {Origin-Host}
       {Origin-Realm}
       1*{Host-IP-Address}
       {Vendor-Id}
       {Product-Name}
       [Origin-State-Id]
       *[Supported-Vendor-Id]
       *[Auth-Application-Id]
       *[Inband-Security-Id]
       *[Acct-Application-Id]
       *[Vendor-Specific-Application-Id]
       [Firmware-Revision]
       *[AVP]
```

Capabilities-Exchange-Answer

The Capabilities-Exchange-Answer (CEA), indicated by the Command-Code set to 257 and the Command Flags' 'R' bit cleared, is sent in response to a CER message.

Message Format:

```
<CEA> ::= < Diameter Header: 257 >
       {Result-Code}
       {Origin-Host}
       {Origin-Realm}
       1*{Host-IP-Address}
       {Vendor-Id}
       {Product-Name}
       [Origin-State-Id]
       [Error-Message]
       *[Failed-AVP]
       *[Supported-Vendor-Id]
       *[Auth-Application-Id]
       *[Inband-Security-Id]
       *[Acct-Application-Id]
       *[Vendor-Specific-Application-Id]
       [Firmware-Revision]
       * [AVP]
```

Device-Watchdog-Request

The Device-Watchdog-Request (DWR), indicated by the Command-Code set to 280 and the Command Flags' 'R' bit set, is sent to a peer when no traffic has been exchanged between two peers.

Message format:

```
<Device-Watchdog-Request> ::= <Diameter Header: 280, REQ>
       {Origin-Host}
       {Origin-Realm}
       [Origin-State-Id]
```

Device-Watchdog-Answer

The Device-Watchdog-Answer (DWA), indicated by the Command-Code set to 280 and the Command Flags' 'R' bit cleared, is sent as a response to the Device-Watchdog-Request message.

Message format:

```
<Device-Watchdog-Answer> ::= <Diameter Header: 280>
       {Result-Code}
       {Origin-Host}
       {Origin-Realm}
       [Error-Message]
       *[Failed-AVP]
       [Original-State-Id]
```

Disconnect-Peer-Request

The Disconnect-Peer-Request (DPR), indicated by the Command-Code set to 282 and the Command Flags' 'R' bit set, is sent to a peer to inform its intentions to shutdown the transport connection.

Message format:

```
<Disconnect-Peer-Request> ::= <Diameter Header: 282, REQ>
       {Origin-Host}
       {Origin-Realm}
       {Disconnect-Cause}
```

Disconnect-Peer-Answer

The Disconnect-Peer-Answer (DPA), indicated by the Command-Code set to 282 and the Command Flags' 'R' bit cleared, is sent as a response to the Disconnect-Peer-Request message. Upon receipt of this message, the transport connection is shutdown.

Message format:

```
<Disconnect-Peer -Answer> ::= <Diameter Header: 282>
       {Result-Code}
       {Origin-Host}
       {Origin-Realm}
       [Error-Message]
       *[Failed-AVP]
```

Accounting-Request

The Accounting-Request (ACR) command, indicated by the Command-Code field set to 271 and the Command Flags' 'R' bit set, is sent by a Diameter node, acting as a client, in order to exchange accounting information with a peer.

Message format:

```
<ACR> ::= <Diameter Header: 271, REQ, PXY>
       <Session-Id>
       {Origin-Host}
       {Origin-Realm}
       {Destination-Realm}
       {Accounting-Record-Type}
       {Accounting-Record-Number}
       [Acct-Application-Id]
       [User-Name]
       [Acct-Interim-Interval]
       [Origin-State-Id]
       [Destination-Host]
       [Event-Timestamp]
       [Service-Context-Id]
       [Service-Information]
       *[AVP]
```

Table 6-6 lists the ACR message format. For a description of each AVP, see Appendix A, Diameter AVP Descriptions. The AVPs with the vendor-specific “V” flag set include a Vendor-ID of 10415 (3GPP).

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
<Session-ID>	263	UTF8String	M
{Origin-Host}	264	DiameterIdentity	M
{Origin-Realm}	296	DiameterIdentity	M
{Destination-Realm}	283	DiameterIdentity	M
{Accounting-Record-Type}	480	Enumerated	M
{Accounting-Record-Number}	485	Unsigned32	M
[Acct-Application-Id]	259	Unsigned32	M
[User-Name]	1	UTF8String	M
[Acct-Interim-Interval]	85	Unsigned32	M
[Origin-State-Id]	278	Unsigned32	M
[Destination-Host]	293	DiameterIdentity	M
[Event-Timestamp]	55	Time	M
[Service-Context-Id]	461	UTF8String	M
[Service-Information]	873	Grouped	M,V
[Subscription-Id]	443	Grouped	M,V
[PS-Information]	874	Grouped	M,V
[3GPP-Charging-Id]	2	OctetString	M,V
[PDN-Connection-Charging-ID]	2050	Unsigned32	M,V
[Node-Id]	2064	UTF8String	M,V
[3GPP-PDP-Type]	3	Enumerated	M,V
*[PDP-Address]	1227	Address	M,V
[Dynamic-Address-Flag]	2051	Enumerated	M,V
[QoS-Information]	1016	Grouped	M,V
[QoS-Class-Identifier]	1028	Enumerated	M,V
[Max-Requested-Bandwidth-UL]	516	Unsigned32	M,V

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
[Max-Requested-Bandwidth-DL]	515	Unsigned32	M,V
[Guaranteed-Bitrate-UL]	1026	Unsigned32	M,V
[Guaranteed-Bitrate-DL]	1025	Unsigned32	M,V
[Bearer-Identifier]	1020	OctetString	M,V
[Allocation-Retention-Priority]	1034	Grouped	V
[APN-Aggregate-Max-Bitrate-UL]	1041	Unsigned32	V
[APN-Aggregate-Max-Bitrate-DL]	1040	Unsigned32	V
[Extended-Max-Requested-BW-DL]	554	Unsigned32	V
[Extended-Max-Requested-BW-UL]	555	Unsigned32	V
[Extended-GBR-DL]	2850	Unsigned32	V
[Extended-GBR-UL]	2851	Unsigned32	V
[Extended-APN-AMBR-DL]	2848	Unsigned32	V
[Extended-APN-AMBR-UL]	2849	Unsigned32	V
[SGSN-Address]	1228	Address	M,V
[GGSN-Address]	847	Address	M,V
[SGW-Address]	2067	Address	M,V
[Serving-Node-Type]	2047	Enumerated	M,V
[SGW-Change]	2065	Enumerated	M,V
[3GPP-IMSI-MCC-MNC]	8	UTF8String	M,V
[IMSI-Unauthenticated-Flag]	2308	Enumerated	M,V
[3GPP-GGSN-MCC-MNC]	9	UTF8String	M,V
[3GPP-NSAPI]	10	OctetString	M,V
[Called-Station-Id]	30	UTF8String	M,V
[3GPP-Selection-Mode]	12	UTF8String	M,V
[3GPP-Charging-Characteristics]	13	UTF8String	M,V
[Charging-Characteristics-Selection-Mode]	2066	Enumerated	M,V
[3GPP-SGSN-MCC-MNC]	18	UTF8String	M,V
[3GPP-MS-TimeZone]	23	OctetString	M,V

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
[Charging-Rule-Base-Name]	1004	UTF8String	M,V
[3GPP-User-Location-Info]	22	OctetString	M,V
[User-CSG-Information]	2319	Grouped	M,V
[UWAN-User-Location-Info]	3918	Grouped	M,V
[3GPP-RAT-Type]	21	OctetString	M,V
[PDP-Context-Type]	1247	Enumerated	M,V
*[Traffic-Data-Volumes]	2046	Grouped	M,V
[QoS-Information]	1016	Grouped	M,V
[QoS-Class-Identifier]	1028	Enumerated	M,V
[Max-Requested-Bandwidth-DL]	515	Unsigned32	M,V
[Max-Requested-Bandwidth-UL]	516	Unsigned32	M,V
[Guaranteed-Bitrate-DL]	1025	Unsigned32	M,V
[Guaranteed-Bitrate-UL]	1026	Unsigned32	M,V
[Bearer-Identifier]	1020	OctetString	M,V
[Allocation-Retention-Priority]	1034	Grouped	V
[APN-Aggregate-Max-Bitrate-DL]	1040	Unsigned32	V
[APN-Aggregate-Max-Bitrate-UL]	1041	Unsigned32	V
[Extended-Max-Requested-BW-DL]	554	Unsigned32	V
[Extended-Max-Requested-BW-UL]	555	Unsigned32	V
[Extended-GBR-DL]	2850	Unsigned32	V
[Extended-GBR-UL]	2851	Unsigned32	V
[Extended-APN-AMBR-DL]	2848	Unsigned32	V
[Extended-APN-AMBR-UL]	2849	Unsigned32	V
[Accounting-Input-Octets]	363	Integer64	M
[Accounting-Input-Packets]	365	Integer64	M
[Accounting-Output-Octets]	364	Integer64	M
[Accounting-Output-Packets]	366	Integer64	M
[Change-Condition]	2037	Integer32	M,V

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
[Change-Time]	2038	Time	M,V
[3GPP-User-Location-Info]	22	OctetString	M,V
[3GPP-RAT-Type]	21	OctetString	M,V
[CP-CIoT-EPS-Optimisation-Indicator]	3930	Enumerated	M,V
[Serving-PLMN-Rate-Control]	4310	Grouped	M,V
[UWAN-User-Location-Info]	3918	Grouped	M,V
*[Service-Data-Container]	2040	Grouped	M,V
[AF-Correlation-Information]	1276	Grouped	M,V
[Charging-Rule-Base-Name]	1004	UTF8String	M,V
[Accounting-Input-Octets]	363	Integer64	M
[Accounting-Output-Octets]	364	Integer64	M
[Accounting-Input-Packets]	365	Integer64	M
[Accounting-Output-Packets]	366	Integer64	M
[Local-Sequence-Number]	2063	Unsigned32	M,V
[QoS-Information]	1016	Grouped	M,V
[QoS-Class-Identifier]	1028	Enumerated	M,V
[Max-Requested-Bandwidth-DL]	515	Unsigned32	M,V
[Max-Requested-Bandwidth-UL]	516	Unsigned32	M,V
[Guaranteed-Bitrate-DL]	1025	Unsigned32	M,V
[Guaranteed-Bitrate-UL]	1026	Unsigned32	M,V
[Bearer-Identifier]	1020	OctetString	M,V
[Allocation-Retention-Priority]	1034	Grouped	V
[APN-Aggregate-Max-Bitrate-DL]	1040	Unsigned32	V
[APN-Aggregate-Max-Bitrate-UL]	1041	Unsigned32	V
[Extended-Max-Requested-BW-DL]	554	Unsigned32	V
[Extended-Max-Requested-BW-UL]	555	Unsigned32	V
[Extended-GBR-DL]	2850	Unsigned32	V
[Extended-GBR-UL]	2851	Unsigned32	V

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
[Extended-APN-AMBR-DL]	2848	Unsigned32	V
[Extended-APN-AMBR-UL]	2849	Unsigned32	V
[Rating-Group]	432	Unsigned32	M
[Change-Time]	2038	Time	M,V
[Service-Identifier]	439	Unsigned32	M,V
[Service-Specific-Info]	1249	Grouped	M,V
[SGSN-Address]	1228	Address	M,V
[Time-First-Usage]	2043	Time	M,V
[Time-Last-Usage]	2044	Time	M,V
[Time-Usage]	2045	Time	M,V
*[Change-Condition]	2037	Integer32	M,V
[3GPP-User-Location-Info]	22	OctetString	M,V
[3GPP-RAT-Type]	21	OctetString	M,V
[Serving-PLMN-Rate-Control]	4310	Grouped	M,V
[APN-Rate-Control]	3933	Grouped	M,V
[UWAN-User-Location-Info]	3918	Grouped	M,V
[User-Equipment-Info]	458	Grouped	M
[Change-Condition]	2037	Integer32	M,V
[3GPP-IMEISV]	20	UTF8String	M,V
[Diagnostics]	2039	Integer32	M,V
[PS-Furnish-Charging-Information]	865	Grouped	M,V
[Low-Priority-Indicator]	2602	Enumerated	M,V
[SGi-PtP-Tunnelling-Method]	3931	Enumerated	M,V
[CP-CIoT-EPS-Optimisation-Indicator]	3930	Enumerated	M,V
[UNI-PDU-CP-Only-Flag]	3932	Enumerated	M,V
[Serving-PLMN-Rate-Control]	4310	Grouped	M,V
[APN-Rate-Control]	3933	Grouped	M,V
[SCS-AS-Address]	3940	Grouped	M,V

Table 6-6 ACR Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
[RAN-Secondary-RAT-Usage-Report]	1302	Grouped	V
[Secondary-RAT-Type]	1304	OctetString	V
[RAN-Start-Timestamp]	1303	Time	V
[RAN-End-Timestamp]	1301	Time	V
[Accounting-Input-Octets]	363	Integer64	M
[Accounting-Output-Octets]	364	Integer64	M
[IMS-Information]	876	Grouped	M,V
[Node-Functionality]	862	Enumerated	M,V

Table 6-7 lists the vendor-specific attributes (VSAs) for ACR messages. For a description of each attribute, see Appendix A, Diameter AVP Descriptions.

Table 6-7 ACR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Format	AVP Flag
[Charging-Gateway-Function-Host]	12951	6068	OctetString	M,V
[Charging-Group-ID]	12951	6069	UTF8String	M,V
[Cumulative-Acct-Input-Octets]	12951	132044	Unsigned64	M,V
[Cumulative-Acct-Output-Octets]	12951	132045	Unsigned64	M,V

Accounting-Answer

The Accounting-Answer (ACA) command, indicated by the Command-Code field set to 271 and the Command Flags' 'R' bit cleared, is used to acknowledge an Accounting-Request command.Message format:

```
<ACA> ::= < Diameter Header: 271, REQ, PXY >
       <Session-Id>
       {Result-Code}
       {Origin-Host}
       {Origin-Realm}
       {Accounting-Record-Type}
       {Accounting-Record-Number}
       [Acct-Application-Id]
       [User-Name]
       [Acct-Interim-Interval]
       [Origin-State-Id]
       [Event-Timestamp]
       *[AVP]
```

Table 6-8 lists the ACA message format. For a description of each AVP, see Chapter A, Diameter AVP Descriptions.

Table 6-8 ACA Message Format

AVP Name	AVP Code	AVP Type	AVP Flag
<Session-ID>	263	UTF8String	M
{Result-Code}	268	Unsigned32	M
{Origin-Host}	264	DiameterIdentity	M
{Origin-Realm}	296	DiameterIdentity	M
{Accounting-Record-Type}	480	Enumerated	M
{Accounting-Record-Number}	485	Unsigned32	M
[Acct-Application-Id]	259	Unsigned32	M
[User-Name]	1	UTF8String	M
[Acct-Interim-Interval]	85	Unsigned32	M
[Origin-State-Id]	278	Unsigned32	M
[Event-Timestamp]	55	Time	M

Appendix A

Diameter AVP Descriptions

This section describes the supported Attribute Value Pair (AVP) definitions for the Gy, Gx, and Gz/Rf Diameter interfaces and therefore the AVPs described in this section do not apply to all interfaces.

Note: *Refer to the 3GPP specifications and RFCs listed in References and Specifications for more details.*

Standard AVP Definitions

This section describes the following standard AVPs used in the messages for DCCA.

Accounting-Input-Octets

AVP Name	Accounting-Input-Octets
AVP Code	363
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Input-Packets

AVP Name	Accounting-Input-Packets
AVP Code	365
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Output-Octets

AVP Name	Accounting-Output-Octets
AVP Code	364
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Output-Packets

AVP Name	Accounting-Output-Packets
AVP Code	366
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Record-Number

AVP Name	Accounting-Record-Number
AVP Code	485
AVP Data Type	Unsigned32
Description	Contains the sequence number of the transferred messages.

Accounting-Record-Type

AVP Name	Accounting-Record-Type
AVP Code	480
AVP Data Type	Enumerated
Description	Defines the type of accounting record: ■ START – Starts an accounting session. ■ INTERIM – Updates an accounting session. ■ STOP – Ends an accounting session. ■ EVENT – Indicates a one-time accounting event.

Acct-Application-Id

AVP Name	Acct-Application-Id
AVP Code	259
AVP Data Type	Unsigned32
Description	Indicates the unique ID of the Diameter Accounting Application and is defined with the value 3.

Acct-Interim-Interval

AVP Name	Acct-Interim-Interval
AVP Code	85
AVP Data Type	Unsigned32
Description	Indicates the number of seconds between each interim update for this session.

Auth-Application-Id

AVP Name	Auth-Application-Id
AVP Code	258
AVP Data Type	Unsigned32
Description	Indicates the unique ID that is used for re-authentication or re-authorization. The AVP contains the following values: ■ Gy interface – 4 (DCCA) ■ Gx interface – 16777238 ■ S6b interface – You can specify the application ID (standard or non-standard) on the S6b interface using the following command. This is any value up to 32 bits. The default is `16777272` (default - 3GPP S6b). `an3000-mcm-slot7cpu1(config)#` **`service-construct interface s6b-interface <name> auth-application-id <value>`**

CC-Input-Octets

AVP Name	CC-Input-Octets
AVP Code	412
AVP Data Type	Unsigned64
Description	The amount of octets to be received.

CC-Output-Octets

AVP Name	CC-Output-Octets
AVP Code	414
AVP Data Type	Unsigned64
Description	The amount of octets to be sent.

CC-Request-Number

AVP Name	CC-Request-Number
AVP Code	415
AVP Data Type	Unsigned32
Description	Indicates the identity of a certain request in a session, starting from 0.

CC-Request-Type

AVP Name	CC-Request-Type
AVP Code	416
AVP Data Type	Enumerated
Description	Indicates the reason for sending a CCR message. This AVP must be included in the CCR message. The CC-Request-Type AVP contains the following values: ■ INITIAL_REQUEST (1) – Indicates that the initial CCR is used to initiate a credit control session. The session contains the credit control information about initiating a session. ■ UPDATE_REQUEST (2) – Indicates that the update CCR contains the credit control information about the sessions of which credit control is set up. When the re-authorization needs to be initiated, an update CCR needs to be initiated. ■ TERMINATION_REQUEST (3) – Indicates an interruption to a credit control session by terminating a credit request.

CC-Session-Failover

AVP Name	CC-Session-Failover
AVP Code	418
AVP Data Type	Enumerated
Description	Indicates whether the OCS supports moving the CCR messages to a secondary server for a CC session. If the OCS supports session failover, the system sends CCR messages to the secondary server when the transport connection with the primary server goes down. The following values are defined: ■ FAILOVER_NOT_SUPPORTED (0) ■ FAILOVER_SUPPORTED (1)

CC-Service-Specific-Units

AVP Name	CC-Service-Specific-Units
AVP Code	417
AVP Data Type	Unsigned64
Description	The amount of service specific units. For example, the number of events for a rating-group.

CC-Time

AVP Name	CC-Time
AVP Code	420
AVP Data Type	Unsigned32
Description	The requested, distributed, or used time of the charging party, in seconds.

CC-Total-Octets

AVP Name	CC-Total-Octets
AVP Code	421
AVP Data Type	Unsigned64
Description	The amount of octets to be sent and received.

Credit-Control-Failure-Handling

AVP Name	Credit-Control-Failure-Handling
AVP Code	427
AVP Data Type	Enumerated
Description	The OCS uses this AVP to instruct clients how to handle the PDN session during a failure condition (transport failure etc.). The following values are defined: ■ TERMINATE (0) ■ CONTINUE (1) ■ RETRY_AND_TERMINATE (2)

Destination-Host

AVP Name	Destination-Host
AVP Code	293
AVP Data Type	DiameterIdentity
Description	Indicates the identity of the destination device. This AVP must be contained in messages that are initiated by an agent. This AVP can be contained in request messages but cannot be contained in response messages.

Destination-Realm

AVP Name	Destination-Realm
AVP Code	283
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the destination device.

Event-Timestamp

AVP Name	Event-Timestamp
AVP Code	55
AVP Data Type	Time
Description	Indicates the time of an event. The value of this AVP is generated by the DCC client. The time is converted from January 1, 1900, 00:00 UTC, in seconds. ***Note:*** *UTC: Coordinated Universal Time.* Gx Interface – Indicates the time when the Create Session Request was received. Gy Interface – Indicates the time when the quota was requested.

Experimental-Result

AVP Name	Experimental-Result
AVP Code	297
AVP Data Type	Grouped
Description	Indicates whether a particular vendor-specific request was completed successfully or whether an error occurred. The detailed syntax is as follows: `Experimental-Result::= <AVP Header: 297>` `{Vendor-Id}` `{Experimental-Result-Code}`

Failed-AVP

AVP Name	Failed-AVP
AVP Code	279
AVP Data Type	Grouped
Description	Provides debugging information in cases where a request is rejected or not fully processed due to erroneous information in a specific AVP. The value of the Result-Code AVP will provide information on the reason for the Failed-AVP.

Filter-ID

AVP Name	Filter-ID
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates the name of the filter list for this user. Used to restrict user access when the FUI AVP (embedded within the MSCC AVP), received in the CCA message from the OCS, contains the Final-Unit-Action AVP set to RESTRICT_ACCESS and one or more Filter-ID AVPs referencing a locally configured Rating Group with an IP address filter list. Multiple Filter-IDs are supported by configuring a Rating Group for each Filter-ID. See “Mapping a Filter-ID to a Rating-Group” in the *Affirmed Networks Gy Interface Specification*.

Final-Unit-Action

AVP Name	Final-Unit-Action
AVP Code	449
AVP Data Type	Enumerated
Description	Indicates the action to be taken once the subscriber has consumed all of the final granted units. The following values are defined: ■ TERMINATE (0) – Upon consumption of the final granted units, the session is terminated. ■ REDIRECT (1) – Upon consumption of the final granted units, the subscriber is redirected to the address specified in the Redirect-Server AVP. ■ RESTRICT-ACCESS (2) – Gy Interface – Upon consumption of the final granted units, user access is restricted according to the IP packet filters identified by the Filter-ID AVP. All of the packets that do not match the filters are dropped. For more information, see “Mapping a Filter-ID to a Rating-Group” in the *Affirmed Networks Gy Interface Specification*. – Gx Interface – If a flow is out of charging credits and when a packet for that flow arrives, the packet is dropped except when an Advice of Charge (AoC) ID is configured for the associated Rating Group on the MCC device. In this case, the subscriber is redirected to the redirect URL configured for the AoC ID.

Final-Unit-Indication

AVP Name	Final-Unit-Indication
AVP Code	430
AVP Data Type	Grouped
Description	The Final-Unit-Indication (FUI) AVP indicates that the Granted-Service-Unit AVP in the CCA contains the final units for the service. After these units have expired, the gateway executes the action indicated in the Final-Unit-Action AVP. Note the following: **Gy Interface** The FUI AVP is embedded within the Multiple-Services-Credit-Control (MSCC) AVP in the CCA message. The MCC supports FUI-based redirection when the MSCC AVP received in the CCA message from the OCS contains: ■ MSCC result code = 2001 (DIAMETER_SUCCESS) or 4012 (CREDIT_LIMIT_REACHED) + FUI AVP ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP, no grant (no GSU AVP) ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP + zero Grant (GSU AVP with zero grant / no grant info) The detailed syntax is as follows: `Final-Unit-Indication ::= < AVP Header: 430 >` `{Final-Unit-Action}` `*[Filter-Id]` `[Redirect-Server]` **Gx Interface** The FUI AVP is included in the CCA message at the command level. The detailed syntax is as follows: `Final-Unit-Indication ::= < AVP Header: 430 >` `{Final-Unit-Action}` `[Redirect-Server]` The FUI AVP is also included in the CCR message within the Charging-Rule-Report AVP to report the status of the PCC rules for which credit is no longer available or credit has been reallocated after the former out-of-credit indication. When reporting an out-of-credit condition, the Final-Unit-Indication AVP indicates the termination action the PCEF applies to the PCC rules as instructed by the OCS. The detailed syntax is as follows: `Final-Unit-Indication ::= < AVP Header: 430 >` `{Final-Unit-Action}` `*[Filter-Id]` `[Redirect-Server]`

Firmware-Revision

AVP Name	Firmware-Revision
AVP Code	267
AVP Data Type	Unsigned32
Description	Used to inform a diameter peer of the firmware revision of the issuing device.

Granted-Service-Unit

AVP Name	Granted-Service-Unit
AVP Code	431
AVP Data Type	Grouped
Description	The total service units that the DCC client can provide for end users.

Host-IP-Address

AVP Name	Host-IP-Address
AVP Code	257
AVP Data Type	Address
Description	Informs a diameter peer of the sender's IP address.

Inband-Security-Id

AVP Name	Inband-Security-Id
AVP Code	299
AVP Data Type	Unsigned32
Description	Indicates support of the security portion of the application.

Multiple-Services-Credit-Control

AVP Name	Multiple-Services-Credit-Control
AVP Code	456
AVP Data Type	Grouped
Description	Indicates all parameters for the CTF quota management and defines the quotas to allow traffic to flow. Each Multiple-Services-Credit-Control (MSCC) carries units for a single rating group.

Multiple-Services-Indicator

AVP Name	Multiple-Services-Indicator
AVP Code	455
AVP Data Type	Enumerated
Description	Indicates whether the system is capable of reporting multiple rating groups within a CC session. It is included in the CCR-Initial message. The following values are defined: ■ MULTIPLE_SERVICES_NOT_SUPPORTED (0) ■ MULTIPLE_SERVICES_SUPPORTED (1)

Origin-Host

AVP Name	Origin-Host
AVP Code	264
AVP Data Type	DiameterIdentity
Description	Contains the identification of the source point of the operation and the realm of the operation originator. Relay agents must not modify this AVP. The value of the Origin-Host AVP is guaranteed to be unique within a single host. ***Note:*** *The Origin-Host AVP may resolve to more than one address as the diameter peer may support more than one address (for example: host1.com).*

Origin-Realm

AVP Name	Origin-Realm
AVP Code	296
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the device that initiates DCC messages.

Origin-State-Id

AVP Name	Origin-State-Id
AVP Code	278
AVP Data Type	Unsigned32
Description	Carries the state associated with an Origin-Host.

Product-Name

AVP Name	Product-Name
AVP Code	269
AVP Data Type	UTF8String
Description	Contains the vendor-assigned name for the product. The Product-Name AVP should remain constant across firmware revisions for the same product.

Rating-Group

AVP Name	Rating-Group
AVP Code	432
AVP Data Type	Unsigned32
Description	The identifier of a rating group. For information on how the Rating-Group AVP is used on the Gx interface to install/activate PCC rules, see “PCC Rules”, in the *Affirmed Networks Gx Interface Specification*.

Re-Auth-Request-Type

AVP Name	Re-Auth-Request-Type
AVP Code	285
AVP Data Type	Enumerated
Description	Included in application-specific authorization answers to inform the client of the action expected upon expiration of the Authorization-Lifetime. The following values are defined: ■ AUTHORIZE_ONLY (0) – An authorization only re-authorization is expected upon expiration of the Authorization-Lifetime. This is the default value if the AVP is not present in answer messages that include the Authorization-Lifetime. ■ AUTHORIZE_AUTHENTICATE (1) – An authentication and authorization re-authorization is expected upon expiration of the Authorization-Lifetime.

Redirect-Address-Type

AVP Name	Redirect-Address-Type
AVP Code	433
AVP Data Type	Enumerated
Description	Defines the type of address given in the Redirect-Server-Address AVP. The address type can be one of the following: ■ URL (2) – The address is in the form of Uniform Resource Locator (URL). ***Note:*** *MCC does not support Redirect-Address-Type SIP URI.*

Redirect-Host

AVP Name	Redirect-Host
AVP Code	292
AVP Data Type	DiameterURI
Description	Contains the host to which the request should be forwarded. If the PGW receives a CCA (Initial, Update or Terminate) message from the PCRF and the Result-Code AVP is set to DIAMETER_REDIRECT_INDICATION, the PGW will forward the CCR message for that session to the host contained in the Redirect-Host AVP. The PCRF can send multiple Redirect-Host AVPs in the CCA message and if one fails, the MCC selects the next one in the list. The Redirect-Host AVP is configurable in CCA data record templates. The default `presentFlag` setting is `yes`, indicating the AVP is processed when received in the CCA message. `an3000-mcm-slot7cpu1(config)#` **`service-construct data-record-template [pgw|ggsn] pcrf diameter cca <`*`template-name`*`> redirecthost presentFlag [yes|no]`**

Redirect-Server

AVP Name	Redirect-Server
AVP Code	434
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) with which the end user is to be connected when the account cannot cover the service cost. This AVP must be present when the Final-Unit-Action AVP is set to REDIRECT. The detailed syntax is as follows: `Redirect-Server ::= < AVP Header: 434 >` `{Redirect-Address-Type}` `{Redirect-Server-Address}` ***Note:*** *The MCC does not support Redirect-Address-Type SIP URI.*

Redirect-Server-Address

AVP Name	Redirect-Server-Address
AVP Code	435
AVP Data Type	UTF8String
Description	Defines the address of the redirect server.

Requested-Action

AVP Name	Requested-Action
AVP Code	436
AVP Data Type	Enumerated
Description	Contains the requested action sent in the CCR where the CC-Request-Type is set to EVENT_REQUEST. The following values are defined: ■ DIRECT_DEBITING (0) – Indicates a request to decrease the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the debited units. ■ REFUND_ACCOUNT (1) – Indicates a request to increase the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the refunded units. ■ CHECK_BALANCE (2) – Indicates a balance check request. In this case, the checking of the account balance is done without any credit reservation from the account. The Check-Balance-Result AVP in the CCA contains the result of the balance check. ■ PRICE_ENQUIRY (3) – Indicates a price inquiry request. In this case, neither checking of the account balance nor reservation from the account is done. Only the price of the service is returned in the Cost-Information AVP in the CCA.

Requested-Service-Unit

AVP Name	Requested-Service-Unit
AVP Code	437
AVP Data Type	Grouped
Description	The amount of requested service units for a particular category, including monetary or non-monetary units.

Result-Code

AVP Name	Result-Code
AVP Code	268
AVP Data Type	Unsigned32
Description	A specified request is successfully fulfilled or fails.

Route-Record

AVP Name	Route-Record
AVP Code	282
AVP Data Type	DiameterIdentity
Description	Contains an identifier inserted by a relay or proxy node to identify the node from which the message is received.

Service-Context-Id

AVP Name	Service-Context-Id
AVP Code	461
AVP Data Type	UTF8String
Description	Indicates the unique ID (may be extended) of a Diameter credit-control (DCC) service-specific document that applies to the request. Use the following CLI parameter to specify the Service-Context-Id AVP sent in ACR messages via the Gz/Rf offline-charging interface to the OFCS. The default service-context-id is `9.32251@3gpp.org`. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface offline-charging`** ***`<name>`*** **`service-context-id`** ***`<service-context-id>`***

Service-Identifier

AVP Name	Service-Identifier
AVP Code	439
AVP Data Type	Unsigned32
Description	Contains the identifier of a service. The specific service the request relates to is uniquely identified by the combination of the Service-Context-Id and Service-Identifier AVPs. Note the following: ■ For dynamic PCC rules, the Service-Identifier AVP sent from the PCRF as part of the Charging-Rule-Definition AVP does not need to be unique. Multiple Charging-Rule-Definition AVPs can have the same Service-Identifier AVP value. ■ For predefined PCC rules, if the MCC gateway receives a CCA message from the PCRF with the Service-Identifier AVP under the Charging-Rule-Definition AVP, the value in the Service-Identifier AVP overrides the Service-ID (`service-construct` –> `service-rule` –> `service-data-flow-id`) value provisioned on the gateway assuming the Charging-Rule-Name (`service-rule`) is valid. ■ If the `reporting-level` of a Rating Group is set to `service-rule`, the PGW sends the Service-Identifier AVP and Rating Group AVP (contained within the MSCC AVP) in CCR messages to the OCS. The Service-Identifier AVP only reports the Service-ID (`service-data-flow-id`) of the first service rule in the Rating Group. If the service rule list contains different Service-IDs, the MSCC AVP does not report all the Service-IDs.

Session-ID

AVP Name	Session-ID
AVP Code	263
AVP Data Type	UTF8String

Description

Identifies a session-ID that uniquely identifies a DCCA session. The default format is as follows:

- <DiameterIdentity>;<high 32 bits>;<low 32 bits>;<optional value>.
- <DiameterIdentity>: the same as filled in the Origin-Host AVP.
- <high 32 bits>: contains the current time of the system.
- <low 32 bits>: contains service instance cookie.
- <optional value>: contains subscriber and bearer information.

For example:

example.com;1450803134;2148532240;8;1450812272;262229

where

- *example.com*: (Diameter identity) represents the Origin-host.
- *1450803134*: (high 32 bits) represents the interface timestamp.
- *2148532240*: (low 32 bits) represents the subscriber ID.
- *8*: (optional value) represents the static Diameter session counter.
- *1450812272*: (optional value) represents the session setup time in decimal format.
- *262229*: (optional value) represents the gateway service ID (formed using Geo-ID, Block-ID, Node-ID, Bearer-ID).

You can configure the format for the Session-ID AVP sent on the Gz/Rf interface using the following command.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface offline-charging <interface-name>
session-id-format [v0|v1]
```

- **v0** – (default) The Session-ID AVP format represents the optional value for the session setup time in decimal format. For example:

  *example.com;1450803134;2148532240;8;***1450812272***;262229*

  where ***1450812272*** represents the session setup time in decimal format.
- **v1** – The Session-ID AVP format represents the optional value for the session setup time in hex format. This format applies to the Gz/Rf interface only. For example:

  01-*example.com;1450803134;2148532240;8;***5679A370-***262229*

where ***01*** is a prefix indicating that this Session-ID format is using version 1 (v1) and ***5679A370*** represents the session setup time in hex format. Note that the delimiter after the timestamp uses a dash (-) instead of a semi-colon (;).

Subscription-Id

AVP Name	Subscription-Id
AVP Code	443
AVP Data Type	Grouped
Description	The information used to identify the end user of the service subscription party. The detailed syntax is as follows: `*Subscription-Id ::= <AVP Header: 443>` `{Subscription-Id-Type}` `{Subscription-Id-Data}` The diameter proxy or agent must route based on this field. In the case of more than one Subscription-Id, the system handles only the first Subscription-Id. **Gz interface** Operators can suppress the Subscription-Id AVP in ACR messages based on the Subscription-Id-Type (IMSI/MSISDN) using ACR data record templates. In the ACR/CCR data record templates, the Subscription-Id options appear as `subscriptionidimsi` and `subscriptionidmsisdn`. **Gx interface** Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/SIP URI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ `subscriptionid` ■ `subscriptionidimsi` ■ `subscriptionidmsisdn` ■ `subscriptionidsipuri` ***Note:*** *The GiGW sends a pseudo IMSI in the Subscription-ID AVP, which needs to be suppressed in PCRF CCR Start data record templates to use the Subscription-Id AVP with type SIP URI. In PCRF CCR Start data record templates, set the subscriptionidimsi* `presentFlag` *to* `no`. **Gy Interface** Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/NAI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ `subscriptionidimsi` ■ `subscriptionidmsisdn` ■ `subscriptionidnai` ***Note:*** *For trusted non-3GPP Wi-Fi access using standard RADIUS/EAP methods for authentication, the end user's Network Access Identifier (NAI), received in the User-Name AVP of the RADIUS Access-Request, will be used in the Subscription-Id AVP in CCR messages to the OCS. In addition to the NAI, if the IMSI/MSISDN are available as part of the end user's identity, the MCC TWAG will encode multiple Subscription-Id AVPs in CCR messages on the Gy interface.*

Subscription-Id-Data

AVP Name	Subscription-Id-Data
AVP Code	444
AVP Data Type	UTF8String
Description	Used to identify the end user. This AVP indicates the number of the charged party when the Subscription-Id-Type is 0. Supports the following format for the mobile number: Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) – Indicates the IMSI number of the charged party when the Subscription-Id-Type is 1.

Subscription-Id-Type

AVP Name	Subscription-Id-Type
AVP Code	450
AVP Data Type	Enumerated
Description	Indicates the type of user's terminal ID. The following values are defined: ■ END_USER_E164 (0) – The identifier is in international E.164 format defined according to the ITU-T E.164 number scheme. For example, MSISDN. ■ END_USER_IMSI (1) – The identifier is in international IMSI format defined according to the ITU-T E.212 number scheme. ■ END_USER_SIP_URI (2) – The identifier is in the form of a SIP URI, as defined in [SIP]. ■ END_USER_NAI (3) – The identifier is in the form of a Network Access Identifier, as defined in RFC 4282 and formatted as defined in 3GPP TS 23.003, clause 19.3.2. Used for Wi-Fi access. Use the following CLI parameter to configure the IMSI format to be sent in the Subscription-Id AVP for non-3gpp access. The default is `imsiformat`. *Applicable to Gx and Gy interfaces.* `an3000-mcm-slot7cpu1(config)#` **`service-construct interface [online-charging|pcrf-charging] <interface-name> subscriptionid-imsi-format-non-3gpp-access [imsiformat|imsinaiprefix0format]`**

Supported-Vendor-Id

AVP Name	Supported-Vendor-Id
AVP Code	265
AVP Data Type	Unsigned32
Description	Is used in the CER and CEA messages to inform the peer that the sender supports the vendor-specific AVPs defined by the vendor identified in this AVP.

Tariff-Change-Usage

AVP Name	Tariff-Change-Usage
AVP Code	452
AVP Data Type	Enumerated
Description	Identifies the reporting period for the used service unit. That is, before, after, or during tariff change. Omission of this AVP means that no tariff change has occurred. The following values are defined: ■ UNIT_BEFORE_TARIFF_CHANGE (0) ■ UNIT_AFTER_TARIFF_CHANGE (1) ■ UNIT_INDETERMINATE (2)

Tariff-Time-Change

AVP Name	Tariff-Time-Change
AVP Code	451
AVP Data Type	Time
Description	Is used when the OCS expects a tariff change to occur within the validity time of the quota it is granting.

Termination-Cause

AVP Name	Termination-Cause
AVP Code	295
AVP Data Type	Enumerated
Description	Shows the reason why the session of the diameter client terminated. The following values are defined: ■ DIAMETER_LOGOUT (1) – Interruption initiated by users. ■ DIAMETER_SERVICE_NOT_PROVIDED (2) – Session breaks before users receive authentication response messages. ■ DIAMETER_BAD_ANSWER (3) – The authentication answer received by the diameter client is not successfully handled. ■ DIAMETER_ADMINISTRATIVE (4) – Users are not granted the access authority or the session is disconnected due to operation management. For example, receiving an Abort-Session-Request. ■ DIAMETER_LINK_BROKEN (5) – Communication with users is suddenly disconnected. ■ DIAMETER_AUTH_EXPIRED (6) – User access ends because the authorized session time is reached. ■ DIAMETER_USER_MOVED (7) – Users are receiving the services from other diameter clients. ■ DIAMETER_SESSION_TIMEOUT (8) – User session is timed out and the service is stopped.

User-Equipment-Info

AVP Name	User-Equipment-Info
AVP Code	458
AVP Data Type	Grouped
Description	Indicates the identity and capability of the terminal that the subscriber is using for the connection to the network. The detailed syntax is as follows: `User-Equipment-Info ::= < AVP Header: 458 >` `{User-Equipment-Info-Type}` `{User-Equipment-Info-Value}` ***Note:*** *For Wi-Fi access, the User-Equipment-Info AVP contains the MAC address of the device if received.*

User-Equipment-Info-Type

AVP Name	User-Equipment-Info-Type
AVP Code	459
AVP Data Type	Enumerated
Description	Defines the type of user equipment information contained in the User-Equipment-Info-Value AVP. Contains the following values: ■ IMEISV (0) – The identifier contains the International Mobile Equipment Identifier and Software Version in the international IMEISV format according to 3GPP TS 23.003 [3GPPIMEI]. ■ MAC (1) – The 48-bit MAC address is formatted as described in [RAD802.1X]. ■ EUI64 (2) – The 64-bit identifier used to identify hardware instance of the product, as defined in [EUI64].

User-Equipment-Info-Value

AVP Name	User-Equipment-Info-Value
AVP Code	460
AVP Data Type	OctetString
Description	Defines which type of identifier is used. ***Note:*** *When the User-Equipment-Info-Type is set to IMEISV(0), the value within the User-Equipment-Info-Value is a UTF-8 encoded decimal.*

User-Name

AVP Name	User-Name
AVP Code	1
AVP Data Type	UTF8String
Description	The user provides the username (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) as provided in the Access-Request message.

Used-Service-Unit

AVP Name	Used-Service-Unit
AVP Code	446
AVP Data Type	Grouped
Description	The amount of used non-monetary service units measured for a rating group. ***Note:*** *The "M" bit flag for this AVP is set to 0 when present in the Usage-Monitoring-Information AVP.* The detailed syntax is as follows: `Used-Service-Unit ::= < AVP Header: 446 >` `[CC-Time]` `[CC-Total-Octets]` `[CC-Input-Octets]` `[CC-Output-Octets]` `*[AVP]`

Validity-Time

AVP Name	Validity-Time
AVP Code	448
AVP Data Type	Unsigned32
Description	Sent from the credit-control server to the credit-control client. This AVP contains the validity time of the granted service units. The measurement of the Validity-Time is started upon receipt of the CCA message containing this AVP. ***Note:*** *The MCC starts the Validity-Time only when the Validity-time AVP is received in the CCA message with the current GSU and does not use Validity-Time (if any) from a previous grant.*

Vendor-Id

AVP Name	Vendor-Id
AVP Code	266
AVP Data Type	Unsigned32
Description	Contains the vendor ID of the diameter application.

Vendor-Specific-Application-Id

AVP Name	Vendor-Specific-Application-Id
AVP Code	259
AVP Data Type	Group
Description	Used to advertise support of a vendor-specific diameter application.

3GPP AVP Definitions

This section describes the 3GPP AVPs sent by the system in CCR, CCA, RAR, ACR, and/or ACA messages. The AVPs include a Vendor-ID of 10415 (3GPP).

Table A-1 3GPP AVP Definitions

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Table A-1 3GPP AVP Definitions

3GPP-Charging-Characteristics

AVP Name	3GPP-Charging-Characteristics
AVP Code	13
AVP Data Type	UTF8String
Description	The charging characteristics for this PDP context received in the Create PDP Context Request message (only available in R99 and later releases).

3GPP-Charging-Id

AVP Name	3GPP-Charging-Id
AVP Code	2
AVP Data Type	OctetString
Description	The Charging ID for this PDP Context. The Charging ID together with the GGSN-Address constitutes a unique identifier for the PDP context.

3GPP-GGSN-Address

AVP Name	3GPP-GGSN-Address
AVP Code	7
AVP Data Type	OctetString
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

3GPP-GGSN-IPv6-Address

AVP Name	3GPP-GGSN-IPv6-Address
AVP Code	16
AVP Data Type	OctetString
Description	The GGSN IPv6 address that is used by the GTP control plane for the context.

3GPP-GGSN-MCC-MNC

AVP Name	3GPP-GGSN-MCC-MNC
AVP Code	9
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) of the network to which the GGSN/PGW belongs. When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the `home-plmnid-list`, the configured PLMN ID is sent in the 3GPP-GGSN-MCC-MNC AVP on all external interfaces (RADIUS, Gx, Gy, Gz/Rf). If an entry is not found in the list, the first entry in the `home-plmnid-list` is used in the 3GPP-GGSN-MCC-MNC AVP.

3GPP-GPRS-Negotiated-QoS-Profile

AVP Name	3GPP-GPRS-Negotiated-QoS-Profile
AVP Code	5
AVP Data Type	UTF8String
Description	This AVP contains the QoS profile.

3GPP-IMEISV

AVP Name	3GPP-IMEISV
AVP Code	20
AVP Data Type	UTF8String
Description	The user International Mobile Equipment Identity of the equipment served (IMEISV).

3GPP-IMSI

AVP Name	3GPP-IMSI
AVP Code	1
AVP Data Type	UTF8String
Description	The International Mobile Subscriber Identity (IMSI) for this user.

3GPP-IMSI-MCC-MNC

AVP Name	3GPP-IMSI-MCC-MNC
AVP Code	8
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).

3GPP-MS-TimeZone

AVP Name	3GPP-MS-TimeZone
AVP Code	23
AVP Data Type	UTF8String
Description	The Time Zone field and the Daylight Saving Time fields are used to indicate the offset between universal time and local time in steps of 15 minutes of where the MS currently resides.

3GPP-NSAPI

AVP Name	3GPP-NSAPI
AVP Code	10
AVP Data Type	UTF8String
Description	Identifies a particular PDP context for the associated PDN and Mobile Subscriber Integrated Services Digital Network-Number (MSISDN)/IMSI from creation to deletion.

3GPP-PDP-Type

AVP Name	3GPP-PDP-Type
AVP Code	3
AVP Data Type	Enumerated
Description	Type of PDP context. For example, IP or PPP. The PDP type may have the following values: ■ 0 = IPv4 ■ 1 = PPP ■ 2 = IPv6 ■ 3 = Ipv4v6 ■ 4 = Non-IP

3GPP-RAT-Type

AVP Name	3GPP-RAT-Type
AVP Code	21
AVP Data Type	OctetString
Description	The Radio Access Technology currently serving the User Equipment (UE).

3GPP-Selection-Mode

AVP Name	3GPP-Selection-Mode
AVP Code	12
AVP Data Type	UTF8String
Description	The selection mode for this PDP context received in the create PDP context request message.

3GPP-Session-Stop-Indicator

AVP Name	3GPP-Session-Stop-Indicator
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates to the AAA server that the last PDP context of a session is released and the IP-CAN session has been terminated.

3GPP-SGSN-Address

AVP Name	3GPP-SGSN-Address
AVP Code	6
AVP Data Type	OctetString
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

3GPP-SGSN-IPv6-Address

AVP Name	3GPP-SGSN-IPv6-Address
AVP Code	15
AVP Data Type	OctetString
Description	Indicates the SGSN IPv6 address used by the GPRS Tunneling Protocol (GTP) control plane for the handling of control messages. It may be used to identify the PLMN to which the user is attached.

3GPP-SGSN-MCC-MNC

AVP Name	3GPP-SGSN-MCC-MNC
AVP Code	18
AVP Data Type	UTF8String
Description	Indicates the information about the international roam-in area of the charged party. On the Gx interface, the 3GPP-SGSN-MCC-MNC AVP can be used in the CCA message to override the PLMN-ID sent in GTP-C messages. See “Overriding the PLMN-ID sent in GTP-C messages” in the *Affirmed Networks Gx Interface Specification* for more information.

3GPP-User-Location-Info

AVP Name	3GPP-User-Location-Info
AVP Code	22
AVP Data Type	OctetString
Description	Indicates details of where the user equipment is currently located (for example, SAI or CGI).

Access-Network-Charging-Address

AVP Name	Access-Network-Charging-Address
AVP Code	501
AVP Data Type	Address
Description	Indicates the IP Address of the network entity within the access network performing charging (for example, the GGSN IP address).

Access-Network-Charging-Identifier-Gx

AVP Name	Access-Network-Charging-Identifier-Gx
AVP Code	1022
AVP Data Type	Grouped
Description	Contains a charging identifier within the Access-Network-Charging-Identifier-Value AVP and the related PCC rule name(s) within the Charging-Rule-Name AVP(s) and/or within the Charging-Rule-Base-Name AVP(s). If the charging identifier applies to the entire IP CAN session, neither the Charging-Rule-Name AVP or Charging-Rule-Base-Name AVP needs to be provided. The detailed syntax is as follows: `Access-Network-Charging-Identifier-Gx ::= < AVP Header: 1022 >` `{Access-Network-Charging-Identifier-Value}` `*[Charging-Rule-Base-Name]` `*[Charging-Rule-Name]`

Access-Network-Charging-Identifier-Value

AVP Name	Access-Network-Charging-Identifier-Value
AVP Code	503
AVP Data Type	OctetString
Description	Contains a charging identifier.

ADC-Rule-Base-Name

AVP Name	ADC-Rule-Base-Name
AVP Code	1095
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of Application Detection and Control (ADC) rules. See "Application Detection and Control" in the *Affirmed Networks Gx Interface Specification* for more information.

ADC-Rule-Definition

AVP Name	ADC-Rule-Definition
AVP Code	1094
AVP Data Type	Grouped
Description	Defines the ADC rule sent by the PCRF. See "Application Detection and Control" in the *Affirmed Networks Gx Interface Specification* for more information. ***Note:*** *The Redirect-Information AVP sent from the PCRF as part of the ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.* The detailed syntax is as follows: `ADC-Rule-Definition ::= < AVP Header: 1094 >` `{ADC-Rule-Name}` `[TDF-Application-Identifier]` `*[Flow-Information]` `[Service-Identifier]` `[Rating-Group]` `[Reporting-Level]` `[Online]` `[Offline]` `[Metering-Method]` `[Flow-Status]` `[QoS-Information]` `[Monitoring-Key]` `[Redirect-Information]` `[Mute-Notification]` `*[AVP]`

ADC-Rule-Install

AVP Name	ADC-Rule-Install
AVP Code	1092
AVP Data Type	Grouped
Description	Used to activate, install or modify Application Detection and Control (ADC) rules as instructed from the PCRF. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information. The detailed syntax is as follows: `ADC-Rule-Install ::= < AVP Header: 1092 >` `*[ADC-Rule-Definition]` `*[ADC-Rule-Name]` `*[ADC-Rule-Base-Name]` `[Monitoring-Flags]` `[Rule-Activation-Time]` `[Rule-Deactivation-Time]` `*[AVP]`

ADC-Rule-Name

AVP Name	ADC-Rule-Name
AVP Code	1096
AVP Data Type	OctetString
Description	Defines a name for an ADC rule. For ADC rules provided by the PCRF, it uniquely identifies an ADC rule within one session. For predefined ADC rules, it uniquely identifies an ADC rule within the PCEF. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information.

ADC-Rule-Remove

AVP Name	ADC-Rule-Remove
AVP Code	1093
AVP Data Type	Grouped
Description	Used to deactivate or remove ADC rules as instructed from the PCRF. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information. The detailed syntax is as follows: `ADC-Rule-Remove ::= < AVP Header: 1093 >` `*[ADC-Rule-Name]` `*[ADC-Rule-Base-Name]` `*[AVP]`

ADC-Rule-Report

AVP Name	ADC-Rule-Report
AVP Code	1097
AVP Data Type	Grouped
Description	Used to report the status of the ADC rules which cannot be installed/activated or enforced. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information. The detailed syntax is as follows: `ADC-Rule-Report ::= < AVP Header: 1097 >` `*[ADC-Rule-Name]` `*[ADC-Rule-Base-Name]` `[PCC-Rule-Status]` `[Rule-Failure-Code]` `[Final-Unit-Indication]` `*[AVP]`

Additional-Exception-Reports

AVP Name	Additional-Exception-Reports
AVP Code	3936
AVP Data Type	Enumerated
Description	Indicates whether additional exception reports are allowed when maximum rate is reached. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The following values are defined: ■ NOT_ALLOWED (0)

AF-Charging-Identifier

AVP Name	AF-Charging-Identifier
AVP Code	505
AVP Data Type	OctetString
Description	Contains the AF Charging Identifier sent by the Application Function (AF). This information may be used for charging correlation with the bearer layer.

AF-Correlation-Information

AVP Name	AF-Correlation-Information
AVP Code	1276
AVP Data Type	Grouped
Description	Contains the AF charging identifier (ICID for IMS) and associated flow identifiers generated by the AF and received by PGW over the Gx interface as defined in TS 29.214 and TS 29.212. The AF-Correlation-Information is defined per Rating Group or per Rating Group and Service Identifier, when Service Identifier level reporting applies. When several AF sessions are conveyed over the same bearer, this AVP may appear several times per MSCC instance. The detailed syntax is as follows: `AF-Correlation-Information:: = < AVP Header: 1276 >` `{AF-Charging-Identifier}` `*[Flows]`

Allocation-Retention-Priority

AVP Name	Allocation-Retention-Priority
AVP Code	1034
AVP Data Type	Grouped
Description	Indicates the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the SDF if provided within the QoS-Information AVP or for the EPS default bearer if provided within the Default-EPS-Bearer-QoS AVP. The detailed syntax is as follows: `Allocation-Retention-Priority ::= < AVP Header: 1034 >` `{Priority-Level}` `[Pre-emption-Capability]` `[Pre-emption-Vulnerability]`

AN-GW-Address

AVP Name	AN-GW-Address
AVP Code	1050
AVP Data Type	Address
Description	Contains the IPv4 and/or IPv6 (if available) address(es) of the SGW for 3GPP networks and access node gateway (AGW) for non-3GPP networks. For 2G/3G based sessions with an IP-CAN-Type of 3GPP-EPS, the MCC sends the IP address of the SGSN in the AN-GW-Address AVP in the CCR-Initial message to the PCRF. If both IPv4 and IPv6 addresses are provided, the MCC sends two instances of the AN-GW-Address AVP.

AN-Trusted

AVP Name	AN-Trusted
AVP Code	1503
AVP Data Type	Enumerated
Description	Indicates whether the access network is trusted or untrusted for the Non-3GPP access network. For example, if there is an LTE-to-Wi-Fi handover and if the PCRF has requested to be notified of an IP-CAN_CHANGE, the PGW includes the AN-Trusted AVP in the CCR-U along with the IP-CAN_Change event-trigger to indicate whether the access network is trusted or untrusted. The following values are defined: ■ TRUSTED (0) – This value is used when the non-3GPP access network is to be handled as trusted. ■ UNTRUSTED (1) – This value is used when the non-3GPP access network is to be handled as untrusted.

Application-Detection-Information

AVP Name	Application-Detection-Information
AVP Code	1098
AVP Data Type	Grouped
Description	Used to report application Start/Stop notifications when application traffic, defined by the TDF-Application-Identifier, has been detected and when the PCRF has subscribed for APPLICATION_START/APPLICATION_STOP Event-Triggers. See "Application Detection and Control" in the *Affirmed Networks Gx Interface Specification* for more information. The detailed syntax is as follows: `Application-Detection-Information:: = <AVP Header: 1098>` `{TDF-Application-Identifier}` `[TDF-Application-Instance-Identifier]` `*[Flow-Information]` `*[AVP]`

Application-Service-Provider-Identity

AVP Name	Application-Service-Provider-Identity
AVP Code	532
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the application service provider.

APN-Aggregate-Max-Bitrate-DL

AVP Name	APN-Aggregate-Max-Bitrate-DL
AVP Code	1040
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per second for the downlink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Aggregate-Max-Bitrate-UL

AVP Name	APN-Aggregate-Max-Bitrate-UL
AVP Code	1041
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Rate-Control

AVP Name	APN-Rate-Control
AVP Code	3933
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink and downlink direction for this APN. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The detailed syntax is as follows: `APN-Rate-Control :: = < AVP Header: 3933 >` `[APN-Rate-Control-Uplink]` `[APN-Rate-Control-Downlink]`

APN-Rate-Control-Downlink

AVP Name	APN-Rate-Control-Downlink
AVP Code	3934
AVP Data Type	Grouped
Description	Indicates the APN-Rate Control parameters applicable to data PDU transfers in the downlink direction for this APN. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The detailed syntax is as follows: `APN-Rate-Control-Downlink :: = < AVP Header: 3934 >` `[Rate-Control-Time-Unit]` `[Rate-Control-Max-Rate]` `[Rate-Control-Max-Message-Size]`

APN-Rate-Control-Uplink

AVP Name	APN-Rate-Control-Uplink
AVP Code	3935
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink direction for this APN. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The detailed syntax is as follows: `APN-Rate-Control-Uplink :: = < AVP Header: 3935>` `[Additional-Exception-Reports]` `[Rate-Control-Time-Unit]` `[Rate-Control-Max-Rate]`

Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	1265
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval for controlling the consumption of time quota, in seconds.

Bearer-Control-Mode

AVP Name	Bearer-Control-Mode
AVP Code	1023
AVP Data Type	Enumerated
Description	Indicates the PCRF selected bearer control mode. The following values are defined: ■ UE_ONLY (0) – Indicates that the UE may request any additional PDP Context establishment. ■ UE_NW (2) – Indicates that both the UE and PCEF may request any additional PDP Context establishment and add own traffic mapping information to a PDP Context.

Bearer-Identifier

AVP Name	Bearer-Identifier
AVP Code	1020
AVP Data Type	OctetString
Description	Indicates the bearer to which specific information refers. When present within the CCR command, subsequent AVPs within the CCR refer to the specific bearer identified by this AVP. The bearer identifier of an IP CAN bearer is unique within the corresponding IP CAN session and is selected by the PCEF.

Bearer-Operation

AVP Name	Bearer-Operation
AVP Code	1021
AVP Data Type	Enumerated
Description	Indicates the bearer event that causes a request for PCC rules. This AVP is supplied if the bearer event relates to an IP CAN bearer initiated by the UE. This AVP is only sent when the IP-CAN-Type AVP is 3GPP-GPRS (2G/3G). The following values are defined: ■ TERMINATION (0) – Indicates that a bearer is being terminated. ■ ESTABLISHMENT (1) – Indicates that a new bearer is being established. ■ MODIFICATION (2) – Indicates that an existing bearer is being modified.

Bearer-Usage

AVP Name	Bearer-Usage
AVP Code	1000
AVP Data Type	Enumerated
Description	Indicates how the bearer is being used. For example, whether it is used as a dedicated IMS signaling context. The following values are defined: ■ GENERAL (0) – Indicates that no specific bearer usage information is available. No TFT filter is associated with the PDP context. ■ IMS_SIGNALLING (1) – Indicates that the bearer is used for IMS signaling only. ■ DEDICATED (2) – Indicates that the bearer is being used for a dedicated purpose and accordingly has an associated TFT filter.

CG-Address

AVP Name	CG-Address
AVP Code	846
AVP Data Type	OctetString
Description	Contains the IP address of the charging gateway.

Change-Condition

AVP Name	Change-Condition
AVP Code	2037
AVP Data Type	Integer32
Description	Indicates the change in charging condition. The following values are defined: ■ 0 = Normal Release – Used to indicate an IP-CAN session termination, an IP-CAN bearer release or a service data flow termination. ■ 1 = Abnormal Release ■ 2 = QoS Change ■ 3 = Volume Limit ■ 4 = Time Limit ■ 8 = RAT Change ■ 14 = CGI-SAI Change ■ 15 = RAI Change ■ 16 = ECGI Change ■ 17 = TAI Change ■ 36 = Change in UP to UE ***Note:*** *If a single accounting record contains multiple changes (for example, an SGSN change with CGI change, RAI change, QoS change), the MCC sends multiple Change-Condition AVPs containing all of the change conditions.*

Change-Time

AVP Name	Change-Time
AVP Code	2038
AVP Data Type	Time
Description	Contains the time in UTC format when the volume counts, associated with the IP-CAN bearer or the service data container, are closed and reported due to a change in charging conditions.

Charging-Characteristics-Selection-Mode

AVP Name	Charging-Characteristics-Selection-Mode
AVP Code	2066
AVP Data Type	Enumerated
Description	Indicates how the applied Charging-Characteristics was selected. The following values are defined: ■ 0 = Serving-Node-Supplied ■ 1 = Subscription-specific ■ 2 = APN-specific ■ 3 = Home-Default ■ 4 = Roaming-Default ■ 5 = Visiting-Default

Charging-Information

AVP Name	Charging-Information
AVP Code	618
AVP Data Type	Grouped
Description	Contains the addresses of the charging functions. ***Note:*** *You can configure the PGW to extract information from the Charging-Information AVP received from the PCRF and use this information during Gy session setup. See "Using Gx Charging-Information AVP for Gy Session Setup" in the Affirmed Networks Gy Interface specification.* The detailed syntax is as follows: `Charging-Information :: = <AVP Header : 618 10415>` `[Primary-Event-Charging-Function-Name]` `[Secondary-Event-Charging-Function-Name]` `*[AVP]`

Charging-Rule-Base-Name

AVP Name	Charging-Rule-Base-Name
AVP Code	1004
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of PCC rules residing at the PCEF. The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured data-profile on the MCC gateway (`workflow` –> `data-profile`). For information on the MCC implementation of this AVP, see “PCC Rules” in the *Affirmed Networks Gx Interface Specification*.

Charging-Rule-Definition

AVP Name	Charging-Rule-Definition
AVP Code	1003
AVP Data Type	Grouped
Description	Defines the dynamic PCC rule for a service data flow sent by the PCRF to the PCEF or used to activate a predefined PCC rule that is pre-configured at the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the *Affirmed Networks Gx Interface Specification*. ***Note:*** *The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.* The detailed syntax is as follows: `Charging-Rule-Definition ::= < AVP Header: 1003 >` `{Charging-Rule-Name}` `[Service-Identifier]` `[Rating-Group]` `*[Flow-Information]` `[Flow-Status]` `[QoS-Information]` `[PS-to-CS-Session-Continuity]` `[Reporting-Level]` `[Online]` `[Offline]` `[Metering-Method]` `[Precedence]` `[Monitoring-Key]` `[Redirect-Information]` `[Mute-Notification]` `[Sponsor-Identity]` `[Application-Service-Provider-Identity]` `[Required-Access-Info]` `*[AVP]`

Charging-Rule-Install

AVP Name	Charging-Rule-Install
AVP Code	1001
AVP Data Type	Grouped
Description	Used to activate, install, or modify PCC rules as instructed from the PCRF to the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the *Affirmed Networks Gx Interface Specification*.

Charging-Rule-Name

AVP Name	Charging-Rule-Name
AVP Code	1005
AVP Data Type	OctetString
Description	For PCC rules provided by the PCRF, it uniquely identifies a PCC rule within one IP CAN session. For PCC rules predefined at the PCEF, it uniquely identifies a PCC rule within the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the *Affirmed Networks Gx Interface Specification*.

Charging-Rule-Remove

AVP Name	Charging-Rule-Remove
AVP Code	1002
AVP Data Type	Grouped
Description	Used to deactivate or remove PCC rules from an IP CAN session. For information on the MCC implementation of this AVP, see “PCC Rules” in the *Affirmed Networks Gx Interface Specification*. The detailed syntax is as follows: `Charging-Rule-Remove ::= < AVP Header: 1002 >` `*[Charging-Rule-Name]` `*[Charging-Rule-Base-Name]`

Charging-Rule-Report

AVP Name	Charging-Rule-Report
AVP Code	1018
AVP Data Type	Grouped
Description	Reports the status of the PCC rules that have been successfully installed, modified or removed or that cannot be installed/activated or enforced at the PCEF. The Rule-Failure-Code indicates the reason that the PCC rules cannot be successfully installed/activated or enforced. The detailed syntax is as follows: `Charging-Rule-Report ::= < AVP Header: 1018 >` `*[Charging-Rule-Name]` `*[Charging-Rule-Base-Name]` `[PCC-Rule-Status]` `[Rule-Failure-Code]` `[Final-Unit-Indication]` `*[RAN-NAS-Release-Cause]`

CP-CIoT-EPS-Optimisation-Indicator

AVP Name	CP-CIoT-EPS-Optimisation-Indicator
AVP Code	3930
AVP Data Type	Enumerated
Description	Indicates whether Control Plane CIoT EPS optimisation is used by this PDN connection for data transfer with the UE. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The following values are defined: ■ NOT_APPLY (0) ■ APPLY (1)

CSG-Access-Mode

AVP Name	CSG-Access-Mode
AVP Code	2317
AVP Data Type	Enumerated
Description	Contains the CSG cell access mode. The following values are defined: ■ CLOSED MODE (0) ■ HYBRID MODE (1)

CSG-Id

AVP Name	CSG-Id
AVP Code	1437
AVP Data Type	Unsigned32
Description	Contains the CSG identification information.

CSG-Information-Reporting

AVP Name	CSG-Information-Reporting
AVP Code	1071
AVP Data Type	Enumerated
Description	Sent from the PCRF to the PCEF to request the PCEF to report the user CSG information change to the OCS. The following values are defined: ■ CHANGE_CSG_CELL (0) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a CSG cell. ■ CHANGE_CSG_SUBSCRIBED_HYBRID_CELL (1) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is a CSG member. ■ CHANGE_CSG_UNSUBSCRIBED_HYBRID_CELL (2) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is not a CSG member.

CSG-Membership-Indication

AVP Name	CSG-Membership-Indication
AVP Code	2318
AVP Data Type	Enumerated
Description	Indicates whether the UE is a member of the accessing CSG cell, if the access mode is Hybrid. The following values are defined: ■ NOT CSG MEMBER (0) ■ CSG MEMBER (1)

Default-EPS-Bearer-QoS

AVP Name	Default-EPS-Bearer-QoS
AVP Code	1049
AVP Data Type	Grouped
Description	Defines the QoS information for the EPS default bearer. When this AVP is sent from the PCEF to the PCRF, it indicates the subscribed QoS for the default EPS bearer. When this AVP is sent from the PCRF to the PCEF, it indicates the authorized QoS for the default EPS bearer.

Diagnostics

AVP Name	Diagnostics
AVP Code	2039
AVP Data Type	Integer32
Description	Provides more information on the reason for sending the ACR Stop message and may contain one of the following values: ■ UNSPECIFIED (0) ■ SESSION_TIMEOUT (1) ■ RESOURCE_LIMITATION (2) ■ ADMIN_DISCONNECT (3) ■ IDLE_TIMEOUT (4) ■ PCRF_UNREACHABLE (5) ■ AAA_UNREACHABLE (6) ■ AAA_INITIATED_SESSION_TERMINATION (7) ■ PCRF_INITIATED_SESSION_TERMINATION (9) ■ PCRF_INITIATED_FLOW_TERMINATION (10) ■ PCRF_ACCOUNTING_PARAMETERS_CHANGED (11) ■ GTP_INITIATED_SESSION_TERMINATION (14) ■ HANDOVER_ERROR (16) ■ REAUTHORICATION_FAILED (18)

Downlink-Rate-Limit

AVP Name	Downlink-Rate-Limit
AVP Code	4312
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for downlink. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

Dynamic-Address-Flag

AVP Name	Dynamic-Address-Flag
AVP Code	2051
AVP Data Type	Enumerated
Description	Indicates whether the PDP context/PDN address is statically or dynamically allocated. If this AVP is not present, the address is statically allocated. The following values are defined: ■ STATIC (0) ■ DYNAMIC (1)

Envelope

AVP Name	Envelope
AVP Code	1266
AVP Data Type	Grouped
Description	Reports the start and end time of a one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299. The detailed syntax is as follows: `Envelope :: = < AVP Header: 1266>` `{Envelope-Start-Time}` `[Envelope-End-Time]` `[CC-Total-Octets]` `[CC-Input-Octets]` `[CC-Output-Octets]` `[CC-Service-Specific-Units]` `*[AVP]`

Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	1267
AVP Data Type	Time
Description	This AVP is set to the time of the end of the time envelope.

Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	1268
AVP Data Type	Enumerated
Description	Used in the CCA-Initial message to indicate whether the client reports the start and end of each time envelope, in cases where the quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2) ■ REPORT_ENVELOPES_WITH_EVENTS (3) ■ REPORT_ENVELOPES_WITH_VOLUME_AND_EVENTS (4)

Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	1269
AVP Data Type	Time
Description	This AVP is set to the time of the packet of user data which caused the time envelope to start.

Event-Trigger

AVP Name	Event-Trigger
AVP Code	1006
AVP Data Type	Enumerated
Description	When sent from the PCRF to the PCEF, this AVP indicates an event that causes a re-request of PCC rules. When sent from the PCEF to the PCRF, this AVP indicates that the corresponding event has occurred at the gateway (PGW). The system supports the following PCRF event-triggers: ■ SGSN_CHANGE (0) ■ QOS_CHANGE (1) ■ RAT_CHANGE (2) ■ PLMN_CHANGE (4) ■ LOSS_OF_BEARER (5) ■ IP-CAN_CHANGE (7) ■ RAI_CHANGE (12) ■ USER_LOCATION_CHANGE (13) ■ OUT_OF_CREDIT (15) ■ REALLOCATION_OF_CREDIT (16) ■ REVALIDATION_TIMEOUT (17) ■ DEFAULT_EPS_BEARER_QOS_CHANGE (20) ■ SUCCESSFUL_RESOURCE_ALLOCATION (22) ■ UE_TIME_ZONE_CHANGE (25) ■ USAGE_REPORT (26) or USAGE_REPORT (33) – MCC supports the following USAGE_REPORT trigger codes, depending on the PCRF's 3GPP TS 29.212 release compliance: – USAGE_REPORT (26) (3GPP TS 29.212 Release 10 or earlier) – USAGE_REPORT (33) (3GPP TS 29.212 Release 11 or later) Specify the PCRF's 3GPP TS 29.212 release compliance when configuring the PCRF charging interface object. `an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance [release-10|release-10-plus]`

AVP Name	Event-Trigger
Description *(Continued)*	■ TAI_CHANGE (26) ■ ECGI_CHANGE (27) The TAI_CHANGE and ECGI_CHANGE event-triggers are applicable for 3GPP 29.212 V10.0.0 or higher. To ensure the PCEF receives and reports the TAI_CHANGE event-trigger with code 26 and the ECGI_CHANGE event-trigger with code 27, set the PCRF's 3GPP TS 29.212 release compliance to `release-10-plus`. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance release-10-plus`** ■ APN-AMBR_MODIFICATION_FAILURE (29) ■ DEFAULT-EPS-BEARER-QOS_MODIFICATION_FAILURE (34) ■ APPLICATION_START (39) ■ APPLICATION_STOP (40) ■ CS_TO_PS_HANDOVER (42) ■ ACCESS_NETWORK_INFO_REPORT (45)

Extended-APN-AMBR-DL

AVP Name	Extended-APN-AMBR-DL
AVP Code	2848
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the downlink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. *Applies to the 5G NSA solution.* See "Enabling 5G NSA QoS AVPs" in the *Affirmed Networks Gx Interface Specification* for more information.

Extended-APN-AMBR-UL

AVP Name	Extended-APN-AMBR-UL
AVP Code	2849
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the uplink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. *Applies to the 5G NSA solution.* See "Enabling 5G NSA QoS AVPs" in the *Affirmed Networks Gx Interface Specification* for more information.

Extended-GBR-DL

AVP Name	Extended-GBR-DL
AVP Code	2850
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate (GBR) in Kpbs for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. *Applies to the 5G NSA solution.* See “Enabling 5G NSA QoS AVPs” in the *Affirmed Networks Gx Interface Specification* for more information.

Extended-GBR-UL

AVP Name	Extended-GBR-UL
AVP Code	2851
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate in Kpbs for an uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. *Applies to the 5G NSA solution.* See “Enabling 5G NSA QoS AVPs” in the *Affirmed Networks Gx Interface Specification* for more information.

Extended-Max-Requested-BW-DL

AVP Name	Extended-Max-Requested-BW-DL
AVP Code	554
AVP Data Type	Unsigned32
Description	Defines the maximum authorized bandwidth in Kpbs for downlink. For more details on this AVP, refer to 3GPP 29.214 Version 15.1.0. *Applies to the 5G NSA solution.* See “Enabling 5G NSA QoS AVPs” in the *Affirmed Networks Gx Interface Specification* for more information.

Extended-Max-Requested-BW-UL

AVP Name	Extended-Max-Requested-BW-UL
AVP Code	555
AVP Data Type	Unsigned32
Description	Defines the maximum authorized bandwidth in Kpbs for uplink.For more details on this AVP, refer to 3GPP 29.214 Version 15.1.0. *Applies to the 5G NSA solution.* See “Enabling 5G NSA QoS AVPs” in the *Affirmed Networks Gx Interface Specification* for more information.

Feature-List

AVP Name	Feature-List
AVP Code	630
AVP Data Type	Unsigned32
Description	Contains a list of supported features of the origin host.

Feature-List-ID

AVP Name	Feature-List-ID
AVP Code	629
AVP Data Type	Unsigned32
Description	Contains the identity of a feature list. If there are multiple feature lists defined for the Gx reference point, the Feature-List-ID AVP differentiates those lists from one another.

Fixed-User-Location-Info

AVP Name	Fixed-User-Location-Info
AVP Code	2825
AVP Data Type	Grouped
Description	Indicates the UE location in a Fixed Access Network. It is populated with the Circuit Id from the DHCP Option 82 field. *Applies to the Gxa interface only*. The detailed syntax is as follows: `Fixed-User-Location-Info::=  < AVP Header: 2825 >` `        [Logical-Access-ID]`

Flows

AVP Name	Flows
AVP Code	510
AVP Data Type	Grouped
Description	Indicates the IP flows via their flow identifiers. The detailed syntax is as follows: `Flows::= < AVP Header: 510 >` `{Media-Component-Number}` `*[Flow-Number]`

Flow-Information

AVP Name	Flow-Information
AVP Code	1058
AVP Data Type	Grouped
Description	Sent from the PCRF to the PCEF and contains the information from a single IP flow packet filter. When included in the Application-Detection-Information AVP, the Flow-Information contains the Flow-Description AVP and Flow-Direction AVP. The detailed syntax is as follows: `Flow-Information ::= < AVP Header: 1058 >` `[Flow-Description]` `[Packet-Filter-Identifier]` `[Packet-Filter-Usage]` `[ToS-Traffic-Class]` `[Security-Parameter-Index]` `[Flow-Label]` `[Flow-Direction]` `[AVP]`

Flow-Status

AVP Name	Flow-Status
AVP Code	511
AVP Data Type	Enumerated
Description	Defines whether the service data flow is enabled or disabled. For predefined PCC rules, the Flow-Status AVP overrides whatever is configured statically on the gateway. The following values are defined: ■ ENABLED (2) – Used to enable all associated IP flow(s) in both directions. ■ DISABLED (3) – Used to disable all associated IP flow(s) in both directions.

Guaranteed-Bitrate-DL

AVP Name	Guaranteed-Bitrate-DL
AVP Code	1025
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the downlink direction, in bps.

Guaranteed-Bitrate-UL

AVP Name	Guaranteed-Bitrate-UL
AVP Code	1026
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the uplink direction, in bps.

GGSN-Address

AVP Name	GGSN-Address
AVP Code	847
AVP Data Type	Address
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

IMS-Information

AVP Name	IMS-Information
AVP Code	876
AVP Data Type	Grouped
Description	Provides additional IMS service-specific information elements. The detailed syntax is as follows: `IMS-Information ::= < AVP Header: 876>` `{Node-Functionality}`

IMSI-Unauthenticated-Flag

AVP Name	IMSI-Unauthenticated-Flag
AVP Code	2308
AVP Data Type	Enumerated
Description	Indicates whether the served IMSI is authenticated. If this AVP is not present, the served IMSI is authenticated. The following values are defined: ■ AUTHENTICATED (0) ■ UNAUTHENTICATED (1)

IP-CAN-Type

AVP Name	IP-CAN-Type
AVP Code	1027
AVP Data Type	Enumerated
Description	Indicates the type of Connectivity Access Network in which the user is connected. The IP-CAN-Type AVP is always present during the IP-CAN session establishment. During an IP-CAN session modification, this AVP is present when there has been a change in the IP-CAN type and the PCRF requested to be informed of this event. The Event-Trigger AVP with value IP-CAN-CHANGE is provided together with the IP-CAN-Type AVP. The following values are defined: ■ 3GPP-GPRS (0) – Indicates that the IP-CAN is associated with a 3GPP GPRS access that is connected to the GGSN based on the Gn/Gp interfaces and is further detailed by the RAT-Type AVP. The RAT-Type AVP includes applicable 3GPP values, except EUTRAN. ■ DOCSIS (1) – Indicates that the IP-CAN is associated with a DOCSIS access. ■ xDSL (2) – Indicates that the IP-CAN is associated with an xDSL access. ■ WiMAX (3) – Indicates that the IP-CAN is associated with a WiMAX access (IEEE 802.16). ■ 3GPP2 (4) – Indicates that the IP-CAN is associated with a 3GPP2 access connected to the 3GPP2 packet core as specified in 3GPP2 X.S0011 [20] and is further detailed by the RAT-Type AVP. ■ 3GPP-EPS (5) – Indicates that the IP-CAN associated with a 3GPP EPS access and is further detailed by the RAT-Type AVP. ■ Non-3GPP-EPS (6) – Indicates that the IP-CAN associated with an EPC based non-3GPP access and is further detailed by the RAT-Type AVP.

Local-Sequence-Number

AVP Name	Local-Sequence-Number
AVP Code	2063
AVP Data Type	Unsigned32
Description	Contains the service data container sequence number, increased by one for each service data container closed.

Logical-Access-Id

AVP Name	Logical-Access-Id
AVP Code	302
AVP Data Type	OctetString
Description	Contains a Circuit ID from the DHCP Option 82 field. *Applies to the Gxa interface only*.

Low-Priority-Indicator

AVP Name	Low-Priority-Indicator
AVP Code	2602
AVP Data Type	Enumerated
Description	Indicates if the PDN connection has a low priority (for example, for Machine Type Communications). *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0*. The following values are defined: ■ NO (0) ■ YES (1)

Max-Requested-Bandwidth-DL

AVP Name	Max-Requested-Bandwidth-DL
AVP Code	515
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the downlink direction.

Max-Requested-Bandwidth-UL

AVP Name	Max-Requested-Bandwidth-UL
AVP Code	516
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the uplink direction.

Metering-Method

AVP Name	Metering-Method
AVP Code	1007
AVP Data Type	OctetString
Description	Defines which parameters are metered for offline charging. The following values are defined: ■ VOLUME (1) – The volume of the service data flow traffic is metered. If the Metering-Method AVP is omitted but has been supplied previously, the previous information remains valid. If the Metering-Method AVP is omitted and has not been supplied previously, the metering method pre-configured at the PCEF is used as the default metering method.

Monitoring-Key

AVP Name	Monitoring-Key
AVP Code	1066
AVP Data Type	Enumerated
Description	Contains the monitoring key that may apply to the PCC rule. This AVP is used for usage monitoring control purposes as an identifier to a usage monitoring control instance. For more information, see “Usage Monitoring and Reporting” in the *Affirmed Networks Gx Interface Specification*.

Mute-Notification

AVP Name	Mute-Notification
AVP Code	2809
AVP Data Type	Enumerated
Description	Used by the PCEF to mute the notification to the PCRF of the detected application's Start/Stop for the specific PCC/ADC rule. The following values are defined: MUTE_REQUIRED (0) – Indicates that the PCEF should not inform the PCRF when the application's Start/Stop for the specific PCC/ADC rule is detected.

Network-Request-Support

AVP Name	Network-Request-Support
AVP Code	1024
AVP Data Type	Enumerated
Description	Indicates whether the UE and access network supports the network requested bearer control mode. The following values are defined: ■ NETWORK_REQUEST NOT SUPPORTED (0) – Indicates that the UE and the access network do not support the network initiated bearer establishment request procedure. ■ NETWORK_REQUEST SUPPORTED (1) – Indicates that the UE and the access network support the network initiated bearer establishment request procedure.

Node-Functionality

AVP Name	Node-Functionality
AVP Code	862
AVP Data Type	Enumerated
Description	Includes the functionality identifier of the node. The functionality identifier can be one of the following: ■ S-CSCF (0) ■ P-CSCF (1) ■ I-CSCF (2) ■ MRFC (3) ■ MGCF (4) ■ BGCF (5) ■ AS (6) ■ IBCF (7) ■ S-GW (8) ■ P-GW (9) ■ HSGW (10) ■ E-CSCF (11) ■ MME (12) ■ TRF (13) ■ TF (14) ■ ATCF (15) ■ Proxy Function (16) ■ ePDG (17)

Node-Id

AVP Name	Node-Id
AVP Code	2064
AVP Data Type	UTF8String
Description	Includes an optional, operator-configurable identifier string for the node that had generated the Accounting-Request (ACR) message.

Offline

AVP Name	Offline
AVP Code	1008
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-definition AVP, this AVP indicates whether the offline charging interface from the PCEF for the associated PCC rule should be enabled. The absence of this AVP within the first provisioning of the Charging-Rule-definition AVP of a new PCC rule indicates that the pre-configured default charging method at the PCEF should be used. When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for offline pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable offline charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. See "Configuring the Default Charging Method" in the *Affirmed Networks Gx Interface Specification* for more information. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. The following values are defined: ■ DISABLE_OFFLINE (0) – Indicates that the offline charging interface for the associated PCC rule is disabled. ■ ENABLE_OFFLINE (1) – Indicates that the offline charging interface for the associated PCC rule is enabled.

Online

AVP Name	Online
AVP Code	1009
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-Definition AVP, this AVP defines whether the online charging interface from the PCEF for the associated PCC rule should be enabled. ***Note:*** *If all the dynamic PCC rules associated with the Rating Group have online charging disabled, the Rating Group is moved to a blocked state for online charging and no longer performs online charging. When the last active Rating Group is moved to a blocked state, the PGW terminates the Gy session by sending the CCR-T message to the PCRF.* When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for online pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable online charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. See “Configuring the Default Charging Method” in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated PCC rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated PCC rule is enabled.

PCC-Rule-Status

AVP Name	PCC-Rule-Status
AVP Code	1019
AVP Data Type	Enumerated
Description	Indicates the status of one PCC rule or a group of PCC rules. The following values are defined: ■ ACTIVE (0) – Used to indicate that the PCC rule(s) are successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF). ■ INACTIVE (1) – Used to indicate that the PCC rule(s) are removed (for those provisioned from PCRF) or inactive (for those pre-provisioned in PCEF). ■ TEMPORARILY INACTIVE (2) – Used to indicate that, for some reason (for example, loss of bearer), already installed or activated PCC rules are temporarily disabled. ■ ACTIVE_WITHOUT_CREDIT_CONTROL (10)

PDN-Connection-Charging-ID

AVP Name	PDN-Connection-Charging-ID
AVP Code	2050
AVP Data Type	Unsigned32
Description	Contains the charging identifier used to identify different records belonging to the same PDN connection. This AVP includes the Charging ID of the first IP-CAN bearer activated within the PDN connection. The PDN-Connection-Charging-ID together with the PGW address uniquely identifies the PDP connection. Applies to the Gy and Gz interfaces.

PDN-Connection-ID

AVP Name	PDN-Connection-ID
AVP Code	1065
AVP Data Type	OctetString
Description	Indicates the PDN connection to which specific information refers. Applies to the Gx interface.

PDP-Address

AVP Name	PDP-Address
AVP Code	1227
AVP Data Type	Address
Description	Contains the IP address associated with the IP CAN bearer session (PDP context/PDN connection).

PDP-Context-Type

AVP Name	PDP-Context-Type
AVP Code	1247
AVP Data Type	Enumerated
Description	Indicates the type of PDP context. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

Precedence

AVP Name	Precedence
AVP Code	1010
AVP Data Type	Unsigned32
Description	Determines the order in which the service data flow templates are applied at service data flow detection at the PCEF. A PCC rule with the Precedence AVP with a lower value is applied before a PCC rule with the Precedence AVP with a a higher value.

Pre-emption-Capability

AVP Name	Pre-emption-Capability
AVP Code	1047
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can get resources that were already assigned to another service data flow with a lower priority level. The following values are defined: ■ PRE-EMPTION_CAPABILITY_ENABLED (0) – Indicates that the service data flow is allowed to get resources that were already assigned to another service data flow with a lower priority level. ■ PRE-EMPTION_CAPABILITY_DISABLED (1) – Indicates that the service data flow is not allowed to get resources that were already assigned to another service data flow with a lower priority level. This is the default value applicable if this AVP is not supplied.

Pre-emption-Vulnerability

AVP Name	Pre-emption-Vulnerability
AVP Code	1048
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can lose the resources assigned to it in order to admit a service data flow with higher priority level. The following values are defined: ■ PRE-EMPTION_VULNERABILITY_ENABLED (0) – Indicates that the resources assigned to the service data flow can be pre-empted and allocated to a service data flow with a higher priority level. This is the default value applicable if this AVP is not supplied. ■ PRE-EMPTION_VULNERABILITY_DISABLED (1) – Indicates that the resources assigned to the service data flow can not be pre-empted and allocated to a service data flow with a higher priority level.

Primary-Event-Charging-Function-Name

AVP Name	Primary-Event-Charging-Function-Name
AVP Code	619
AVP Data Type	DiameterURI
Description	Defines the address of the primary online charging system. The protocol definition in the DiameterURI is either omitted or supplied with a value of "Diameter".

Priority-Level

AVP Name	Priority-Level
AVP Code	1046
AVP Data Type	Unsigned32
Description	Indicates whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations (typically used for admission control of GBR traffic). The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. ■ Values 1 to 15 are defined, with value 1 as the highest level of priority. ■ Values 1 to 8 should only be assigned for services that are authorized to receive prioritized treatment within an operator domain. ■ Values 9 to 15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming.

PS-Furnish-Charging-Information

AVP Name	PS-Furnish-Charging-Information
AVP Code	865
AVP Data Type	Grouped
Description	Contains online charging session specific information. The detailed syntax is as follows: `PS-Furnish-Charging-Information :: = < AVP Header: 865>` `{3GPP-Charging-Id}` `{PS-Free-Format-Data}` `[PS-Append-Free-Format-Data]`

PS-Information

AVP Name	PS-Information
AVP Code	874
AVP Data Type	Grouped
Description	Allows the transmission of additional PS service specific information elements. For a list of the supported AVPs within the PS-Information AVP on the Gz/Rf interface, see the *Affirmed Networks Charging Data Records Specification*.

PS-to-CS-Session-Continuity

AVP Name	PS-to-CS-Session-Continuity
AVP Code	1099
AVP Data Type	Enumerated
Description	Indicates whether the service data flow is a candidate for PS-to-CS session continuity as specified in 3GPP TS 23.216 [40]. The following value is defined: VIDEO_PS2CS_CONT_CANDIDATE (0) – Indicates that the service data flow carries video and is a candidate for PS to CS session continuity.

QoS-Class-Identifier

AVP Name	QoS-Class-Identifier
AVP Code	1028
AVP Data Type	Enumerated
Description	Identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bit rates and ARP for the IP-CAN bearer or service flow. The allowed values for the nine standard QoS Class Identifiers (QCIs) are defined in Table 6.1.7 of 3GPP TS 23.203 [7].

QoS-Information

AVP Name	QoS-Information
AVP Code	1016
AVP Data Type	Grouped
Description	Defines the QoS information for resources requested by the UE, an IP-CAN bearer, PCC rule, QCI or APN. *Gx-Interface* – When the PCEF sends this AVP to the PCRF, it indicates the requested QoS information associated with the resources requested by the UE, and IP-CAN bearer or the subscribed QoS information at the Access Point Name (APN) level.

Quota-Consumption-Time

AVP Name	Quota-Consumption-Time
AVP Code	881
AVP Data Type	Unsigned32
Description	Contains an idle traffic threshold time in seconds.

Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	871
AVP Data Type	Unsigned32
Description	Contains the quota holding time in seconds and is contained in the Multiple-Services-Credit-Control AVP.

RAN-End-Timestamp

AVP Name	RAN-End-Timestamp
AVP Code	1301
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the end time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. *Applies to 5G NSA.*

RAN-NAS-Release-Cause

AVP Name	RAN-NAS-Release-Cause
AVP Code	2819
AVP Data Type	OctetString
Description	Indicates the Radio Access Network (RAN) or Non-Access Stratum (NAS) release cause code information. The RAN-NAS-Release-Cause AVP is sent from the PCEF to the PCRF.

RAN-Secondary-RAT-Usage-Report

AVP Name	RAN-Secondary-RAT-Usage-Report
AVP Code	1302
AVP Data Type	Grouped
Description	Contains the volume count as reported by the RAN for the secondary RAT (separated for uplink and downlink) including the time of the report. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. *Applies to 5G NSA.* The detailed syntax is as follows: `RAN-Secondary-RAT-Usage-Report :: =   < AVP Header: 1302>` `[Secondary-RAT-Type]` `[RAN-Start-Timestamp]` `[RAN-End-Timestamp]` `[Accounting-Input-Octets]` `[Accounting-Output-Octets]`

RAN-Start-Timestamp

AVP Name	RAN-Start-Timestamp AVP
AVP Code	1303
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the start time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. *Applies to 5G NSA.*

RAT-Type

AVP Name	RAT-Type
AVP Code	1032
AVP Data Type	Enumerated
Description	The Radio Access Technology (RAT) currently serving the UE. Values 0-999 are used for generic radio access technologies that can apply to different IP-CAN types and are not IP-CAN specific. Notes: ■ Values 1000-1999 are used for 3GPP specific radio access technology types. ■ Values 2000-2999 are used for 3GPP2 specific radio access technology types. ■ The informative Annex C presents a mapping between the code values for different access network types. The following values are defined: ■ WLAN (0) – Indicates that the RAT is WLAN. ■ UTRAN (1000) – Indicates that the RAT is UTRAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GERAN (1001) – Indicates that the RAT is GERAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GAN (1002) – Indicates that the RAT is GAN. For further details, refer to 3GPP TS 29.060 [18] and 3GPP TS 43.318 [29]. ■ HSPA_EVOLUTION (1003) – Indicates that the RAT is HSPA Evolution. For further details, refer to 3GPP TS 29.060 [18]. ■ EUTRAN (1004) – Indicates that the RAT is EUTRAN. For further details, refer to 3GPP TS 29.274 [22] ■ EUTRAN-NB-IoT (1005) – Indicates that the RAT is NB-IoT. For further details refer to 3GPP TS 29.274 [22]. ■ CDMA2000_1X (2000) – Indicates that the RAT is CDMA2000 1X. For further details, refer to 3GPP2 X.S0011 [20]. ■ HRPD (2001) – Indicates that the RAT is HRPD. For further details, refer to 3GPP2 X.S0011 [20]. ■ UMB (2002) – Indicates that the RAT is UMB. For further details, refer to 3GPP2 X.S0011 [20]. ■ EHRPD (2003) – Indicates that the RAT is eHRPD. For further details, refer to 3GPP2 X.S0057 [24].

Rate-Control-Max-Message-Size

AVP Name	Rate-Control-Max-Message-Size
AVP Code	3937
AVP Data Type	Unsigned32
Description	Indicates the maximum data PDU size in octets. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

Rate-Control-Max-Rate

AVP Name	Rate-Control-Max-Rate
AVP Code	3938
AVP Data Type	Unsigned32
Description	Indicates the maximum number of data PDUs per time unit indicated in the Rate-Control-Time-Unit AVP. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

Rate-Control-Time-Unit

AVP Name	Rate-Control-Time-Unit
AVP Code	3939
AVP Data Type	Unsigned32
Description	Indicates a time unit for rate control. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The following values are defined: ■ UNRESTRICTED (0) – This value indicates the number of unrestricted data PDUs. ■ MINUTE (1) ■ HOUR (2) ■ DAY (3) ■ WEEK (4)

Redirect-Information

AVP Name	Redirect-Information
AVP Code	1085
AVP Data Type	Grouped
Description	Indicates whether the detected application traffic should be redirected to another controlled address. ***Note:*** *The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition/ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.* The detailed syntax is as follows: `Redirect-Information ::= < AVP Header: 1085 >` `[Redirect-Support]` `[Redirect-Address-Type]` `[Redirect-Server-Address]` `*[AVP]`

Redirect-Support

AVP Name	Redirect-Support
AVP Code	1086
AVP Data Type	Enumerated
Description	Indicates whether redirection is disabled or enabled for a predefined PCC rule. ***Note:*** *The Redirect-Support AVP sent from the PCRF as part of the Redirection-Information AVP applies to predefined PCC rules only and is currently not supported for dynamic PCC rules.* The following values are defined: ■ REDIRECTION_DISABLED (0) – Indicates that redirection is disabled for a predefined PCC rule. ■ REDIRECTION_ENABLED (1) – Indicates that redirection is enabled for a predefined PCC rule. This is the default value applicable if a Redirect-Information AVP is provided for the first time and if this AVP is not supplied.

Reporting-Level

AVP Name	Reporting-Level
AVP Code	1011
AVP Data Type	Enumerated
Description	Defines on what level the PCEF reports the usage for the related PCC rule. The following values are defined: ■ SERVICE_IDENTIFIER_LEVEL (0) – Indicates that the usage is reported at the service-id and rating-group level and is applicable when the Service-Identifier and Rating-Group are provisioned within the Charging-Rule-Definition AVP. ■ RATING_GROUP_LEVEL (1) – Indicates that the usage is reported at the rating-group level and is applicable when the Rating-Group is provisioned within the Charging-Rule-Definition AVP.

Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	872
AVP Data Type	Enumerated
Description	Specifies the reason for usage reporting for one or more types of quota for a particular category. It can occur directly in the Multiple-Services-Credit-Control AVP or in the Used-Service-Units AVP within a Credit Control Request command reporting credit usage.

Required-Access-Info

AVP Name	Required-Access-Info
AVP Code	536
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the access network information required for that Application Function (AF) session. The following values are defined: ■ USER_LOCATION (0) – Indicates that the PCEF should report the user location information to the PCRF. If the user location information is available, the PCEF reports the user location information within the 3GPP-User-Location-Info AVP and the time when it was last known in the User-Location-Info-Time AVP (if available). If the user location information is not available, the PCEF provides the serving PLMN identifier within the 3GPP-SGSN-MCC-MNC AVP. ■ MS_TIME_ZONE (1) – Indicates that the PCEF should report the user timezone information to the PCRF. The PCEF reports the user timezone information within the 3GPP-MS-TimeZone AVP.

Resource-Allocation-Notification

AVP Name	Resource-Allocation-Notification
AVP Code	1063
AVP Data Type	Enumerated
Description	Included within the Charging-Rule-Install AVP and defines whether the rules included within the Charging-Rule-Install AVP need notification. The following value is defined: ■ ENABLE_NOTIFICATION (0) – Indicates that the allocation of resources for the related PCC rules are confirmed.

Revalidation-Time

AVP Name	Revalidation-Time
AVP Code	1042
AVP Data Type	Time
Description	Sent from the PCRF to the GGSN/PGW in CCA and RAR messages to indicate the NTP time before which the GGSN/PGW should issue a new request for PCC rules from the PCRF. The Revalidation-Time AVP is always sent with the REVALIDATION_TIMEOUT event trigger. ***Note:*** *The PCRF can disable the revalidation timer on the GGSN/PGW by not including the REVALIDATION_TIMEOUT event-trigger in the CCA or RAR message.*

Rule-Activation-Time

AVP Name	Rule-Activation-Time
AVP Code	1043
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCC/ADC rule has to be enforced. The AVP is included in the Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Deactivation-Time

AVP Name	Rule-Deactivation-Time
AVP Code	1044
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCEF has to stop enforcing the PCC/ADC rule. The AVP is included in Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Failure-Code

AVP Name	Rule-Failure-Code
AVP Code	1031
AVP Data Type	Enumerated
Description	Sent by the PCEF to the PCRF within a Charging-Rule-Report AVP to identify the reason a PCC Rule is being reported. The following values are defined: UNKNOWN_RULE_NAME (1) – Indicates that the pre-provisioned PCC rule could not be successfully activated because the Charging-Rule-Name or Charging-Rule-Base-Name is unknown to the PCEF. RATING_GROUP_ERROR (2) – Indicates that the PCC rule could not be successfully installed or enforced because the Rating-Group specified within the Charging-Rule-Definition AVP by the PCRF is unknown or invalid. SERVICE_IDENTIFIER_ERROR (3) – Indicates that the PCC rule could not be successfully installed or enforced because the Service-Identifier specified within the Charging-Rule-Definition AVP by the PCRF is invalid, unknown, or not applicable to the service being charged. GW/PCEF_MALFUNCTION (4) – Indicates that the PCC rule could not be successfully installed (for those provisioned from the PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to GW/PCEF malfunction. RESOURCES_LIMITATION (5) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to a limitation of resources at the PCEF. MAX_NR_BEARERS_REACHED (6) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to the fact that the maximum number of bearers has been reached for the IP-CAN session. UNKNOWN_BEARER_ID (7) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id specified within the Charging-Rule-Install AVP by the PCRF is unknown or invalid. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_BEARER_ID (8) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id is not specified within the Charging-Rule-Install AVP by the PCRF. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_FLOW_DESCRIPTION (9) – Indicates that the PCC rule could not be successfully installed or enforced because the Flow-Information AVP is not specified within the Charging-Rule-Definition AVP by the PCRF during the first install request of the PCC rule. RESOURCE_ALLOCATION_FAILURE (10) – Indicates that the PCC rule could not be successfully installed or maintained since the bearer establishment/modification failed, or the bearer was released.

AVP Name	Rule-Failure-Code
Description *(Continued)*	UNSUCCESSFUL_QOS_VALIDATION (11) – Indicates that the QoS validation has failed. PS_TO_CS_HANDOVER (13) – Indicates that the PCC rule could not be maintained because of PS to CS handover. Applicable only for 3GPP-GPRS and 3GPP-EPS.

SCS-Address

AVP Name	SCS-Address
AVP Code	3941
AVP Data Type	Address
Description	Contains the IP address of the SCS/AS. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

SCS-AS-Address

AVP Name	SCS-AS-Address
AVP Code	3940
AVP Data Type	Grouped
Description	Indicates the address of the Service Capability Server/Application Server (SCS/AS). *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The detailed syntax is as follows: `SCS-AS-Address :: = < AVP Header: 3940>` `[SCS-Realm]` `[SCS-Address]`

SCS-Realm

AVP Name	SCS-Realm
AVP Code	3942
AVP Data Type	DiameterIdentity
Description	Indicates the Diameter Realm Identity of the SCS/AS. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

Secondary-Event-Charging-Function-Name

AVP Name	Secondary-Event-Charging-Function-Name
AVP Code	620
AVP Data Type	DiameterURI
Description	Defines the address of the secondary online charging system for the bearer. The protocol definition in the DiameterURI is either omitted or supplied with a value of “Diameter”.

Secondary-RAT-Type

AVP Name	Secondary-RAT-Type
AVP Code	1304
AVP Data Type	OctetString
Description	Contains the Secondary RAT Type value, as provided by the RAN. The field is provided by the RAN and transferred to the SGW/PGW in the RAN Traffic Volume element. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. *Applies to 5G NSA.* The following values are defined: ■ Reserved (0) ■ 5G NR (1)

Service-Data-Container

AVP Name	Service-Data-Container
AVP Code	2040
AVP Data Type	Grouped
Description	Provides a list of changes in charging conditions reported for flow-based charging. When there is a change in charging conditions, this container identifies the volume count (separated for uplink and downlink), elapsed time or number of events, per service data flow identified per rating group or combination of the rating group and service ID within an IP-CAN bearer. The detailed syntax is as follows: `Service-Data-Container:: = < AVP Header: 2040>` `[AF-Correlation-Information]` `[Charging-Rule-Base-Name]` `[Accounting-Input-Octets]` `[Accounting-Output-Octets]` `[Accounting-Input-Packets]` `[Accounting-Output-Packets]` `[Local-Sequence-Number]` `[QoS-Information]` `[Rating-Group]` `[Change-Time]` `[Service-Identifier]` `[Service-Specific-Info]` `[SGSN-Address]` `[Time-First-Usage]` `[Time-Last-Usage]` `[Time-Usage]` `*[Change-Condition]` `[3GPP-User-Location-Info]` `[3GPP-RAT-Type]` `[Serving-PLMN-Rate-Control]` `[APN-Rate-Control]`

Service-Information

AVP Name	Service-Information
AVP Code	873
AVP Data Type	Grouped
Description	Contains service-specific parameters. The format and the contents of the fields inside the Service-Information AVP are specified in the middle-tier documents which are applicable for the specific service.

Service-Specific-Info

AVP Name	Service-Specific-Info
AVP Code	1249
AVP Data Type	Grouped
Description	Contains service-specific data if provided by an Application Server or a PCEF only for predefined PCC rules. The detailed syntax is as follows: `Service-Specific-Info::=   < AVP Header: 1249 >` `        [Service-Specific-Data]` `        [Service-Specific-Type]`

Serving-Node-Type

AVP Name	Serving-Node-Type
AVP Code	2047
AVP Data Type	Enumerated
Description	Indicates the type of serving node. The following values are defined: ■ SGSN (0) ■ PMIPSGW (1) ■ GTPSGW (2) ■ ePDG (3) ■ hSGW (4) ■ MME (5)

Serving-PLMN-Rate-Control

AVP Name	Serving-PLMN-Rate-Control
AVP Code	4310
AVP Data Type	Grouped
Description	Indicates the Serving PLMN Rate Control used by the MME. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The detailed syntax is as follows: `Serving-PLMN-Rate-Control::=<AVP header: 4310 10415>` `[Uplink-Rate-Limit]` `[Downlink-Rate-Limit]` A Downlink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for downlink messages is deactivated in the MME. If the Serving PLMN Rate Control is activated, the value of the Downlink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25]. An Uplink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for uplink messages is deactivated in the MME. If Serving PLMN Rate Control is activated, the value of the Uplink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25].

SGi-PtP-Tunnelling-Method

AVP Name	SGi-PtP-Tunnelling-Method
AVP Code	3931
AVP Data Type	Enumerated
Description	Indicates whether the SGi PtP tunnelling method is based on UDP/IP or other methods for a non-IP PDN type PDN connection. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The following values are defined: ■ UDP_IP_BASED (0) ■ OTHERS (1)

Session-Release-Cause

AVP Name	Session-Release-Cause
AVP Code	1045
AVP Data Type	Enumerated
Description	Indicates the reason for terminating the IP-CAN session by the PCRF. The following values are defined: ■ UNSPECIFIED_REASON (0) – Used for unspecified reasons. ■ UE_SUBSCRIPTION_REASON (1) – Indicates that the subscription of the UE has changed (for example, has been removed) and the session needs to be terminated. ■ INSUFFICIENT_SERVER_RESOURCES (2) – Indicates that the server is overloaded and needs to abort the session.

SGSN-Address

AVP Name	SGSN-Address
AVP Code	1228
AVP Data Type	Address
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

SGW-Address

AVP Name	SGW-Address
AVP Code	2067
AVP Data Type	Address
Description	Indicates the IP-address of the SGW Node.

SGW-Change

AVP Name	SGW-Change
AVP Code	2065
AVP Data Type	Enumerated
Description	Indicates this is the first ACR Start message after an SGW change. If this AVP is not present in the ACR Start message, this ACR Start is not due to an SGW change. The following values are defined: ■ ACR START NOT DUE TO SGW CHANGE (0) ■ ACR START DUE TO SGW CHANGE (1)

Sponsor-Identity

AVP Name	Sponsor-Identity
AVP Code	531
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the sponsor.

Sponsored-Connectivity-Data

AVP Name	Sponsored-Connectivity-Data
AVP Code	530
AVP Data Type	Grouped
Description	Contains the data associated with the sponsored data connectivity. The detailed syntax is as follows: `Sponsored-Connectivity-Data::= < AVP Header: 530 >` `[Sponsor-Identity]` `[Application-Service-Provider-Identity]` `[Granted-Service-Unit]` `*[AVP]`

Supported-Features

AVP Name	Supported-Features
AVP Code	628
AVP Data Type	Grouped
Description	Informs the destination host about the features that the origin host requires to successfully complete this command exchange. ***Note:*** *When a Supported-Features AVP is used to identify features that have been defined by 3GPP, the Vendor-Id AVP contains the vendor ID of 3GPP (10415).* The detailed syntax is as follows: `Supported-Features ::= < AVP header: 628 10415 >` `{Vendor-Id}` `{Feature-List-ID}` `{Feature-List}`

TDF-Application-Identifier

AVP Name	TDF-Application-Identifier
AVP Code	1088
AVP Data Type	OctetString
Description	References the application detection filter to which the PCC rule for Application Detection and Control (ADC) in the PCEF applies. This AVP also identifies the application when reporting application Start/Stop notifications to the PCRF. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information.

TDF-Application-Instance-Identifier

AVP Name	TDF-Application-Instance-Identifier
AVP Code	2802
AVP Data Type	OctetString
Description	This value is dynamically assigned by the PCEF to correlate the application Start/Stop events for use with the Application Detection and Control (ADC) functionality. See “Application Detection and Control” in the *Affirmed Networks Gx Interface Specification* for more information.

Time-First-Usage

AVP Name	Time-First-Usage
AVP Code	2043
AVP Data Type	Time
Description	Contains the time in UTC format for the first IP packet to be transmitted and mapped to the current service data container.

Time-Last-Usage

AVP Name	Time-Last-Usage
AVP Code	2044
AVP Data Type	Time
Description	Contains the time in UTC format for the last IP packet to be transmitted and mapped to the current service data container.

Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	1270
AVP Data Type	Grouped
Description	The OCS may include this AVP in the Multiple-Services-Credit-Control AVP, when granting time quota. The detailed syntax is as follows: `Time-Quota-Mechanism ::= < AVP Header: 1270>` `{Time-Quota-Type}` `{Base-Time-Interval}`

Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	868
AVP Data Type	Unsigned32
Description	Contains a threshold value in seconds.

Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	1271
AVP Data Type	Enumerated
Description	Indicates the time quota consumption mechanism used for the associated rating group. The following values are defined: ■ DISCRETE_TIME_PERIOD (0) ■ CONTINUOUS_TIME_PERIOD (1)

Time-Usage

AVP Name	Time-Usage
AVP Code	2045
AVP Data Type	Time
Description	Contains the effective used time within the service data container reporting interval.

Traffic-Data-Volumes

AVP Name	Traffic-Data-Volumes
AVP Code	2046
AVP Data Type	Grouped
Description	When a change in charging conditions occurs for this IP-CAN bearer, this container reports the volume count (separated for uplink and downlink). The detailed syntax is as follows: `Traffic-Data-Volumes :: = < AVP Header: 2046>` `[QoS-Information]` `[Accounting-Input-Octets]` `[Accounting-Input-Packets]` `[Accounting-Output-Octets]` `[Accounting-Output-Packets]` `[Change-condition]` `[Change-Time]` `[3GPP-User-Location-Info]` `[3GPP-RAT-Type]` `[CP-CIoT-EPS-Optimisation-Indicator]` `[Serving-PLMN-Rate-Control]`

Trigger

AVP Name	Trigger
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: `Trigger :: = < AVP Header: 1264 >` `*[Trigger-Type]`

Trigger-Type

AVP Name	Trigger-Type
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gy interface, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. When included in the CCA on the Gx interface, the Trigger-Type VSA is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. For more information, see Trigger-Type (VSA) (page A-129).

TWAN-User-Location-Info

AVP Name	TWAN-User-Location-Info
AVP Code	2714
AVP Data Type	Grouped
Description	Contains the UE location in a Trusted WLAN Access Network (TWAN) as TWAN Identifier defined in TS 29.274 [226] (TS 32.299 v15.3). Sent from the TWAG/TWAP to the OCS via the Gy interface. The detailed syntax is as follows: `TWAN-User-Location-Info :: = < AVP Header: 2714 >` `{SSID}` `[BSSID]` The Twan-User-Location-Info AVP is configurable in Gy CCR data record templates and appears as **`psinfotwanuserlocationinfo`**. The RADIUS Attribute Mapping template must be configured so that the TWAG extracts the SSID/BSSID received in the Called-Station-ID AVP (within the RADIUS Access-Request) and includes it in the Twan-User-Location-Info AVP on the Gy interface. For example: **1** Configure the attributes in the incoming RADIUS messages. `an3000-mcm-slot7cpu1(config)#` **`service-construct radius-mapping template`** ***`<template-name>`*** **`attribute calledstationid`** **`to-format (.*):(.*)`** **`to-list [apbssid apssid]`** **2** Associate the RADIUS Attribute Mapping Template with the Session Learner interface. `an3000-mcm-slot7cpu1(config)#` **`service-construct interface gw-session-learner [non-3gpp-access-diameter-client|radius-proxy|twag-radius-client]`** ***`<interface-name>`*** **`radius-server-trigger radius-mapping access-request`** ***`<template-name>`*** ***Note:*** *For more information on configuring the RADIUS Attribute Mapping template, see the Affirmed Networks TWAG/TWAP Administration Guide.*

UNI-PDU-CP-Only-Flag

AVP Name	UNI-PDU-CP-Only-Flag
AVP Code	3932
AVP Data Type	Enumerated
Description	Indicates whether this PDN connection is applied with the "Control Plane Only flag". For example, using only the S11-U from the SGW when Control Plane CIoT EPS Optimization is enabled. If this AVP is not present, then both the user plane and control plane can be used in UNI for the PDU transfer. For example, using both S1-U and S11-U respectively from the SGW when Control Plane CIoT EPS Optimization is enabled. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.* The following values are defined: ■ UNI-PDU-both-UP-CP (0) ■ UNI-PDU-CP-Only (1)

Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	1226
AVP Data Type	Unsigned32
Description	Contains a threshold value in service specific units.

Uplink-Rate-Limit

AVP Name	Uplink-Rate-Limit
AVP Code	4311
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for uplink. *Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.*

Usage-Monitoring-Information

AVP Name	Usage-Monitoring-Information
AVP Code	1067
AVP Data Type	Grouped
Description	Contains the usage monitoring control information. The detailed syntax is as follows: `Usage-Monitoring-Information::= < AVP Header: 1067 >` `[Monitoring-Key]` `[Granted-Service-Unit]` `[Used-Service-Unit]` `[Usage-Monitoring-Level]` `[Usage-Monitoring-Report]` `[Usage-Monitoring-Support]` `*[AVP]`

Usage-Monitoring-Level

AVP Name	Usage-Monitoring-Level
AVP Code	1068
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether the usage monitoring instance applies to the IP-CAN session or to one or more PCC rules. The following values are defined: ■ SESSION_LEVEL (0) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to the entire IP-CAN session. ■ PCC_RULE_LEVEL (1) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to one or more PCC rules. ***Note:*** *The absence of the Usage-Monitoring-Level AVP indicates the value PCC_RULE_LEVEL (1).*

Usage-Monitoring-Report

AVP Name	Usage-Monitoring-Report
AVP Code	1069
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the accumulated usage for the usage monitoring key included in the Monitoring-Key AVP (included within the Usage-Monitoring-Information AVP). The following value is defined: USAGE_MONITORING_REPORT_REQUIRED (0) – This value, if provided within an RAR or CCA command, indicates that the PCEF should report accumulated usage regardless of whether a usage threshold is reached for a certain usage monitoring key.

Usage-Monitoring-Support

AVP Name	Usage-Monitoring-Support
AVP Code	1070
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether usage monitoring is disabled for a certain Monitoring Key. The following value is defined: USAGE_MONITORING_DISABLED (0) – This value indicates that usage monitoring is disabled for a monitoring key.

User-CSG-Information

AVP Name	User-CSG-Information
AVP Code	2319
AVP Data Type	Grouped
Description	Contains the Closed Subscriber Group (CSG) information associated with CSG cell access, including the CSG ID within the PLMN, the access mode and the CSG membership indication for the user when hybrid access applies. The detailed syntax is as follows: `User-CSG-Information :: = < AVP Header: 2319>` `{CSG-Id}` `[CSG-Access-Mode]` `[CSG-Membership-Indication]`

User-Location-Info-Time

AVP Name	User-Location-Info-Time
AVP Code	2812
AVP Data Type	Time
Description	Contains the NTP time at which the UE was last known to be in the location, which is reported during bearer deactivation or IP-CAN session termination. The User-Location-Info-Time AVP is sent from the PCEF to the PCRF.

UWAN-User-Location-Info

AVP Name	UWAN-User-Location-Info
AVP Code	3918
AVP Data Type	Grouped
Description	Contains the UE location in an Untrusted Wireless Access Network (UWAN). The UWAN User Location Information includes the UE local IP address and optionally UDP source port number (if NAT is detected). The detailed syntax is as follows: `UWAN-User-Location-Info :: = < AVP Header: 3918>` `{UE-Local-IP-Address}` `[UDP-Source-Port]`

Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	869
AVP Data Type	Unsigned32
Description	Contains a threshold value in octets.

NASREQ/RADIUS AVP Definitions

This section describes the following NASREQ/RADIUS AVPs in the CCR and CCA messages.

- Called-Station-Id (page A-102)
- Framed-Interface-Id (page A-102)
- Framed-IP-Address (page A-102)
- Framed-IPv6-Prefix (page A-103)

Called-Station-Id

AVP Name	Called-Station-Id
AVP Code	30
AVP Data Type	UTF8String
Description	The gateway support node of the GPRS. The identifier for the target network: APN.

Framed-Interface-Id

AVP Name	Framed-Interface-Id
AVP Code	96
AVP Data Type	Unsigned64
Description	The user IPv6 interface identifier.

Framed-IP-Address

AVP Name	Framed-IP-Address
AVP Code	8
AVP Data Type	OctetString
Description	The IP address allocated for this user.

Framed-IPv6-Prefix

AVP Name	Framed-IPv6-Prefix
AVP Code	97
AVP Data Type	OctetString
Description	The IPv6 address prefix allocated for this user.

Vodafone AVP Definitions

This section describes the following Vodafone AVPs in the CCR and CCA messages. The AVPs include Vendor-ID of 12645 (Vodafone).

vf-Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	276
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval, for controlling the consumption of time quota in seconds.

vf-Context-Type

AVP Name	Context-Type
AVP Code	256
AVP Data Type	Enumerated
Description	This AVP indicates whether the PDP context corresponding to the CC session is a primary or secondary. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

vf-Envelopes

AVP Name	Envelopes
AVP Code	269
AVP Data Type	Grouped
Description	The Envelope AVP reports the start and end time of one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299.

vf-Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	271
AVP Data Type	Time
Description	The Envelope-End-Time AVP is set to the time of the end of the time envelope.

vf-Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	272
AVP Data Type	Enumerated
Description	This AVP is of type Enumerated and is used in the initial CCA to indicate whether the client reports the start and end of each time envelope, in those cases in which quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2)

vf-Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	270
AVP Data Type	Time
Description	The Envelope-Start-Time AVP is set to the time of the packet of user data which caused the time envelope to start.

vf-Quota-Consumption-Time

AVP Name	Quota-Consumption-Tim
AVP Code	257
AVP Data Type	Unsigned32
Description	This AVP contains an idle traffic threshold time in seconds. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing a CC-Time AVP (for example, when the granted quota is a time quota).

vf-Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	258
AVP Data Type	Unsigned32
Description	Contains the length of time (in seconds) for which the system can hold quota during a period of inactivity. A quota expires when no traffic associated with the quota is observed for the value indicated by this AVP. The system stops the timer when it sends a CCR message and resets the timer each time it receives a CCA with a Quota-Holding-Time value.

vf-Radio-Access-Technology

AVP Name	Radio-Access-Technology
AVP Code	260
AVP Data Type	Enumerated
Description	This AVP specifies the radio access technology being used by the user. The following values are defined: ■ UTRAN (1) ■ GERAN (2) ■ WLAN (3)

vf-Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	261
AVP Data Type	Enumerated
Description	The Reporting-Reason specifies the reason for usage reporting for one or more types of quota for a rating group. It can occur directly in the Multiple-Services-Credit-Control AVP, or in the Used-Service-Units AVP. The following values apply to all quota types and are used in the Multiple-Services-Credit-Control AVP: ■ QHT ■ FINAL ■ VALIDITY_TIME ■ RATING_CONDITION_CHANGE The following values apply to one particular quota type and are used in the Used-Service-Units AVP: ■ THRESHOLD ■ QUOTA_EXHAUSTED ■ OTHER_QUOTA_TYPE The following values are defined for the Reporting-Reason AVP: ■ THRESHOLD 0 ■ QHT 1 ■ FINAL 2 ■ QUOTA_EXHAUSTED 3 ■ VALIDITY_TIME 4 ■ OTHER_QUOTA_TYPE 5 ■ RATING_CONDITION_CHANGE 6 ■ FORCED_REAUTHORISATION 7

vf-Rulebase-Id

AVP Name	Rulebase-Id
AVP Code	262
AVP Data Type	UTF8String
Description	Contains the identifier of the currently selected rule base. ***Note:*** *All charging instances configured under* `zone <name> gateway apn <name> charging` *are included in the CCR-I to the OCS. If no charging instances are configured and the system receives charging instances from external interfaces, those charging instances are included in the CCR-I. If no charging instances are available, the data profile charging instance is used.*

vf-Time-Of-First-Usage

AVP Name	Time-Of-First-Usage
AVP Code	263
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the first packet of usage of the quota being reported for the rating group.

vf-Time-Of-Last-Usage

AVP Name	Time-Of-Last-Usage
AVP Code	264
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the last packet of usage of the quota being reported for the rating group.

vf-Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	274
AVP Data Type	Grouped
Description	This AVP may be used by the OCS for granting time quota. Time-Quota-Mechanism AVP takes precedence over Quota-Consumption-Time.

vf-Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	259
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for time quota in seconds.

vf-Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	275
AVP Data Type	Enumerated
Description	The Time-Quota-Type is used to indicate which time quota consumption mechanism is used for the associated rating group.

vf-Trigger

AVP Name	Trigger
AVP Code	265
AVP Data Type	Grouped
Description	This AVP is used to arm the triggers in the client and to identify the triggers which caused re-authorization in usage. On receipt of the AVP, the OCS arms all known triggers present in the AVP and resets all known triggers missing from the AVP. The presence of the Trigger AVP without any Trigger-Type AVP in a CCA allows the OCS to disable all the triggers. The presence of the Trigger AVP in the CCR identifies the event(s) triggering the CCR.

vf-Trigger-Type

AVP Name	Trigger-Type
AVP Code	266
AVP Data Type	Enumerated
Description	When included in the CCA, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. The system supports the following triggers between the system and the OCS. ■ CHANGE_IN_QOS (2) – Specifies whether a QoS change triggers an update. ■ CHANGE_IN_RAT (4) – Specifies whether a RAT Type change triggers an update. ■ CHANGE_IN_SERVING_NODE (61) – Specifies whether a Serving Node change triggers an update.

vf-Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	273
AVP Data Type	Unsigned32
Description	This AVP contains a threshold value in service specific units. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing CC-Service-Specific-Units AVP (that is, when the granted quota is service specific).

vf-User-Location-Information

AVP Name	User-Location-Information
AVP Code	267
AVP Data Type	OctetString
Description	Indicates details of where the UE is currently located (for example, SAI or CGI).

vf-Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	268
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for volume quota in bytes.

Vendor-Specific AVP Definitions

This section describes the following supported Vendor-Specific AVPs.

ACW-Entitlement

AVP Name	acwentitlement
Vendor-ID	9
AVP Code	131501
AVP Data Type	OctetString
Description	Used to send tethering entitlement information from the PCRF to the PGW for the subscriber session. This VSA can be sent in CCA-I/CCA-U and RAR messages on the Gx interface. The following values for this VSA are defined: GTETH/MHS1/TETH/GVC/GVC1/MHSM/TOGGL/TGLSB/TGLSI. GTETH/MHS1/TETH – If one or more of these values are present in the acwentitlement VSA, the MCC will enable tethering (entitlement) for the session, irrespective of the configured qos-flows. However, the session-level AMBR will still be applied. If the VSA contains any value other than these three values or if the PGW does not receive the acwentitlement VSA in the CCR-I, the MCC will disable tethering (entitlement) for the session and the configured qos-flow will take effect. ***Note:*** *You can control tethering for an individual APN, based on the presence of configurable strings in the ACW- Entitlement VSA received from the PCRF. For more information, see the Affirmed Networks Gateway Administration Guide.*

ACW-Sponsor-URL

AVP Name	ACW-Sponsor-URL
Vendor-ID	9
AVP Code	131506
AVP Data Type	UTF8String
Description	Contains the URL of the manifest server.

AF-Msisdn

AVP Name	AF-Msisdn
Vendor-ID	37963
AVP Code	26
AVP Data Type	String
Description	Contains an MSISDN value used to set the MSISDN for the specific UE.

Assume-Positive-Treatment

AVP Name	Assume-Positive-Treatment
Vendor-ID	12951
AVP Code	7001
AVP Data Type	OctetString
Description	Used on the Gx interface in the CCA-Initial message to indicate a subscriber credit control group type (RTR, UC or Prepaid). The MCC PGW uses the credit control group received in the Assume-Positive-Treatment AVP from the PCRF as a key to select a Subscriber Analyzer. Use the following CLI command to configure the Subscriber Analyzer key for the credit control group. `an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <name> key <name> credit-control-group <name>`

Charging-Gateway-Function-Host

AVP Name	Charging-Gateway-Function-Host
Vendor-ID	12951
AVP Code	6068
AVP Data Type	OctetString
Description	This AVP is sent from the PGW to the OFCS and contains the Fully Qualified Domain Name (FQDN) of the Charging Data Function (CDF). When the PGW receives the Charging-Gateway-Function-Host AVP via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. *Supported on the Gz/Rf Interface only.*

Charging-Group-ID

AVP Name	Charging-Group-ID
Vendor-ID	12951
AVP Code	6069
AVP Data Type	UTF8String
Description	This AVP is sent from the PGW to the OFCS and identifies the subscriber charging group. When the PGW receives the Charging-Group-ID via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. *Supported on the Gz/Rf Interface only.*

Cisco-CC-Failure-Type

AVP Name	Cisco-CC-Failure-Type
Vendor-ID	9
AVP Code	132077
AVP Data Type	Unsigned32
Description	Indicates the OCS failure reasons to the PCRF. *Supported on the Gx Interface*. The following values are defined: ■ CC_CONECTION_FAILURE (0) ■ CC_RESPONSE_TIMEOUT (1) ■ CC_UNKNOWN_FAILURE (5) ■ CC_RESULT_CODE – Result code value received over the Gy interface

Cisco-Event

AVP Name	Cisco-Event
Vendor-ID	9
AVP Code	131195
AVP Data Type	Grouped
Description	Indicates the OCS failure result code. *Supported on the Gx Interface*. The detailed syntax is as follows: `Cisco-Event::= < AVP Header: 131195>` `[Cisco-Event-Trigger-Type]` `[Cisco-CC-Failure-Type]`

Cisco-Event-Trigger

AVP Name	Cisco-Event-Trigger
Vendor-ID	9
AVP Code	131193
AVP Data Type	Grouped
Description	Indicates the events that cause the OCS failure. *Supported on the Gx Interface*. The detailed syntax is as follows: `Cisco-Event-Trigger::= < AVP Header: 131193>` `        [Cisco-Event-Trigger-Type]`

Cisco-Event-Trigger-Type

AVP Name	Cisco-Event-Trigger-Type
Vendor-ID	9
AVP Code	131192
AVP Data Type	Enumerated
Description	Indicates the events that cause the OCS failure. *Supported on the Gx Interface*. The following values are defined: ■ NO_CISCO_EVENT_TRIGGERS (0) ■ CREDIT_CONTROL_FAILURE (5)

Cumulative-Acct-Input-Octets

AVP Name	Cumulative-Acct-Input-Octets
Vendor-ID	12951
AVP Code	132044
AVP Data Type	Unsigned64
Description	Contains the total number of input octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. *Supported on the Gz/Rf Interface*.

Cumulative-Acct-Output-Octets

AVP Name	Cumulative-Acct-Output-Octets
Vendor-ID	12951
AVP Code	132045
AVP Data Type	Unsigned64
Description	Contains the total number of output octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. *Supported on the Gz/Rf Interface*.

Gx-TMO-Append-IMEI

AVP Name	Gx-TMO-Append-IMEI
Vendor-ID	29168
AVP Code	117
AVP Data Type	Enumerated
Description	Defines whether the IMEI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMEI. ■ Enable (1) – Append the IMEI.

Gx-TMO-Append-IMSI

AVP Name	Gx-TMO-Append-IMSI
Vendor-ID	29168
AVP Code	116
AVP Data Type	Enumerated
Description	Defines whether the IMSI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMSI. ■ Enable (1) – Append the IMSI.

Gx-TMO-Append-MSIP

AVP Name	Gx-TMO-Append-MSIP
Vendor-ID	29168
AVP Code	118
AVP Data Type	Enumerated
Description	Defines whether the IP address assigned to the mobile station is appended to the redirect URL. See "Processing the Gx-TMO-Redirect-Server VSA" in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MS IP address. ■ Enable (1) – Append the MS IP address. If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. For IPv6 IP addresses, the MCC uses the preferred format.

Gx-TMO-Append-MSISDN

AVP Name	Gx-TMO-Append-MSISDN
Vendor-ID	29168
AVP Code	115
AVP Data Type	Enumerated
Description	Defines whether the MSISDN of the mobile station is appended to the redirect URL. See "Processing the Gx-TMO-Redirect-Server VSA" in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MSISDN. ■ Enable (1) – Append the MSISDN.

Gx-TMO-Append-Original-URL

AVP Name	Gx-TMO-Append-Original-URL
Vendor-ID	29168
AVP Code	114
AVP Data Type	Enumerated
Description	Defines whether the original URL is appended to the redirect URL. See "Processing the Gx-TMO-Redirect-Server VSA" in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – (disable) Do not append the original URL. ■ Enable (1) – Append the original URL.

Gx-TMO-Deactivate-By-Redirect

AVP Name	Gx-TMO-Deactivate-By-Redirect
Vendor-ID	29168
AVP Code	119
AVP Data Type	Enumerated
Description	Defines whether the system performs a one-time redirection. After the one-time redirection, the rule is deactivated. See "Processing the Gx-TMO-Redirect-Server VSA" in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ Disable (0) – The rule is not deactivated once enforced. ■ Enable (1) – The rule is deactivated once enforced.

Gx-TMO-Redirect-Address-Type

AVP Name	Gx-TMO-Redirect-Address-Type
Vendor-ID	29168
AVP Code	112
AVP Data Type	Enumerated
Description	Defines the address type of the address given in the TMO-Redirect-Server-Address VSA. See “Processing the Gx-TMO-Redirect-Server VSA” in the *Affirmed Networks Gx Interface Specification* for more information. The following values are defined: ■ IPv4 Address (0) ■ IPv6 Address (1) ■ URL (2)

Gx-TMO-Redirect-Server

AVP Name	Gx-TMO-Redirect-Server
Vendor-ID	29168
AVP Code	111
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) to which the end user is redirected. See “Processing the Gx-TMO-Redirect-Server VSA” in the *Affirmed Networks Gx Interface Specification* for more information. The detailed syntax is as follows: `Gx-TMO-Redirect-Server::= < AVP Header: 111>` `[Gx-TMO-Redirect-Address-Type]` `[Gx-TMO-Redirect-Server-Address]` `[Gx-TMO-Append-Original-URL]` `[Gx-TMO-Append-MSISDN]` `[Gx-TMO-Append-IMSI]` `[Gx-TMO-Append-IMEI]` `[Gx-TMO-Append-MSIP]` `[Gx-TMO-Deactivate-By-Redirect]` ***Note:*** *If both the Redirect-Information AVP and Gx-TMO Redirect-Server VSA are received, the MCC will use the Gx-TMO Redirect-Server VSA for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server VSA is enabled in the CCA data record template.*

Gx-TMO-Redirect-Server-Address

AVP Name	Gx-TMO-Redirect-Server-Address
Vendor-ID	29168
AVP Code	113
AVP Data Type	UTF8String
Description	Defines the address of the redirect server (for example, HTTP redirect server) with which the end user is to be connected. See “Processing the Gx-TMO-Redirect-Server VSA” in the *Affirmed Networks Gx Interface Specification* for more information.

Max-Wait-Time

AVP Name	Max-Wait-Time
Vendor-ID	12951
AVP Code	7103
AVP Data Type	Unsigned32
Description	Indicates the maximum amount of time (in milliseconds) that the originator is willing to wait for a response since the Origination-TimeStamp. This AVP is only provided with the initial authorization request. *Supported on the Gx and Gy Interfaces only.*

MVNO-Reseller-Id

AVP Name	MVNO-Reseller-Id (`affmnvostringid`)
Vendor-ID	9
AVP Code	131507
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) name or identifier. *Supported on the Gx and Gy Interfaces only.*

MVNO-Sub-Class-Id

AVP Name	MVNO-Sub-Class-Id (`affmvnosubclassid`)
Vendor-ID	9
AVP Code	131508
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) sub-name or sub-identifier. *Supported on the Gx and Gy Interfaces only.*

Origination-Time-Stamp

AVP Name	Origination-Time-Stamp
Vendor-ID	12951
AVP Code	7102
AVP Data Type	Unsigned64
Description	Indicates the time that the message was sent, representing the number of seconds since January 1, 1900 00:00 UTC. This AVP is only provided with the initial authorization request. *Supported on the Gx and Gy Interfaces only.*

Override-Charging-Action-Exclude-Rule

AVP Name	Override-Charging-Action-Exclude-Rule
Vendor-ID	9
AVP Code	132021
AVP Data Type	Octet String
Description	Indicates the name of the service rule to which override control will not be applied. Service rules that are specified in this VSA will still be associated with the rating group based on the workflow configuration. This VSA may be included more than once if more than one rule needs to be excluded.

Override-Charging-Action-Parameters

AVP Name	Override-Charging-Action-Parameters
Vendor-ID	9
AVP Code	132019
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: `Override-Charging-Action-Parameters ::= < AVP Header: 132019>` `*[Override-Charging-Action-Exclude-Rule]` `[Override-Charging-Parameters]`

Override-Charging-Parameters

AVP Name	Override-Charging-Parameters
Vendor-ID	9
AVP Code	132022
AVP Data Type	Grouped
Description	Contains the override charging parameters to be applied to predefined and static rules on the PCEF. The detailed syntax is as follows: `Override-Charging-Parameters ::= < AVP Header: 132022>` `[Override-Rating-Group]` `[Override-Online]` `[Override-Offline]`

Override-Control

AVP Name	Override-Control
Vendor-ID	9
AVP Code	132017
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: `Override-Control ::= < AVP Header: 132017>` `[Override-Charging-Action-Parameters]` `*[Override-Charging-Action-Exclude-Rule]` `[Override-Charging-Parameters]` `[Override-Rating-Group]` `[Override-Online]` `[Override-Offline]`

Override-Offline

AVP Name	Override-Offline
Vendor-ID	9
AVP Code	132027
AVP Data Type	Enumerated
Description	This VSA overrides the offline value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the offline charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the offline charging interface for the associated service rule is enabled.

Override-Online

AVP Name	Override-Online
Vendor-ID	9
AVP Code	132026
AVP Data Type	Enumerated
Description	This VSA overrides the online value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated service rule is enabled.

Override-Rating-Group

AVP Name	Override-Rating-Group
Vendor-ID	9
AVP Code	132024
AVP Data Type	Unsigned32
Description	This VSA is used to override the value of the rating group configured for a static/predefined rule. The rating group identifier sent in this VSA must correspond to a predefined rating group quota-id on the MCC (`services` –> `charging` –> `rating-group` –> `quota-id`).

QoS-Group-Rule-Definition

AVP Name	QoS-Group-Rule-Definition
Vendor-ID	9
AVP Code	132003
AVP Data Type	Grouped
Description	The PCRF uses this VSA to define the parameters for a pre-configured service rule whose name matches the QoS-Group-Rule-Name. For example, if the received QoS-Group-Rule-Definition VSA has a redirect server address, the MCC will apply redirection to the associated flows. ***Note:*** *The Redirect-Server AVP sent from the PCRF as part of the QoS-Group-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules.* The detailed syntax is as follows: `QoS-Group-Rule-Definition ::= <AVP header: 132003>` `{QoS-Group-Rule-Name}` `[Monitoring-Key]` `[Flow-Status]` `[Redirect-Server]`

QoS-Group-Rule-Install

AVP Name	QoS-Group-Rule-Install
Vendor-ID	9
AVP Code	132001
AVP Data Type	Grouped
Description	The PCRF uses this VSA to activate predefined service rules on the MCC gateway. The detailed syntax is as follows: `QoS-Group-Rule-Install ::= <AVP header: 132001 >` `*[QoS-Group-Rule-Definition]`

QoS-Group-Rule-Name

AVP Name	QoS-Group-Rule-Name
Vendor-ID	9
AVP Code	132004
AVP Data Type	OctetString
Description	The MCC tries to match the service rule name, received in the QoS-Group-Rule-Name VSA, to a service rule name configured locally on the MCC gateway (`service-construct` –> `service-rule`). ■ If a match is found, the MCC activates the matching service rule and all services associated with the received QoS-Rule-Group-Name. ■ If a match is not found, the MCC sends the Charging-Rule-Report AVP with the QoS-Group-Rule-Name VSA to the PCRF.

QoS-Group-Rule-Remove

AVP Name	QoS-Group-Rule-Remove
Vendor-ID	9
AVP Code	132002
AVP Data Type	Grouped
Description	The PCRF uses this VSA to deactivate predefined service rules on the MCC gateway. The detailed syntax is as follows: `QoS-Group-Rule-Remove ::= <AVP header: 132002 >` `*[QoS-Group-Rule-Name]`

TMO-Transparent-Data

AVP Name	TMO-Transparent-Data
Vendor-ID	29168
AVP Code	1
AVP Data Type	Octet String
Description	Contains information from the TMO-Transparent-Data VSA received from the RADIUS server. The MCC gateway includes this VSA in the CCR message to the OCS when the RADIUS server returns a transparent data value in the Access-Accept message. If the RADIUS Access-Accept message does not return a transparent data value, you can configure the MCC gateway to assign a transparent data value and the MCC gateway will include the configured transparent data value in this VSA to the OCS. For more information on how to configure the MCC gateway to assign a transparent data value, see the *Affirmed Networks Gateway Administration Guide.*

TMO-Virtual-Gi-ID

AVP Name	TMO-Virtual-Gi-ID
Vendor-ID	29168
AVP Code	15
AVP Data Type	32 Bit Integer
Description	Contains the value of the virtual-ID. See “Processing the TMO-Virtual-Gi-ID VSA” in the *Affirmed Networks RADIUS Interface Specification* for more information.

Trigger (VSA)

AVP Name	Trigger
Vendor-ID	12951
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: `Trigger :: = < AVP Header: 1264 >` `       1*4 [Trigger-Type]`

Trigger-Type (VSA)

AVP Name	Trigger-Type
Vendor-ID	12951
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gx interface, the Trigger-Type AVP is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. On the Gx interface, the Trigger-Type AVP contains the following values: ■ CHANGE_IN_UE_TIMEZONE (5) – Indicates a change in the UE time zone. ■ CHANGEINLOCATION_MCC (30) – Indicates a change in the MCC of the serving network, which represents the first half of the PLMN. ■ CHANGEINLOCATION_MNC (31) – Indicates a change in the MNC of the serving network, which represents the second half of the PLMN. ■ CHANGE_IN_LOCATION (3) – Indicates a change in the user location. If this Trigger-Type is received, the PGW will send the OCS an updated ULI for all changes, including the PLMN; therefore, this value should not be used in conjunction with the MCC and MNC location change Trigger-Types. Upon receiving the Trigger AVP –> Trigger-Type AVP from the PCRF and if the subscriber does not have a Gy session, the PGW sends a CCR-Event message to the OCS via the Gy interface that includes the following AVPs: CC-Request-Type: EVENT_REQUEST (4) Requested-Action: Location Update (4) Service-Information PS-Information 3GPP-MS-Timezone 3GPP-SGSN-MCC-MNC 3GPP-User-Location-Info ***Note:*** *If the subscriber already has a Gy session, the PGW sends a CCR-U message if a quota request is needed for any of the rating groups.*

ASN.1 Definition

The following is a complete ASN.1 definition of the GPRS records. It is based on the ASN.1 definition in 3GPP TS 32.298 with imported types from 3GPP TS 29.002.

```
AFFIRMEDNETWORKS
DEFINITIONS IMPLICIT TAGS ::=

BEGIN


GPRSRecord::= CHOICE
{
       sgsnPDPRecord[20] SGSNPDPRecord,
       ggsnPDPRecord[21] GGSNPDPRecord,
       sgsnMMRecord[22] SGSNMMRecord,
       sgsnSMORecord[23] SGSNSMORecord,
       sgsnSMTRecord[24] SGSNSMTRecord,
       sgsnLCTRecord[25] SGSNLCTRecord,
       sgsnLCORecord[26] SGSNLCORecord,
       sgsnLCNRecord[27] SGSNLCNRecord,
--
-- pGWRecordRel6 is for REQS-6185 Batelco
--
       pGWRecordRel6[28] PGWRecord,
       egsnPDPRecord[70] EGSNPDPRecord,
       sgsnMBMSRecord[76] SGSNMBMSRecord,
       ggsnMBMSRecord[77] GGSNMBMSRecord,
       sGWRecord [78] SGWRecord,
       pGWRecord [79] PGWRecord

}

SGWRecord ::= SET
```

```
{
       recordType[0] RecordType,
       servedIMSI[3] IMSI OPTIONAL,
       s-GWAddress[4] GSNAddress,
       chargingID[5] ChargingID OPTIONAL,
       servingNodeAddress[6] SEQUENCE OF GSNAddress,
       accessPointNameNI[7] AccessPointNameNI OPTIONAL,
       pdpPDNType[8] PDPType OPTIONAL,
       servedPDPPDNAddress[9] PDPAddress OPTIONAL,
       dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
       listOfTrafficVolumes[12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
       recordOpeningTime[13] TimeStamp,
       duration  [14] CallDuration,
       causeForRecClosing[15] CauseForRecClosing,
       diagnostics[16] Diagnostics OPTIONAL,
       recordSequenceNumber[17] INTEGER OPTIONAL,
       nodeID    [18] NodeID OPTIONAL,
       recordExtensions[19] ManagementExtensions OPTIONAL,
       localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
       apnSelectionMode[21] APNSelectionMode OPTIONAL,
       servedMSISDN[22] MSISDN OPTIONAL,
       chargingCharacteristics[23] ChargingCharacteristics,
       chChSelectionMode[24] ChChSelectionMode OPTIONAL,
       iMSsignalingContext[25] NULL OPTIONAL,
       servingNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
       servedIMEISV[29] IMEI OPTIONAL,
       rATType   [30] RATType OPTIONAL,
       mSTimeZone [31] MSTimeZone OPTIONAL,
       userLocationInformation[32] OCTET STRING OPTIONAL,
       sGWChange [34] SGWChange OPTIONAL,
       servingNodeType[35] SEQUENCE OF ServingNodeType,
       p-GWAddressUsed[36] GSNAddress OPTIONAL,
       p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
       startTime [38] TimeStamp OPTIONAL,
       stopTime  [39] TimeStamp OPTIONAL,
       pDNConnectionChargingID[40] ChargingID OPTIONAL,
       iMSIunauthenticatedFlag [41] NULL OPTIONAL,
       userCSGInformation[42] UserCSGInformation OPTIONAL,
       servedPDPPDNAddressExt [43] PDPAddress OPTIONAL,
    lowPriorityIndicator            [44] NULL OPTIONAL,
    dynamicAddressFlagExt           [47] DynamicAddressFlag OPTIONAL,
    s-GWiPv6Address                         [48] GSNAddress OPTIONAL,
```

```
    servingNodeiPv6Address          [49] SEQUENCE OF GSNAddress
OPTIONAL,
    p-GWiPv6AddressUsed                     [50] GSNAddress OPTIONAL,
    retransmission                          [51] NULL OPTIONAL,
    userLocationInfoTime        [52] TimeStamp OPTIONAL,
   cPCIoTEPSOptimisationIndicator [59] CPCIoTEPSOptimisationIndicator
OPTIONAL,
    uNIPDUCPOnlyFlag            [60] UNIPDUCPOnlyFlag OPTIONAL,
    servingPLMNRateControl      [61] ServingPLMNRateControl OPTIONAL,
    pDPPDNTypeExtension         [62] PDPPDNTypeExtension OPTIONAL,
    listOfRANSecondaryRATUsageReports                    [64] SEQUENCE
OF RANSecondaryRATUsageReport OPTIONAL

}

SGWRecordCust1 ::= SET
{
       recordType[0] RecordType,
       servedIMSI[3] IMSI OPTIONAL,
       s-GWAddress[4] GSNAddress,
       chargingID[5] ChargingID OPTIONAL,
       servingNodeAddress[6] SEQUENCE OF GSNAddress,
       accessPointNameNI[7] AccessPointNameNI OPTIONAL,
       pdpPDNType[8] PDPType OPTIONAL,
       servedPDPPDNAddress[9] PDPAddress OPTIONAL,
       dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
       listOfTrafficVolumes[12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
       recordOpeningTime[13] TimeStamp,
       duration  [14] CallDuration,
       causeForRecClosing[15] CauseForRecClosing,
       diagnostics[16] Diagnostics OPTIONAL,
       recordSequenceNumber[17] INTEGER OPTIONAL,
       nodeID    [18] NodeID OPTIONAL,
    recordExtensions[19] SEQUENCE OF ContentInfo OPTIONAL,
    localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
       apnSelectionMode[21] APNSelectionMode OPTIONAL,
       servedMSISDN[22] MSISDN OPTIONAL,
       chargingCharacteristics[23] ChargingCharacteristics,
       chChSelectionMode[24] ChChSelectionMode OPTIONAL,
       iMSsignalingContext[25] NULL OPTIONAL,
       servingNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
       servedIMEISV[29] IMEI OPTIONAL,
       rATType   [30] RATType OPTIONAL,
```

```
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        sGWChange [34] SGWChange OPTIONAL,
        servingNodeType[35] SEQUENCE OF ServingNodeType,
        p-GWAddressUsed[36] GSNAddress OPTIONAL,
        p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
        startTime [38] TimeStamp OPTIONAL,
        stopTime  [39] TimeStamp OPTIONAL,
        pDNConnectionChargingID[40] ChargingID OPTIONAL,
        iMSIunauthenticatedFlag [41] NULL OPTIONAL,
        userCSGInformation[42] UserCSGInformation OPTIONAL,
        servedPDPPDNAddressExt [43] PDPAddress OPTIONAL,
    lowPriorityIndicator            [44] NULL OPTIONAL,
    dynamicAddressFlagExt           [47] DynamicAddressFlag OPTIONAL,
    s-GWiPv6Address                         [48] GSNAddress OPTIONAL,
    servingNodeiPv6Address          [49] SEQUENCE OF GSNAddress
OPTIONAL,
    p-GWiPv6AddressUsed                     [50] GSNAddress OPTIONAL,
    retransmission                          [51] NULL OPTIONAL,
    userLocationInfoTime        [52] TimeStamp OPTIONAL,
   cPCIoTEPSOptimisationIndicator [59] CPCIoTEPSOptimisationIndicator
OPTIONAL,
    uNIPDUCPOnlyFlag            [60] UNIPDUCPOnlyFlag OPTIONAL,
    servingPLMNRateControl      [61] ServingPLMNRateControl OPTIONAL,
    pDPPDNTypeExtension         [62] PDPPDNTypeExtension OPTIONAL,
    listOfRANSecondaryRATUsageReports                   [64] SEQUENCE
OF RANSecondaryRATUsageReport OPTIONAL
}

PGWRecord ::= SET
{
        recordType[0] RecordType,
        servedIMSI[3] IMSI OPTIONAL,
        p-GWAddress[4] GSNAddress,
        chargingID[5] ChargingID OPTIONAL,
        servingNodeAddress[6] SEQUENCE OF GSNAddress,
        accessPointNameNI[7] AccessPointNameNI OPTIONAL,
        pdpPDNType[8] PDPType OPTIONAL,
        servedPDPPDNAddress[9] PDPAddress OPTIONAL,
        dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
         listOfTrafficVolumes[12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
         recordOpeningTime[13] TimeStamp,
        duration  [14] CallDuration,
```

```
        causeForRecClosing[15] CauseForRecClosing,
        diagnostics[16] Diagnostics OPTIONAL,
        recordSequenceNumber[17] INTEGER OPTIONAL,
        nodeID    [18] NodeID OPTIONAL,
        recordExtensions[19] ManagementExtensions OPTIONAL,
        localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[21] APNSelectionMode OPTIONAL,
        servedMSISDN[22] MSISDN OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        iMSsignalingContext[25] NULL OPTIONAL,
        externalChargingID[26] OCTET STRING OPTIONAL,
        servinggNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
        pSFurnishChargingInformation[28] PSFurnishChargingInformation
OPTIONAL,
        servedIMEISV[29] IMEI OPTIONAL,
        rATType   [30] RATType OPTIONAL,
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        cAMELChargingInformation[33] OCTET STRING OPTIONAL,
        listOfServiceData[34] SEQUENCE OF ChangeOfServiceCondition
OPTIONAL,
        servingNodeType[35] SEQUENCE OF ServingNodeType,
        servedMNNAI[36] SubscriptionID OPTIONAL,
        p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
        startTime [38] TimeStamp OPTIONAL,
        stopTime  [39] TimeStamp OPTIONAL,
        served3gpp2MEID[40] OCTET STRING OPTIONAL,
        pDNConnectionChargingID[41] ChargingID OPTIONAL,
        iMSIunauthenticatedFlag [42] NULL OPTIONAL,
        userCSGInformation[43] UserCSGInformation OPTIONAL,
        threeGPP2UserLocationInformation[44] OCTET STRING OPTIONAL,
        servedPDPPDNAddressExt [45] PDPAddress OPTIONAL,
    lowPriorityIndicator                    [46] NULL OPTIONAL,
    dynamicAddressFlagExt                   [47] DynamicAddressFlag
OPTIONAL,
    servingNodeiPv6Address                [49] SEQUENCE OF GSNAddress
OPTIONAL,
    p-GWiPv6AddressUsed                             [50] GSNAddress
OPTIONAL,
    tWANUserLocationInformation             [51] TWANUserLocationInfo
OPTIONAL,
    retransmission                                 [52] NULL OPTIONAL,
    userLocationInfoTime                    [53] TimeStamp OPTIONAL,
```

```
    uWANUserLocationInformation             [62] UWANUserLocationInfo
OPTIONAL,
    sGiPtPTunnellingMethod          [64] SGiPtPTunnellingMethod
OPTIONAL,
    uNIPDUCPOnlyFlag                [65] UNIPDUCPOnlyFlag OPTIONAL,
    servingPLMNRateControl          [66] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                  [67] APNRateControl OPTIONAL,
    pDPPDNTypeExtension             [68] PDPPDNTypeExtension OPTIONAL,
    sCSASAddress                    [71] SCSASAddress OPTIONAL,
    listOfRANSecondaryRATUsageReports    [73] SEQUENCE OF
RANSecondaryRATUsageReport OPTIONAL
}

PGWRecordRel84 ::= SET
{
       recordType[0] RecordType,
       servedIMSI[3] IMSI OPTIONAL,
       p-GWAddress[4] GSNAddress,
       chargingID[5] ChargingID OPTIONAL,
       servingNodeAddress[6] SEQUENCE OF GSNAddress,
       accessPointNameNI[7] AccessPointNameNI OPTIONAL,
       pdpPDNType[8] PDPType OPTIONAL,
       servedPDPPDNAddress[9] PDPAddress OPTIONAL,
       dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
       recordOpeningTime[13] TimeStamp,
       duration  [14] CallDuration,
       causeForRecClosing[15] CauseForRecClosing,
       diagnostics[16] Diagnostics OPTIONAL,
       recordSequenceNumber[17] INTEGER OPTIONAL,
       nodeID    [18] NodeID OPTIONAL,
       recordExtensions[19] ManagementExtensions OPTIONAL,
       localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
       apnSelectionMode[21] APNSelectionMode OPTIONAL,
       servedMSISDN[22] MSISDN OPTIONAL,
       chargingCharacteristics[23] ChargingCharacteristics,
       chChSelectionMode[24] ChChSelectionMode OPTIONAL,
       iMSsignalingContext[25] NULL OPTIONAL,
       externalChargingID[26] OCTET STRING OPTIONAL,
       servinggNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
       pSFurnishChargingInformation[28] PSFurnishChargingInformation
OPTIONAL,
       servedIMEISV[29] IMEI OPTIONAL,
       rATType   [30] RATType OPTIONAL,
```

```
       mSTimeZone [31] MSTimeZone OPTIONAL,
       userLocationInformation[32] OCTET STRING OPTIONAL,
       cAMELChargingInformation[33] OCTET STRING OPTIONAL,
       listOfServiceDataRel84[34] SEQUENCE OF
ChangeOfServiceConditionRel84 OPTIONAL,
       servingNodeType[35] SEQUENCE OF ServingNodeType,
       servedMNNAI[36] SubscriptionID OPTIONAL,
       p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
       startTime [38] TimeStamp OPTIONAL,
       stopTime  [39] TimeStamp OPTIONAL,
       served3gpp2MEID[40] OCTET STRING OPTIONAL,
       pDNConnectionChargingID[41] ChargingID OPTIONAL,
       iMSIunauthenticatedFlag [42] NULL OPTIONAL,
       userCSGInformation[43] UserCSGInformation OPTIONAL,
       threeGPP2UserLocationInformation[44] OCTET STRING OPTIONAL,
       servedPDPPDNAddressExt [45] PDPAddress OPTIONAL,
    lowPriorityIndicator                    [46] NULL OPTIONAL,
        dynamicAddressFlagExt                    [47] DynamicAddressFlag
OPTIONAL,
        servingNodeiPv6Address                    [49] SEQUENCE OF
GSNAddress OPTIONAL,
        p-GWiPv6AddressUsed                               [50] GSNAddress
OPTIONAL,
        tWANUserLocationInformation             [51]
TWANUserLocationInfo OPTIONAL,
        retransmission                                   [52] NULL
OPTIONAL,
        userLocationInfoTime                     [53] TimeStamp OPTIONAL,
       uWANUserLocationInformation             [62]
UWANUserLocationInfo OPTIONAL,
    sGiPtPTunnellingMethod          [64] SGiPtPTunnellingMethod
OPTIONAL,
    uNIPDUCPOnlyFlag                [65] UNIPDUCPOnlyFlag OPTIONAL,
    servingPLMNRateControl          [66] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                  [67] APNRateControl OPTIONAL,
    pDPPDNTypeExtension             [68] PDPPDNTypeExtension OPTIONAL,
    sCSASAddress                    [71] SCSASAddress OPTIONAL,
    listOfRANSecondaryRATUsageReports    [73] SEQUENCE OF
RANSecondaryRATUsageReport OPTIONAL
}

PGWRecordCust1 ::= SET
{
       recordType[0] RecordType,
```

```
        servedIMSI[3] IMSI OPTIONAL,
        p-GWAddress[4] GSNAddress,
        chargingID[5] ChargingID OPTIONAL,
        servingNodeAddress[6] SEQUENCE OF GSNAddress,
        accessPointNameNI[7] AccessPointNameNI OPTIONAL,
        pdpPDNType[8] PDPType OPTIONAL,
        servedPDPPDNAddress[9] PDPAddress OPTIONAL,
        dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
    listOfTrafficVolumes    [12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
    recordOpeningTime[13] TimeStamp,
        duration  [14] CallDuration,
        causeForRecClosing[15] CauseForRecClosing,
        diagnostics[16] Diagnostics OPTIONAL,
        recordSequenceNumber[17] INTEGER OPTIONAL,
        nodeID    [18] NodeID OPTIONAL,
        recordExtensions[19] SEQUENCE OF ContentInfo OPTIONAL,
        localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[21] APNSelectionMode OPTIONAL,
        servedMSISDN[22] MSISDN OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        iMSsignalingContext[25] NULL OPTIONAL,
        externalChargingID[26] OCTET STRING OPTIONAL,
        servinggNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
        pSFurnishChargingInformation[28] PSFurnishChargingInformation
OPTIONAL,
        servedIMEISV[29] IMEI OPTIONAL,
        rATType   [30] RATType OPTIONAL,
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        cAMELChargingInformation[33] OCTET STRING OPTIONAL,
        listOfServiceData[34] SEQUENCE OF ChangeOfServiceCondition
OPTIONAL,
        servingNodeType[35] SEQUENCE OF ServingNodeType,
        servedMNNAI[36] SubscriptionID OPTIONAL,
        p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
        startTime [38] TimeStamp OPTIONAL,
        stopTime  [39] TimeStamp OPTIONAL,
        served3gpp2MEID[40] OCTET STRING OPTIONAL,
        pDNConnectionChargingID[41] ChargingID OPTIONAL,
        iMSIunauthenticatedFlag [42] NULL OPTIONAL,
        userCSGInformation[43] UserCSGInformation OPTIONAL,
```

```
        threeGPP2UserLocationInformation[44] OCTET STRING OPTIONAL,
        servedPDPPDNAddressExt [45] PDPAddress OPTIONAL,
    lowPriorityIndicator                    [46] NULL OPTIONAL,
    dynamicAddressFlagExt                   [47] DynamicAddressFlag
OPTIONAL,
    servingNodeiPv6Address                 [49] SEQUENCE OF GSNAddress
OPTIONAL,
    p-GWiPv6AddressUsed                                [50] GSNAddress
OPTIONAL,
    tWANUserLocationInformation             [51] TWANUserLocationInfo
OPTIONAL,
    retransmission                                 [52] NULL OPTIONAL,
    userLocationInfoTime                    [53] TimeStamp OPTIONAL,
    uWANUserLocationInformation             [62] UWANUserLocationInfo
OPTIONAL,
    sGiPtPTunnellingMethod          [64] SGiPtPTunnellingMethod
OPTIONAL,
    uNIPDUCPOnlyFlag                [65] UNIPDUCPOnlyFlag OPTIONAL,
    servingPLMNRateControl          [66] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                  [67] APNRateControl OPTIONAL,
    pDPPDNTypeExtension             [68] PDPPDNTypeExtension OPTIONAL,
    sCSASAddress                    [71] SCSASAddress OPTIONAL,
    listOfRANSecondaryRATUsageReports    [73] SEQUENCE OF
RANSecondaryRATUsageReport OPTIONAL
}

PGWRecordRel84Cust1 ::= SET
{
        recordType[0] RecordType,
        servedIMSI[3] IMSI OPTIONAL,
        p-GWAddress[4] GSNAddress,
        chargingID[5] ChargingID OPTIONAL,
        servingNodeAddress[6] SEQUENCE OF GSNAddress,
        accessPointNameNI[7] AccessPointNameNI OPTIONAL,
        pdpPDNType[8] PDPType OPTIONAL,
        servedPDPPDNAddress[9] PDPAddress OPTIONAL,
        dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
        recordOpeningTime[13] TimeStamp,
        duration  [14] CallDuration,
        causeForRecClosing[15] CauseForRecClosing,
        diagnostics[16] Diagnostics OPTIONAL,
        recordSequenceNumber[17] INTEGER OPTIONAL,
        nodeID    [18] NodeID OPTIONAL,
        recordExtensions[19] SEQUENCE OF ContentInfo OPTIONAL,
```

```
        localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[21] APNSelectionMode OPTIONAL,
        servedMSISDN[22] MSISDN OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        iMSsignalingContext[25] NULL OPTIONAL,
        externalChargingID[26] OCTET STRING OPTIONAL,
        servinggNodePLMNIdentifier[27] PLMN-Id OPTIONAL,
        pSFurnishChargingInformation[28] PSFurnishChargingInformation
OPTIONAL,
        servedIMEISV[29] IMEI OPTIONAL,
        rATType   [30] RATType OPTIONAL,
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        cAMELChargingInformation[33] OCTET STRING OPTIONAL,
        listOfServiceDataRel84[34] SEQUENCE OF
ChangeOfServiceConditionRel84 OPTIONAL,
        servingNodeType[35] SEQUENCE OF ServingNodeType,
        servedMNNAI[36] SubscriptionID OPTIONAL,
        p-GWPLMNIdentifier[37] PLMN-Id OPTIONAL,
        startTime [38] TimeStamp OPTIONAL,
        stopTime  [39] TimeStamp OPTIONAL,
        served3gpp2MEID[40] OCTET STRING OPTIONAL,
        pDNConnectionChargingID[41] ChargingID OPTIONAL,
        iMSIunauthenticatedFlag [42] NULL OPTIONAL,
        userCSGInformation[43] UserCSGInformation OPTIONAL,
        threeGPP2UserLocationInformation[44] OCTET STRING OPTIONAL,
        servedPDPPDNAddressExt [45] PDPAddress OPTIONAL,
    lowPriorityIndicator                    [46] NULL OPTIONAL,
        dynamicAddressFlagExt                   [47] DynamicAddressFlag
OPTIONAL,
        servingNodeiPv6Address                   [49] SEQUENCE OF
GSNAddress OPTIONAL,
        p-GWiPv6AddressUsed                             [50] GSNAddress
OPTIONAL,
        tWANUserLocationInformation             [51]
TWANUserLocationInfo OPTIONAL,
        retransmission                                    [52] NULL
OPTIONAL,
        userLocationInfoTime                    [53] TimeStamp OPTIONAL,
        uWANUserLocationInformation             [62]
UWANUserLocationInfo OPTIONAL,
    sGiPtPTunnellingMethod          [64] SGiPtPTunnellingMethod
OPTIONAL,
    uNIPDUCPOnlyFlag                [65] UNIPDUCPOnlyFlag OPTIONAL,
```

```
    servingPLMNRateControl          [66] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                  [67] APNRateControl OPTIONAL,
    pDPPDNTypeExtension             [68] PDPPDNTypeExtension OPTIONAL,
    sCSASAddress                    [71] SCSASAddress OPTIONAL,
    listOfRANSecondaryRATUsageReports    [73] SEQUENCE OF
RANSecondaryRATUsageReport OPTIONAL
}

SGSNMMRecord::= SET
{
       recordType[0] RecordType,
       servedIMSI[1] IMSI,
       servedIMEI[2] IMEI OPTIONAL,
       sgsnAddress[3] GSNAddress OPTIONAL,
       msNetworkCapability[4] MSNetworkCapability OPTIONAL,
       routingArea[5] RoutingAreaCode OPTIONAL,
       locationAreaCode[6] LocationAreaCode OPTIONAL,
       cellIdentifier[7] CellId OPTIONAL,
       changeLocation[8] SEQUENCE OF ChangeLocation OPTIONAL,
       recordOpeningTime[9] TimeStamp,
       duration  [10] CallDuration OPTIONAL,
       sgsnChange[11] SGSNChange OPTIONAL,
       causeForRecClosing[12] CauseForRecClosing,
       diagnostics[13] Diagnostics OPTIONAL,
       recordSequenceNumber[14] INTEGER OPTIONAL,
       nodeID    [15] NodeID OPTIONAL,
       recordExtensions[16] ManagementExtensions OPTIONAL,
       localSequenceNumber[17] LocalSequenceNumber OPTIONAL,
       servedMSISDN[18] MSISDN OPTIONAL,
       chargingCharacteristics[19] ChargingCharacteristics,
       cAMELInformationMM [20] CAMELInformationMM OPTIONAL,
       rATType   [21] RATType OPTIONAL,
       chChSelectionMode[22] ChChSelectionMode OPTIONAL,
       cellPLMNId[23] PLMN-Id OPTIONAL
}

SGSNPDPRecord ::= SET
{
       recordType[0] RecordType,
       networkInitiation[1] NetworkInitiatedPDPContext OPTIONAL,
       servedIMSI[3] IMSI OPTIONAL,
       servedIMEI[4] IMEI OPTIONAL,
```

```
        sgsnAddress[5] GSNAddress OPTIONAL,
        msNetworkCapability[6] MSNetworkCapability OPTIONAL,
        routingArea[7] RoutingAreaCode OPTIONAL,
        locationAreaCode[8] LocationAreaCode OPTIONAL,
        cellIdentifier[9] CellId OPTIONAL,
        chargingID[10] ChargingID,
        ggsnAddressUsed[11] GSNAddress,
        accessPointNameNI[12] AccessPointNameNI OPTIONAL,
        pdpType   [13] PDPType OPTIONAL,
        servedPDPAddress[14] PDPAddress OPTIONAL,
        listOfTrafficVolumes[15] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
        recordOpeningTime[16] TimeStamp,
        duration  [17] CallDuration,
        sgsnChange[18] SGSNChange OPTIONAL,
        causeForRecClosing[19] CauseForRecClosing,
        diagnostics[20] Diagnostics OPTIONAL,
        recordSequenceNumber[21] INTEGER OPTIONAL,
        nodeID    [22] NodeID OPTIONAL,
        recordExtensions[23] ManagementExtensions OPTIONAL,
        localSequenceNumber[24] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[25] APNSelectionMode OPTIONAL,
        accessPointNameOI[26] AccessPointNameOI OPTIONAL,
        servedMSISDN[27] MSISDN OPTIONAL,
        chargingCharacteristics[28] ChargingCharacteristics,
        rATType   [29] RATType OPTIONAL,
        cAMELInformationPDP  [30] CAMELInformationPDP OPTIONAL,
        rNCUnsentDownlinkVolume[31] DataVolumeGPRS OPTIONAL,
        chChSelectionMode[32] ChChSelectionMode OPTIONAL,
        dynamicAddressFlag[33] DynamicAddressFlag OPTIONAL,
        iMSIunauthenticatedFlag [34] NULL OPTIONAL,
        userCSGInformation[35] UserCSGInformation OPTIONAL,
        servedPDPPDNAddressExt [36] PDPAddress OPTIONAL
}

SGSNSMORecord::= SET
{
        recordType[0] RecordType,
        servedIMSI[1] IMSI,
        servedIMEI[2] IMEI OPTIONAL,
        servedMSISDN[3] MSISDN OPTIONAL,
        msNetworkCapability[4] MSNetworkCapability OPTIONAL,
        serviceCentre[5] AddressString OPTIONAL,
```

```
        recordingEntity[6] RecordingEntity OPTIONAL,
        locationArea[7] LocationAreaCode OPTIONAL,
        routingArea[8] RoutingAreaCode OPTIONAL,
        cellIdentifier[9] CellId OPTIONAL,
        messageReference[10] MessageReference,
        eventTimeStamp[11] TimeStamp,
        smsResult [12] SMSResult OPTIONAL,
        recordExtensions[13] ManagementExtensions OPTIONAL,
        nodeID    [14] NodeID OPTIONAL,
        localSequenceNumber[15] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[16] ChargingCharacteristics,
        rATType   [17] RATType OPTIONAL,
        destinationNumber[18] SmsTpDestinationNumber OPTIONAL,
        cAMELInformationSMS[19] CAMELInformationSMS OPTIONAL,
        chChSelectionMode[20] ChChSelectionMode OPTIONAL
}

SGSNSMTRecord::= SET
{
        recordType[0] RecordType,
        servedIMSI[1] IMSI,
        servedIMEI[2] IMEI OPTIONAL,
        servedMSISDN[3] MSISDN OPTIONAL,
        msNetworkCapability[4] MSNetworkCapability OPTIONAL,
        serviceCentre[5] AddressString OPTIONAL,
        recordingEntity[6] RecordingEntity OPTIONAL,
        locationArea[7] LocationAreaCode OPTIONAL,
        routingArea[8] RoutingAreaCode OPTIONAL,
        cellIdentifier[9] CellId OPTIONAL,
        eventTimeStamp[10] TimeStamp,
        smsResult [11] SMSResult OPTIONAL,
        recordExtensions[12] ManagementExtensions OPTIONAL,
        nodeID    [13] NodeID OPTIONAL,
        localSequenceNumber[14] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[15] ChargingCharacteristics,
        rATType   [16] RATType OPTIONAL,
        chChSelectionMode[17] ChChSelectionMode OPTIONAL,
        cAMELInformationSMS[18] CAMELInformationSMS OPTIONAL
}

SGSNMTLCSRecord::= SET
{
        recordType[0] RecordType,
```

```
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType,
        lcsClientIdentity[3] LCSClientIdentity,
        servedIMSI[4] IMSI,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        locationType[7] LocationType,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString,
        eventTimeStamp[11] TimeStamp,
        measurementDuration[12] CallDuration OPTIONAL,
        notificationToMSUser[13] NotificationToMSUser OPTIONAL,
        privacyOverride[14] NULL OPTIONAL,
        location  [15] LocationAreaAndCell OPTIONAL,
        routingArea[16] RoutingAreaCode OPTIONAL,
        locationEstimate[17] Ext-GeographicalInformation OPTIONAL,
        positioningData[18] PositioningData OPTIONAL,
        lcsCause  [19] LCSCause OPTIONAL,
        diagnostics[20] Diagnostics OPTIONAL,
        nodeID    [21] NodeID OPTIONAL,
        localSequenceNumber[22] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        rATType   [25] RATType OPTIONAL,
        recordExtensions[26] ManagementExtensions OPTIONAL,
        causeForRecClosing[27] CauseForRecClosing
}

SGSNMOLCSRecord::= SET
{
        recordType[0] RecordType,
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType OPTIONAL,
        lcsClientIdentity[3] LCSClientIdentity OPTIONAL,
        servedIMSI[4] IMSI,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        locationMethod[7] LocationMethod,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString OPTIONAL,
        eventTimeStamp[11] TimeStamp,
```

```
        measurementDuration[12] CallDuration OPTIONAL,
        location  [13] LocationAreaAndCell OPTIONAL,
        routingArea[14] RoutingAreaCode OPTIONAL,
        locationEstimate[15] Ext-GeographicalInformation OPTIONAL,
        positioningData[16] PositioningData OPTIONAL,
        lcsCause  [17] LCSCause OPTIONAL,
        diagnostics[18] Diagnostics OPTIONAL,
        nodeID    [19] NodeID OPTIONAL,
        localSequenceNumber    [20] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[21] ChargingCharacteristics,
        chChSelectionMode[22] ChChSelectionMode OPTIONAL,
        rATType   [23] RATType OPTIONAL,
        recordExtensions[24] ManagementExtensions OPTIONAL,
        causeForRecClosing[25] CauseForRecClosing
}

SGSNNILCSRecord::= SET
{
        recordType[0] RecordType,
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType OPTIONAL,
        lcsClientIdentity[3] LCSClientIdentity OPTIONAL,
        servedIMSI[4] IMSI OPTIONAL,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        servedIMEI[7] IMEI OPTIONAL,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString OPTIONAL,
        eventTimeStamp[11] TimeStamp,
        measurementDuration[12] CallDuration OPTIONAL,
        location  [13] LocationAreaAndCell OPTIONAL,
        routingArea    [14] RoutingAreaCode OPTIONAL,
        locationEstimate[15] Ext-GeographicalInformation OPTIONAL,
        positioningData[16] PositioningData OPTIONAL,
        lcsCause  [17] LCSCause OPTIONAL,
        diagnostics[18] Diagnostics OPTIONAL,
        nodeID    [19] NodeID OPTIONAL,
        localSequenceNumber    [20] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[21] ChargingCharacteristics,
        chChSelectionMode[22] ChChSelectionMode OPTIONAL,
        rATType   [23] RATType OPTIONAL,
        recordExtensions[24] ManagementExtensions OPTIONAL,
```

```
        causeForRecClosing[25] CauseForRecClosing
}

SGSNMBMSRecord ::= SET
{
        recordType[0] RecordType,
        ggsnAddress[1] GSNAddress,
        chargingID[2] ChargingID,
        listofRAs [3] SEQUENCE OF RAIdentity OPTIONAL,
        accessPointNameNI[4] AccessPointNameNI OPTIONAL,
        servedPDPAddress[5] PDPAddress OPTIONAL,
        listOfTrafficVolumes[6] SEQUENCE OF ChangeOfMBMSCondition
OPTIONAL,
        recordOpeningTime[7] TimeStamp,
        duration  [8] CallDuration,
        causeForRecClosing[9] CauseForRecClosing,
        diagnostics[10] Diagnostics OPTIONAL,
        recordSequenceNumber[11] INTEGER OPTIONAL,
        nodeID    [12] NodeID OPTIONAL,
        recordExtensions[13] ManagementExtensions OPTIONAL,
        localSequenceNumber[14] LocalSequenceNumber OPTIONAL,
        sgsnPLMNIdentifier[15] PLMN-Id OPTIONAL,
        numberofReceivingUE[16] INTEGER OPTIONAL,
        mbmsInformation[17] MBMSInformation OPTIONAL
}

GGSNMBMSRecord ::= SET
{
        recordType[0] RecordType,
        ggsnAddress[1] GSNAddress,
        chargingID[2] ChargingID,
        listofDownstreamNodes[3] SEQUENCE OF GSNAddress,
        accessPointNameNI[4] AccessPointNameNI OPTIONAL,
        servedPDPAddress[5] PDPAddress OPTIONAL,
        listOfTrafficVolumes[6] SEQUENCE OF ChangeOfMBMSCondition
OPTIONAL,
        recordOpeningTime[7] TimeStamp,
        duration  [8] CallDuration,
        causeForRecClosing[9] CauseForRecClosing,
        diagnostics[10] Diagnostics OPTIONAL,
        recordSequenceNumber[11] INTEGER OPTIONAL,
        nodeID    [12] NodeID OPTIONAL,
        recordExtensions[13] ManagementExtensions OPTIONAL,
```

```
        localSequenceNumber[14] LocalSequenceNumber OPTIONAL,
        mbmsInformation[15] MBMSInformation OPTIONAL
}
GWMBMSRecord ::= SET
{
        recordType[0] RecordType,
        mbmsGWAddress[1] GSNAddress,
        chargingID[2] ChargingID,
        listofDownstreamNodes[3] SEQUENCE OF GSNAddress,
        accessPointNameNI[4] AccessPointNameNI OPTIONAL,
        pdpPDNType[5] PDPType OPTIONAL,
        servedPDPPDNAddress[6] PDPAddress OPTIONAL,
        listOfTrafficVolumes[7] SEQUENCE OF ChangeOfMBMSCondition
OPTIONAL,
        recordOpeningTime[8] TimeStamp,
        duration  [9] CallDuration,
        causeForRecClosing[10] CauseForRecClosing,
        diagnostics[11] Diagnostics OPTIONAL,
        recordSequenceNumber[12] INTEGER OPTIONAL,
        nodeID    [13] NodeID OPTIONAL,
        recordExtensions[14] ManagementExtensions OPTIONAL,
        localSequenceNumber[15] LocalSequenceNumber OPTIONAL,
        mbmsInformation[16] MBMSInformation OPTIONAL,
        commonTeid   [17] CTEID OPTIONAL,
        iPMulticastSourceAddress[18] PDPAddress OPTIONAL
}


GGSNPDPRecord ::= SET
{
        recordType[0] CallEventRecordType,
        networkInitiation[1] NetworkInitiatedPDPContext OPTIONAL,
        servedIMSI[3] IMSI,
        ggsnAddress[4] GSNAddress,
        chargingID[5] ChargingID,
        sgsnAddress[6] SEQUENCE OF GSNAddress,
        accessPointNameNI[7] AccessPointNameNI OPTIONAL,
        pdpType   [8] PDPType OPTIONAL,
        servedPDPAddress[9] PDPAddress OPTIONAL,
        dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
        listOfTrafficVolumes[12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
        recordOpeningTime[13] TimeStamp,
```

```
        duration  [14] CallDuration,
        causeForRecClosing[15] CauseForRecClosing,
        diagnostics[16] Diagnostics OPTIONAL,
        recordSequenceNumber[17] INTEGER OPTIONAL,
        nodeID    [18] NodeID OPTIONAL,
        recordExtensions[19] ContentInfo OPTIONAL,
        localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[21] APNSelectionMode OPTIONAL,
        servedMSISDN[22] MSISDN OPTIONAL,
        chargingCharacteristics    [23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        iMSsignalingContext[25] NULL OPTIONAL,
        externalChargingID[26] OCTET STRING OPTIONAL,
        sgsnPLMNIdentifier[27] PLMN-Id OPTIONAL,
        servedIMEISV[29] IMEI OPTIONAL,
        rATType   [30] RATType OPTIONAL,
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        cAMELChargingInformation[33] OCTET STRING OPTIONAL

}

EGSNPDPRecord ::= SET
{
        recordType[0] CallEventRecordType,
        networkInitiation[1] NetworkInitiatedPDPContext OPTIONAL,
        servedIMSI[3] IMSI,
        ggsnAddress[4] GSNAddress,
        chargingID[5] ChargingID,
        sgsnAddress[6] SEQUENCE OF GSNAddress,
        accessPointNameNI[7] AccessPointNameNI OPTIONAL,
        pdpType   [8] PDPType OPTIONAL,
        servedPDPAddress[9] PDPAddress OPTIONAL,
        dynamicAddressFlag[11] DynamicAddressFlag OPTIONAL,
        listOfTrafficVolumes[12] SEQUENCE OF ChangeOfCharCondition
OPTIONAL,
        recordOpeningTime[13] TimeStamp,
        duration  [14] CallDuration,
        causeForRecClosing[15] CauseForRecClosing,
        diagnostics[16] Diagnostics OPTIONAL,
        recordSequenceNumber[17] INTEGER OPTIONAL,
        nodeID    [18] NodeID OPTIONAL,
        recordExtensions[19] ContentInfo OPTIONAL,
```

```
        localSequenceNumber[20] LocalSequenceNumber OPTIONAL,
        apnSelectionMode[21] APNSelectionMode OPTIONAL,
        servedMSISDN[22] MSISDN OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        iMSsignalingContext[25] NULL OPTIONAL,
        externalChargingID[26] OCTET STRING OPTIONAL,
        sgsnPLMNIdentifier[27] PLMN-Id OPTIONAL,
        pSFurnishChargingInformation[28] PSFurnishChargingInformation
OPTIONAL,
        servedIMEISV[29] IMEI OPTIONAL,
        rATType   [30] RATType OPTIONAL,
        mSTimeZone [31] MSTimeZone OPTIONAL,
        userLocationInformation[32] OCTET STRING OPTIONAL,
        cAMELChargingInformation[33] OCTET STRING OPTIONAL,
        listOfServiceData[34] SEQUENCE OF ChangeOfServiceCondition
OPTIONAL
}

SGSNLCTRecord::= SET
{
        recordType[0] CallEventRecordType,
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType,
        lcsClientIdentity[3] LCSClientIdentity,
        servedIMSI[4] IMSI,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        locationType[7] LocationType,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString,
        eventTimeStamp[11] TimeStamp,
        measurementDuration[12] CallDuration OPTIONAL,
        notificationToMSUser[13] NotificationToMSUser OPTIONAL,
        privacyOverride[14] NULL OPTIONAL,
        location  [15] LocationAreaAndCell OPTIONAL,
        routingArea[16] RoutingAreaCode OPTIONAL,
        locationEstimate[17] Ext-GeographicalInformation OPTIONAL,
        positioningData[18] PositioningData OPTIONAL,
        lcsCause  [19] LCSCause OPTIONAL,
        diagnostics[20] Diagnostics OPTIONAL,
        nodeID    [21] NodeID OPTIONAL,
```

```
        localSequenceNumber    [22] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[23] ChargingCharacteristics,
        chChSelectionMode[24] ChChSelectionMode OPTIONAL,
        rATType   [25] RATType OPTIONAL,
        recordExtensions[26] ManagementExtensions OPTIONAL,
        causeForRecClosing[27] CauseForRecClosing
}

SGSNLCORecord::= SET
{
        recordType[0] CallEventRecordType,
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType OPTIONAL,
        lcsClientIdentity[3] LCSClientIdentity OPTIONAL,
        servedIMSI[4] IMSI,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        locationMethod[7] LocationMethod,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString OPTIONAL,
        eventTimeStamp[11] TimeStamp,
        measurementDuration[12] CallDuration OPTIONAL,
        location  [13] LocationAreaAndCell OPTIONAL,
        routingArea[14] RoutingAreaCode OPTIONAL,
        locationEstimate[15] Ext-GeographicalInformation OPTIONAL,
        positioningData[16] PositioningData OPTIONAL,
        lcsCause  [17] LCSCause OPTIONAL,
        diagnostics[18] Diagnostics OPTIONAL,
        nodeID    [19] NodeID OPTIONAL,
        localSequenceNumber    [20] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[21] ChargingCharacteristics,
        chChSelectionMode[22] ChChSelectionMode OPTIONAL,
        rATType   [23] RATType OPTIONAL,
        recordExtensions[24] ManagementExtensions OPTIONAL,
        causeForRecClosing[25] CauseForRecClosing
}

SGSNLCNRecord::= SET
{
        recordType[0] CallEventRecordType,
        recordingEntity[1] RecordingEntity,
        lcsClientType[2] LCSClientType OPTIONAL,
```

```
        lcsClientIdentity[3] LCSClientIdentity OPTIONAL,
        servedIMSI[4] IMSI OPTIONAL,
        servedMSISDN[5] MSISDN OPTIONAL,
        sgsnAddress[6] GSNAddress OPTIONAL,
        servedIMEI[7] IMEI OPTIONAL,
        lcsQos    [8] LCSQoSInfo OPTIONAL,
        lcsPriority[9] LCS-Priority OPTIONAL,
        mlcNumber [10] ISDN-AddressString OPTIONAL,
        eventTimeStamp[11] TimeStamp,
        measurementDuration[12] CallDuration OPTIONAL,
        location  [13] LocationAreaAndCell OPTIONAL,
        routingArea[14] RoutingAreaCode OPTIONAL,
        locationEstimate[15] Ext-GeographicalInformation OPTIONAL,
        positioningData[16] PositioningData OPTIONAL,
        lcsCause  [17] LCSCause OPTIONAL,
        diagnostics[18] Diagnostics OPTIONAL,
        nodeID    [19] NodeID OPTIONAL,
        localSequenceNumber    [20] LocalSequenceNumber OPTIONAL,
        chargingCharacteristics[21] ChargingCharacteristics,
        chChSelectionMode[22] ChChSelectionMode OPTIONAL,
        rATType   [23] RATType OPTIONAL,
        recordExtensions[24] ManagementExtensions OPTIONAL,
        causeForRecClosing[25] CauseForRecClosing
}


----------------------------------------------------------------------
--------
--
--  GPRS DATA TYPES
--
----------------------------------------------------------------------
--------

AddressString ::= OCTET STRING (SIZE (1..maxAddressLength))
        -- This type is used to represent a number for addressing
        -- purposes. It is composed of
        -- a)  one octet for nature of address, and numbering plan
        --     indicator.
        -- b)  digits of an address encoded as TBCD-String.

        -- a)The first octet includes a one bit extension indicator, a
        --     3 bits nature of address indicator and a 4 bits numbering
```

```
        --     plan indicator, encoded as follows:

        -- bit 8: 1  (no extension)

        -- bits 765: nature of address indicator
        -- 000  unknown
        -- 001  international number
        -- 010  national significant number
        -- 011  network specific number
        -- 100  subscriber number
        -- 101  reserved
        -- 110  abbreviated number
        -- 111  reserved for extension

        -- bits 4321: numbering plan indicator
        -- 0000  unknown
        -- 0001  ISDN/Telephony Numbering Plan (Rec CCITT E.164)
        -- 0010  spare
        -- 0011  data numbering plan (CCITT Rec X.121)
        -- 0100  telex numbering plan (CCITT Rec F.69)
        -- 0101  spare
        -- 0110  land mobile numbering plan (CCITT Rec E.212)
        -- 0111  spare
        -- 1000  national numbering plan
        -- 1001  private numbering plan
        -- 1111  reserved for extension

        -- all other values are reserved.

        -- b)The following octets representing digits of an address
        --     encoded as a TBCD-STRING.

BCDDirectoryNumber::= OCTET STRING
        -- This type contains the binary coded decimal representation of
        -- a directory number e.g. calling/called/connected/translated
number.
        -- The encoding of the octet string is in accordance with the
        -- the elements "Calling party BCD number", "Called party BCD
number"
        -- and "Connected number" defined in TS GSM 04.08.
        -- This encoding includes type of number and number plan
information
        -- together with a BCD encoded digit string.
```

```
       -- It may also contain both a presentation and screening
indicator
       -- (octet 3a).
       -- For the avoidance of doubt, this field does not include
       -- octets 1 and 2, the element name and length, as this would be
       -- redundant.

CallEventRecordType ::= INTEGER
{
--
--     Record values 18..22 are GPRS specific.
--     The contents are defined in TS 32.015
--
       sgsnPDPRecord(18),
       ggsnPDPRecord(19),
       sgsnMMRecord(20),
       sgsnSMORecord(21),
       sgsnSMTRecord(22),

--
--   Record values 26..28 are PS-LCS specific.
--     The contents are defined in  TS 32.215
--

       sgsnMtLCSRecord(26),
       sgsnMoLCSRecord     (27),
       sgsnNiLCSRecord     (28),

--
--   Record values 70 is for Flow based Charging
--   The contents are defined in TS 32.251 [11]
--
       egsnPDPRecord(70),

--
--   Record values 76..79 are MBMS specific.
--   The contents are defined in TS 32.251 [11]
--   Record values 76 and 77 are MBMS bearer context specific
--
       sgsnMBMSRecord(76),
       ggsnMBMSRecord(77)

}
```

```
CallingNumber ::= BCDDirectoryNumber

CallReferenceNumber ::= OCTET STRING (SIZE (1..8))

Diagnostics      ::= CHOICE
{
       gsm0408Cause[0] INTEGER,
       -- See TS 24.008 [64]

       gsm0902MapErrorValue[1] INTEGER,
       -- Note: The value to be stored here corresponds to
       -- the local values defined in the MAP-Errors and
       -- MAP-DialogueInformation modules, for full details
       -- see TS 29.002 [60].

       itu-tQ767Cause[2] INTEGER,
       -- See ITU-T Q.767 [67]

       networkSpecificCause[3] ManagementExtensions,
       -- To be defined by network operator

       manufacturerSpecificCause[4] INTEGER,
       -- To be defined by manufacturer

       positionMethodFailureCause[5] PositionMethodFailure-Diagnostic,
       -- see TS 29.002 [60]
       unauthorizedLCSClientCause[6] UnauthorizedLCSClient-Diagnostic
       -- see TS 29.002 [60]
}

MessageReference::= OCTET STRING

RecordingEntity ::= AddressString

CellId ::= OCTET STRING (SIZE(2))
       --
       -- Coded according to TS GSM 04.08
       --

LocationAreaCode::= OCTET STRING (SIZE(2))
       --
```

```
	-- See TS GSM 04.08
	--

MCC-MNC   ::= OCTET STRING (SIZE(3))
	--
	-- See TS 24.008 [64]
	--

CallDuration ::= INTEGER
	--
	-- The call duration in seconds.
	-- For successful calls this is the chargeable duration.
	-- For call attempts this is the call holding time.
	--

ManagementExtensions::= SET OF ManagementExtension


TimeStamp    ::= OCTET STRING (SIZE(9))
	--
	-- The contents of this field are a compact form of the UTCTime
format
	-- containing local time plus an offset to universal time.
Binary coded
	-- decimal encoding is employed for the digits to reduce the
storage and
	-- transmission overhead
	-- e.g. YYMMDDhhmmssShhmm
	-- where
	-- YY = Year 00 to 99BCD encoded
	-- MM = Month 01 to 12 BCD encoded
	-- DD= Day 01 to 31BCD encoded
	-- hh= hour 00 to 23BCD encoded
	-- mm= minute 00 to 59BCD encoded
	-- ss= second 00 to 59BCD encoded
	-- S=  Sign 0 = "+", "-"ASCII encoded
	-- hh= hour 00 to 23BCD encoded
	-- mm= minute 00 to 59BCD encoded
	--

LevelOfCAMELService::= BIT STRING
{
	basic     (0),
```

```
        callDurationSupervision(1),
        onlineCharging(2)
}

DefaultGPRS-Handling ::= ENUMERATED
{
        continueTransaction (0) ,
        releaseTransaction (1)
}
-- exception handling:
-- reception of values in range 2-31 shall be treated as
"continueTransaction"
-- reception of values greater than 31 shall be treated as
"releaseTransaction"

DefaultSMS-Handling ::= ENUMERATED
{
        continueTransaction (0) ,
        releaseTransaction (1)
}
--      exception handling:
--      reception of values in range 2-31 shall be treated as
"continueTransaction"
--      reception of values greater than 31 shall be treated as
"releaseTransaction"

DataVolumeMBMS::= INTEGER
        --
        -- The volume of data transferred in octets.
        --

DeferredLocationEventType ::= BIT STRING
{
        msAvailable(0)
} (SIZE (1..16))

        -- exception handling
        -- a ProvideSubscriberLocation-Arg containing other values than
listed above in
        -- DeferredLocationEventType shall be rejected by the receiver
with a return error cause of
        -- unexpected data value.

ETSIAddress::= AddressString
```

```
	--
	-- First octet for nature of address, and numbering plan 
indicator (3 for X.121)
	-- Other octets TBCD
	-- See TS 29.002 [60]
	--

Ext-GeographicalInformation ::= OCTET STRING (SIZE 
(1..maxExt-GeographicalInformation))
	-- Refers to geographical Information defined in 3G TS 23.032.
	-- This is composed of 1 or more octets with an internal 
structure according to
	-- 3G TS 23.032
	-- Octet 1: Type of shape, only the following shapes in 3G TS 
23.032 are allowed:
	--	(a) Ellipsoid point with uncertainty circle
	--	(b) Ellipsoid point with uncertainty ellipse
	--	(c) Ellipsoid point with altitude and uncertainty 
ellipsoid
	--	(d) Ellipsoid Arc
	--	(e) Ellipsoid Point
	-- Any other value in octet 1 shall be treated as invalid
	-- Octets 2 to 8 for case (a) Â¨C Ellipsoid point with 
uncertainty circle
	--	Degrees of Latitude3 octets
	--	Degrees of Longitude3 octets
	--	Uncertainty code1 octet
	-- Octets 2 to 11 for case (b) Â¨C Ellipsoid point with 
uncertainty ellipse:
	--	Degrees of Latitude3 octets
	--	Degrees of Longitude3 octets
	--	Uncertainty semi-major axis1 octet
	--	Uncertainty semi-minor axis1 octet
	--	Angle of major axis1 octet
	--	Confidence1 octet
	-- Octets 2 to 14 for case (c) Â¨C Ellipsoid point with altitude 
and uncertainty ellipsoid
	--	Degrees of Latitude3 octets
	--	Degrees of Longitude3 octets
	--	Altitude2 octets
	--	Uncertainty semi-major axis1 octet
	--	Uncertainty semi-minor axis1 octet
	--	Angle of major axis1 octet
	--	Uncertainty altitude1 octet
	--	Confidence1 octet
```

```
-- Octets 2 to 13 for case (d) Â¨C Ellipsoid Arc
--     Degrees of Latitude3 octets
--     Degrees of Longitude3 octets
--     Inner radius2 octets
--     Uncertainty radius1 octet
--     Offset angle1 octet
--     Included angle1 octet
--     Confidence1 octet
-- Octets 2 to 7 for case (e) Â¨C Ellipsoid Point
--     Degrees of Latitude3 octets
--     Degrees of Longitude3 octets

--
-- An Ext-GeographicalInformation parameter comprising more than one octet and
-- containing any other shape or an incorrect number of octets or coding according
-- to 3G TS 23.032 shall be treated as invalid data by a receiver.
--
-- An Ext-GeographicalInformation parameter comprising one octet shall be discarded
-- by the receiver if an Add-GeographicalInformation parameter is received
-- in the same message.
--
-- An Ext-GeographicalInformation parameter comprising one octet shall be treated as
-- invalid data by the receiver if an Add-GeographicalInformation parameter is not
-- received in the same message.

IMEI ::= TBCD-STRING (SIZE (8))
    -- Refers to International Mobile Station Equipment Identity
    -- and Software Version Number (SVN) defined in TS GSM 03.03.
    -- If the SVN is not present the last octet shall contain the
    -- digit 0 and a filler.
    -- If present the SVN shall be included in the last octet.


IMSI ::= TBCD-STRING (SIZE (3..8))
    -- digits of MCC, MNC, MSIN are concatenated in this order.

DiameterIdentity        ::= OCTET STRING
```

```
IPAddress::= CHOICE
{
	iPBinaryAddress IPBinaryAddress,
	iPTextRepresentedAddress IPTextRepresentedAddress
}

IPBinaryAddress::= CHOICE
{
	iPBinV4Address[0] OCTET STRING (SIZE(4)),
	iPBinV6Address[1] OCTET STRING (SIZE(16))
}

IPTextRepresentedAddress::= CHOICE
{	--
	-- IP address in the familiar "dot" notation
	--
	iPTextV4Address[2] IA5String (SIZE(7..15)),
	iPTextV6Address[3] IA5String (SIZE(15..45))
}

ISDN-AddressString ::=
		AddressString (SIZE (1..maxISDN-AddressLength))
	-- This type is used to represent ISDN numbers.

LCSCause		::= OCTET STRING (SIZE(1))
	--
	-- See LCS Cause Value, 3GPP TS 49.031
	--

LCSClientIdentity ::= SEQUENCE
{
	lcsClientExternalID[0] LCSClientExternalID OPTIONAL,
	lcsClientDialedByMS[1] AddressString OPTIONAL,
	lcsClientInternalID[2] LCSClientInternalID OPTIONAL
}

LCSQoSInfo		::= OCTET STRING (SIZE(4))
	--
	-- See LCS QoS IE, 3GPP TS 49.031
	--

LCSClientExternalID ::= SEQUENCE
```

```
{
       externalAddress[0] AddressStringOPTIONAL
--     extensionContainer[1] ExtensionContainerOPTIONAL
}

LCSClientInternalID ::= ENUMERATED
{
       broadcastService(0),
       o-andM-HPLMN(1),
       o-andM-VPLMN(2),
       anonymousLocation(3),
       targetMSsubscribedService(4)
}
       -- for a CAMEL phase 3 PLMN operator client, the value
targetMSsubscribedService shall be used


LCS-Priority ::= OCTET STRING (SIZE (1))
       -- 0 = highest priority
       -- 1 = normal priority
       -- all other values treated as 1


LocalSequenceNumber ::= INTEGER
       --
       -- Sequence number of the record in this node
       -- 0.. 4294967295 is equivalent to 0..2**32-1, unsigned integer
in four octets

LocationMethod ::= ENUMERATED
{
       msBasedEOTD(0),
       msAssistedEOTD(1),
       assistedGPS(2),
       msBasedOTDOA(3),
       msAssistedOTDOA(4)
}
       -- exception handling:
       -- When this parameter is received with value msBasedEOTD or
msAssistedEOTD and the MS
       -- is camped on an UMTS Service Area then the receiver shall
reject it
       -- with a return error cause of unexpected data value.
```

```
	-- When this parameter is received with value msBasedOTDOA or msAssistedOTDOA and the MS
	-- is camped on a GSM Cell then the receiver shall reject it with a return error cause of
	-- unexpected data value.
	-- an unrecognized value shall be rejected by the receiver with a return error cause of
	-- unexpected data value.

LocationAreaAndCell::= SEQUENCE
{
	locationAreaCode[0] LocationAreaCode,
	cellId    [1] CellId,
	mCC-MNC                 [2] MCC-MNC OPTIONAL
}

MSISDN ::= ISDN-AddressString
	--
	-- See TS GSM 03.03
	--


CauseInformation  ::=  SEQUENCE
{
	 msgTimeStamp             [1] TimeStamp OPTIONAL,
	 msgType                  [2] INTEGER OPTIONAL,
	 msgSourceIp              [3] IPAddress OPTIONAL,
	 msgCause                 [4] OCTET STRING OPTIONAL
}

CauseInfoList ::= SEQUENCE
{
	causelist           SEQUENCE OF CauseInformation
}

MvnoInformation  ::= SEQUENCE
{
	 mvnoResellerId           [0] OCTET STRING,
	 mvnoSubclassId           [1] OCTET STRING OPTIONAL
}

ManagementExtension ::= SEQUENCE
{
```

```
        identifierOBJECT IDENTIFIER,
        significance[1] BOOLEAN DEFAULT FALSE,
        information[2] ANY DEFINED BY identifier
}

maxAddressLength  INTEGER ::= 20

maxISDN-AddressLength  INTEGER ::= 9

maxExt-GeographicalInformation  INTEGER ::= 20
        -- the maximum length allows for further shapes in 3G TS 23.032 
to be included in later
        -- versions of 3G TS 29.002

NotificationToMSUser ::= ENUMERATED
{
        notifyLocationAllowed(0),
        notifyAndVerify-LocationAllowedIfNoResponse(1),
        notifyAndVerify-LocationNotAllowedIfNoResponse(2),
        locationNotAllowed (3)
}
        -- exception handling:
        -- At reception of any other value than the ones listed the 
receiver shall ignore
        -- NotificationToMSUser.

LCSClientType ::= ENUMERATED
{
        emergencyServices(0),
        valueAddedServices(1),
        plmnOperatorServices(2),
        lawfulInterceptServices(3)
}
        --  exception handling:
        --  unrecognized values may be ignored if the LCS client uses the 
privacy override
        --  otherwise, an unrecognized value shall be treated as 
unexpected data by a receiver
        --  a return error shall then be returned if received in a MAP 
invoke

LocationType ::= SEQUENCE
{
        locationEstimateType[0] LocationEstimateType,
```

```
	deferredLocationEventType[1] DeferredLocationEventTypeOPTIONAL
}

LocationEstimateType ::= ENUMERATED
{
	currentLocation(0),
	currentOrLastKnownLocation(1),
	initialLocation(2),
	activateDeferredLocation(3),
	cancelDeferredLocation(4)
}
	-- exception handling:
	-- a ProvideSubscriberLocation-Arg containing an unrecognized 
LocationEstimateType
	-- shall be rejected by the receiver with a return error cause 
of unexpected data value

--ExtensionContainer ::= SEQUENCE
--{
--	privateExtensionList[0] PrivateExtensionListOPTIONAL,
--	pcs-Extensions[1] PCS-ExtensionsOPTIONAL
--}

--PrivateExtensionList ::= SEQUENCE SIZE 
(1..maxNumOfPrivateExtensions) OF PrivateExtension

--PrivateExtension ::= SEQUENCE
--{
--	extId  MAP-EXTENSION.&extensionId({ExtensionSet}),
--	extTypeMAP-EXTENSION.&ExtensionType({ExtensionSet}{@extId}) 
OPTIONAL
--}

--maxNumOfPrivateExtensions  INTEGER ::= 10

--ExtensionSetMAP-EXTENSION ::=
--{
	 -- ExtensionSet is the set of all defined private extensions
--}
	-- Unsupported private extensions shall be discarded if 
received.

--PCS-Extensions ::= SEQUENCE
--{
```

```
--}

--MAP-EXTENSION  ::= CLASS
--{
--     &ExtensionTypeOPTIONAL,
--     &extensionId OBJECT IDENTIFIER
--}
       -- The length of the Object Identifier shall not exceed 16
octets and the
       -- number of components of the Object Identifier shall not
exceed 16

ServiceKey ::= INTEGER (0..2147483647)

SMSResult        ::= Diagnostics

TBCD-STRING ::= OCTET STRING
       -- This type (Telephony Binary Coded Decimal String) is used to
       -- represent several digits from 0 through 9, *, #, a, b, c, two
       -- digits per octet, each digit encoded 0000 to 1001 (0 to 9),
       -- 1010 (*), 1011 (#), 1100 (a), 1101 (b) or 1110 (c); 1111 used
       -- as filler when there is an odd number of digits.

       -- bits 8765 of octet n encoding digit 2n
       -- bits 4321 of octet n encoding digit 2(n-1) +1

PositionMethodFailure-Diagnostic ::= ENUMERATED {
       congestion  (0),
       insufficientResources  (1),
       insufficientMeasurementData  (2),
       inconsistentMeasurementData  (3),
       locationProcedureNotCompleted  (4),
       locationProcedureNotSupportedByTargetMS  (5),
       qoSNotAttainable  (6),
       positionMethodNotAvailableInNetwork(7),
       positionMethodNotAvailableInLocationArea(8)
}
--     exception handling:
--     any unrecognized value shall be ignored

UnauthorizedLCSClient-Diagnostic ::= ENUMERATED {
       noAdditionalInformation (0),
       clientNotInMSPrivacyExceptionList (1),
```

```
        callToClientNotSetup (2),
        privacyOverrideNotApplicable (3),
        disallowedByLocalRegulatoryRequirements (4)
}
--      exception handling:
--      any unrecognized value shall be ignored

PositioningData              ::= OCTET STRING (SIZE(1..33))
    --
    -- See Positioning Data IE (octet 3..n), 3GPP TS 49.031

MSTimeZone::= OCTET STRING (SIZE (2))
        --
        -- 1.Octet: Time Zone and 2. Octet: Daylight saving time, see TS 
29.060 [75]
        --

-- Trigger and cause values for IP flow level recording are defined for 
support of independent
-- online and offline charging and also for tight interworking between 
online and offline charging.
-- Unused bits will always be zero.
-- Some of the values are non-exclusive (e.g. PDP context modification 
reasons).

SmsTpDestinationNumber ::= OCTET STRING
        --
        -- This type contains the binary coded decimal representation of
        -- the SMS address field the encoding of the octet string is in
        -- accordance with the definition of address fields in TS 
23.040.
        -- This encoding includes type of number and numbering plan 
indication
        -- together with the address value range.
        --


RAIdentity ::= OCTET STRING (SIZE (6))
-- Routing Area Identity is coded in accordance with 3GPP TS 29.060 
[105].
-- It shall contain the value part defined in 3GPP TS 29.060 only. I.e. 
the 3GPP TS 29.060
-- type identifier octet shall not be included.
```

```
MBMSInformation::= SET
{
       tmgi       [1] TMGI,
       mbmsSessionIdentity [2] MBMSSessionIdentity OPTIONAL,
       mbmsServiceType [3] MBMSServiceType,
       mbmsUserServiceType [4] MBMSUserServiceType,
       mbms2G3GIndicator [5] MBMS2G3GIndicator OPTIONAL,
       fileRepairSupported [6] BOOLEAN OPTIONAL,
       rAI        [7] RoutingAreaCode OPTIONAL,
       mbmsServiceArea [8] MBMSServiceArea OPTIONAL,
       requiredMBMSBearerCaps [9] RequiredMBMSBearerCapabilities
OPTIONAL
}
MBMS2G3GIndicator::= ENUMERATED
{
       for-2G    (0),-- For GERAN access only
       for-3G    (1),-- For UTRAN access only
       for-2G-AND-3G(2)-- For both UTRAN and GERAN access
}

MBMSServiceType::= ENUMERATED
{
       multicast(0),
       broadcast(1)
}

MBMSUserServiceType::= ENUMERATED
{
       download(0),
       streaming(1)
}

RequiredMBMSBearerCapabilities::= OCTET STRING (SIZE (3..14))
       --
       -- This  octet string
       -- is a 1:1 copy of the contents (i.e. starting with octet 5) of
the "Quality of
       -- service Profile" information element specified in 3GPP TS
29.060.

MBMSSessionIdentity::= OCTET STRING (SIZE (1))
       --
       -- This octet string is a 1:1 copy of the contents of the
MBMS-Session-Identity
```

```
	-- AVP specified in 3GPP TS 29.061
TMGI	::= OCTET STRING
	--
	-- This  octet string
	-- is a 1:1 copy of the contents (i.e. starting with octet 4) of
the "TMGI"
	-- information element specified in 3GPP TS 29.060.
MBMSServiceArea::= OCTET STRING
	--
	-- Editor's Note: The structure of this octet string is subject
to discussions
	-- in other working groups.

ContentInfo  ::= SEQUENCE
{
	extensionType[0]INTEGER,
	length    [1]INTEGER,
	serviceList     [2] SEQUENCE --<UNBOUNDED>-- OF ServiceEvent,
	changeTimeList[3]     SEQUENCE OF ChangeTimeExtension OPTIONAL,
	recordOpeningTime[4]TimeStampExtension OPTIONAL,
	duration    [5]  CallDurationExtension OPTIONAL,
	transparentVSA    [6] OCTET STRING (SIZE(3..255)) OPTIONAL
}

ServiceEvent ::= SEQUENCE
{
    serviceCode   [0]  INTEGER OPTIONAL,
    uplinkVolume[1]  INTEGER OPTIONAL,
    downlinkVolume[2]  INTEGER OPTIONAL,
    usageduration[3]  INTEGER OPTIONAL,
    url          [5]  OCTET STRING (SIZE(1..120)) OPTIONAL,
    chargingRuleBaseName[8]  ChargingRuleBaseName OPTIONAL,
    ratingGroup [9] RatingGroupId OPTIONAL,
    serviceIdentifier[10] ServiceIdentifier OPTIONAL,
    localSequenceNumber[11] LocalSequenceNumber OPTIONAL,
    envelopeStartTime[12] TimeStampExtension OPTIONAL,
    envelopeEndTime    [13] TimeStampExtension OPTIONAL,
    duration  [14] CallDurationExtension OPTIONAL
}

CallDurationExtension::= OCTET STRING (SIZE(3..12))
	-- The contents of this field are 2^32 - 1. Variable with one
digit after the comma
```

```
        -- Value is in seconds with one digit after the comma

ChangeTimeExtension ::= SEQUENCE
{
        changeTime  [1]  TimeStampExtension OPTIONAL
}

TimeStampExtension::= OCTET STRING (SIZE(10))
        --
        -- The contents of this field are a compact form of the UTCTime
format
        -- containing local time plus an offset to universal time.
Binary coded
        -- decimal encoding is employed for the digits to reduce the
storage and
        -- transmission overhead
        -- e.g. yymmddhhmmsss Suuvv
        -- where
        -- YY = Year 00 to 99BCD encoded
        -- MM = Month 01 to 12 BCD encoded
        -- DD= Day 01 to 31BCD encoded
        -- hh= hour 00 to 23BCD encoded
        -- mm= minute 00 to 59BCD encoded
        -- ss= second 00 to 59BCD encoded
        -- s=  1/ 10 of a second: 0 to 9 BCD encoded
        -- S=  Sign 0 = "+", "-"ASCII encoded
        -- hh= hour 00 to 23BCD encoded
        -- mm= minute 00 to 59BCD encoded



RecordType ::= INTEGER
{--
--      Record values 0..17 and 87  are CS specific.
--      The contents are defined in TS 32.250 [10]

        moCallRecord(0),
        mtCallRecord(1),
        roamingRecord(2),
        incGatewayRecord(3),
        outGatewayRecord(4),
        transitCallRecord(5),
        moSMSRecord(6),
        mtSMSRecord(7),
```

```
        moSMSIWRecord(8),
        mtSMSGWRecord(9),
        ssActionRecord(10),
        hlrIntRecord(11),
        locUpdateHLRRecord(12),
        locUpdateVLRRecord(13),
        commonEquipRecord(14),
        moTraceRecord(15),--- used in earlier releases
        mtTraceRecord(16),--- used in earlier releases
        termCAMELRecord    (17),
--
--      Record values 18..22 are GPRS specific.
--      The contents are defined in TS 32.251 [11]
--
        sgsnPDPRecord(18),
        sgsnMMRecord(20),
        sgsnSMORecord(21),
        sgsnSMTRecord(22),
--
--   Record values 23..25 are CS-LCS specific.
--      The contents are defined in TS 32.250 [10]
--
        mtLCSRecord(23),
        moLCSRecord(24),
        niLCSRecord(25),
--
--   Record values 26..28 are GPRS-LCS specific.
--      The contents are defined in TS 32.251 [11]
--
        sgsnMTLCSRecord(26),
    sgsnMOLCSRecord(27),
    sgsnNILCSRecord(28),
--
--   Record values 30..62 are MMS specific.
--   The contents are defined in TS 32.270 [30]
--
        mMO1SRecord(30),
        mMO4FRqRecord(31),
        mMO4FRsRecord(32),
        mMO4DRecord(33),
        mMO1DRecord(34),
        mMO4RRecord(35),
        mMO1RRecord(36),
```

```
        mMOMDRecord(37),
        mMR4FRecord(38),
        mMR1NRqRecord(39),
        mMR1NRsRecord(40),
        mMR1RtRecord(41),
        mMR1AFRecord(42),
        mMR4DRqRecord(43),
        mMR4DRsRecord(44),
        mMR1RRRecord(45),
        mMR4RRqRecord(46),
        mMR4RRsRecord(47),
        mMRMDRecord(48),
        mMFRecord (49),
        mMBx1SRecord(50),
        mMBx1VRecord(51),
        mMBx1URecord(52),
        mMBx1DRecord(53),
        mM7SRecord(54),
        mM7DRqRecord(55),
        mM7DRsRecord(56),
        mM7CRecord(57),
        mM7RRecord(58),
        mM7DRRqRecord(59),
        mM7DRRsRecord(60),
        mM7RRqRecord(61),
        mM7RRsRecord(62),
--
--  Record values 63..69, 70, 82 are IMS specific.
--  The contents are defined in TS 32.260 [20]
--
        sCSCFRecord(63),
        pCSCFRecord(64),
        iCSCFRecord(65),
        mRFCRecord(66),
        mGCFRecord(67),
        bGCFRecord(68),
        aSRecord  (69),
        eCSCFRecord(70),
        iBCFRecord(82),
--
--  Record values 71..75 are LCS specific.
--  The contents are defined in TS 32.271 [31]
--
```

```
        lCSGMORecord(71),
        lCSRGMTRecord(72),
        lCSHGMTRecord(73),
        lCSVGMTRecord(74),
        lCSGNIRecord(75),
--
--  Record values 76..79,86 are MBMS specific.
--  The contents are defined in TS 32.251 [11]
--  Record values 76,77 and 86 are MBMS bearer context specific
--
        sgsnMBMSRecord(76),
        ggsnMBMSRecord(77),
        gwMBMSRecord(86),

--
--  And TS 32.273 [33]
--  Record values 78 and 79 are MBMS service specific
--  and defined in TS 32.273 [33]
--
        sUBBMSCRecord(78),
        cONTENTBMSCRecord(79),
--
--  Record Values 80..81 are PoC specific.
--  The contents are defined in TS 32.272 [32]
--
        pPFRecord (80),
        cPFRecord (81),

--  Record values 84..85 are EPC specific.
--  The contents are defined in TS 32.251 [11]
--
        sGWRecord (84),
        pGWRecord (85),
--
--  Record Value 83 is MMTel specific.
--  The contents are defined in TS 32.275 [35]
--
        mMTelRecord(83),

--  Record value  87 is CS specific.
--  The contents are defined in TS 32.250 [10]
--
```

```
        mSCsRVCCRecord(87)


}

SubscriptionID::= SET
{
        subscriptionIDType[0]SubscriptionIDType,
        subscriptionIDData[1]UTF8String
}


SubscriptionIDType::= ENUMERATED
{
        eND-USER-E164(0),
        eND-USER-IMSI(1),
        eND-USER-SIP-URI(2),
        eND-USER-NAI(3),
        eND-USER-PRIVATE(4)
}

----------------------------------------------------------------------
--------
--
--  PS DATA TYPES
--
----------------------------------------------------------------------
--------

AccessPointNameNI::= IA5String (SIZE(1..63))
--
-- Network Identifier part of APN in  dot representation.
-- For example, if the complete APN is
'apn1a.apn1b.apn1c.mnc022.mcc111.gprs'
-- NI is 'apn1a.apn1b.apn1c' and is presented in this form in the CDR..
--

AccessPointNameOI::= IA5String (SIZE(1..37))
--
-- Operator Identifier part of APN in dot representation.
-- In the 'apn1a.apn1b.apn1c.mnc022.mcc111.gprs' example, the OI
portion is 'mnc022.mcc111.gprs'
-- and is presented in this form in the CDR.
--
```

```
AdditionalExceptionReports      ::= ENUMERATED
{
    notAllowed      (0),
    allowed         (1)
}

AFChargingIdentifier ::= OCTET STRING
--
-- see AF-Charging-Identifier AVP as defined in TS 29.214
--

AFRecordInformation::= SEQUENCE
{
       aFChargingIdentifier[1] AFChargingIdentifier,
       flows     [2] Flows OPTIONAL
}

APNRateControl      ::= SEQUENCE
--
-- See TS 24.008 [208] for more information
--
{
    aPNRateControlUplink    [0] APNRateControlParameters OPTIONAL,
    aPNRateControlDownlink  [1] APNRateControlParameters OPTIONAL
}

APNRateControlParameters        ::= SEQUENCE
{
    additionalExceptionReports  [0] AdditionalExceptionReports
OPTIONAL,
    rateControlTimeUnit         [1] RateControlTimeUnit OPTIONAL,
    rateControlMaxRate          [2] INTEGER OPTIONAL,
    rateControlMaxMessageSize   [3] DataVolumeGPRS OPTIONAL --
aPNRateControlDownlink only
}

APNSelectionMode::= ENUMERATED
--
-- See Information Elements TS 29.060 [215], TS 29.274 [223] or TS
29.275 [224]
--
{
```

```
        mSorNetworkProvidedSubscriptionVerified(0),
        mSProvidedSubscriptionNotVerified(1),
        networkProvidedSubscriptionNotVerified(2)
}

CAMELAccessPointNameNI::= AccessPointNameNI

CAMELAccessPointNameOI::= AccessPointNameOI

CAMELInformationMM::= SET
{
        sCFAddress[1] SCFAddress OPTIONAL,
        serviceKey[2]ServiceKey OPTIONAL,
        defaultTransactionHandling[3]DefaultGPRS-Handling OPTIONAL,
        numberOfDPEncountered  [4] NumberOfDPEncountered OPTIONAL,
        levelOfCAMELService[5] LevelOfCAMELService OPTIONAL,
        freeFormatData[6] FreeFormatData OPTIONAL,
        fFDAppendIndicator[7]FFDAppendIndicator OPTIONAL
}

CAMELInformationPDP::= SET
{
        sCFAddress[1] SCFAddress OPTIONAL,
        serviceKey[2]ServiceKey OPTIONAL,
        defaultTransactionHandling[3]DefaultGPRS-Handling OPTIONAL,
        cAMELAccessPointNameNI[4] CAMELAccessPointNameNI OPTIONAL,
        cAMELAccessPointNameOI[5] CAMELAccessPointNameOI OPTIONAL,
        numberOfDPEncountered[6] NumberOfDPEncountered OPTIONAL,
        levelOfCAMELService[7] LevelOfCAMELService OPTIONAL,
        freeFormatData[8]FreeFormatData OPTIONAL,
        fFDAppendIndicator[9]FFDAppendIndicator OPTIONAL
}

CAMELInformationSMS::= SET
{
        sCFAddress[1] SCFAddress OPTIONAL,
        serviceKey[2]ServiceKey OPTIONAL,
        defaultSMSHandling[3]DefaultSMS-Handling OPTIONAL,
        cAMELCallingPartyNumber[4] CallingNumber OPTIONAL,
        cAMELDestinationSubscriberNumber[5] SmsTpDestinationNumber
OPTIONAL,
        cAMELSMSCAddress[6] AddressString OPTIONAL,
        freeFormatData[7]FreeFormatData OPTIONAL,
```

```
	smsReferenceNumber[8] CallReferenceNumberOPTIONAL
}

CauseForRecClosing::= INTEGER
--
-- In PGW-CDR and SGW-CDR the value servingNodeChange is used for
partial record
-- generation due to Serving Node Address list Overflow
-- In SGSN servingNodeChange indicates the SGSN change
--
-- LCS related causes belong to the MAP error causes acc. TS 29.002
[214]
--
-- cause codes 0 to 15 are defined 'CauseForTerm' (cause for
termination)
--
{
	normalRelease(0),
	abnormalRelease(4),
	cAMELInitCallRelease(5),
	volumeLimit(16),
	timeLimit (17),
	servingNodeChange(18),
	maxChangeCond(19),
	managementIntervention(20),
	intraSGSNIntersystemChange(21),
	rATChange (22),
	mSTimeZoneChange(23),
	sGSNPLMNIDChange (24),
	sGWChange       (25),
	unauthorizedRequestingNetwork(52),
	unauthorizedLCSClient(53),
	positionMethodFailure(54),
	unknownOrUnreachableLCSClient(58),
	listofDownstreamNodeChange(59)
}

ChangeCondition::= ENUMERATED
{
	qoSChange (0),
	tariffTime(1),
	recordClosure(2),
	cGI-SAICHange(6), -- bearer modification. â€œCGI-SAI Changeâ€?
```

```
        rAIChange (7), -- bearer modification. â€œRAI Changeâ€?
        dT-Establishment(8),
        dT-Removal(9),
        eCGIChange(10), -- bearer modification. â€œECGI Changeâ€?
        tAIChange (11), -- bearer modification. â€œTAI Changeâ€?
        userLocationChange(12),-- bearer modification. â€œUser Location
Changeâ€?
    userPlaneToUEChange                     (18),   -- bearer
modification. "Change of user plane to UE"
    servingPLMNRateControlChange            (19)    -- bearer
modification â€œServing PLMN Rate Control Change"
}

ChangeOfCharCondition::= SEQUENCE
--
-- qosRequested and qosNegotiated are used in S-CDR only
-- ePCQoSInformation used in SGW-CDR only
-- cPCIoTOptimisationIndicator is used in SGW-CDR only
--
{
        qosRequested[1] QoSInformation OPTIONAL,
        qosNegotiated[2] QoSInformation OPTIONAL,
        dataVolumeGPRSUplink[3] DataVolumeGPRS OPTIONAL,
        dataVolumeGPRSDownlink[4] DataVolumeGPRS OPTIONAL,
        changeCondition[5] ChangeCondition,
        changeTime[6] TimeStamp,
        userLocationInformation[8] OCTET STRING OPTIONAL,
        ePCQoSInformation[9] EPCQoSInformation OPTIONAL,
    rATType                         [15] RATType OPTIONAL,
    uWANUserLocationInformation     [17] UWANUserLocationInfo OPTIONAL,
    cPCIoTEPSOptimisationIndicator  [19]
CPCIoTEPSOptimisationIndicator OPTIONAL,
    servingPLMNRateControl          [20] ServingPLMNRateControl
OPTIONAL
}

ChangeOfMBMSCondition::= SEQUENCE
--
-- Used in MBMS record
--
{
        qosRequested[1] QoSInformation OPTIONAL,
        qosNegotiated[2] QoSInformation OPTIONAL,
        dataVolumeMBMSUplink[3] DataVolumeMBMS OPTIONAL,
```

```
        dataVolumeMBMSDownlink[4] DataVolumeMBMS,
        changeCondition[5] ChangeCondition,
        changeTime[6] TimeStamp,
        failureHandlingContinue[7] FailureHandlingContinue OPTIONAL
}

ChangeOfServiceConditionRel84::= SEQUENCE
--
-- Used for Flow based Charging service data container
--
{
        ratingGroup [1] RatingGroupId,
        chargingRuleBaseName        [2] ChargingRuleBaseName OPTIONAL,
        resultCode[3] ResultCode OPTIONAL,
        localSequenceNumber[4] LocalSequenceNumber OPTIONAL,
        timeOfFirstUsage[5] TimeStamp OPTIONAL,
        timeOfLastUsage[6] TimeStamp OPTIONAL,
        timeUsage [7] CallDuration OPTIONAL,
        serviceConditionChange        [8] ServiceConditionChange,
        qoSInformationNeg[9] OCTET STRING OPTIONAL,
        servingNodeAddress [10] GSNAddress OPTIONAL,
        datavolumeFBCUplink[12] DataVolumeGPRS OPTIONAL,
        datavolumeFBCDownlink        [13] DataVolumeGPRS OPTIONAL,
        timeOfReport[14] TimeStamp,
        failureHandlingContinue        [16] FailureHandlingContinue
OPTIONAL,
        serviceIdentifier[17] ServiceIdentifier OPTIONAL,
        pSFurnishChargingInformation                [18]
PSFurnishChargingInformation OPTIONAL,
        aFRecordInformation[19] SEQUENCE OF AFRecordInformation
OPTIONAL,
        userLocationInformation        [20] OCTET STRING OPTIONAL,
        eventBasedChargingInformation                [21]
EventBasedChargingInformation OPTIONAL,
        timeQuotaMechanism[22] TimeQuotaMechanism OPTIONAL,
        serviceSpecificInfo[23] SEQUENCE OF ServiceSpecificInfo
OPTIONAL,
        threeGPP2UserLocationInformation        [24] OCTET STRING
OPTIONAL,
         sponsorIdentity                              [25] OCTET STRING
OPTIONAL,
          applicationServiceProviderIdentity              [26] OCTET
STRING OPTIONAL,
    rATType                             [30] RATType OPTIONAL,
```

```
    uWANUserLocationInformation        [32] UWANUserLocationInfo
OPTIONAL,
    servingPLMNRateControl              [35] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                      [36] APNRateControl OPTIONAL

}
ChangeOfServiceCondition::= SEQUENCE
--
-- Used for Flow based Charging service data container
--
{
       ratingGroup [1] RatingGroupId,
       chargingRuleBaseName        [2] ChargingRuleBaseName OPTIONAL,
       resultCode[3] ResultCode OPTIONAL,
       localSequenceNumber[4] LocalSequenceNumber OPTIONAL,
       timeOfFirstUsage[5] TimeStamp OPTIONAL,
       timeOfLastUsage[6] TimeStamp OPTIONAL,
       timeUsage [7] CallDuration OPTIONAL,
       serviceConditionChange        [8] ServiceConditionChange,
       qoSInformationNeg[9] EPCQoSInformation OPTIONAL,
       servingNodeAddress [10] GSNAddress OPTIONAL,
       datavolumeFBCUplink[12] DataVolumeGPRS OPTIONAL,
       datavolumeFBCDownlink        [13] DataVolumeGPRS OPTIONAL,
       timeOfReport[14] TimeStamp,
       failureHandlingContinue        [16] FailureHandlingContinue
OPTIONAL,
       serviceIdentifier[17] ServiceIdentifier OPTIONAL,
       pSFurnishChargingInformation                [18]
PSFurnishChargingInformation OPTIONAL,
       aFRecordInformation[19] SEQUENCE OF AFRecordInformation
OPTIONAL,
       userLocationInformation        [20] OCTET STRING OPTIONAL,
       eventBasedChargingInformation                [21]
EventBasedChargingInformation OPTIONAL,
       timeQuotaMechanism[22] TimeQuotaMechanism OPTIONAL,
       serviceSpecificInfo[23] SEQUENCE OF ServiceSpecificInfo
OPTIONAL,
       threeGPP2UserLocationInformation        [24] OCTET STRING
OPTIONAL,
        sponsorIdentity                               [25] OCTET STRING
OPTIONAL,
    applicationServiceProviderIdentity              [26] OCTET STRING
OPTIONAL,
    rATType                             [30] RATType OPTIONAL,
```

```
    uWANUserLocationInformation        [32] UWANUserLocationInfo
OPTIONAL,
    servingPLMNRateControl              [35] ServingPLMNRateControl
OPTIONAL,
    aPNRateControl                      [36] APNRateControl OPTIONAL

}

ChangeLocation::= SEQUENCE
--
-- used in SGSNMMRecord only
--
{
       locationAreaCode[0] LocationAreaCode,
       routingAreaCode[1] RoutingAreaCode,
       cellId    [2] CellId OPTIONAL,
       changeTime[3] TimeStamp,
       mCC-MNC   [4] PLMN-Id OPTIONAL
}

ChargingCharacteristics::= OCTET STRING (SIZE(2))

ChargingID::= INTEGER
--
-- Generated in P-GW, part of IP CAN bearer
-- 0..4294967295 is equivalent to 0..2**32-1
--

CivicAddressInformation     ::= OCTET STRING
--
-- as defined in subclause 3.1 of IETF RFC 4776 [409] excluding the
first 3 octets.
--

ChargingRuleBaseName ::= IA5String (SIZE(1..16))
--
-- identifier for the group of charging rules
-- see Charging-Rule-Base-Name AVP as desined in TS 29.212 [220]
--

ChChSelectionMode::= ENUMERATED
{
       servingNodeSupplied(0),-- For S-GW/P-GW
```

```
        subscriptionSpecific(1),-- For SGSN only
        aPNSpecific(2),-- For SGSN only
        homeDefault(3),-- For SGSN, S-GW and P-GW
        roamingDefault(4),-- For SGSN, S-GW and P-GW
        visitingDefault(5)-- For SGSN, S-GW and P-GW
}

CPCIoTEPSOptimisationIndicator ::= BOOLEAN

CSGAccessMode ::= ENUMERATED
{
        closedMode  (0),
        hybridMode  (1)
}

CSGId  ::= OCTET STRING (SIZE(4))
        --
        -- Defined in 23.003[200]. Coded according to TS 29.060[215] for
GTP, and in TS 29.274 [223] for
        -- 24.008 [208]
        --

CTEID  ::=OCTET STRING (SIZE(4))

  --
  -- Defined in 32.251[11] for MBMS-GW-CDR. Common Tunnel Endpoint
Identifier of MBMS GW for user --
  -- plane, defined in TS23.246 [207].
  --

DataVolumeGPRS::= INTEGER
--
-- The volume of data transferred in octets.
--

SecondaryRATType        ::= INTEGER
{
        reserved                (0),
        nR                      (1)             -- New Radio 5G
}


DynamicAddressFlag::= BOOLEAN
```

```
EPCQoSInformation::= SEQUENCE
--
-- See TS 29.212 [220] for more information
--
{
       qCI       [1] INTEGER OPTIONAL,
       maxRequestedBandwithUL[2] INTEGER OPTIONAL,
       maxRequestedBandwithDL[3] INTEGER OPTIONAL,
       guaranteedBitrateUL[4] INTEGER OPTIONAL,
       guaranteedBitrateDL[5] INTEGER OPTIONAL,
       aRP       [6] INTEGER OPTIONAL,
       aPNAggregateMaxBitrateUL [7] INTEGER OPTIONAL,
       aPNAggregateMaxBitrateDL     [8] INTEGER OPTIONAL,
        extendedMaxRequestedBandwithUL      [9] INTEGER OPTIONAL,
        extendedMaxRequestedBandwithDL      [10] INTEGER OPTIONAL,
        extendedGBRUL                       [11] INTEGER OPTIONAL,
        extendedGBRDL                       [12] INTEGER OPTIONAL,
        extendedAPNAMBRUL                   [13] INTEGER OPTIONAL,
        extendedAPNAMBRDL                   [14] INTEGER OPTIONAL

}

EventBasedChargingInformation  ::= SEQUENCE
{
       numberOfEvents[1] INTEGER,
       eventTimeStamps  [2] SEQUENCE OF TimeStamp OPTIONAL
}

FailureHandlingContinue  ::= BOOLEAN
--
-- This parameter is included when the failure handling procedure has
been executed and new
-- containers are opened. This parameter shall be included in the first
and subsequent
-- containers opened after the failure handling execution.
--

FFDAppendIndicator::= BOOLEAN

Flows  ::= SEQUENCE
--
-- See Flows AVP as defined in TS 29.214 [221]
```

```
--
{
	mediaComponentNumber[1] INTEGER,
	flowNumber  [2] SEQUENCE OF INTEGER OPTIONAL
}

FreeFormatData::=OCTET STRING (SIZE(1..160))
--
-- Free formatted data as sent in the FurnishChargingInformationGPRS
-- see TS 29.078 [217]
--

GSNAddress::= IPAddress

MSNetworkCapability::= OCTET STRING (SIZE(1..8))
--
-- see TS 24.008 [208]
--

NetworkInitiatedPDPContext::= BOOLEAN
--
-- Set to true if PDP context was initiated from network side
--

NodeID ::= IA5String (SIZE(1..20))

NumberOfDPEncountered  ::= INTEGER

PDPAddress::= CHOICE
{
	iPAddress [0] IPAddress
--	eTSIAddress   [1] ETSIAddress : has only been used in earlier
releases for X.121 format
}

PDPType   ::= OCTET STRING (SIZE(2))
--
-- OCTET 1: PDP Type Organization
-- OCTET 2: PDP/PDN Type Number
-- See TS 29.060 [215] for encoding details.
--

PDPPDNTypeExtension ::= INTEGER
```

```
--
-- This integer is 1:1 copy of the PDP type value as defined in TS
29.061 [215].
--

PLMN-Id   ::= OCTET STRING (SIZE (3))
--
--     This is a 1:1 copy from the Routing Area Identity (RAI) IE
specified in TS 29.060 [215]
--  as follows:
--     OCTET 1 of PLMN-Id = OCTET 2 of RAI
--     OCTET 2 of PLMN-Id = OCTET 3 of RAI
--     OCTET 3 of PLMN-Id = OCTET 4 of RAI
--



PSFurnishChargingInformation  ::= SEQUENCE
{
       pSFreeFormatData[1] FreeFormatData,
       pSFFDAppendIndicator  [2] FFDAppendIndicator OPTIONAL
}

RateControlTimeUnit ::= INTEGER
{   unrestricted    (0),
    minute          (1),
    hour            (2),
    day             (3),
    week            (4)
}

QoSInformation::= OCTET STRING (SIZE (4..255))
--
-- This  octet string
-- is a 1:1 copy of the contents (i.e. starting with octet 5) of the
"Bearer Quality of
-- Service" information element specified in TS 29.274 [223].
--

RANSecondaryRATUsageReport  ::= SEQUENCE
{
        dataVolumeUplink                                [1]
DataVolumeGPRS,
        dataVolumeDownlink                              [2]
DataVolumeGPRS,
```

```
        rANStartTime                                        [3] TimeStamp,
        rANEndTime                                          [4] TimeStamp,
        secondaryRATType                                    [5]
SecondaryRATType OPTIONAL
}

RatingGroupId ::= INTEGER
--
-- IP service flow identity (DCCA), range of 4 byte (0...4294967295)
-- see Rating-Group AVP as used in TS 32.299 [50]
--

RATType::= INTEGER (0..255)
       --
       -- This integer is 1:1 copy of the RAT type value as defined in
TS 29.061 [215].
       --

ResultCode ::= INTEGER
--
-- charging protocol return value, range of 4 byte (0...4294967295)
-- see Result-Code AVP as used in 32.299 [40]
--

RoutingAreaCode::= OCTET STRING (SIZE(1))
--
-- See TS 24.008 [208]
--

ServiceConditionChange::= BIT STRING
{
       qoSChange  (0),-- bearer modification
       sGSNChange  (1), -- bearer modification
       sGSNPLMNIDChange  (2), -- bearer modification
       tariffTimeSwitch  (3), -- tariff time change
       pDPContextRelease  (4), -- bearer release
       rATChange  (5),-- bearer modification
       serviceIdledOut  (6),-- IP flow idle out, DCCA QHT expiry
       reserved   (7),-- old: QCTexpiry is no report event
       configurationChange  (8),-- configuration change
       serviceStop  (9), -- IP flow termination.From â€Service Stopâ€?
in
       dCCATimeThresholdReached (10),-- DCCA quota reauthorization
```

```
        dCCAVolumeThresholdReached (11), -- DCCA quota reauthorization
        dCCAServiceSpecificUnitThresholdReached(12), -- DCCA quota
reauthorization
        dCCATimeExhausted (13), -- DCCA quota reauthorization
        dCCAVolumeExhausted (14), -- DCCA quota reauthorization
        dCCAValidityTimeout (15), -- DCCA quota validity time (QVT
expiry)
        reserved1 (16),-- reserved due to no use case,
        dCCAReauthorisationRequest (17), -- DCCA quota reauthorization
request by OCS
        dCCAContinueOngoingSession (18), -- DCCA failure handling
(CCFH),
        dCCARetryAndTerminateOngoingSession(19), -- DCCA failure
handling (CCFH),
        dCCATerminateOngoingSession (20), -- DCCA failure handling,
        cGI-SAIChange(21), -- bearer modification. â€œCGI-SAI Changeâ€?
        rAIChange (22), -- bearer modification. â€œRAI Changeâ€?
        dCCAServiceSpecificUnitExhausted(23), -- DCCA quota
reauthorization
        recordClosure(24),-- PGW-CDR closure
        timeLimit (25),-- intermediate recording. From â€œService Data
Time Limitâ€?Change-Condition AVP value
        volumeLimit(26),-- intermediate recording.From â€œService Data
Volume Limitâ€?Change-Condition AVP value
        serviceSpecificUnitLimit(27),-- intermediate recording
        envelopeClosure (28),
        eCGIChange(29), -- bearer modification. â€œECGI Changeâ€?
        tAIChange (30), -- bearer modification. â€œTAI Changeâ€?
        userLocationChange(31),-- bearer modification. â€œUser Location
Changeâ€?
    servingPLMNRateControlChange            (36),    -- bearer
modification. "Serving PLMNRate
                                                     -- Control Change"
    aPNRateControlChange                    (37)     -- bearer
modification. "APN Rate ControlChange


}
--
-- Trigger and cause values for IP flow level recording are defined for
support of independent
-- online and offline charging and also for tight interworking between
online and offline charging.
-- Unused bits will always be zero.
-- Some of the values are non-exclusive (e.g. bearer modification
reasons).
        --
```

```
SCFAddress::= AddressString
--
-- See TS 29.002 [214]
--

ServiceIdentifier::= INTEGER
--
-- The service identifier is used to identify the service or the
service component
-- the service data flow relates to. See Service-Identifier AVP as
defined in TS 29.212 [220]
--

ServingNodeType::= ENUMERATED
{
       sGSN   (0),
       pMIPSGW(1),
gTPSGW    (2),
       ePDG   (3),
       hSGW   (4),
          mME    (5)
}

ServingPLMNRateControl      ::= SEQUENCE
--
-- See TS 29.128 [244] for more information
--
{
    sPLMNDLRateControlValue [0] INTEGER,
    sPLMNULRateControlValue [1] INTEGER
}

SGiPtPTunnellingMethod      ::= ENUMERATED
{
    uDPIPbased      (0),
    others          (1)
}

SGSNChange::= BOOLEAN
--
-- present if first record after inter SGSN routing area update in new
SGSN
```

```
--

SGWChange::= BOOLEAN
--
-- present if first record after inter S-GW change
--

TimeQuotaMechanism::= SEQUENCE
{
       timeQuotaType[1] TimeQuotaType,
       baseTimeInterval[2] INTEGER
}

TimeQuotaType::= ENUMERATED
{
       discretetimeperiod(0),
       continuoustimeperiod(1)
}

UNIPDUCPOnlyFlag    ::= BOOLEAN

UserCSGInformation::= SEQUENCE
{
       cSGId     [0] CSGId,
       cSGAccessMode[1] CSGAccessMode,
       cSGMembershipIndication [2] NULL OPTIONAL
}

UWANUserLocationInfo  ::= SEQUENCE
{
       uELocalIPAddress [0] IPAddress,
       uDPSourcePort [1] OCTET STRING (SIZE(2)) OPTIONAL,
       sSID                  [2] OCTET STRING OPTIONAL, -- see format
in IEEE Std 802.11-2012 [408]
       bSSID                 [3] OCTET STRING OPTIONAL, -- see format
in IEEE Std 802.11-2012 [408]
        tCPSourcePort           [4] OCTET STRING (SIZE(2)) OPTIONAL,
        civicAddressInformation [5] CivicAddressInformation OPTIONAL,
        wLANOperatorId          [6] WLANOperatorId OPTIONAL,
        logicalAccessID         [7] OCTET STRING OPTIONAL

}
```

```
WLANOperatorId ::= SEQUENCE
{
        wLANOperatorName    [0] OCTET STRING,
        wLANPLMNId          [1] PLMN-Id
}

SCSASAddress        ::= SET
--
--
--
{
    sCSAddress  [1] IPAddress,
    sCSRealm    [2] DiameterIdentity
}

ServiceSpecificInfo  ::=  SEQUENCE
{
       serviceSpecificData[0] GraphicString OPTIONAL,
       serviceSpecificType[1] INTEGER OPTIONAL
}
TWANUserLocationInfo  ::= SEQUENCE
{
       sSID   [0] OCTET STRING,-- see format in IEEE Std 802.11-2012
[408]
       bSSID  [1] OCTET STRING OPTIONAL-- see format in IEEE Std
802.11-2012 [408]
}

END
```